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Life Cycle Inventories of Petro-chemical Solvents

Data v2.0 (2007)

ETH

Jürgen Sutter
ETH S + U, Zurich

ecoinvent report No. 22

Uster, December 2007

Project "ecoinvent data v2.0"

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Participating consultants:	Basler & Hofmann, Zürich Bau- und Umweltchemie, Zürich Carbotech AG, Basel Chudacoff Oekoscience, Zürich Doka Life Cycle Assessments, Zürich Dr. Werner Environment & Development, Zürich Ecointsys - Life Cycle Systems Sarl. ENERS Energy Concept, Lausanne ESU-services Ltd., Uster Infras AG, Bern
Software Support:	ifu Hamburg GmbH
Project leader:	Rolf Frischknecht, ecoinvent Centre, Empa, Dübendorf
Marketing and Sales:	Annette Köhler, ecoinvent Centre, Empa, Dübendorf

Life Cycle Inventories of Petrochemical Solvents

Author:	Jürgen Sutter
Reviewer:	Roland Hischier, EMPA St. Gallen; Mirelle Faist Emmenegger, ESU Services, Uster
Contact address:	ecoinvent Centre Empa P.O. Box CH-8600 Dübendorf http://www.ecoinvent.org/ frischknecht@ecoinvent.org
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Citation:

Sutter, J. (2007) Life Cycle Inventories of Petrochemical Solvents. ecoinvent report No. 22. Swiss Centre for Life Cycle Inventories, Dübendorf, 2007

Zusammenfassung

Für die Anwendung von Lebenszyklusanalysen (Ökobilanz, LCA) in der chemischen Industrie werden Ökoinventare zur Herstellung der verwendeten Chemikalien benötigt. Zu den wichtigsten Basischemikalien in der chemischen Industrie gehören Lösungsmittel, da diese in grossen Mengen gebraucht werden. Inventare der verwendeten Lösungsmittel stehen aber noch nicht in ausreichendem Masse zur Verfügung, wodurch die Anwendung von Ökobilanzen in der chemischen Industrie erschwert wird.

Im Projekt „Ökoinventare von petrochemischen Lösungsmitteln“ wurden Ökoinventare für die Herstellung der wichtigsten in der Schweizerischen Pharma- und Spezialitätenchemie verwendeten Lösungsmittel erstellt werden, gemäss den Qualitätsrichtlinien von ecoinvent.

Die Auswahl der wichtigsten Lösungsmittel wurde nach folgenden Kriterien gemacht:

- Die Lösungsmittel müssen mengenmässig bedeutend sein für die Schweizerische Pharma- und Spezialitätenchemie
- Es müssen unterschiedliche Substanzklassen vertreten sein (Aliphate, Alkohole, Aldehyde, Ketone, Ether, Ester, Amine, chlorierte Kohlenwasserstoffe und höherwertige Heterozyklen)
- Es sollen Lösungsmittel von stark unterschiedlichem ökonomischen Wert vertreten sein.
- Lösungsmittel, die als Hilfsstoffe speziell bei Destillationen eingesetzt werden, sollen vertreten sein.

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Part I

Methodological Remarks

Author: Jürgen Sutter, ETH S+U, Zurich



Introduction

Background

Organic solvents are used in large amounts by a wide variety of industrial sectors. In Europe, the annual consumption of fresh solvents is estimated to be 4 million tonnes in 2004 (European Solvents Industry Group 2007). The largest demand for solvents comes from the paint and coatings industry which relies on almost 2 million tonnes every year. The next largest use is in chemical industry (pharmaceuticals, agricultural chemicals, specialty chemicals) followed by the use of solvents in metal cleaning and degreasing. Other important end markets include use in rubber and polymer manufacture, detergents, and personal care products (European Solvents Industry Group 2007, Whim et al. 1996).

Most of the organic solvents are manufactured in the petrochemical industry. Thus, they are mainly derived from either crude oil or natural gas. Aliphatic hydrocarbons are produced by refining and distilling various cuts and fractions and aromatics either by cracking or reforming naphtha streams. The production of oxygenated solvents requires methane, ethylene, propylene, butylenes, air and water as main building blocks to most of the important ones (Whim et al. 1996, Wypych 2001).

The large amounts of organic solvents used annually combined with the facts that their production causes the depletion of non-renewable resources and the solvents itself show hazardous properties (e.g. toxicity, high volatility) makes the manufacturing of solvents to an environmental issue. To quantify potential environmental impacts arising from the petrochemical production of solvents environmental assessment tools such as LCA should be applied. In spite of this importance, only few LCI data of solvents have been published in databases. Additionally, the existing inventories of solvents have not been analyzed so far in the context of full production routes in order to identify key production steps that contribute substantially to environmental impacts.

Goal and Scope

The aim of this project is to present a comprehensive environmental assessment of the main petrochemical solvent production routes using the LCA method. The assessment also comprises the analysis of the environmental impact of the single unit operations applied in these production routes.

Solvents under study

Plentiful solvents are used in the various industries. For example, the number of solvents generally available to chemists is between 250 and 300 (Reichardt 2003). And of course, there is an infinite number of solvent mixtures. Since it is impossible to include all solvents within a study 50 important solvents have been selected for our work according the following criteria:

The selection covers

- (1) solvents of various chemical groups (alcohols, aldehydes, aliphatic hydrocarbons, amides, aromatic hydrocarbons, esters, ethers, halogenated hydrocarbons, and ketones).
 - (2) quantitatively important representatives of all groups according to technical literature (Flick 1998, Marcus 1998, Stoye 2000, Whim et al. 1996, Wypych 2001).
 - (3) solvents used particularly in the chemical industry where a large variety of solvents are employed. This selection is based on results of a comprehensive survey on the Swiss chemical industry (Seyler et al. 2006).
-

The 50 organic solvents under study are presented in Tab. 0.1. The selection covers the quantitatively most important solvent groups. With regard to 11 solvents, inventory data have already been published in the ecoinvent database. For the other 39 solvents new inventories have been established or existing inventories have been improved with better data.

Tab. 0.1 The 50 most important solvents.

No.	Name	Synonym(s), Abbreviation	ecoinvent 1.2	CAS-no.
Alcohols				
1	amyl-alcohol, tert.-	2-methyl-2-butanol		75-85-4
2	benzyl alcohol			100-51-6
3	butanol, 1-	n-butyl alcohol	x	71-36-3
4	butanol, 2-	s-butyl alcohol		78-92-2
5	butanol, iso-	2-methyl-1-propanol		78-83-1
6	butylene glycol, 1,4-	1,4-butanediol		1100-63-4
7	ethanol	ethyl alcohol		64-17-5
8	methanol	methyl alcohol	x	67-56-1
9	pentanol	n-pentyl alcohol		71-41-0
10	propanol, 1-	n-propyl alcohol		71-23-8
11	propanol, iso-	propanol, 2-, isopropyl alcohol	x	67-63-0
Aldehydes				
12	benzaldehyde			100-52-7
13	formaldehyde		x	50-00-0
14	propionaldehyde	propanal		123-38-6
Aliphatic hydrocarbons				
15	cyclohexane			110-82-7
16	heptane, n-			142-85-5
17	hexane, iso-			96-14-0
18	hexane, n-			110-54-3
19	methyl cyclohexane			108-87-2
20	pentane, n-		x	109-66-0
Aliphatic carbon acids				
21	acetic acid	ethanoic acid	x	64-19-7
22	formic acid			64-18-6
Aromatic hydrocarbons				
23	ethyl benzene		x	100-41-4
24	toluene	methyl benzene	x	108-88-3
25	xylene	dimethyl benzene	x	1330-20-7
Chlorinated hydrocarbons				
26	chlorobenzene	monochlorobenzene		180-90-7
27	dichloromethane	methylene chloride	x	75-09-2
Esters				
28	butyl acetate			123-86-4
29	ethyl acetate			141-78-6
30	isoamyl acetate			628-63-7
31	isobutyl acetate			110-19-0
32	isopropyl acetate			108-21-4
33	methyl acetate			79-20-9
34	methyl formate			592-84-7
Ethers and glycol ethers				
35	dioxane	diethylene oxide, diethylene ether		123-91-1
36	ethyl ether	ether, ethoxy ethane		60-29-7
37	ethylene glycol diethyl ether	diethoxy ethane		629-14-4
38	ethylene glycol dimethyl ether	dimethoxy ethane, DME		110-71-4

39	ethylene glycol monoethyl ether	cellosolve		110-80-5
40	tert-butyl methyl ether	MTBE	x	1634-04-4
41	tetrahydrofuran	THF		109-99-9
Ketones				
42	acetone	2-propanone, dimethyl ketone		67-64-1
43	cyclohexanone			108-94-1
44	methyl ethyl ketone ^a	MEK	x	78-93-3
45	methyl isobutyl ketone	methyl pentanone, MIK		108-10-1
Miscellaneous solvents				
46	acetic anhydride ^a			108-24-7
47	acetonitrile	methylcyanide, ethanenitrile		75-05-8
48	dimethyl sulfoxide			67-68-5
49	dimethyl formamide			68-12-2
50	methyl pyrrolidone			872-50-4

Petrochemical solvent production routes

The petrochemical production of solvents can be divided in four main production routes that are of importance on an industrial scale. The routes differ with regard to their starting material and key processes (Asinger 1971, Chauvel et al. 1989a, Chauvel et al. 1989b, Ullmanns 2006, Wells 1991):

- (A) In the methanol route, methanol from natural gas is used as starting material. In this route, important solvents such as acetic acid, formaldehyde or tetrahydrofuran are produced.
 - (B) In the naphtha/steam cracking route, naphtha from crude oil is treated in a steam cracking process that results in a large variety of alkenes and alkanes. Based on these intermediates, many alcohols such as ethanol, butanol or propanol are produced and further processed to acetates.
 - (C) In the BTX/naphtha route, naphtha from crude oil or BTX reformat is separated in a molecular sieve to obtain alkanes such as hexane and heptane.
 - (D) In the BTX splitting route, BTX reformat or pyrolysis gasoline are splitted to obtain aromatic solvents such as toluene or xylene. Toluene and the intermediate benzene are further processed to cyclohexane, acetone, and benzaldehyde.
-

General methodology

The methodology for the investigation of life cycle inventory data of different unit processes has been described in detail in the methodology report of the ecoinvent 2000 project (Frischknecht et al. 2007) and was used in this study. Thus, all unit processes are compatible with the ecoinvent datasets. For readers, who are not familiar with this methodology, it is recommended to read the methodology report before working with the unit process datasets shown in this report.

Life-cycle inventories of solvents

Life-cycle inventories for some organic solvents have already been published in the ecoinvent database. In order to comply with the quality standards of the ecoinvent database, all solvent inventories are established according to the guidelines presented in Frischknecht et al. (2007) and Hischier et al. (2004). Thus, all data are presented on a unit process level. Whenever possible, actual production data is used to determine inventory flows. In case such data is not available, basic information from technical reference books together with additional assumptions and simplifications was used, as described in Hischier et al. (2004). With this procedure, missing inventory data of energy consumption, process efficiency, water consumption, emissions to air and water, transports, and infrastructure can be estimated. Additionally, inventory flows can be estimated accurately taking advantage from the fact that the petrochemical production of solvents is mostly based on few different operation steps such as esterification, chlorination or carbonylation for example. Thus, if process data for such operation steps is available for a specific solvent it was used as approximation for other solvents.

Tab. 0.1 shows the data quality specifically for input material (amount of educts, ancillaries), water consumption (process water and cooling water), energy consumption (process energy and feedstock energy), emissions to air and water. If process data have been available from literature or manufacturers it is indicated with "1". In case no information was found the inventory data has been estimated according to Hischier et al. (2004), indicated as "3" or known inventory data of good quality have been used as approximation for solvents requiring the same unit processes, indicated as "2".

Tab. 0.1 The solvents in this study with corresponding data quality indicators.

Selected solvents			Available inventory information				
Name	Name of dataset	CAS-no.	input materia	water cons.	energy cons.	emissions to air	emissions to water
amyl-alcohol, tert.-	2-methyl-2-butanol	75-85-4	3	3	2	3	3
benzyl alcohol	benzyl alcohol	100-51-6	1	1	1	1	1
butanol, 2-	2-butanol	78-92-2	1	1	1	3	3
butanol, iso-	2-methyl-1-propanol	78-83-1	1	1	1	3	3
butylene glycol, 1,4-	butane-1,4-diol	1100-63-4	1	1	1	1	1
ethanol	ethanol from ethylene	64-17-5	1	1	1	3	3
pentanol	1-pentanol	71-41-0	3	3	2	3	3
propanol, 1-	propanol	71-23-8	1	1	1	1	1
benzaldehyde	benzaldehyde	100-52-7	3	3	1	3	3
propionaldehyde	propanal	123-38-6	3	3	2	3	3
cyclohexane	cyclohexane	110-82-7	1	1	1	3	3
hexane, iso-	2-methylpentane	96-14-0	3	3	1	3	3
methylcyclohexane	methylcyclohexane	108-87-2	3	3	2	3	3
formic acid	formic acid	64-18-6	1	1	1	3	3
chlorobenzene	monochlorobenzene	180-90-7	1	1	1	1	1
butyl acetate	butyl acetate	123-86-4	3	3	2	3	3
ethyl acetate	ethyl acetate	141-78-6	1	1	1	3	3
isoamyl acetate	3-methyl-1-butyl acetate	628-63-7	3	3	2	3	3
isobutyl acetate	isobutyl acetate	110-19-0	3	3	2	3	3
isopropyl acetate	isopropyl acetate	108-21-4	3	3	2	3	3
methyl acetate	methyl acetate	79-20-9	1	1	1	3	3
methyl formate	methyl formate	592-84-7	1	3	1	3	3
dioxane	1,4-dioxane	123-91-1	3	3	1	3	3
ethyl ether	diethyl ether	60-29-7	1	1	1	3	3
ethylene glycol diethyl ether	ethylene glycol diethyl ether	629-14-4	3	3	3	3	3
ethylene glycol dimethyl ether	ethylene glycol dimethyl ether	110-71-4	3	3	3	3	3
ethylene glycol monoethyl ether	ethylene glycol monoethyl ether	110-80-5	3	3	3	3	3
tetrahydrofuran	tetrahydrofuran	109-99-9	3	3	1	3	3
cyclohexanone	cyclohexanone	108-94-1	1	1	1	3	3
methyl isobutyl ketone	4-methyl-2-pentanone	108-10-1	1	1	1	1	1
acetic anhydride ^a	acetic anhydride	108-24-7	1	1	1	3	3
acetonitrile	acetonitrile	75-05-8	1	1	1	3	3
dimethyl sulfoxide	dimethyl sulfoxide	67-68-5	3	3	3	3	3
dimethyl formamide	N,N-dimethylformamide	68-12-2	3	3	1	3	3
methyl pyrrolidone	N-dimethyl-2-pyrrolidone	872-50-4	3	3	3	3	3

^a the existing ecoinvent process has been improved with better data.

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Part II

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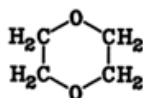
1 1,4-Dioxane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

1.1 Introduction

1,4-Dioxane (CAS-No. 123-91-1, $C_4H_8O_2$, synonyms: p-dioxane, diethylene oxide, diethylene dioxide, diethylene ether, 1,4-dioxacyclohexane, dioxyethylene ether) is an extraordinary solvent, capable of solubilizing most organic compounds, water in all proportions, and many inorganic compounds. At ambient temperature, 1,4-dioxane is a clear liquid with an ether-like odor. Its relative stability is similar to other aliphatic ethers, and it forms peroxides on exposure to air as they do.

Dioxane is a cyclic diether forming a six-membered ring with the following structure:



For this inventory the functional unit is 1 kg of pure 1,4-dioxane. The most important chemical and physical properties of 1,4-dioxane used in this inventory are given here:

Tab. 1.1 Chemical and physical properties of 1,4-dioxane (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	88.11	$g\ mol^{-1}$	
Specific gravity	1035.6	$kg\ m^{-3}$	at 20 °C
Boiling point	101.3	°C	at normale pressure
Melting point	11.8	°C	at normale pressure

1.2 Reserves and Resources

Most of the 1,4-dioxane manufactured is prepared by dehydration and ring closure of diethylene glycol.

1.3 Production Technologies and Use

1.3.1 Production Technologies

1,4-Dioxane is manufactured commercially by dehydration and ring closure of diethylene glycol. Concentrated sulfuric acid (ca. 5 %) is used as a catalyst, although phosphoric acid, p-toluenesulfonic acid, and strongly acidic ion-exchange resins are recognized alternatives. Operating conditions vary; temperatures range from 130 to 200 °C and pressures range from a slight pressure to a partial vacuum (25 –110 kPa). A favorable temperature is reported to be 160 °C. The process is continuous, with 1,4-dioxane vaporized from the reaction vessel. The vapors are passed through an acid trap and two distillation columns to remove water and to purify the product. Yields of ca. 90 % are possible. 2-Methyl-1,3-dioxolane, acetaldehyde, crotonaldehyde, and polyglycol are unwanted byproducts.

1.3.2 Use

The largest single use of 1,4-dioxane is the stabilization of 1,1,1-trichloroethane against chemical attack by aluminum. The aluminum oxide film prevents the metal from reacting with 1,1,1-trichloroethane until the film is disturbed. When the bare metal contacts 1,1,1-trichloroethane, chlorine is extracted from the solvent to form AlCl_3 , which causes dehydrohalogenation of 1,1,1-trichloroethane to produce HCl and vinylidene chloride ($\text{CH}_2 = \text{CCl}_2$). Dioxane is the foremost inhibitor of these reactions. An insoluble 1,4-dioxane–aluminum chloride complex is formed, which deactivates the catalyst and seals any openings in the Al_2O_3 film.

Dioxane is used as a solvent in the formulation of inks, coatings, and adhesives, and as solvent for extracting animal and vegetable oils. As a chemical intermediate, 1,4-dioxane reaction products are useful as insecticides, herbicides, plasticizers, and monomers. The oxonium complexes of 1,4-dioxane with salts, mineral acids, halogens, and sulfur trioxide are used as catalysts and as reagents for anhydrous acid reactions, brominations, and sulfonations. In the laboratory, 1,4-dioxane is useful as a cryoscopic solvent for molecular mass determinations and as a stable reaction medium for diverse reactions. Historically, 1,4-dioxane is reported to be valuable in polymer manufacture and as a solvent for natural and synthetic resins, including cellulose derivatives, polyvinyl acetal resins, acrylonitrile–methyl vinyl pyridine copolymers, and others.

1.4 System Characterisation

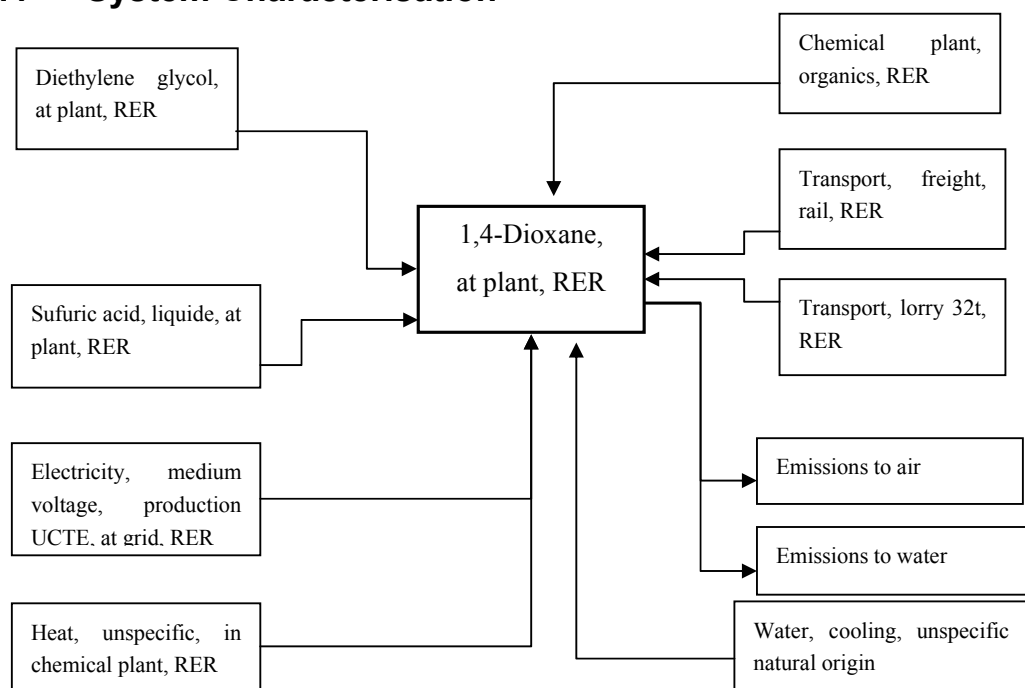


Fig. 1.1 Process flow chart for the production of 1,4-dioxane

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg 1,4-dioxane, at plant, in Europe**. For this study the production of 1,4-dioxane by means of the dehydration and ring closure of diethylene glycol is considered. All data in the present report are referred to 1 kg 1,4-dioxane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or

shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

1.5 Raw materials and auxiliaries

The amount for the diethylene glycol is calculated with an assumed yield of 90 %. The other data are taken from Geisler (2004), who modelled processes for several chemicals. The data for auxiliaries and energy inputs, which are given by Geisler (2004) are modelled with basic knowledge about the processes and expert estimations.

Tab. 1.2 Consumption of raw materials and auxiliaries for the production of 1 kg 1,4-dioxane

Input	Values	Remarks
Diethylen glycol (kg) ¹	1.339	Calculated with a yield of 90 %
Nitrogen (kg) ²	1.113E-01	Geisler 2004
Cooling water (kg) ²	298	Geisler 2004

1.6 Energy

Tab. 1.3 Energy consumption for the production of 1 kg 1,4-dioxane

Input	Geisler 2004
Steam (MJ)	15.125
Electricity (kWh)	0.361

1.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and 50 km by lorry 32t and 100 km by train for the nitrogen. Tab. 1.4 summarises the total transport amounts for the production of 1 kg 1,4-dioxane.

Tab. 1.4 Total transport amounts for the production of 1 kg 1,4-dioxane

tkm kg ⁻¹ 1,4-dioxane	lorry	train
Raw materials	1.394E-01	8.142E-01

1.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of 1,4-dioxane plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

1.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of 1,4-dioxane. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 1 % of the raw material input. The assumed values for the emissions to air are given in Tab. 1.5.

Tab. 1.5 Process emissions to air from the production of 1 kg 1,4-dioxane

Output	Value	Remarks
Diethylene glycol, to air (kg)	1.338E-02	Estimated as 1 % of input
Carbon dioxide, fossil, to air (kg)	1.8E-1	from waster water treatment
Waste heat (MJ)	1.3	calculated from electricity input

1.10 Emissions to water

The remaining amount of unreacted diethylene glycol (9 % of the input) was assumed to leave the production process with the waste water.

Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for diethylene glycol was assumed with 90%. The carbon contained in the removed diethylene glycol was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of diethylene glycol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 1.6.

Tab. 1.6 Process emissions to water from the production 1 kg 1,4-dioxane

Output	Value	Remarks
Diethylene glycol, to water (kg)	1.205E-02	calculated from mass balance
COD, BOD (kg)	1.7E-02	calculated from water emissions
TOC, DOC (kg)	5.5E-03	calculated from water emissions

1.11 Life cycle inventory of 1,4-dioxane production and data Quality Considerations

Tab. 1.7 shows the life cycle inventory and the data quality indicators of the production of 1,4-dioxane. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 1.7 Unit process raw data for the production of 1,4-dioxane

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dioxane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	dioxane, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.98E-1	1	1.38	(4,5,1,1,1,5); Geisler 2004
	5	-	diethylene glycol, at plant	RER	0	kg	1.34E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	nitrogen, liquid, at plant	RER	0	kg	1.11E-1	1	1.38	(4,5,1,1,1,5); Geisler 2004
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.39E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	8.14E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.61E-1	1	1.38	(4,5,1,1,1,5); Geisler 2004
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.51E+1	1	1.38	(4,5,1,1,1,5); Geisler 2004
emission air, high population density	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.30E+0	1	1.38	(4,5,1,1,1,5); Calculated from the electricity input
	-	4	Diethylene glycol	-	-	kg	1.34E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg	1.80E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	Diethylene glycol	-	-	kg	1.20E-2	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.70E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.70E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	5.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	5.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

1.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 1.8 shows selected LCI results and the cumulative energy demand for the production of 1,4-dioxane.

Tab. 1.8 Selected LCI results and the cumulative energy demand for the production of 1,4-dioxane

Name		dioxane, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	70.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	8.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.4E-1
CO ₂ , fossil	air	kg	3.3E+0
NM VOC	air	kg	1.6E-2
nitrogen oxides	air	kg	5.1E-3
sulphur dioxide	air	kg	6.7E-3
particulates, <2.5 um	air	kg	4.7E-4
BOD	water	kg	2.0E-2
cadmium	soil	kg	1.1E-9

1.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dioxane, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 1,4-dioxane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dioxan, ab Werk
	491	Synonyms	p-dioxane, diethylene oxide, diethylene dioxide, diethylene ether, 1,4-dioxacyclohexane, dioxyethylene ether
	492	GeneralComment	The process "1,4-dioxane, at plant, RER" is modelled for the production of 1,4-dioxane from diethylene glycol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H8O2
	501	StatisticalClassification	
	502	CASNumber	123-91-1
TimePeriod	601	StartDate	2003
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Dehydration of diethylene glycol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents
ProofReading	5616	Validator	11
	5615	Details	automatic validation in Excel
	5619	OtherDetails	none

1.14 References

Friskhnecht et al. 2007 Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007)

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2 2-Methyl-2-butanol

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier
 Last changes: 2007

2.1 Introduction

2-Methyl-2-butanol (CAS-No. 75-85-4, $C_5H_{11}OH$, Synonym: tert-amyl alcohol, tert-pentanol) is a colourless liquid of characteristic odor. It is only sparingly miscible with water at 20 °C. However, it is almost completely miscible with most organic solvents.

For this inventory the functional unit is 1 kg of pure 2-methyl-2-butanol. The most important chemical and physical properties of 2-methyl-2-butanol used in this inventory are given here:

Tab. 2.1 Chemical and physical properties of 2-methyl-2-butanol (Ulmans 2004)

Property	Value	Unit	Remarks
Molecular weight	88.2	$g\ mol^{-1}$	
Specific gravity	809	$kg\ m^{-3}$	at 20 °C
Boiling point	101.8	°C	at normale pressure
Melting point	- 11.9	°C	at normale pressure

2.2 Reserves and Resources

Only a few industrial processes are known for the production of pentanols, namely, hydroformylation of butenes, hydrolysis of chloropentanes, and hydration of pentenes. Production of C5-alcohols from fusel oils by fermentation has been replaced mainly by the oxo synthesis and currently has little industrial importance. Production of pentanols from pentane by the Sharples process (chlorination of pentanes and subsequent hydrolysis) has also been largely replaced by oxo synthesis.

2.3 Production Technologies and Use

2.3.1 Production Technologies

Hydration of pentenes

2-Methyl-2-butanol is produced mainly by hydration of 2-methyl-2-butene (isopentene). There are several processes which use different catalysts, e. g. acid ion exchangers, zeolites, layer silicates, formic acid, HCl, $SbCl_3$, $ZnCl_2$, or acetic acid.

2.3.2 Use

2-Methyl-2-butanol is employed as solvent for coating materials based on epoxy resins and polyurethanes, in the production of aryl pyruvic acids, and in the oxidation of olefins to the corresponding carbonyl compounds. It is also used as a stabilizing agent for 1,1,1-trichloroethane, chlorinated C3 – C4-hydrocarbons, and N,N,- dialkylallyl amines. The reaction of 2,2,6,6-tetramethyl-4-cyano-4-hydroxypiperidine with 2-methyl-2-butanol yields highly active light stabilizers for polymers.

2.4 System Characterisation

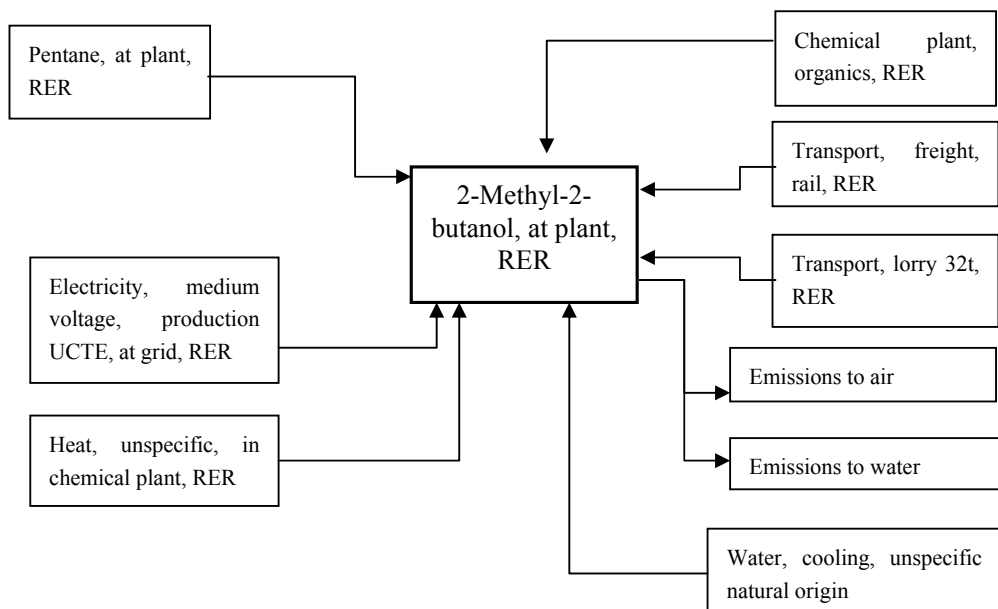


Fig. 2.1 Process flow chart for the production of 2-methyl-2-butanol

This report corresponds to the module in the ecoinvent database for the production of **1 kg 2-methyl-2-butanol, at plant, in Europe**. For this study the production of 2-methyl-2-butanol by means of the hydration of 2-methyl-2-butene is considered. All data in the present report are referred to 1 kg 2-methyl-2-butanol 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

2.5 Raw materials and auxiliaries

As there was no data available for the production of 2-methyl-2-butanol, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries was estimated with data from Chauvel 1989, who gives data for the production of isopropanol from the hydration of isopropylene. This process is considered to be similar to the hydration of 2-methyl-2-butene.

For the 2-methyl-2-butene input the dataset “pentane, at plant, RER” is used.

Tab. 2.2 Consumption of raw materials and auxiliaries for the production of 1 kg 2-methyl-2-butanol

Input	Values	Remarks
2-methyl-2-butene (kg)	0.837	Calculated with a yield of 95%
Process water (kg)	1	Estimation with data from similar process
Cooling water (kg)	190	Estimation with data from similar process

2.6 Energy

As no data for the production of 2-methyl-2-butanol are available, the values for the energy use have been estimated with data from Chauvel 1989, who gives data for the production of isopropanol from the hydration of isopropylene. This process is considered to be similar to the hydration of 2-methyl-2-butene.

Tab. 2.3 Energy consumption for the production of 1 kg 2-methyl-2-butanol

Input	Values	Remarks
Steam (MJ)	12.65	Estimation with data from similar process
Electricity (kWh)	0.09	Estimation with data from similar process
Fuel (MJ)	9.5	Estimation with data from similar process

In this study the use of the fuel is modelled with the dataset “light fuel oil, burned in industrial furnace 1 MW, non-modulating” with a energetic efficiency of 74 %, and the use of steam is modelled with the dataset “heat, unspecific, in chemical plant, RER”.

2.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 2.4 summarises the total transport amounts for the production of 1 kg 2-methyl-2-butanol.

Tab. 2.4 Total transport amounts for the production of 1 kg 2-methyl-2-butanol

tkm kg ⁻¹ 2-methyl-2-butanol	lorry	train
Raw materials	8.37E-02	5.02E-01

2.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of 2-methyl-2-butanol plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

2.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of 2-methyl-2-butanol. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 2.5 Process emissions to air from the production of 1 kg 2-methyl-2-butanol

Output	Value	Remarks
2-Methyl-2-butene, to air (kg)	1.67E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.14E-1	from waster water treatment
Waste heat (MJ)	3.24E-01	calculated from electricity input

2.10 Emissions to water

The remaining amount of unreacted 2-methyl-2-butene (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for 2-methyl-2-butene was assumed with 90%. The carbon contained in the removed 2-methyl-2-butene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of 2-methyl-2-butene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 2.6.

Tab. 2.6 Process emissions to water from the production 1 kg 2-methyl-2-butanol

Output	Value	Remarks
2-Methyl-2-butene, to water (kg)	4.02E-03	calculated from mass balance
COD, BOD (kg)	1.63E-02	calculated from water emissions
TOC, DOC (kg)	2.83E-03	calculated from water emissions

2.11 Life cycle inventory and Data Quality Considerations

Tab. 2.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of 2-methyl-2-butanol. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 2.7 Unit process raw data for the production of 2-methyl-2-butanol

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	2-methyl-2-butanol, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	2-methyl-2-butanol, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	1.90E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	water, deionised, at plant	CH	0	kg	1.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	pentane, at plant	RER	0	kg	8.37E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	8.37E-2	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	5.02E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	9.00E-2	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.27E+1	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	9.50E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	3.24E-1	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
emission water, river	-	4	2-Methyl-2-butene	-	-	kg	1.67E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.14E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	2-Methyl-2-butene	-	-	kg	4.02E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.63E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.63E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.83E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.83E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

2.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 2.8 shows selected LCI results and the cumulative energy demand for the production of 2-methyl-2-butanol.

Tab. 2.8 Selected LCI results and the cumulative energy demand for the production of 2-methyl-2-butanol

Name		2-methyl-2-butanol, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	104.9
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	9.7E-2
CO ₂ , fossil	air	kg	3.1E+0
NM ₂ OC	air	kg	5.3E-3
nitrogen oxides	air	kg	4.8E-3
sulphur dioxide	air	kg	6.1E-3
particulates, <2.5 µm	air	kg	3.0E-4
BOD	water	kg	2.1E-2
cadmium	soil	kg	3.7E-10

2.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	2-methyl-2-butanol, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 2-methyl-2-butanol including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	2-Methyl-2-butanol, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "2-methyl-2-butanol, at plant, RER" is modelled for the production of 2-methyl-2-butanol from hydration of 2-methyl-2-pentane in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C5H12O
	501	StatisticalClassification	
	502	CASNumber	75-85-4
TimePeriod	601	StartDate	1989
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydration of 2-methyl-2-pentane
Representativeness	722	Percent	10
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard methods.
DataGeneratorAn	727	UncertaintyAdjustments	none
	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

2.14 References

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- Ullmann 2004 Lappe, P., Hoffmann, T.: Pentanols. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebaum D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/ueic/articles>

3 2-Methylpentane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

3.1 Introduction

2-Methylpentane (CAS-No. 107-83-5, C_6H_{14} ; Synonym: isohexane) is a colourless liquid with a characteristic odour. It is completely miscible with common organic solvents. Its vapor has an irritant effect on mucous membranes and a narcotic effect in higher concentrations.

For this inventory the functional unit is 1 kg of pure 2-methylpentane. The most important chemical and physical properties of 2-methylpentane used in this inventory are given here:

Tab. 3.1 Chemical and physical properties of 2-methylpentane (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	86.2	$g\ mol^{-1}$	
Specific gravity	653.15	$kg\ m^{-3}$	at 20 °C
Boiling point	60.3	°C	at normale pressure
Melting point	- 153.7	°C	at normale pressure

3.2 Reserves and Resources

2-Methylpentane is usually produced by isomerization of n-hexane.

3.3 Production Technologies and Use

3.3.1 Production Technologies

Isomerization converts n-butane, n-pentane and n-hexane into their respective isoparaffins of substantially higher octane number. The straight-chain paraffins are converted to their branched-chain counterparts whose component atoms are the same but are arranged in a different geometric structure. Isomerization is important for the conversion of n-butane into isobutane, to provide additional feedstock for alkylation units, and the conversion of normal pentanes and hexanes into higher branched isomers for gasoline blending. Isomerization is similar to catalytic reforming in that the hydrocarbon molecules are rearranged, but unlike catalytic reforming, isomerization just converts normal paraffins to isoparaffins. There are two distinct isomerization processes, butane (C4) and pentane/hexane (C5/C6).

Pentane/hexane isomerization increases the octane number of the light gasoline components n-pentane and n-hexane, which are found in abundance in straight-run gasoline. In a typical C5/C6 isomerization process, dried and desulfurized feedstock is mixed with a small amount of organic chloride and recycled hydrogen, and then heated to reactor temperature. It is then passed over supported-metal catalyst in the first reactor where benzene and olefins are hydrogenated. The feed next goes to the isomerization reactor where the paraffins are catalytically isomerized to isoparaffins. The reactor effluent is then cooled and subsequently separated in the product separator into two streams: a liquid product (isomerate) and a recycle hydrogen-gas stream. The isomerate is washed (caustic and water), acid stripped, and stabilized before going to storage.

3.3.2 Use

2-Methylpentane is used as a solvent for example in rubber and related industries.

3.4 System Characterisation

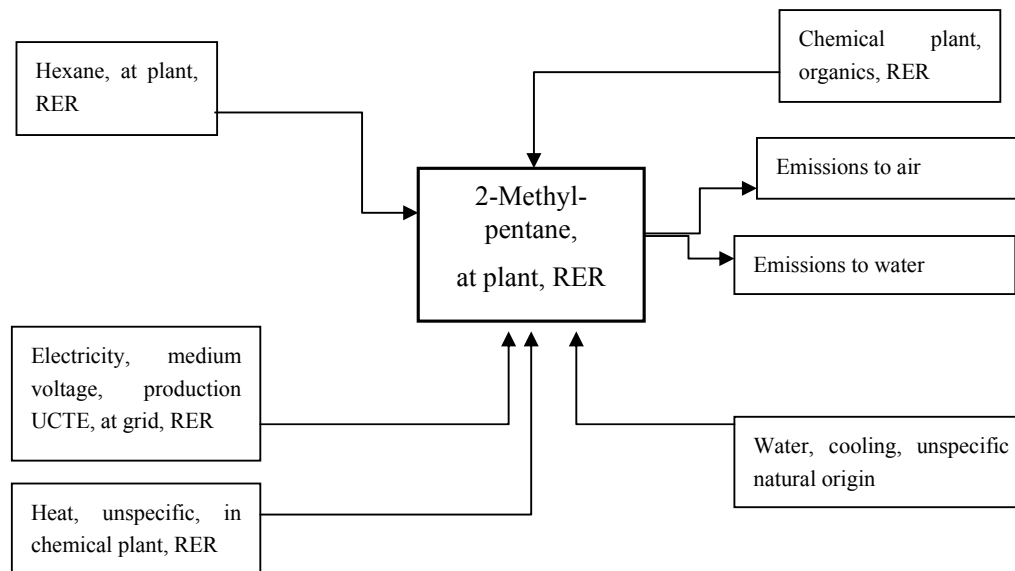


Fig. 3.1 Process flow chart for the production of 2-methylpentane

This report corresponds to the module in the ecoinvent database for the production of **1 kg 2-methylpentane, at plant, in Europe**. For this study the production of 2-methylpentane by means of the esterification of ethanol with acetic acid is considered. All data in the present report are referred to 1 kg 2-methylpentane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

3.5 Raw materials and auxiliaries

As there was no data available for the production of 2-methylpentane, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the 2-methylpentane production.

Tab. 3.2 Consumption of raw materials and auxiliaries for the production of 1 kg 2-methylpentane

Input	Values	Inventory ³
Hexane (kg)	1.05	Calculated with a yield of 95%
Cooling water (kg)	24	Estimation

3.6 Energy

The energy demand is calculated with data from DOE (1998), which gives data for the hexane isomerization in US refineries.

Tab. 3.3 Energy consumption for the production of 1 kg 2-methylpentane

Input	DOE1998
Steam (MJ)	0.2
Electricity (kWh)	0.056

3.7 Transportation

No transports are considered for the raw material input, because the isomerization of hexane to 2-methylpentane is usually done at the refinery.

3.8 Infrastructure and land use

No infrastructure and land use are considered, because the isomerization of hexane to 2-methylpentane is usually done at the refinery. The infrastructure is included in the hexane dataset.

3.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of 2-methylpentane. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 3.4.

Tab. 3.4 Process emissions to air from the production of 1 kg 2-methylpentane

Output	Value	Remarks
Hexane, to air (kg)	2.11E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.4E-01	from waster water treatment
Waste heat (MJ)	2.02E-01	calculated from electricity input

3.10 Emissions to water

The remaining amount of unreacted hexane (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste

water plant. The removal efficiency for hexane was assumed with 90%. The carbon contained in the removed hexane was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of hexane in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 3.5.

Tab. 3.5 Process emissions to water from the production 1 kg 2-methylpentane

Output	Value	Remarks
Hexane, to water (kg)	5.05E-03	calculated from mass balance
COD, BOD (kg)	1.71E-02	calculated from water emissions
TOC, DOC (kg)	4.23E-03	calculated from water emissions

3.11 Life cycle inventory and Data Quality Considerations

Tab. 3.6 shows the life cycle inventory and the data quality indicators for the inventory of the production of 2-methylpentane. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 3.6 Unit process raw data for the production of 2-methylpentane

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	isohexane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
401	-	0	isohexane, at plant	RER	0	kg	1.00E+0			
662			Location							
493			InfrastructureProcess							
403			Unit							
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	hexane, at plant	RER	0	kg	1.05E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	5.60E-2	1	1.16	(1,4,1,5,1,3); DOE 1998
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.00E-1	1	1.16	(1,4,1,5,1,3); DOE 1998
	-	4	Heat, waste	-	-	MJ	2.02E-1	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
emission air, high population density	-	4	Hexane	-	-	kg	2.11E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.40E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Hexane	-	-	kg	5.05E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.71E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.71E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4.23E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	4.23E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

3.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 3.7 shows selected LCI results and the cumulative energy demand for the production of 2-methylpentane.

Tab. 3.7 Selected LCI results and the cumulative energy demand for the production of 2-methylpentane

Name		isohexane, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	65.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.7
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.9E-1
CO ₂ , fossil	air	kg	1.2E+0
NM VOC	air	kg	1.9E-2
nitrogen oxides	air	kg	2.9E-3
sulphur dioxide	air	kg	5.9E-3
particulates, <2.5 um	air	kg	3.1E-4
BOD	water	kg	5.4E-2
cadmium	soil	kg	7.1E-10

3.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	isohexane, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of isohexane including materials, energy uses, and emissions.
	404	Amount	1
	490	LocalName	Isohexan, ab Werk
	491	Synonyms	2-methyl pentane
	492	GeneralComment	The process "isohexane, at plant, RER" is modelled for the production of isohexane from Hexane in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H14O2
	501	StatisticalClassification	
	502	CASNumber	96-14-0
TimePeriod	601	StartDate	1998
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Isomerization of n-hexane
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	49
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

3.14 References

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4 3-Methyl-1-butyl acetate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

4.1 Introduction

3-Methyl-1-butyl acetate (CAS-No. 123-92-2, $\text{CH}_3\text{COO}(\text{CH}_2)_2\text{CH}(\text{CH}_3)_2$, $\text{C}_7\text{H}_{14}\text{O}_2$, Synonym: isopentyl acetate, isoamyl acetate) is a liquid with a strong fruity odour and has been identified in many fruit aromas. It is the main component of banana aroma and is, therefore, also used in banana flavors. It is almost insoluble in water and soluble in ether and the most common organic solvents.

For this inventory the functional unit is 1 kg of pure 3-methyl-1-butyl acetate. The most important chemical and physical properties of 3-methyl-1-butyl acetate used in this inventory are given here:

Tab. 4.1 Chemical and physical properties of 3-methyl-1-butyl acetate (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	130	g mol^{-1}	
Specific gravity	876	kg m^{-3}	at 20 °C
Boiling point	142	°C	at normale pressure
Melting point	- 78	°C	at normale pressure

4.2 Reserves and Resources

Most of the 3-methyl-1-butyl acetate manufactured is prepared by the esterification of isoamyl alcohol with acetic acid.

4.3 Production Technologies and Use

4.3.1 Production Technologies

The reaction of an alcohol and an organic acid produces an ester in the presence of an acid catalyst. The reaction, esterification, is the most common process for manufacturing acetate esters .

Commercial processes for esterification are usually continuous and are performed in the liquid phase. The reaction requires sulfuric acid. Another type of catalyst is ion-exchange resins. In contrast, gas-phase technology is available with sulfonated polymer or supported phosphoric acid as catalyst.

Esterification involves an equilibrium between reactants and product. Adding excess reactants or removal of water drive the equilibrium to favor the formation of ester. Azeotroping agents eliminate water and force the reaction to completion. Common agents are cyclohexane, benzene, and toluene.

Esterification with acetic anhydride offers an alternative route for making acetates. Due to its high reactivity, acetic anhydride readily forms esters with alcohols. The reaction is irreversible. The esterification goes to completion without eliminating products or water.

Additions of acetic acid to olefins present an alternative way to make acetates, especially for branched esters. Synthesis of these hindered esters by esterification is not efficient. The equilibrium does not favor the desirable products. Rhône-Poulenc uses this method to prepare isopropyl acetate from acetic

acid and propylene. Catalysts can be strong protic or Lewis acids, like sulfuric acid or $\text{BF}_3 \cdot \text{OEt}_2$. Ethyl acetate can also be produced directly from acetaldehyde.

Adding hydrogen and carbon monoxide to acetic acid or acetic anhydride can produce acetates. Alternatively, hydrogenation of acetic acid yields ethanol and ethyl acetate. Many studies relate to supported heterogeneous catalysts with molybdenum and group VIII metals (Pd, Rh, or Ru). Additionally, homogeneous catalysis with ruthenium can result in high selectivity.

4.3.2 Use

3-Methyl-1-butyl acetate finds use as a fruit flavor (banana, pear, apple, berry) for foods, candy, butter, coconut, cola, rum, as a flavor for paint and insecticides, and as a solvent.

4.4 System Characterisation

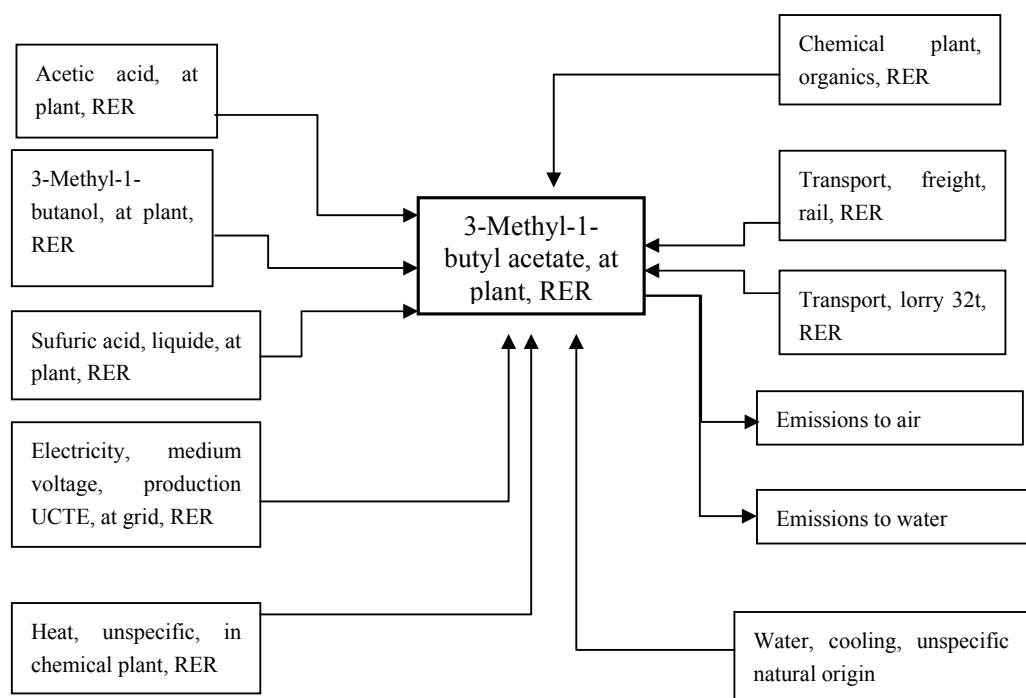


Fig. 4.1 Process chain for the production of butyl acetate

This report corresponds to the module in the ecoinvent database for the production of **1 kg 3-methyl-1-butyl acetate, at plant, in Europe**. For this study the production of 3-methyl-1-butyl acetate by means of the esterification of 3-methyl-1-butanol with acetic acid is considered. All data in the present report are referred to 1 kg 3-methyl-1-butyl acetate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

4.5 Raw materials and auxiliaries

As there was no data available for the production of 3-methyl-1-butyl acetate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries, i. e. sulfuric acid was estimated with data from Wells 1991, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of 3-methyl-1-butanol.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the 3-methyl-1-butyl acetate production.

Tab. 4.2 Consumption of raw materials and auxiliaries for the production of 1 kg 3-methyl-1-butyl acetate

Input	Values	Remarks
Isoamyl alcohol (kg)	7.126E-01	Calculated with a yield of 95 %
Acetic acid (kg)	4.858E-01	Calculated with a yield of 95 %
Sulfuric acid (kg)	1.31E-02	estimated
Cooling water (kg)	24	estimated

4.6 Energy

As no data for the production of 3-methyl-1-butyl acetate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of 3-methyl-1-butanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 4.3 Energy consumption for the production of 1 kg 3-methyl-1-butyl acetate

Input	Esterification ¹	This study
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

¹ Source: Kim 2003

4.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 4. Error! Bookmark not defined. summarises the total transport amounts for the production of 1 kg 3-methyl-1-butyl acetate.

Tab. 4.Error! Bookmark not defined. **Total transport amounts for the production of 1 kg 3-methyl-1-butyl acetate**

tkm kg ⁻¹ ethyl acetate	lorry	train
Raw materials	1.21E-01	7.27E-01

4.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of 3-methyl-1-butyl acetate plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

4.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-03 kWh/kg 3-methyl-1-butyl acetate is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of 3-methyl-1-butyl acetate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in **Error! Unknown switch argument.**

Tab. 4.4 **Process emissions to air from the production of 1 kg 3-methyl-1-butyl acetate**

Output	Value	Remarks
3-Methyl-1-butanol, to air (kg)	1.425E-03	Estimated as 0.2 % of input
Acetic acid, to air (kg)	9.717E-04	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	7.696E-2	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

4.10 Emissions to water

The remaining amount of unreacted acetic acid and 3-methyl-1-butanol (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for 3-methyl-1-butanol and acetic acid was assumed with 90%. The carbon contained in the removed 3-methyl-1-butanol and acetic acid was accounted as CO₂ emissions to air. The sulphate emissions are calculated from the sulphuric acid input.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of 3-methyl-1-butanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 4.5.

Tab. 4.5 **Process emissions to water from the production 1 kg 3-methyl-1-butyl acetate**

Output	Value	Remarks
3-Methyl-1-butanol, to water (kg)	3.42E-03	calculated from mass balance
Acetic acid, to water (kg)	2.33E-03	calculated from mass balance

Sulphate, to water (kg)	1.283E-02	calculated from mass balance
COD, BOD (kg)	1.13E-02	calculated from water emissions
TOC, DOC (kg)	3.26E-03	calculated from water emissions

4.11 Life cycle inventory and Data Quality Considerations

Tab. 4.6 shows the life cycle inventory and the data quality indicators for the inrror! **Style not defined.**entoryrror! **Bookmark not defined.**of the production of 3-methyl-1-butyl acetate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 4.6 Unit process raw data for the production of 3-methyl-1-butyl acetate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	3-methyl-1-butyl acetate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401	0	3-methyl-1-butyl acetate, at plant	RER	0	kg	1.00E+0			
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	3-methyl-1-butanol, at plant	RER	0	kg	7.13E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	4.86E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	1.31E-2	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.21E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.27E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	8.84E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
emission air, high population density	-	4	3-Methyl-1-butanol	-	-	kg	1.43E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	9.72E-4	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	7.70E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	3-Methyl-1-butanol	-	-	kg	3.42E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.33E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.13E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.13E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.26E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.26E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
emission water, river	-	4	3-Methyl-1-butanol	-	-	kg	3.42E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.33E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.13E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.13E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.26E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.26E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	3-Methyl-1-butanol	-	-	kg	3.42E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.33E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation

4.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 4.7 shows selected LCI results and the cumulative energy demand for the production 3-methyl-1-butyl acetate.

Tab. 4.7 Selected LCI results and the cumulative energy demand for the production of 3-methyl-1-butyl acetate

Name		3-methyl-1-butyl acetate, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	105.0
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	8.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.3E-1
CO ₂ , fossil	air	kg	4.7E+0
NM VOC	air	kg	1.0E-2
nitrogen oxides	air	kg	6.8E-3
sulphur dioxide	air	kg	9.8E-3
particulates, <2.5 µm	air	kg	5.6E-4
BOD	water	kg	2.8E-2
cadmium	soil	kg	1.1E-9

4.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	3-methyl-1-butyl acetate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 3-methyl-1-butyl acetate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	3-Methyl-1-butylazetat, ab
	491	Synonyms	isopentyl acetate
	492	GeneralComment	The process "3-methyl-1-butyl acetate, at plant, RER" is modelled for the production of 3-methyl-1-butyl acetate from 3-methyl-1-butanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H14O2
	501	StatisticalClassification	
	502	CASNumber	123-92-2
TimePeriod	601	StartDate	1989
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Esterification of 3-methyl-1-butanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
DataGeneratorAnd	727	UncertaintyAdjustments	none
	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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5 4-Methyl-2-pentanone

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Acknowledgement

We thank Michael Overcash for the information provided in a personal communication.

5.1 Introduction

4-Methyl-2-pentanone (CAS-No. 108-10-1, $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$, $\text{C}_6\text{H}_{12}\text{O}$, Synonyms: methyl isobutyl ketone, MIBK) is a colorless liquid with a less pronounced ketone-like odor. It is poorly soluble in water but is miscible with common organic solvents. It forms an azeotrope with water and with a large number of solvents. For example, the azeotrope with water (bp 87.9 °C) contains 75.7 wt % 4-methyl-2-pentanone, and that with n-butanol (bp 114.4 °C) 70 wt % 4-methyl-2-pentanone.

For this inventory the functional unit is 1 kg of pure 4-methyl-2-pentanone. The most important chemical and physical properties of 4-methyl-2-pentanone used in this inventory are given in Methyl cyclohexane (CAS-No. 108-87-2, C_7H_{14} , Synonyms: hexahydrotoluene, cyclohexyl methane) is a colorless transparent liquid with a faint, benzene-like odor. It is soluble in organic solvents as methanol, ethanol, and chloroform. The most important physical properties of methylcyclohexane are given here:

Tab. 31.1 Chemical and physical properties of methyl cyclohexane (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	98.183	g mol ⁻¹	
Specific gravity	769.39	kg m ⁻³	at 20 °C
Boiling point	100.934	°C	at normale pressure
Melting point	- 126.593	°C	at normale pressure
Heat of combustion, gross caloric value (HHV)	4.568	MJ kg ⁻¹	

Tab. 5.1 Chemical and physical properties of 4-methyl-2-pentanone (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	100.16	g mol ⁻¹	
Specific gravity	800.4	kg m ⁻³	at 20 °C
Boiling point	116	°C	at normale pressure
Melting point	- 80	°C	at normale pressure

5.2 Reserves and Resources

4-Methyl-2-pentanone can be produced industrially from acetone or isopropyl alcohol according to the following four processes:

1. In three steps from acetone via diacetone alcohol and mesityl oxide, with subsequent hydro-

genation.

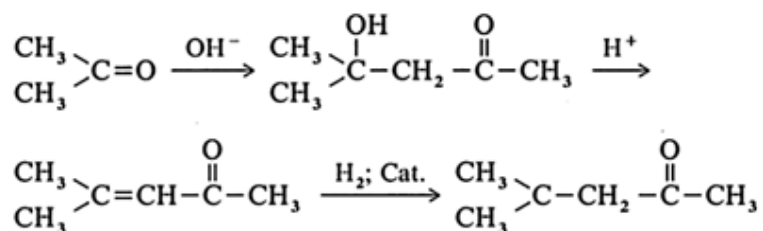
2. In two steps from acetone via mesityl oxide.
3. In one step from acetone and hydrogen.
4. As a byproduct in the dehydrogenation of isopropyl alcohol to acetone.

5.3 Production Technologies and Use

5.3.1 Production Technologies

In the *three-step process* acetone undergoes alkali-catalyzed condensation in the first step to form diacetone alcohol. Dehydration of the latter to mesityl oxide takes place in the liquid phase at 90 – 130 °C over an acid catalyst, such as phosphoric or sulfuric acid, with high selectivity. Selective hydrogenation of the carbon – carbon double bond in mesityl oxide to give 4-methyl-2-pentanone may be carried out in both the liquid phase and the gas phase with various catalysts, preferably palladium. The selectivity is excellent.

Dehydration and hydrogenation can also be combined into a single process step.



The *two-step process* is less important industrially. Acetone is converted in a single step to mesityl oxide, which is then hydrogenated to 4-methyl-2-pentanone in a second step. Catalysts such as copper chromite or zirconium phosphate are used for the condensation, and palladium on aluminum oxide for the hydrogenation. 4-Methyl-2-pentanol [108-11-2] is formed as a byproduct.

For the *single-step production* of 4-methyl-2-pentanone from acetone and hydrogen a combination of a cation exchanger with palladium is generally used as catalyst. The acetone is fed with hydrogen over the palladium-charged catalyst at 130 °C and 0.5 – 5.0 MPa. A selectivity of > 95 % at a conversion of < 50 % is thereby attained. As a result of the increasing production of acetone by dehydrogenation of isopropyl alcohol, the production of 4-methyl-2-pentanone as a byproduct in this process is becoming more important.

5.3.2 Use

The major application of 4-methyl-2-pentanone is as a solvent for vinyl, epoxy, and acrylic resins, for natural resins, and for nitrocellulose. A further use is as a solvent for dyes in the printing industry. 4-Methyl-2-pentanone is also a versatile extracting agent, e.g., for the production of antibiotics, or the removal of paraffins from mineral oil for the production of lubricating oils. The importance of 4-methyl-2-pentanone as an intermediate for synthesis is relatively low. The most important product is 4-methyl-2-pentanol, obtained by hydrogenation of the ketone. 4-Methyl-2-pentanone peroxide has some importance as a polymerization initiator for ethylene and for the hardening of unsaturated polyester resins.

5.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg 4-methyl-2-pentanone, at plant, in Europe**. For this study the production of 4-methyl-2-pentanone by means of the one step-hydrogenation of acetone is considered. All data in the present report are referred to 1 kg 4-methyl-2-pentanone 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

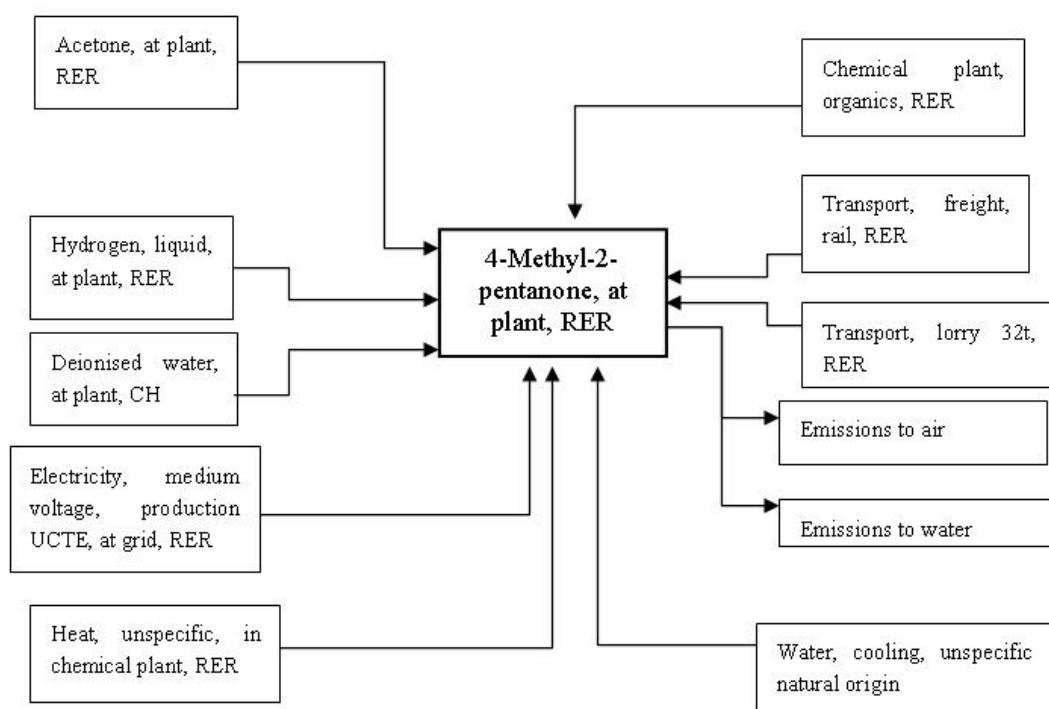


Fig. 5.1 Process flow chart for the production of 4-methyl-2-pentanone

5.5 Data sources

The data for the 4-methyl-2-pentanone production are taken from a database, which reports data from US industry production of the years 1998 to 2004 (Overcash 1998-2004). These data are taken from several US production plants and are representative for US production.

5.6 Raw materials and auxiliaries

Tab. 5.2 Consumption of raw materials and auxiliaries for the production of 1 kg 4-methyl-2-pentanone

Input	Overcash 1998 - 2004
Acetone(kg)	1.4229
H ₂ (kg)	2.16E-02
Process water (kg)	2.1208E-01
Cooling water (kg)	63.3922

5.7 Energy

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 5.3 Energy consumption for the production of 1 kg 4-methyl-2-pentanone

Input	Overcash 1998 - 2004	This study
Steam (MJ)	6.1	4.35
Electricity (kWh)	0.725	7.25E-1
Potential recovered energy (MJ)	- 1.75	
Sum of consumed energy (MJ) ²	6.96	

5.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the acetone and 50 km by lorry 32t and 100 km by train for the hydrogen. Tab. 5.4 summarises the total transport amounts for the production of 1 kg 4-methyl-2-pentanone.

Tab. 5.4 Total transport amounts for the production of 1 kg 4-methyl-2-pentanone

tkm kg ⁻¹ 4-methyl-2-pentanone	lorry	train
Total	1.4337E-01	8.559E-01

5.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of 4-methyl-2-pentanone plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

5.10 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-01 kWh/kg 4-methyl-2-pentanone is

converted to waste heat and that 100% of the waste heat is released to the air.

Overcash 1998 – 2004 report several air emissions from the production of 4-methyl-2-pentanone.

Tab. 5.5 Process emissions to air from the production of 1 kg 4-methyl-2-pentanone

Output	Value	Remarks
Acetone, to air (kg)	7.578E-02	Overcash 1998 - 2004
H ₂ , to air (kg)	2.5E-04	Overcash 1998 - 2004
Isopropanol, to air (kg)	1.0E-05	Overcash 1998 - 2004
Methyl pentane, to air (kg)	5.0E-05	Overcash 1998 - 2004
4-Methyl-2-pentanone, to air (kg)	1.074E-02	Overcash 1998 - 2004
Waste heat (MJ)	2.61	calculated from electricity input

5.11 Emissions to water

Overcash 1998 – 2004 report several water emissions from the production of 4-methyl-2-pentanone.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of the emissions in the waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 5.6.

Tab. 5.6 Process emissions to water from the production 1 kg 4-methyl-2-pentanone

Output	Value	Remarks
Acetone, to water (kg)	1.2807E-01	Overcash 1998 - 2004
Diisobutyl ketone, to water (kg)	2.462E-02	Overcash 1998 - 2004
Isopropanol, to water (kg)	1.14E-03	Overcash 1998 – 2004
Methyl pentane, to water (kg)	4.55E-03	Overcash 1998 – 2004
Methyl isobutyl carbinol, to water (kg)	3.8E-04	Overcash 1998 - 2004
4-Methyl-2-pentanone, to water (kg)	7.368E-02	Overcash 1998 - 2004
BOD, COD (kg)	5.51E-01	calculated from the water emissions
TOC, DOC (kg)	1.56E-01	calculated from the water emissions

5.12 Life cycle inventory and Data Quality Considerations

Tab. 5.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of 4-methyl-2-pentanone. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 5.7 Unit process raw data for the production of 4-methyl-2-pentanone

	InputGroup	OutputGroup	Name	Location	Infrastructure	Process	Unit	4-methyl-2-pentanone, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401											
662			Location								
493			InfrastructureProcess								
403			Unit								
	-	0	4-methyl-2-pentanone, at plant	RER	0	kg	1.00E+0				
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-	m3	6.34E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	water, deionised, at plant	CH	0	kg	2.12E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	acetone, liquid, at plant	RER	0	kg	1.42E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	hydrogen, liquid, at plant	RER	0	kg	2.16E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.43E-1	1	2.09	(4,5,na,na,na,na); Standard distances	
	5	-	transport, freight, rail	RER	0	tkm	8.56E-1	1	2.09	(4,5,na,na,na,na); Standard distances	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	4.45E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation	
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E+0	1	1.07	(1,2,1,1,1,3); Calculated from the electricity input	
	-	4	Acetone	-	-	kg	7.58E-2	1	1.51	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	Hydrogen	-	-	kg	2.50E-4	1	1.51	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	2-Propanol	-	-	kg	1.00E-5	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	2-Methyl pentane	-	-	kg	5.00E-5	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	4-Methyl-2-pentanone	-	-	kg	1.07E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
emission water, river	-	4	Acetone	-	-	kg	1.28E-1	1	3.00	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	Diisobutyl ketone	-	-	kg	2.46E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	2-Propanol	-	-	kg	1.14E-3	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	Methyl pentane	-	-	kg	4.55E-3	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	4-Methyl-2-pentanol	-	-	kg	3.80E-4	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	4-Methyl-2-pentanone	-	-	kg	7.37E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004	
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	5.51E-1	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions	
	-	4	COD, Chemical Oxygen Demand	-	-	kg	5.51E-1	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions	
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.56E-1	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions	
	-	4	TOC, Total Organic Carbon	-	-	kg	1.56E-1	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions	

5.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 5.8 shows selected LCI results and the cumulative energy demand for the production of 4-methyl-2-pentanone.

Tab. 5.8 Selected LCI results and the cumulative energy demand for the production of 4-methyl-2-pentanone

Name	Location	Unit	4-methyl-2-pentanone, at plant
	Unit Infrastructure	Unit	RER kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	107.0
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	7.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.7
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	9.8E-2
CO ₂ , fossil	air	kg	3.5E+0
NM VOC	air	kg	9.2E-2
nitrogen oxides	air	kg	8.3E-3
sulphur dioxide	air	kg	1.2E-2
particulates, <2.5 µm	air	kg	4.0E-4
BOD	water	kg	5.5E-1
cadmium	soil	kg	5.1E-10

5.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	4-methyl-2-pentanone, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 4-methyl-2-pentanone including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	4-Methyl-2-Pentanone, ab Werk
	491	Synonyms	methyl isobutyl ketone
	492	GeneralComment	The process "4-methyl-2-pentanone, at plant, RER" is modelled for the production of 4-methyl-2-pentanone from acetone in Europe. Raw materials, auxiliaries, energy consumptions and emissions are modelled with data from US production. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H12O
	501	StatisticalClassification	
	502	CASNumber	108-10-1
TimePeriod	601	StartDate	1998
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrogenation of acetone
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 380.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

5.15 References

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6 Acetic anhydride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

6.1 Introduction

Acetic anhydride (CAS-No. 108-24-7, $(\text{CH}_3\text{CO})_2\text{O}$) is a colourless liquid with a pungent odor and is strongly lachrymatory. It is miscible with polar solvents and dissolves in cold alcohol with slow decomposition. The solubility of acetic anhydride in water at 20 °C is 2.6 wt %, with slow decomposition; the solubility of water in acetic anhydride at 15 °C is 10.7 wt %, with gradual decomposition. For this inventory the functional unit is 1 kg of pure acetic anhydride. The most important chemical and physical properties of acetic anhydride used in this inventory are given here:

Tab. 6.1 Chemical and physical properties of acetic anhydride (Ullmann 2005)

Property	Value	Unit	Remarks
Molecular weight	102.09	g mol^{-1}	
Specific gravity	1083.8	kg m^{-3}	at 20 °C
Boiling point	139.5	°C	at normale pressure
Melting point	- 73.1	°C	at normale pressure

6.2 Reserves and Resources

Today, acetic anhydride is made mostly by either the ketene process or the oxidation of acetaldehyde. Production by another process, the carbonylation of methyl acetate (Halcon process), was begun in 1983. In Western Europe, 77 % of acetic anhydride is made by the ketene process and 23 % by the oxidation of acetaldehyde. Since production by the Halcon process began at the Tennessee-Eastman plant, 25 % of acetic anhydride in the United States has been made by this process and 75 % by the ketene process.

6.3 Use

Acetic anhydride is used chiefly as an acetylating and dehydrating agent; it is used on a large scale for the acetylation of cellulose. Other areas of application for acetic anhydride are the production of poly(methylacrylimide) hard foam, acetylated plastic auxiliaries, explosives, particularly hexogen production, the production of certain types of brake fluids, the production of auxiliaries for drilling fluids, the detergent industry, for the production of cold – bleaching activators, the dyeing industry, in the preparation of organic intermediates, in the production of pharmaceuticals, in the food industry, flavors and fragrances, and herbicides such as metolachlor (Dual) and alachlor (Lasso).

6.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg acetic anhydride, at plant, in Europe**. Two processes for the acetic anhydride production were considered: the ketene process and the oxidation of acetaldehyde. The following process were modelled:

- Acetic anhydride production, ketene process, RER
- Acetaldehyde oxidation, RER. Multi output process with acetic anhydride, from acetaldehyde,

- at plant, RER and acetic acid, from acetaldehyde, at plant, RER as allocated products
- Acetic anhydride, at plant, RER. Supply mix from the ketene process and the acetaldehyde oxidation process

All data in the present report are referred to 1 kg acetic anhydride 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

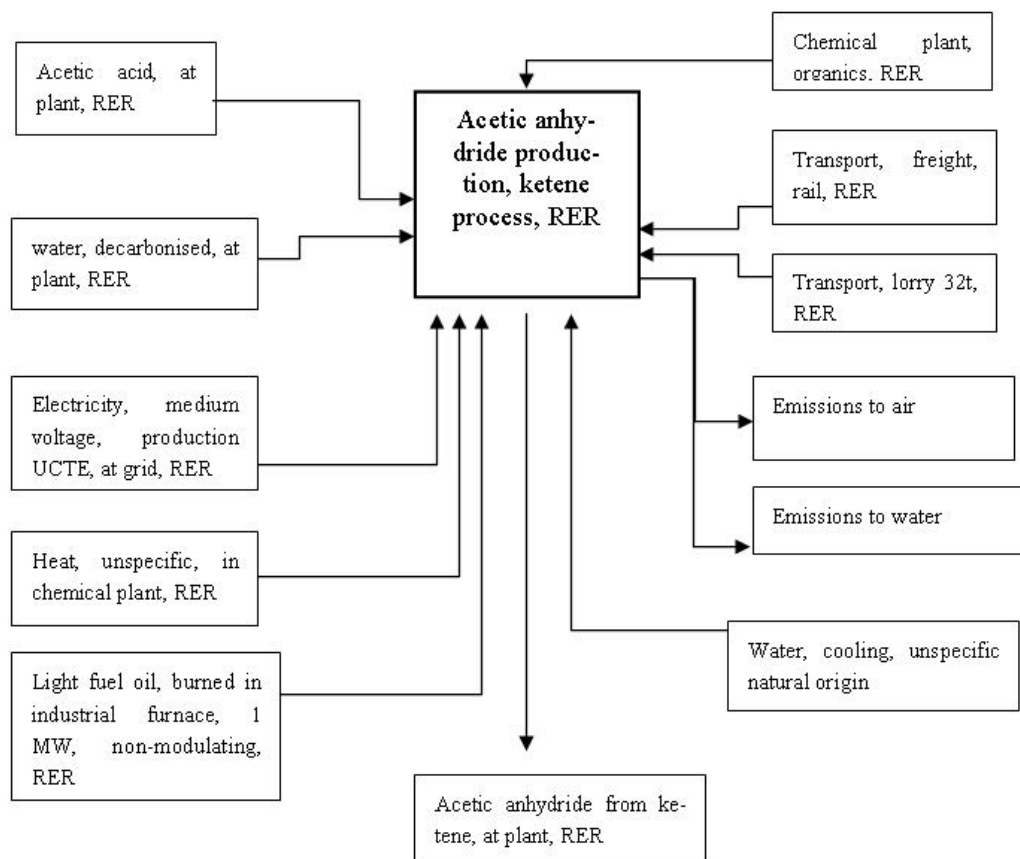


Figure 6.1 Process chain for the production of acetic anhydride (ketene process)

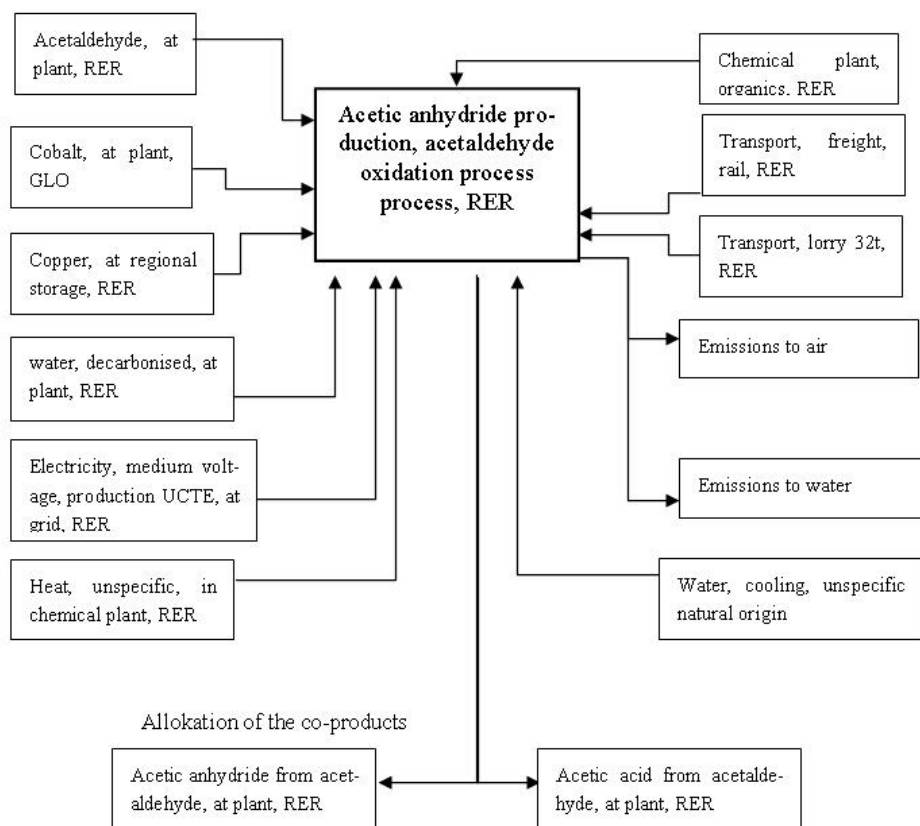


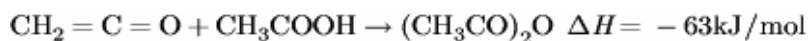
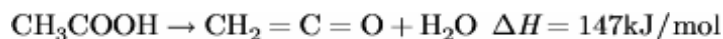
Figure 6.2 Process chain for the production of acetic anhydride (acetaldehyde oxidation process)

6.5 Acetic anhydride production, ketene process, RER

6.5.1 Production Technologies

Ketene process (Ullmann 2005)

The ketene process for the production of acetic anhydride proceeds in two steps: the thermal cleavage of acetic acid to form ketene and the reaction of ketene with acetic acid:



Production of ketene (Ullmann 2005)

Thermal Cleavage of Acetic Acid. Hot acetic acid vapor is broken into ketene and water at 700 – 750 °C in the presence of traces of phosphoric acid catalyst. The pressure in the reactor is generally reduced so that the ketene can be isolated before it reacts with acetic acid or with water. The cleavage takes place in a multicool reactor with coils made of highly heat-resistant steel alloys. The alloy Sicro-mal, containing 25 % Cr, 20 % Ni, and 2 % Si, is particularly suitable. Triethyl phosphate is employed as the catalyst.

Figure 6.3 shows how the vacuum process operates. Acetic acid is fed continuously into the evapora-

tor (a). The acetic acid vapor leaving the evaporator passes over the catalyst evaporator (b), which is supplied continuously with catalyst, and thence into the ketene oven (c). The acetic acid is heated (d) and then cleaved (e). As soon as the cracked gases have left the oven, ammonia is added to prevent the ketene from reacting with the water or with the remaining acetic acid. For the same reason, and also to remove condensable gases, the hot gas is cooled in combined water and brine condensers (f), (g). A 40 – 48 % aqueous solution of acetic acid condenses at about 0 °C and is separated (h) from the cleavage gas (0 – 10 °C). About 4 – 8 % of this acetic acid results from the reaction of the condensed acetic anhydride with water. This process has been developed by the Consortium für Elektrochemische Industrie, the research institute of Wacker-Chemie.

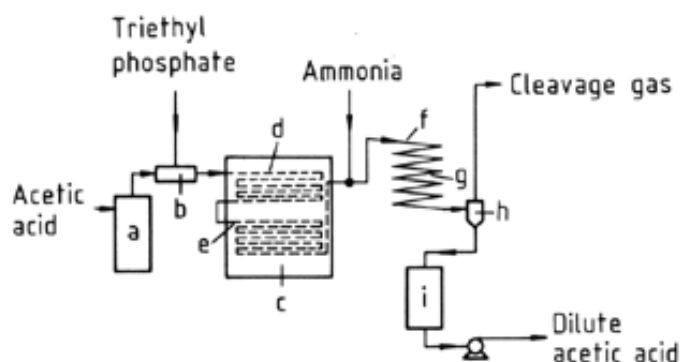


Figure 6.3 Cleavage of acetic acid by the vacuum process

a) Acetic acid evaporator; b) Catalyst evaporator; c) Ketene oven; d) Convection zone; e) Cleavage zone; f) Water cooler; g) Brine cooler; h) Separator; i) Receiver

The construction of the ketene oven depends strongly on the power required. In small ovens (ca. 50 t/month), separate preheating and cleaving ovens are preferred. The acetic acid is cleaved in a spiral tube which is usually electrically heated. Ovens of medium size (up to ca. 700 t/month), however, are often constructed as single-chamber ovens, generally heated with gas or oil. For capacities of 700 t/month, product loss is to be expected. This is because of the unsatisfactory pressure conditions in the cleavage zone. Ovens with much greater capacities work satisfactorily with three- and four-chamber systems and partial gas flows. It is also possible to operate several ovens with acetic acid supplied from a central evaporator.

Thermal Cleavage of Acetone. This variation of the ketene process (cleavage of acetone into ketene and methane) has no economic significance for the manufacture of acetic anhydride today.

Reaction of ketene with acetic acid (Ullmann 2005)

Two processes are available for the reaction of ketene with acetic acid. The classical scrubber process is being replaced gradually by the Wacker process with liquid-ring pumping. A further process, in which the reaction occurs at higher temperatures, has not yet gained industrial importance.

Scrubber Process (see Figure 6.4). The ketene emerging from the separator is absorbed by glacial acetic acid circulating in scrubbers (a) and (b), which are filled with Raschig rings. The circulation is maintained by centrifugal pumps and the heat of reaction removed by coolers (f). Most of the ketene is absorbed in scrubber (a), the rest in scrubber (b). The liquid mixture leaving scrubber (a) contains 85 – 90 % raw anhydride and is collected in (g), cooled and used in scrubber (c) to wash the off-gas from (b). The off-gas from (c) is finally washed with brine-cooled, dilute acetic acid in scrubber (d) in order to remove acid.

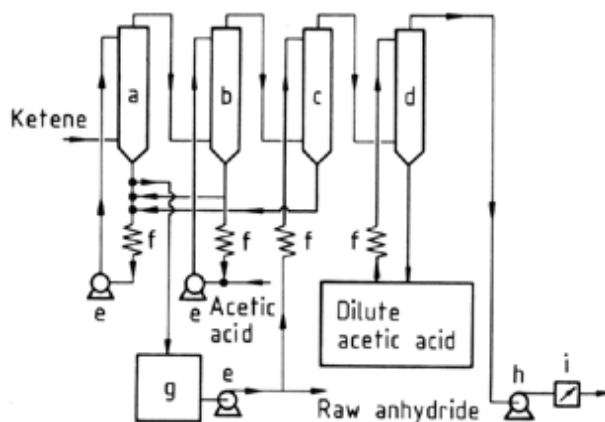


Figure 6.4 Scrubber process

a), b), c), d) Scrubbers; e) Centrifugal pumps; f) Coolers; g) Collecting vessel; h) Vacuum pump; i) Ring – balance manometer

Wacker Process (Liquid-ring pump process) (see Figure 6.5). The ketene emerging from the separator is pumped through a Nash Hytor liquid-ring pump (a) at about 13 – 20 kPa. It reacts in the pump, at about 45 – 55°C, with acetic acid. At the same time, the liquid-ring pump provides the vacuum required for the acetic acid cleavage. Reaction and compression heat are removed by passing the raw anhydride through a cooler (c). This serves simultaneously as separator for the off-gas from the cleavage reactor. The off-gas is burnt in the ketene oven after passing the measuring point (d) or is led to a flare system. The raw anhydride (90 %) is collected continuously. Part of the anhydride is returned to the reaction pump, where it is used as the reaction medium. Advantages of this process stem from the impressive simplicity of the apparatus and operation, which lead to high reliability and the possibility of producing raw anhydride of higher purity.

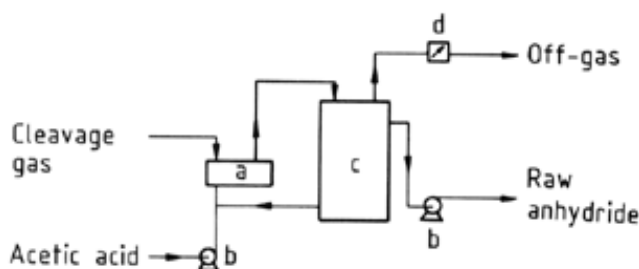


Figure 6.5 Wacker process (Liquid – ring pump process)

a) Liquid – ring pump; b) Pump; c) Cooler; d) Measuring point

Pure anhydride distillation (Ullmann 2005)

The raw anhydride can be distilled either continuously or discontinuously. In discontinuous distillation, three fractions are obtained:

Forerun (1 %) is about 40 % acetic acid; the rest consists of volatile liquids, particularly methyl acetate and acetone.

Middle run (about 10 %). Depending on the quality of the column, it contains 30 – 60 % acetic anhydride; the rest is acetic acid.

Pure Anhydride. Using discontinuous distillation at normal pressures, the purity of the anhydride can-

not exceed 99 % (the remainder is acetic acid) because, at the still temperature required, a certain amount of decomposition occurs. These difficulties can be overcome by operating under reduced pressure.

For continuous distillation, two or three columns are used, depending on the required number of fractions (forerun and middle run are separated). The middle run portion is returned to the reactor as reagent acid. The residue of the distillation contains small amounts of tarry products and is worked up continuously in a Sambay evaporator.

A newer continuous and energy-saving vacuum process works with two evaporators and only one column.

The Wacker process, described above, is also particularly suitable as a workup process. Waste acids, such as those from cellulose acetylation, can be processed after concentration without difficulty. No auxiliary materials are necessary, only energy in the form of gas, oil, or electric current. The process has been adopted by many firms. The yield is over 96 % at about 75 % cleavage.

6.5.2 Raw materials and auxiliaries

The values for the input of the raw materials and the auxiliaries are calculated as average values of the available data.

Tab. 6.2 Consumption of raw materials and auxiliaries for the production of 1 kg acetic anhydride from ketene

Input	Rudd 1981	Chauvel 1989	Wells 1991	Naran Lala 2004	This study
Acetic acid (kg)	1.25	1.3	1.25	1.28	1.27
Process water (kg)		1		0.24	0.62
Cooling water (kg)		240		505	373

6.5.3 Energy

The values for the energy inputs are calculated as average values of the available data.

In this study the use of the fuel is modelled with the modul "light fuel oil, burned in industrial furnace 1 MW, non-modulating"

Tab. 6.3 Energy consumption for the production of 1 kg acetic anhydride from ketene

Input	Chauvel 1989	Kim 2003	Naran Lala 2004	This study
Steam (MJ)	11	4.35	15.125	10.2
Electricity (kWh)	0.39	0.0695	0.175	2.83E-01
Light fuel oil (kg)	0.1643	0.042	0.2064	1.24E-01

6.5.4 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 6.4 summarises the total transport amounts for the production of 1 kg acetic anhydride.

Tab. 6.4 Total transport amounts for the production of 1 kg acetic anhydride from ketene

tkm kg ⁻¹ acetic anhydride	lorry	train
Raw materials	1.27E-01	7.62E-01

6.5.5 Infrastructure and land use

No information was readily available about infrastructure and land-use of acetic anhydride plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

6.5.6 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of acetic anhydride. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 6.5.

Tab. 6.5 Process emissions to air from the production of 1 kg acetic anhydride from ketene

Output	Value	Remarks
Acetic acid, to air (kg)	2.5E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	5.34E-1	from waster water treatment
Waste heat (MJ)	1.02	calculated from electricity input

6.5.7 Emissions to water

The remaining amount of unreacted acetic acid was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for acetic acid was assumed with 90%. The carbon contained in the removed acetic acid was accounted as CO₂ emissions to air.

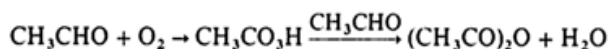
The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 6.6.

Tab. 6.6 Process emissions to water from the production 1 kg acetic anhydride from ketene

Output	Value	Remarks
Acetic acid, to water (kg)	4.05E-02	calculated from mass balance
COD, BOD (kg)	4.14E-02	calculated from water emissions
TOC, DOC (kg)	1.62E-02	calculated from water emissions

6.6 Acetaldehyde oxidation, RER

Acetic anhydride can be obtained directly by liquid-phase oxidation of acetaldehyde. The peracetic acid formed from oxygen and acetaldehyde reacts under suitable conditions with a second molecule of acetaldehyde to form acetic anhydride and water :



Rapid removal of the reaction water and the use of suitable catalysts are essential in this process. Mixtures of acetic acid and acetic anhydride are always obtained; their ratio can be varied within wide limits by changing the reaction conditions. Generally, the highest possible anhydride yield is sought.

Because of the rapid hydrolysis of acetic anhydride above 60 °C, the process is operated preferably between 40 °C and 60 °C. Suitable catalysts are combinations of metal salts. Particularly important are mixtures of manganese acetate and copper acetate, of cobalt acetate and nickel acetate, and of cobalt and copper salts of higher fatty acids. Manganese acetate should hinder the formation of explosive amounts of peracetic acid during the oxidation of acetaldehyde. For increasing the rate of oxidation, the use of pure oxygen at a pressure of several hundred kilopascals instead of air has been proposed.

The strongly exothermic reaction requires efficient cooling. For this purpose, the addition of low-boiling solvents has been found to be of assistance. Methyl and ethyl acetates are favored because they form azeotropic mixtures with water (but not with acetic acid or acetic anhydride) and hence allow a rapid, continuous separation of the water formed in the reaction. The ratio of acetic anhydride to acetic acid in the product depends on the ratio of ethyl acetate to acetaldehyde in the initial mixture.

In practice, a 1 : 2 mixture of acetaldehyde and ethyl acetate is oxidized with the addition of 0.05 to 0.1 % cobalt acetate and copper acetate at 40 °C; the ratio of Co:Cu is 1:2. The ratio of acetic anhydride to acetic acid obtained is 56:44, whereas on oxidizing in the absence of ethyl acetate this ratio is only 20:80. The optimization of other reaction conditions can also lead to an increase in the acetic anhydride-acetic acid ratio. For example, at 55 °C and atmospheric pressure, a ratio of 80:20 was achieved. At a higher temperature (62 – 90 °C, 200 – 300 kPa, acetaldehyde concentration in the final mixture of up to 40 %) a ratio of 75:25 was obtained at high aldehyde conversion.

Other suitable low-boiling solvents are methylene chloride, diisopropyl ether, cyclohexanone, or ethylidene diacetate. Nonvolatile esters also can be used as diluents, provided they do not have to be removed from the reaction zone. These include alkyl benzoates and alkyl phthalates.

The acetaldehyde oxidation is illustrated in Figure 6.6 by the process of Usines de Melle. The gas mixture containing oxygen and acetaldehyde is pumped into the reactor (a). The oxidation takes place in the liquid phase and in the presence of catalysts. The reactor effluent is sent through a water-cooled condenser (b) constructed as a separator; non-condensable gases are sent to the packed column (c). Fresh acetaldehyde is introduced at the top of this column. The condensates from both the cooler (b) and the column (c) are distilled to obtain the product. Acetaldehyde is recovered from the branch stream (d) of the non-condensable gas. The other part of the gas flow is supplemented with air and returned to the reactor.

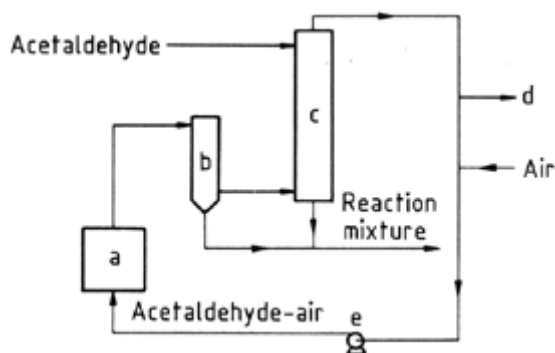


Figure 6.6 Acetaldehyde oxidation

a) Reactor; b) Condenser; c) Column; d) Branch stream (acetaldehyde recovery from non-condensable gas); e) Pump

Both towers and vessels are suitable as reactors if the heat of reaction can be dissipated. The process of Distillers Co. is shown in Figure 6.7 as an example. The off-gas contains combustible low-boiling products, such as acetaldehyde, and solvents, such as methyl acetate and ethyl acetate. These can be flared off.

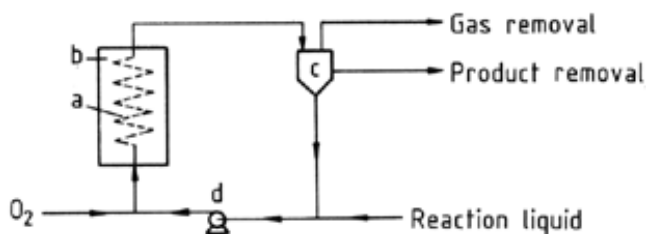


Figure 6.7 Reactor for the acetaldehyde oxidation (Distillers Co.)

a) Reactor tubing; b) Cooling bath; c) Separator; d) Circulation pump

6.6.1 Raw materials and auxiliaries

The values for the input of the raw materials and the auxiliaries are calculated as average values of the available data.

The catalyst assumed is a mixture of copper acetate and cobalt acetate with an input of 2.13E-04 kg copper from the dataset “Copper, ta regional storage, RER” and 1.07E-04 kg cobalt from the dataset “Cobalt, at plant, GLO”.

Tab. 6.7 Consumption of raw materials and auxiliaries for the acetaldehyde oxidation

Input	Rudd 1981 (per kg acetic anhydride)	Chauvel 1989 (per kg acetic anhydride)	Wells 1991 (per kg acetic anhydride)	This study (per kg acetaldehyde)
Acetaldehyde (kg)	1.08	1.1	1.2	1
Catalysts (kg)			0.001	
Copper (kg)				2.06E-04

Cobalt (kg)				1.48E-04
Ethyl acetate (kg)	0.01			8.88E-03
Process water (kg)		2		1.78
Cooling water (kg)		300		266

6.6.2 Energy

The data for the input of raw materials and water are taken from Chauvel (1989), which reports data from French production plants in the late 1980ies.

Tab. 6.8 Energy consumption for the acetaldehyde oxidation

Input	Chauvel 1989 (per kg acetic anhydride)	This study (per kg acetaldehyde)
Steam (MJ)	5.5	4.88
Electricity (kWh)	1	8.88E-01

6.6.3 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and 100 km by lorry 32t and 200 km by train for the metals used as catalysts. Tab. 6.9 summarises the total transport amounts for the acetaldehyde oxidation.

Tab. 6.9 Total transport amounts for the acetaldehyde oxidation

t km kg ⁻¹ acetaldehyde	lorry	train
Raw materials	2.78E-01	1.67

6.6.4 Infrastructure and land use

No information was readily available about infrastructure and land-use of acetaldehyde oxidation plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

6.6.5 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the acetaldehyde oxidation. As approximation the air emissions occurring from the purge vent, the disillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 6.10.

Tab. 6.10 Process emissions to air from the acetaldehyde oxidation

Output	Value	Remarks
Acetaldehyde, to air (kg)	2E-03	Estimated as 0.2 % of input
Ethyl acetate, to air (kg)	1.78E-05	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.84E-1	from waster water treatment
Waste heat (MJ)	3.2	calculated from electricity input

6.6.6 Emissions to water

The remaining amount of unreacted acetic aldehyde and the remaining ethyl acetate was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for acetaldehyde and ethyl acetate was assumed with 90%. The carbon contained in the removed acetaldehyde and ethyl acetate was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of acetaldehyde and ethyl acetate in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Other emissions to water are copper and cobalt, which are calculated from the catalysts input.

A summary of the values used in this inventory is given in Tab. 6.11.

Tab. 6.11 Process emissions to water from the acetaldehyde oxidation

Output	Value	Remarks
Acetaldehyde, to water (kg)	1.02E-02	calculated from mass balance
Ethyl acetate, to water (kg)	7.1E-06	calculated from mass balance
Copper, to water (kg)	2.06E-04	calculated from mass balance
Cobalt, to water (kg)	1.48E-04	calculated from mass balance
COD, BOD (kg)	1.78E-02	calculated from water emissions
TOC, DOC (kg)	5.57E-03	calculated from water emissions

6.6.7 Allocation of acetic anhydride and acetic acid

Within the acetaldehyde oxidation, acetic anhydride and acetic acid are obtained as co-products. According to the data from Chauvel 1989, it was assumed that in average an amount of 0.2 kg acetic acid per kg acetic anhydride is produced. As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 6.12.

Tab. 6.12 Allocation of the co-products of the acetaldehyde oxidation

Process	Acetaldehyde oxidation, RER		
	Unit	per kg acetaldehyde	Allocation factor
Acetic anhydride from acetaldehyde,	kg	0.888	83.3

at plant, RER			
Acetic acid from acetaldehyde, at plant, RER	kg	0.178	16.7

6.7 Acetic anhydride, at plant, RER

In Western Europe, 77 % of acetic anhydride is made by the ketene process and 23 % by the oxidation of acetaldehyde (Ulmann 2005). These values are applied in order to assess a European mix for the acetic anhydride production

Tab. 6.13 Acetic anhydride, at plant, RER. Processes used in the supply mix (RER)

Production process	Unit	Acetic anhydride, at plant, RER
Acetic anhydride from ketene, at plant, RER	kg	0.77
Acetic anhydride from acetaldehyde, at plant, RER	kg	0.23
Total	kg	1

6.8 Life cycle inventory of acetic anhydride production and data quality Considerations

Tab. 6.14 shows the life cycle inventory and the data quality indicators for the ketene process.

Tab. 6.15 shows the life cycle inventory and the data quality indicators for the acetaldehyde oxidation process.

Tab. 6.16 shows the life cycle inventory and the data quality indicators for the production mix of acetic anhydride.

The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 6.14 Unit process raw data for the ketene process

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	acetic anhydride from ketene, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER			
							0			
							kg			
401										
662			Location							
493			InfrastructureProcess							
403			Unit							
resource, in water technosphere	-	0	acetic anhydride from ketene, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.73E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	1.27E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	water, deionised, at plant	CH	0	kg	6.20E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.27E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.62E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.83E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.02E+1	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	1.24E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.02E+0	1	1.51	(1,3,5,3,1,3); Calculated from the electricity input
	-	4	Acetic acid	-	-	kg	2.54E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	5.34E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Acetic acid	-	-	kg	4.05E-2	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	4.14E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	4.14E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.62E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	1.62E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

Tab. 6.15 Unit process raw data for the acetaldehyde oxidation

	401	InputGroup	OutputGro	Name	Location	Infrastructu	reProcess	Unit	acetaldehyde oxidation process	Uncertainty	StandardD	eviation95	%	GeneralComment	acetic	acetic
															anhydride from	acid
															acetaldehyde, from	from
															at plant	acetaldehyde, at plant
	662			Location					RER						RER	RER
	493			InfrastructureProcess					0						0	0
	403			Unit					kg						kg	kg
allocated products resource, in water technosphere	-	2		acetic anhydride from acetaldehyde, at plant	RER	0		kg	8.88E-1						100	0
	-	2		acetic acid from acetaldehyde, at plant	RER	0		kg	1.78E-1						0	100
	4	-		Water, cooling, unspecified natural origin	-	-		m3	2.66E-1	1	1.51		(1,3,5,3,1,3); Chauvel 1989	83.333	16.667	
	5	-		acetaldehyde, at plant	RER	0		kg	1.00E+0	1	1.51		(1,3,5,3,1,3); Wells 1991	83.333	16.667	
	5	-		water, deionised, at plant	CH	0		kg	1.78E+0	1	1.51		(1,3,5,3,1,3); Chauvel 1989	83.333	16.667	
	5	-		ethyl acetate, at plant	RER	0		kg	8.88E-3	1	1.51		(1,3,5,3,1,3); Wells 1991	83.333	16.667	
	5	-		copper, at regional storage	RER	0		kg	2.06E-4	1	1.51		(1,3,5,3,1,3); Wells 1991	83.333	16.667	
	5	-		cobalt, at plant	GLO	0		kg	1.48E-4	1	1.51		(1,3,5,3,1,3); Wells 1991	83.333	16.667	
	5	-		transport, lorry >16t, fleet average	RER	0		tkm	2.78E-1	1	2.09		(4,5,na,na,na,na); Standard distance 100km	83.333	16.667	
	5	-		transport, freight, rail	RER	0		tkm	1.67E+0	1	2.09		(4,5,na,na,na,na); Standard distance 600km	83.333	16.667	
	5	-		electricity, medium voltage, production UCTE, at grid	UCTE	0		kWh	8.88E-1	1	1.51		(1,3,5,3,1,3); Chauvel 1989	83.333	16.667	
	5	-		heat, unspecific, in chemical plant	RER	0		MJ	4.88E+0	1	1.51		(1,3,5,3,1,3); Chauvel 1989	83.333	16.667	
	5	-		chemical plant, organics	RER	1		unit	4.26E-10	1	3.09		(4,5,na,na,na,na); Estimation	83.333	16.667	
	emission air, high population density	-	4		Heat, waste	-	-		MJ	3.20E+0	1	1.51		(1,3,5,3,1,3); Calculated from the electricity input	83.333	16.667
		-	4		Carbon dioxide, fossil	-	-		kg	1.84E-1	1	1.30		(4,5,na,na,na,na); from water treatment	83.333	16.667
-		4		Acetaldehyde	-	-		kg	2.00E-3	1	1.62		(4,5,na,na,na,na); Estimation	83.333	16.667	
-		4		Ethyl acetate	-	-		kg	1.78E-5	1	1.30		(4,5,na,na,na,na); Estimation	83.333	16.667	
emission water, river	-	4		Acetaldehyde	-	-		kg	1.02E-2	1	1.30		(4,5,na,na,na,na); Estimation	83.333	16.667	
	-	4		Ethyl acetate	-	-		kg	7.10E-6	1	1.30		(4,5,na,na,na,na); Estimation	83.333	16.667	
	-	4		Copper, ion	-	-		kg	2.06E-4	1	3.09		(4,5,na,na,na,na); Estimation	83.333	16.667	
	-	4		Cobalt	-	-		kg	1.48E-4	1	3.09		(4,5,na,na,na,na); Estimation	83.333	16.667	
	-	4		BOD5, Biological Oxygen Demand	-	-		kg	1.78E-2	1	1.62		(4,5,na,na,na,na); from water emissions	83.333	16.667	
	-	4		COD, Chemical Oxygen Demand	-	-		kg	1.78E-2	1	1.62		(4,5,na,na,na,na); from water emissions	83.333	16.667	
	-	4		DOC, Dissolved Organic Carbon	-	-		kg	5.57E-3	1	1.62		(4,5,na,na,na,na); from water emissions	83.333	16.667	
	-	4		TOC, Total Organic Carbon	-	-		kg	5.57E-3	1	1.62		(4,5,na,na,na,na); from water emissions	83.333	16.667	

Tab. 6.16 Unit process raw data for the production mix of acetic anhydride

technosphere	401	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	acetic anhydride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
	662			Location				RER			
	493			InfrastructureProcess				0			
	403			Unit				kg			
		-	0	acetic anhydride, at plant	RER	0	kg	1.00E+0			
		5	-	acetic anhydride from acetaldehyde, at plant	RER	0	kg	2.30E-1	1	1.06	(1,1,2,1,1,1); Ullmann 2005
		5	-	acetic anhydride from ketene, at plant	RER	0	kg	7.70E-1	1	1.06	(1,1,2,1,1,1); Ullmann 2005

6.9 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 6.17 shows selected LCI results and the cumulative energy demand for the production acetic anhydride.

Tab. 6.17 Selected LCI results and the cumulative energy demand for the production of acetic anhydride

Name	Location Unit Infrastructure	Unit	acetic acid from acetaldehyde, at plant	acetic anhydride from acetaldehyde, at plant	acetic anhydride from ketene, at plant	acetic anhydride, a plant
			RER kg 0	RER kg 0	RER kg 0	RER kg 0
LCIA results						
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	62.4	62.4	79.0	75.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.1	6.1	10.0	9.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8	0.8	1.2	1.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8	0.8	1.4	1.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1	0.2	0.2
selected LCI results						
land occupation	resource	m2a	1.9E-1	1.9E-1	2.8E-1	2.6E-1
CO2, fossil	air	kg	2.3E+0	2.3E+0	3.5E+0	3.2E+0
NM VOC	air	kg	1.4E-2	1.4E-2	1.4E-2	1.4E-2
nitrogen oxides	air	kg	4.7E-3	4.7E-3	5.4E-3	5.2E-3
sulphur dioxide	air	kg	5.5E-3	5.5E-3	7.9E-3	7.3E-3
particulates, <2.5 um	air	kg	4.0E-4	4.0E-4	5.6E-4	5.2E-4
BOD	water	kg	2.0E-2	2.0E-2	5.0E-2	4.3E-2
cadmium	soil	kg	8.9E-10	8.9E-10	1.4E-9	1.3E-9

6.10 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	acetic anhydride from ketene, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of acetic anhydride from ketene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Essigsäureanhydrid aus Keten, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "acetic anhydride from ketene, at plant, RER" is modelled for the production of acetic anhydride from ketene in Europe. Raw materials and energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H6O3
	501	StatisticalClassification	
	502	CASNumber	108-24-7
TimePeriod	601	StartDate	1981
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Reaction of ketene with acetic acid
Representative	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

ReferenceFunction	401	Name	acetaldehyde oxidation process
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Direct hydration of ethylene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Azetalddehyd Oxidierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "acetaldehyde oxidation process" delivers the co-products acetic anhydride from acetaldehyde and acetic acid from acetaldehyde. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1981
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Oxidation of acetaldehyde
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

ReferenceFunction	401	Name	acetic anhydride, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Acetic anhydride production mix
	404	Amount	1
	490	LocalName	Essigsäureanhydrid, ab Werk
	491	Synonyms	
	492	GeneralComment	Acetic anhydride production mix in Europe
	494	InfrastructureIncluded	
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H6O3
	501	StatisticalClassification	
	502	CASNumber	108-24-7
TimePeriod	601	StartDate	2005
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Production mix
Representative	722	Percent	10
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	No extrapolations
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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7 Benzal chloride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

7.1 Introduction

Benzal chloride (CAS-No. 98-87-3, $C_6H_5CHCl_2$; Synonyms: dichloromethylbenzene, dichlorotoluene, benzylidene chloride) is a liquid which fumes in moist air and which has a pungent odor and a strong irritant effect on the mucous membranes and eyes. Benzal chloride is freely soluble in alcohol, ether, chloroform, and carbon tetrachloride, but only slightly soluble in water (0.05 g/L at 5 °C; 0.25 g/L at 39 °C).

For this inventory the functional unit is 1 kg of pure benzal chloride. The most important chemical and physical properties of benzal chloride used in this inventory are given here:

Tab. 7.1 Chemical and physical properties of benzal chloride (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	161.03	$g\ mol^{-1}$	
Specific gravity	1253.6	$kg\ m^{-3}$	at 20 °C
Boiling point	205.2	°C	at normale pressure
Melting point	- 16.2	°C	at normale pressure

7.2 Reserves and Resources

Benzal chloride is produced exclusively by the side-chain chlorination of toluene.

7.3 Production Technologies and Use

7.3.1 Production Technologies

Benzal chloride is produced exclusively by side-chain chlorination of toluene. The preferred chlorination processes are the same as for benzal chloride.

Substitution of chlorine for hydrogen in the aliphatic side-chain occurs by way of a radical chain mechanism; by contrast, chlorine substitution on the aromatic ring takes place according to an electrophilic polar mechanism.

This chlorination is highly exothermic (96 – 105 kJ/mol chlorine). In view of the high rate of chlorine radical formation and of hydrogen displacement, it is not surprising that, depending on the substrate and reaction conditions, radical chain lengths of 103 to 106 have been found. Because the mechanisms of side-chain chlorination and nuclear chlorination are fundamentally different, selectivity can be readily achieved. The prerequisites for high side-chain chlorination efficiency are as follows:

- 1) Achievement and maintenance of an optimal radical concentration
- 2) Elimination of components that might impart an electrophilic course to the reaction
- 3) Elimination of components capable of terminating the radical chains

4) Elimination of components conducive to other side reactions

5) In general, taking those precautions, which would encourage radical reactions or suppress electrophilic reactions

The discussion which follows considers each of these points separately.

1) Chlorine radical formation can be promoted by the addition of a radical-forming agent such as 2,2'-azobis(isobutyronitrile) (AIBN), benzoyl peroxide, or hexaphenylethane. Such compounds are consumed in the reaction, however, and thus have to be added repeatedly. For this reason, two other methods, singly or in combination, are of greater importance, particularly in industrial chlorinations: irradiation (ultraviolet light, γ -radiation) and the use of an elevated temperature (100 – 200 °C). In both cases, the effect is a result of excitation of chlorine molecules.

Irradiation is generally carried out with mercury vapor lamps that emit light in the wavelength range 300 – 500 nm, corresponding to the region in which the absorption bands of chlorine lie (hence, its yellowish-green color). The effects of varying the wavelength have been investigated in detail. A solution containing 0.1 % chlorine absorbs 90 % of the available light energy over a distance of only 12 cm. At a chlorine concentration of 3 – 4 mol/L, the corresponding distance is reduced to 0.01 cm. Consequently, thorough mixing of the reactor contents is advantageous, as is the use of reactors of relatively small diameter. These considerations apply regardless of whether the sources of radiation are inside the reactor or external to it. In the case of bubble column reactors, the height of the reactor is also important in view of the velocity of the rising gas bubbles, which is ca. 20 cm/s.

2) Friedel-Crafts catalysts favor nuclear chlorination. For this reason, particular attention must be paid to ensure, for example, that the starting toluene is free of dissolved iron salts and that all rust particles have been filtered. By extension, steel, including stainless steel, is unsuitable as a reactor material. The preferred reactor materials are glass, enamel, and polytetrafluoroethylene.

Extensive purification of technical-grade starting materials is very expensive, however. For this reason, numerous suggestions have been made for additives that might inhibit the undesirable effects of heavy metals. Most such additives are assumed to complex with, e.g., iron cations. Some additives are reported to accelerate the side-chain chlorination, thereby inhibiting in a relative sense nuclear chlorination. The following are examples of additives which have been recommended at concentrations of 0.1 – 2 %: pyridine and phenylpyridines, alkylene polyamines, hexamethylenetetramine, phosphoramides, acid amides, alkyl and dialkyl acid amides, ureas, phosphorus chlorides, alkyl phosphites and phosphines, phosphorus trichloride and trialkyl phosphates, cyclic thioureas, lactams, red phosphorus, and aminoethanol.

3) Oxygen, a well-known radical scavenger, greatly reduces the consumption of chlorine; its presence is therefore undesirable. Toluene can be freed of oxygen, for example, purging with an inert gas. The use of distilled chlorine, likewise, is an effective means of preventing oxygen from being present at significant concentrations.

It has also been reported, however, that even chlorine which contains up to 2 % oxygen is suitable for side-chain chlorination at relatively low temperatures (40 – 50 °C) provided that phosphorus trichloride (1 – 3 %) is present.

4) The presence of water leads to the formation of aqueous hydrochloric acid (indicated by cloudiness) and possibly to the hydrolysis of the chlorinated toluene; in addition, hydrolysis products which arise may be further altered by chlorination. The hydrogen chloride that is inevitably formed during chlorination is thought to be responsible for the fact that nuclear chlorination can never be entirely suppressed.

5) Various authors have drawn attention to the adverse effects caused by excessive chlorine concentrations (especially nuclear chlorination, both addition and substitution). Besides ensuring that the level of irradiation is adequate for rapid reaction, it may be useful to introduce an inert gas (N₂, HCl) into the chlorine stream. Dividing the reactor into smaller units is apparently also advantageous, particularly in the case of a continuous process. Much attention has also been paid to the development of ki-

netic models, to the design of reactors, to the effects of agitators, to the distribution of radiation, and to the effects of inhibitors.

In the course of radical chlorination of toluene, all three hydrogen atoms of the side-chain are successively replaced by chlorine. As a result, mixtures of the three expected compounds are obtained: benzyl chloride, benzal chloride, and benzotrichloride.

Today, most side-chain chlorination of toluene to produce benzal chloride is apparently carried out in a continuous fashion. The silver- or lead-lined reaction columns described in the past represent an obsolete technology since they are subject to considerable corrosion. For photochlorination, it is now customary to use reactors of enamel or glass, e.g., of the loop type. The operating techniques have remained fundamentally unchanged, however.

7.3.2 Use

Benzal chloride is used almost exclusively to produce benzaldehyde.

7.4 System Characterisation

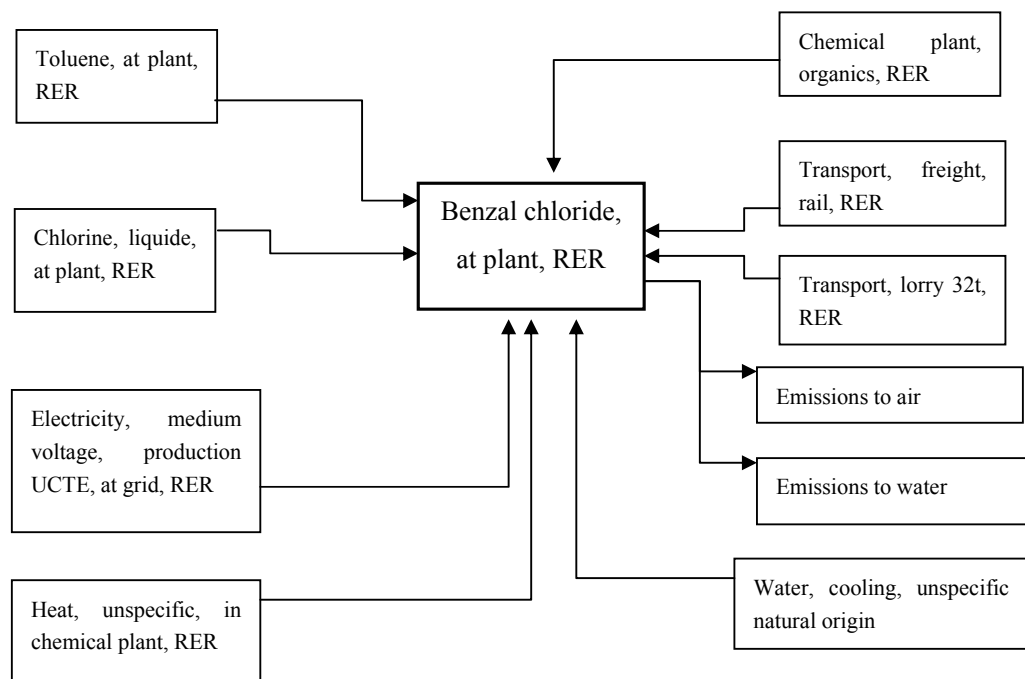


Fig. 7.1 Process flow chart for the production of benzal chloride

This report corresponds to the module in the ecoinvent database for the production of **1 kg benzal chloride, at plant, in Europe**. For this study the production of benzal chloride by means of the chloriantion of toluene is considered. All data in the present report are referred to 1 kg benzal chloride 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

7.5 Raw materials and auxiliaries

As there was no data available for the consumption of the raw materials for the production of benzyl chloride, the values are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the benzal chloride production.

Tab. 7.2 Consumption of raw materials and auxiliaries for the production of 1 kg benzal chloride

Input	Values	Remarks
Toluene (kg)	0.605	calculated with a yield of 95 %
Chlorine (kg)	0.921	calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

7.6 Energy

Gwalior (2005) gives data for the energy consumption of benzal chloride production in an Indian plant.

Tab. 7.3 Energy consumption for the production of 1 kg benzal chloride

Input	Gwalior 2005
Steam (MJ)	4.4825
Electricity (kWh)	2.027E-01

7.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the toluene and 50 km by lorry 32t and 100 km by train for the chlorine. Tab. 7.4 summarises the total transport amounts for the production of 1 kg benzal chloride.

Tab. 7.4 Total transport amounts for the production of 1 kg benzal chloride

tkm kg ⁻¹ benzal chloride	lorry	train
Total	1.07E-01	4.6E-01

7.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of benzal chloride plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of

fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

7.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of benzal chloride. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 7.5.

Tab. 7.5 Process emissions to air from the production of 1 kg benzal chloride

Output	Value	Remarks
Toluene, to air (kg)	1.21E-03	Estimated as 0.2 % of input
Chlorine, to air (kg)	1.84E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	8.75E-2	from waster water treatment
Waste heat (MJ)	7.297E-01	calculated from electricity input

7.10 Emissions to water

The remaining amount of unreacted toluene (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for was assumed with 90%. The carbon contained in the removed toluene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of toluene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in The remailing amount of the unreacted chlorine (4.8 % of the input) and the chloride contained in the hydrogen chloride byproduct was assumed to leave the the production process as chloride emission in the waste water. The value for AOX is calculated from the chloride emissions.

Tab. 7.6.

The remailing amount of the unreacted chlorine (4.8 % of the input) and the chloride contained in the hydrogen chloride byproduct was assumed to leave the the production process as chloride emission in the waste water. The value for AOX is calculated from the chloride emissions.

Tab. 7.6 Process emissions to water from the production 1 kg benzal chloride

Output	Value	Remarks
Toluene, to water (kg)	2.91E-03	calculated from mass balance
Chloride, to water (kg)	4.82E-01	calculated from mass balance
COD, BOD (kg)	8.7E-03	calculated from water emissions
TOC, DOC (kg)	2.7E-03	calculated from water emissions
AOX (kg)	4.82E-01	calculated from water emissions

7.11 Life cycle inventory and Data Quality Considerations

Tab. 7.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of benzal chloride. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 7.7 Unit process raw data for the production of benzal chloride

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	benzal chloride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	benzal chloride, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	toluene, liquid, at plant	RER	0	kg	6.05E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	chlorine, liquid, production mix, at plant	RER	0	kg	9.21E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.07E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	4.55E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.03E-1	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	4.48E+0	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	7.30E-1	1	1.14	(1,3,1,5,1,3); Calculated from the electricity input
	-	4	Toluene	-	-	kg	1.21E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Chlorine	-	-	kg	1.84E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	8.75E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Toluene	-	-	kg	2.91E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Chlorine	-	-	kg	4.82E-1	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	8.70E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	8.70E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.70E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.70E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	4.82E-1	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

7.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 7.8 shows selected LCI results and the cumulative energy demand for the production of benzal chloride.

Tab. 7.8 Selected LCI results and the cumulative energy demand for the production of benzal chloride

Name		benzal chloride, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	59.6
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	9.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	2.1E-1
CO ₂ , fossil	air	kg	2.4E+0
NM VOC	air	kg	3.0E-3
nitrogen oxides	air	kg	4.3E-3
sulphur dioxide	air	kg	6.0E-3
particulates, <2.5 µm	air	kg	4.5E-4
BOD	water	kg	1.0E-2
cadmium	soil	kg	1.4E-9

7.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	benzal chloride, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of benzal chloride including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Benzalchlorid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "benzal chloride, at plant, RER" is modelled for the production of benzal chloride from toluene in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H6Cl2
	501	StatisticalClassification	
	502	CASNumber	98-87-3
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Chlorination of toluene
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

7.14 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007)

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8 Benzaldehyde

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

8.1 Introduction

Benzaldehyde (CAS-No. 100-52-7, C₇H₆O) is a colorless and highly refractive liquid with an odor of bitter almonds. Like many other aromatic aldehydes, it is volatile in steam.

For this inventory the functional unit is 1 kg of pure benzaldehyde. The most important chemical and physical properties of benzaldehyde used in this inventory are given here:

Tab. 8.1 Chemical and physical properties of benzaldehyde (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	106.13	g mol ⁻¹	
Specific gravity	1063	kg m ⁻³	at 20 °C
Boiling point	179	°C	at normale pressure
Melting point	-56	°C	at normale pressure

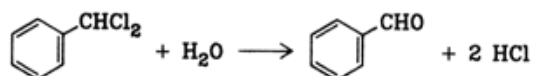
8.2 Reserves and Resources

Benzaldehyde is produced principally by the hydrolysis of benzal chloride or the partial oxidation of toluene. There are various other manufacturing processes, but at present they have no industrial importance.

8.3 Production Technologies and Use

8.3.1 Production Technologies

The hydrolysis of benzal chloride, which is readily obtainable by side-chain chlorination of toluene, is among the oldest industrial processes for the production of benzaldehyde. It can be carried out either in an alkaline or in an acidic medium.



Hydrolysis under basic conditions can be carried out with calcium hydroxide, calcium carbonate, sodium hydrogencarbonate, or sodium carbonate. Excessive alkalinity increases the probability of side reactions. The preferred alkaline saponifying agent is sodium carbonate.

According to a fairly old process, benzal chloride is saponified with a small excess of 15 % sodium carbonate solution at 138 °C. The reactor is a carbon steel agitator vessel lined internally with Hertsite (a phenolic resin) and having an agitator made of a copper – silicon alloy. The cooler, separator, and pumps are made of Karbate (a graphitic material). On completion of the saponification, the benzaldehyde is isolated from the reaction mixture by steam distillation and distilled in a stainless steel column at 2 – 3 kPa (20 – 30 mbar). The chlorine content of the distilled product is less than 0.01 %.

Instead of pure benzal chloride, a mixture of benzal chloride and benzotrichloride, as obtained industrially in the side-chain chlorination of toluene, can be hydrolyzed. Under the reaction conditions, benzotrichloride is saponified to sodium benzoate, which remains in the aqueous alkaline solution. It

is converted separately to benzoic acid.

In a new continuous process, benzal chloride and the alkaline hydrolyzing agent are reacted as countercurrents in a flow reactor in an unreactive organic solvent (e.g., toluene, xylene). A temperature of 125 – 145 °C and a pressure of 1200 – 1800 kPa (12 – 18 bar) are maintained. In the extraction zone the dissolved aldehyde is extracted from the aqueous alkaline phase with an unreactive organic solvent flowing in the opposite direction. Water, likewise as a countercurrent, washes the solution of the crude benzaldehyde in the washing zone. The flow reactor, extraction zone, and washing zone are integrated in a special apparatus. The wash water and extract are returned to the hydrolysis zone in order to recover benzaldehyde.

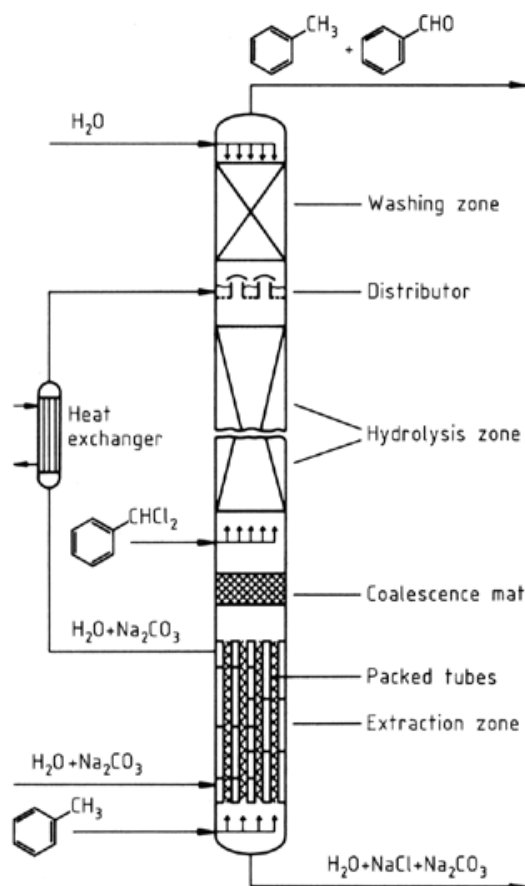


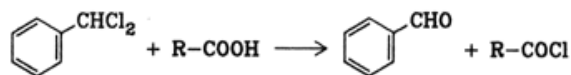
Fig. 8.1 Hydrolysis of benzal chloride to benzaldehyde in a flow-through reactor

Benzal chloride can also be converted into benzaldehyde by boiling with aqueous solutions of hexamethylenetetramine. Because benzyl chloride also reacts with hexamethylenetetramine to give benzaldehyde (Sommelet reaction), industrial mixtures of benzyl chloride and benzal chloride can be used to form benzaldehyde in this way.

The acid hydrolysis of benzal chloride is carried out in the presence of acids and with metal salts as catalysts. It gives very high yields of benzaldehyde (more than 90 %). Hydrogen chloride is formed simultaneously and is usually passed through an absorption unit to be recovered as concentrated hydrochloric acid.

The hydrolysis of benzal chloride was at one time commonly carried out in the presence of concentrated sulfuric acid. This process has the disadvantage that large amounts of dilute sulfuric acid are formed as a waste product. It can be used for the hydrolysis of certain substituted benzaldehydes that are difficult to saponify in other ways. Benzal chloride is also saponified in the presence of phosphoric

acid or sulfonic acids, hydrochloric acid, and formic acid. The reaction of benzal chloride with organic acids, such as acetic acid, in the presence of a tin(II) or tin(IV) chloride catalyst, gives benzaldehyde and the corresponding acid chloride.



R = alkyl, aryl, or cycloalkyl

The hydrolysis of benzal chloride is catalyzed by metal salts, preferably those of iron or zinc. Water should not accumulate in the reaction mixture during the reaction because it would reduce the activity of the catalyst.

Two parts by weight of anhydrous zinc chloride are added to 700 parts by weight of distilled benzal chloride, stirred, and heated to 105 – 110 °C. The mixture is maintained at 100 – 120 °C, while 85 parts by weight of water are added at the rate at which water is consumed. After all the water has been added, the reaction temperature is maintained for one hour to ensure removal of the liberated hydrogen chloride. The yield of crude benzaldehyde is 95 %, and the purity of the undistilled product is greater than 96.5 %.

The catalyst is not deactivated by a momentary excess of water if insoluble zinc catalysts, such as zinc phosphate or zinc laurate, are used. Tin chlorides and copper(II) chloride are also catalytic. The hydrolysis of benzal chloride can be carried out batchwise or continuously, e.g., in a cascade consisting of several reactors.

A new continuous process hydrolyzes benzal chloride in the vapor phase at temperatures of 100 – 300 °C. The reaction is catalyzed by activated carbon which has been treated with an acid, such as sulfuric acid, or impregnated with a metal chloride, such as iron(III) chloride, or with a sulfate, such as copper(II) sulfate. The benzaldehyde yield is 97 %. This process is particularly suitable for the hydrolysis of trifluoromethyl-substituted benzal chlorides, which are otherwise difficult to convert into the corresponding benzaldehydes. In a similar process, benzal chloride in the vapor phase at 300 °C is hydrolyzed to benzaldehyde using a silicon dioxide or aluminum oxide catalyst.

Oxidation of Toluene (Ullmann 2004)

The partial oxidation of toluene with oxygen to give benzaldehyde can be carried out in either the gas phase or the liquid phase. Benzaldehyde itself is easily further oxidized to benzoic acid and other products. Conditions must therefore be carefully chosen to favor only partial oxidation.

In the gas phase, the oxidation is carried out by passing toluene vapor, together with oxygen in a gaseous mixture such as air, through a catalyst bed in a tube-bundle or fluidized-bed reactor at a temperature of 250 – 650 °C. The reaction is highly exothermic, and effective cooling is necessary. It is advisable to dilute the mixture of toluene vapor and oxygen-containing gas with an unreactive gas, such as water vapor, nitrogen, or carbon dioxide. The benzaldehyde yield is favored by a low conversion rate (10 – 20 %) per pass, a short residence time (0.1 – 1.0 s), and a precise adjustment of the amount of oxygen. Even then, it is only 40 – 60 % of the theoretical yield based on toluene. The oxidation of toluene in the gas phase produces maleic anhydride, citraconic anhydride, phthalic anhydride, anthraquinone, cresol, acetic acid, and other compounds, in addition to substantial quantities of benzoic acid, carbon monoxide, and carbon dioxide. The complete combustion of the toluene is retarded by applying high pressure or by adding potassium sulfate or sodium fluoride to the catalyst.

Oxides of the elements of Groups V and VI of the periodic table are frequently used as catalysts, sometimes with other oxides. These oxide catalysts contain molybdenum and at least one additional element from among iron, nickel, cobalt, antimony, bismuth, vanadium, phosphorus, samarium, tantalum, tin, and chromium. Some have palladium and phosphoric acid on activated carbon. Others are mixed oxides of silver and transition metals, such as cobalt nickel, vanadium, platinum, or palladium,

and one is a mixture of silver vanadate and iron vanadate.

Another process uses a mixed-oxide catalyst that has silicon dioxide as a carrier and contains uranium, copper, iron, phosphorus, tellurium, and lead, in addition to molybdenum. At a toluene consumption of 35 – 50 % per pass, the selectivity of the benzaldehyde formation at reaction temperatures of 475 – 550 °C is 40 – 70 %. Catalysts based only on mixed oxides of molybdenum and uranium require reaction temperatures of around 600 °C.

More important than gas-phase oxidation for the production of benzaldehyde is the oxidation of toluene in the liquid phase by oxygen in the form of air or other gaseous mixtures. This is carried out at 80 – 250 °C, preferably in the presence of cobalt, nickel, manganese, iron, or chromium compounds (alone or in combination) as catalysts. Lead compounds, ruthenium compounds, and thallium salts of organic acids are also catalytic. A pressure at which the reaction medium remains liquid is chosen. Alkali, alkaline earth, and tertiary alkyl hypochlorites have been used as oxidation promoters in place of the usual bromides. These promoters may cause corrosion, however.

Toluene can also be oxidized in an alcohol, e.g., methanol, an aliphatic aldehyde, e.g., acetaldehyde, or an acid, e.g., benzoic acid or acetic acid. Water can also be added to the reaction, but one process improves the benzaldehyde selectivity by continuously removing the water formed in the oxidation reaction. When a cobalt catalyst is used and conversion rates of up to about 25 % are achieved, this process gives aldehyde selectivities of 40 – 80 %.

In another process, the selectivity of the benzaldehyde formation is improved by liquid-phase oxidation of the toluene with oxygen in the presence of a phosphoric acid – palladium catalyst that also contains certain phosphorus, sulfur, or nitrogen modifiers. For an oxygen consumption of 63 %, the yield of benzaldehyde is 41 %.

In the large-scale production of phenol by the Dow process and of ϵ -caprolactam by the Snia Viscosa process via the toluene – benzoic acid synthesis route, considerable amounts of benzaldehyde are formed as a by-product. Because a large proportion of this byproduct is worked up into pure benzaldehyde, oxidation processes intended exclusively to produce benzaldehyde are seldom used industrially at the present time.

Even when catalytic liquid-phase oxidation of toluene is carried out for the express purpose of manufacturing benzaldehyde, substantial quantities of benzoic acid are formed. Depending on the reaction conditions, carbon dioxide, carbon monoxide, formaldehyde, formic acid, benzyl alcohol, benzyl hydroperoxide, biphenyl, methylbiphenyls, and other compounds are formed also as byproducts. The crude benzaldehyde is normally refined by distillation at reduced pressure in a stainless steel column.

Benzaldehyde produced by oxidation of toluene often contains small amounts of impurities that are difficult to remove by distillation and which discolor the product. To remove these impurities the benzaldehyde can be treated with aqueous alkali or with water and a base metal such as zinc powder. A color-stable product can be obtained by treatment with oxidizing agents, e.g., hydrogen peroxide, followed by distillation.

Those processes in which toluene is oxidized with other agents, such as manganese dioxide in sulfuric acid, sodium persulfate, chromium(VI) oxide in acetic anhydride, or chromyl chloride (Etard reaction), cause waste water disposal problems and are without industrial importance.

8.3.2 Use

Benzaldehyde is an important starting material for the manufacture of odorants and flavors. The compound is responsible for the odor of natural bitter almond oil and is incorporated directly in perfumes, soaps, foods, drinks, and other products. Substantial amounts are used in the production of derivatives that are also employed in the perfume and flavor industries. Another application of benzaldehyde is the production of triphenylmethane dyes. For example, the leuco base of Malachite Green is obtained by condensation of benzaldehyde with dimethylaniline. The acridine dye benzoflavin is obtained from

benzaldehyde and m-toluenediamine. In the pharmaceutical industry benzaldehyde is used as an intermediate in the manufacture of chloramphenicol, ephedrin, ampicillin, diphenylhydantoin, and other products.

Other important chemical intermediates obtained from benzaldehyde are benzoin, benzylamine, benzyl alcohol, mandelic acid, and benzylideneacetone. Benzaldehyde is used in photochemistry, as a corrosion inhibitor and dyeing auxiliary, in the electroplating industry, and in the production of agricultural chemicals. The use of benzaldehyde in the extractive separation of isomeric amines has been proposed. Flavours and essences

8.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg benzaldehyde, at plant, in Europe**. For this study the production of benzaldehyde by means of the hydrolysis of benzal chloride is considered. All data in the present report are referred to 1 kg benzaldehyde 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

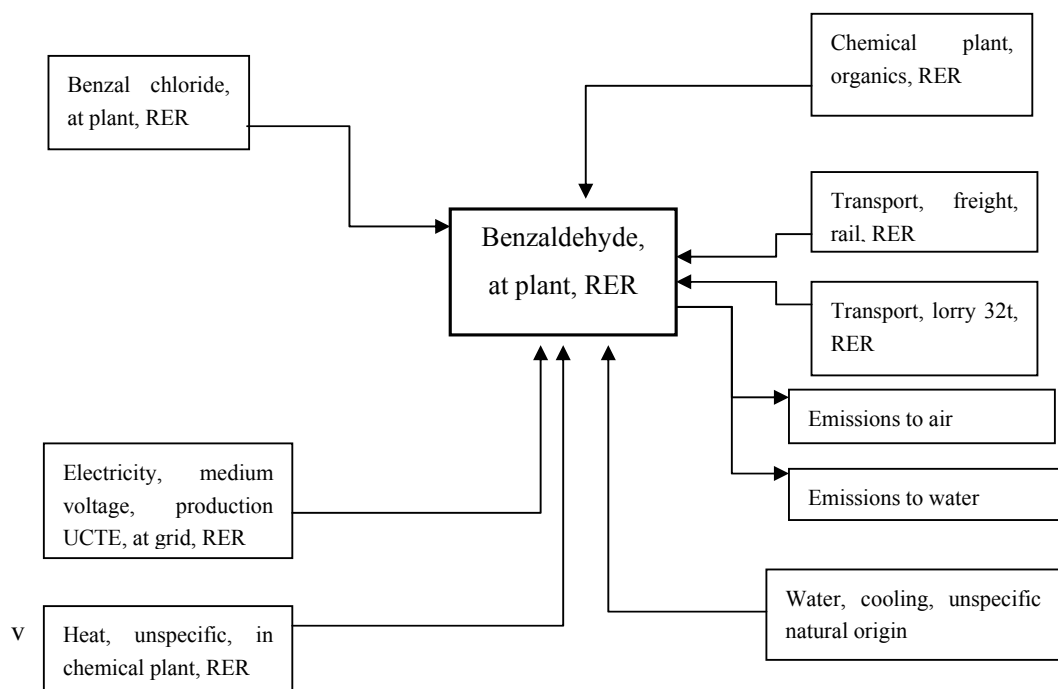


Fig. 8.2 Process flow chart for the production of benzaldehyde

8.5 Raw materials and auxiliaries

As there was no data available for the consumption of the raw materials for the production of benzaldehyde, the values are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the benzaldehyde production.

Tab. 8.2 Consumption of raw materials and auxiliaries for the production of 1 kg benzaldehyde

Input	Values	Remarks
Benzal chloride (kg)	1.589	calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

8.6 Energy

Gwalior (2005) gives data for the energy consumption of benzaldehyde production in an Indian plant.

Tab. 8.3 Energy consumption for the production of 1 kg benzaldehyde

Input	Gwalior 2005
Steam (MJ)	5.58
Electricity (kWh)	0.0297

8.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 8.4 summarises the total transport amounts for the production of 1 kg benzaldehyde.

Tab. 8.4 Total transport amounts for the production of 1 kg benzaldehyde

tkm kg ⁻¹ benzaldehyde	lorry	train
Raw materials	1.589E-01	9.533E-01

8.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of benzaldehyde plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

8.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of benzaldehyde. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission

sources were estimated to 0.2 % of the raw material input. The assumed values for the emissions to air are given in Tab. 8.5.

Tab. 8.5 Process emissions to air from the production of 1 kg benzaldehyde

Output	Value	Remarks
Benzal chloride, to air (kg)	3.18E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.32E-1	from waste water treatment
Waste heat (MJ)	1.07E-01	calculated from electricity input

8.10 Emissions to water

The remaining amount of unreacted benzal chloride (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for benzal chloride was assumed with 90% leading to emissions of 0.639 g benzal chloride per kg product in the treated water. The carbon contained in the removed benzal chloride was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC, DOC, and AOX used in this inventory were calculated from the amount of benzal chloride in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 8.6.

The chloride contained in the removed benzal chloride and in the hydrogen chloride byproduct was assumed to leave the production process as chloride emission in the waste water. The value for AOX is calculated from the chloride and the benzal chloride emissions.

Tab. 8.6 Process emissions to water from the production 1 kg benzaldehyde

Output	Value	Remarks
benzal chloride, to water (kg)	7.63E-03	calculated from mass balance
chloride, to water (kg)	6.75E-01	calculated from mass balance
COD, BOD (kg)	3.73E-03	calculated from water emissions
TOC, DOC (kg)	2.55E-03	calculated from water emissions
AOX (kg)	6.78E-01	calculated from water emissions

8.11 Life cycle inventory and Data Quality Considerations

Tab. 8.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of benzaldehyde. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 8.7 Unit process raw data for the production of benzaldehyde

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	benzaldehyde, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	benzaldehyde, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	benzal chloride, at plant	RER	0	kg	1.59E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.59E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	9.53E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.97E-2	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	5.58E+0	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.07E-1	1	1.14	(1,3,1,5,1,3); Calculated from the electricity input
	-	4	Benzal chloride	-	-	kg	3.18E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.32E-1	1	1.30	(4,5,na,na,na,na); Calculated from the carbon balance
emission water, river	-	4	Benzal chloride	-	-	kg	7.63E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Chloride	-	-	kg	6.75E-1	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	6.78E-1	1	1.62	(4,5,na,na,na,na); Calculated from mass balance
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	3.73E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	3.73E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.55E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.55E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

8.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 8.8 shows selected LCI results and the cumulative energy demand for the production of benzaldehyde.

Tab. 8.8 Selected LCI results and the cumulative energy demand for the production of benzaldehyde

Name		benzaldehyde, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	105.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	16.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.3
selected LCI results			
land occupation	resource	m2a	4.3E-1
CO ₂ , fossil	air	kg	4.6E+0
NM VOC	air	kg	8.4E-3
nitrogen oxides	air	kg	8.0E-3
sulphur dioxide	air	kg	1.1E-2
particulates, <2.5 µm	air	kg	8.2E-4
BOD	water	kg	2.1E-2
cadmium	soil	kg	2.6E-9

8.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	benzaldehyde, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of benzaldehyde including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Benzaldehyd, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "benzaldehyde, at plant, RER" is modelled for the production of benzaldehyde from benzal chloride in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H6O
	501	StatisticalClassification	
	502	CASNumber	100-52-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrolysis of benzal chloride
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

8.14 References

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9 Benzene chlorination

Author: Jürgen Sutter, ETH S+U, Zurich
Review: Roland Hischier, EMPA, St. Gallen
Last changes: 2007

Acknowledgement

We thank Michael Overcash for the information provided in a personal communication.

9.1 Introduction

Benzene chlorination is the most important process to produce monochlorobenzene.

9.2 Production Technologies

Chlorobenzenes are prepared industrially by reaction of liquid benzene with gaseous chlorine in the presence of a catalyst at moderate temperature and atmospheric pressure. Hydrogen chloride is formed as a byproduct. Generally, mixtures of isomers and compounds with varying degrees of chlorination are obtained, because any given chlorobenzene can be further chlorinated up to the stage of hexachlorobenzene. Because of the directing influence exerted by chlorine, the unfavored products 1,3-dichlorobenzene, 1,3,5-trichlorobenzene, and 1,2,3,5-tetrachlorobenzene are formed to only a small extent if at all. The velocity of chlorination for an individual chlorine compound depends on the compound's structure and, because of this, both the degree of chlorination and also the isomer ratio change continuously during the course of a reaction. Sets of data on the composition of products from different reactions are only comparable with one another if they refer to identical reaction conditions and materials having the same degree of chlorination. By altering the reaction conditions and changing the catalyst, one can vary the ratios of the different chlorinated products within certain limits. Lewis acids (FeCl_3 , AlCl_3 , SbCl_3 , MnCl_2 , MoCl_3 , SnCl_4 , TiCl_4) are used as principal catalysts. Elevated temperatures in substitution reactions favor the introduction of a second chlorine atom in the ortho and meta positions, whereas para substitution is favored if cocatalysts and lower temperatures are used. As a further example of the influence of catalysts on the composition of the product, attention is drawn to the formation of 1,2,4,5-tetrachlorobenzene. The optimal reaction temperature depends on the desired degree of chlorination. Mono- and dichlorination are carried out at 20 – 80 °C. In the production of hexachlorobenzene by chlorination of benzene, however, temperatures of about 250 °C are needed toward the end of the reaction.

The usual catalyst employed in large scale production is ferric chloride, with or without the addition of sulfur compounds. Ferric chloride complexed with 1 mol of water is claimed to have the best catalytic effect. Benzene and chlorine of technical purity always contain some water, however, thus, it may be that this hydrate compound is always present in iron-catalyzed industrial reactions.

The ratio of resulting chlorobenzenes to one another is also influenced by the benzene : chlorine ratio. For this reason, the highest selectivity is achieved in batch processes. If the same monochlorobenzene: dichlorobenzene ratio expected from a batch reactor is to result from continuous operation in a single-stage reactor, then a far lower degree of benzene conversion must be accepted (as a consequence of a low benzene : chlorine starting ratio). The selectivity of a continuous reactor approaches that of a discontinuous reactor as the number of reaction stages is increased. Mathematical models analyzing and interpreting benzene chlorination will be found in and elsewhere.

Solvents can also influence the chlorination rate as well as the selectivity of the reaction, although solvents are not used in industrial chlorination.

Continuous chlorination

Benzene is treated with chlorine gas in a suitable reactor in the presence of dissolved ferric chloride. The reactants must be mixed as intensively as possible. The catalyst can be introduced along with the substrate or it can be allowed to form during the reaction on the surface of iron rings in the reactor. The reaction is highly exothermic.



Unwanted heat of reaction can be dissipated either by circulating some of the reactor liquid through an external heat exchanger or by permitting evaporative cooling to occur at the boiling temperature. Circulation cooling has the advantage of enabling the reaction temperature to be varied in accordance with the requirements of a given situation. Evaporative cooling is more economical, however.

The reactor must be designed to ensure that the liquid within it has a suitable residence spectrum, since this favors high chlorination selectivity. As noted above, the quantity ratio of the chlorobenzenes to one another is determined by the benzene : chlorine starting ratio.

Almost quantitative conversion of chlorine is achieved in continuously operated plants for the manufacture of mono- and dichlorobenzenes under normal operating conditions.

Cast iron, steel, nickel, and glass-lined steel can be used as construction materials. However, all starting materials must be substantially free from water; otherwise, severe corrosion is caused by the hydrochloric acid formed. Intrusion of water is dangerous for another reason: water causes the ferric chloride catalyst to be inactivated. If this occurs, chlorine collects in the reactor and exhaust system, and local overheating may ensue, causing spontaneous combustion of the aromatic hydrocarbon with chlorine in a highly exothermic reaction to form carbon and hydrogen chloride (2 mol of HCl/mol of Cl₂). Furthermore, above 280 °C metallic iron begins to burn in the chlorine stream. It is, therefore, advisable to monitor the reaction continuously by observing the heat production rate and the chlorine content of the waste gas.

On leaving the reactor, the liquid and gas portions of the reaction mixture are separated. The waste gas contains hydrogen chloride and — in proportions corresponding to their vapor pressures at the temperature of the waste gas — benzene and chlorobenzenes. If the chlorine conversion is incomplete, chlorine may be present as well. Concerning the treatment of the waste gas, see Section.

The liquid phase contains benzene, chlorobenzenes, hydrogen chloride, and iron catalyst. Production processes exist in which the product mixture is neutralized with sodium hydroxide solution or soda before it is subjected to fractional distillation. In modern continuous distillation trains, the mixture of products can be distilled without preliminary treatment, however. The separated fractions consist of benzene, monochlorobenzene, dichlorobenzenes, trichlorobenzenes, and higher chlorobenzenes. Iron catalyst is removed along with the distillation residue, disposal of which is discussed in Section. Dissolved hydrogen chloride is removed during the benzene distillation and combined with the waste gas. Unreacted benzene is recycled to the reactor.

Whereas chlorobenzene and 1,2-dichlorobenzene are obtainable as pure distillates, the other fractions are mixtures of close-boiling polychlorobenzene isomers. A further separation, insofar as this is economically justifiable, would consist of combined crystallization/distillation processes.

9.3 System Characterisation

In this study the chlorination of benzene is described. All data in the present report are referred to 1 kg benzene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population

density area. The emissions into water are assumed to be emitted into rivers. Within the chlorination of benzene, monochlorobenzene, o-dichlorobenzene, p-dichlorobenzene, and hydrochloric acid are derived as co-products. Therefore an allocation between the four products was made. The following processes are modelled:

- Benzene chlorination, RER. Multi output process with monochlorobenzene, at plant, RER, o-dichlorobenzene, at plant, RER, p-dichlorobenzene, at plant, RER, and hydrochloric acid, from benzene chlorination, at plant, RER as allocated products

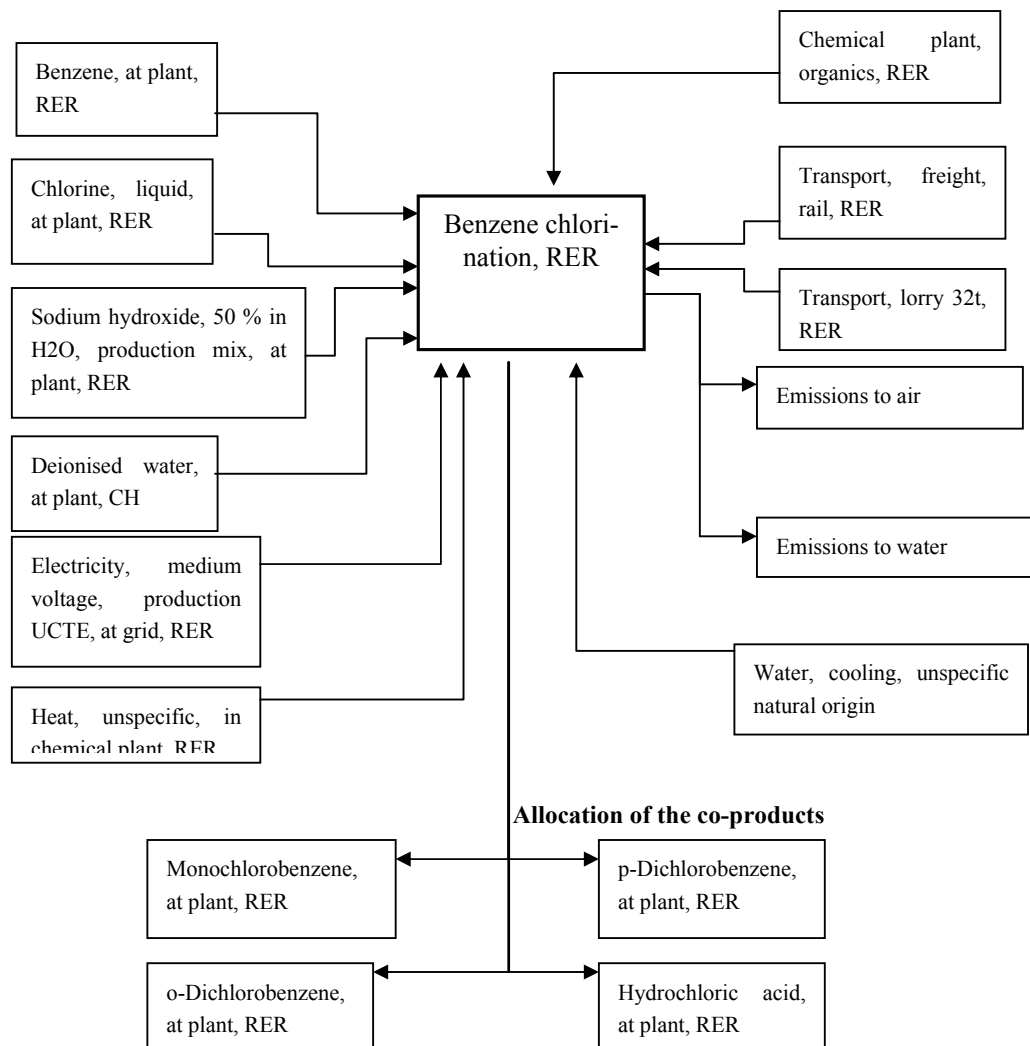


Fig. 9.1 Process flow chart for the benzene chlorination

9.4 Data sources

The data for the benzene chlorination are taken from a database, which reports data from US industry production of the years 1998 to 2004 (Overcash 1998-2004). These data are taken from several US production plants and are representative for US production.

9.5 Raw materials and auxiliaries

Tab. 9.1 Consumption of raw materials and auxiliaries for the benzene chlorination per kg benzene

Input	Overcash 1998-2004 ³
Benzene (kg)	1
Chlorine (kg)	8.38E-02
Sodium hydroxide (kg)	2.37E-02
Process water (kg)	9.52E-02
Cooling water (kg)	2.88

9.6 Energy

Tab. 9.2 Energy consumption for the benzene chlorination per kg benzene

Input	Overcash 1998-2004
Steam (MJ)	1.26E-01
Electricity (kWh)	5.03E-03

9.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the benzene and the sodium hydroxide, and 50 km by lorry 32t and 100 km by train for the chlorine. Tab. 9.3 summarises the total transport amounts for the benzene chlorination.

Tab. 9.3 Total transport amounts for the benzene chlorination per kg benzene

tkm kg ⁻¹ benzene	lorry	train
Raw materials	1.66E-02	8.26E-02

9.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of benzene chlorination plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

9.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

Overcash 1998–2004 report air emissions from the benzene chlorination.

Tab. 9.4 Process emissions to air from the benzene chlorination per kg benzene

Output	Value	Remarks
Benzene, to air (kg)	2.796E-03	Overcash 1998-2004
Waste heat (MJ)	1.81E-03	calculated from electricity input

9.10 Emissions to water

Overcash 1998–2004 report several water emissions from the benzene chlorination.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of benzene, monochlorobenzene, and o-dichlorobenzene in the waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 9.5.

Tab. 9.5 Process emissions to water from the benzene chlorination per kg benzene

Output	Value	Remarks
Benzene, to water (kg)	1.1E-02	Overcash 1998-2004
Monochlorobenzene, to water (kg)	1.02E-02	Overcash 1998-2004
Sodium, to water (kg)	1.36E-03	Overcash 1998-2004
Chloride, to water (kg)	2.095E-03	Overcash 1998-2004
o-Dichlorobenzene, to water (kg)	4.934E-04	Overcash 1998-2004
COD, BOD (kg)	5.34E-02	calculated from water emissions
TOC, DOC (kg)	1.69E-02	calculated from water emissions

9.11 Allocation of the co-products of the benzene chlorination

Within the benzene chlorination, monochlorobenzene, o-dichlorobenzene, p-dichlorobenzene, and hydrochloric acid are obtained as co-products. As allocation factor the weight of the products was used. The yields of the diverse co-products and the used allocation factors are given in Tab. 9.6.

Tab. 9.6 Yields and allocation factors of the co-products from the benzene chlorination

Process	Benzene chlorination, RER		
	Unit	per kg benzene	Allocation factor
Monochlorobenzene, at plant, RER	kg	1.05E-01	67.27
o-dichlorobenzene, at plant, RER	kg	4.39E-03	2.8
p-dichlorobenzene, at plant, RER	kg	5.95E-03	3.8
Hydrochloric acid, at plant, RER	kg	4.09E-02	26.13

9.12 Monochlorobenzene, at plant, RER

Monochlorobenzene (CAS-No. 108-90-7, C_6H_5Cl) is a colorless liquid which is volatile with steam and is a good solvent. It is miscible with all commonly used organic solvents and forms many azeotropes. Monochlorobenzene is flammable and has an aromatic odor. Upon addition of 24 % benzene, it forms a eutectic mixture with a solidification point of $-60.5\text{ }^{\circ}\text{C}$.

The most important chemical and physical properties of monochlorobenzene used in this inventory are given in Tab. 9.7.

Tab. 9.7 Chemical and physical properties of monochlorobenzene (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	112.56	g mol^{-1}	
Specific gravity	1106	kg m^{-3}	at $20\text{ }^{\circ}\text{C}$
Boiling point	132.2	$^{\circ}\text{C}$	at normale pressure
Melting point	- 45.2	$^{\circ}\text{C}$	at normale pressure

9.12.1 Reserves and Resources

Most monochlorobenzene is now produced from benzene and chlorine in continuously operated plants. Depending on the ratio of benzene to chlorine chosen, one can achieve either a low rate of benzene conversion and little dichlorobenzene formation, or almost complete conversion of the benzene with a higher degree of dichlorobenzene formation. Which of the two alternatives is favored depends on a profitability calculation, in which the distillation costs occasioned by the dichlorobenzenes need to be taken into account. The composition of a chlorination mixture containing the highest possible proportion of monochlorobenzene has been given as 4 – 5 % unreacted benzene, 73 % monochlorobenzene, and 22 – 23 % dichlorobenzene.

9.12.2 Use

Monochlorobenzene is nitrated in large quantities, the product subsequently being converted via such intermediates as nitrophenol, nitroanisole, nitrophenetole, chloroaniline, and phenylenediamine into dyes, crop protection products, pharmaceuticals, rubber chemicals, etc. The production of phenol, aniline, and DDT from monochlorobenzene, formerly carried out on a large scale, has been almost entirely discontinued due to the introduction of new processes and legislation forbidding the use of DDT.

9.13 1,2-Dichlorobenzene, at plant, RER

1,2-Dichlorobenzene (CAS-No. 95-50-1, $C_6H_4Cl_2$; synonym: o-dichlorobenzene) is a colorless, mobile liquid which is miscible with the commonly used organic solvents. It is difficult to ignite and has an unpleasant odor.

The most important chemical and physical properties of 1,2-dichlorobenzene used in this inventory are given in Tab. 9.8.

Tab. 9.8 Chemical and physical properties of 1,2-dichlorobenzene (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	147	g mol^{-1}	

Specific gravity	1306	kg m ⁻³	at 20 °C
Boiling point	179	°C	at normale pressure
Melting point	- 16.7	°C	at normale pressure

9.13.1 Use

1,2-Dichlorobenzene after conversion to 1,2-dichloro-4-nitrobenzene, is used mainly in the production of dyes and pesticides. It is also used to produce disinfectants and deodorants and on a small scale as a heat transfer fluid.

9.14 1,4-Dichlorobenzene, at plant, RER

1,4-Dichlorobenzene (CAS-No. 106-46-7, C₆H₄Cl₂; synonym: p-dichlorobenzene) is a white, volatile, crystalline compound soluble in many organic solvents. It has a strong camphor odor. 1,4-Dichlorobenzene occurs as a stable monoclinic α -modification, which is transformed into the triclinic β -form at 30.8 °C.

The most important chemical and physical properties of 1,4-dichlorobenzene used in this inventory are given in Tab. 9.9.

Tab. 9.9 Chemical and physical properties of 1,4-dichlorobenzene (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	147	g mol ⁻¹	
Boiling point	173.9	°C	at normale pressure
Melting point	54	°C	at normale pressure

9.14.1 Use

1,4-Dichlorobenzene is used mainly in the production of disinfectant blocks and room deodorants and as a moth control agent. After conversion into 2,5-dichloronitrobenzene, it finds application in the production of dyes. It is also used in the production of insecticides and, more recently, of poly-phenylene-sulfide-based plastics, materials with excellent thermal stability.

9.15 Life cycle inventory and Data Quality Considerations

Tab. 9.10 shows the life cycle inventory and the data quality indicators for the inventory of the production of monochlorobenzene. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 9.10 Unit process raw data for the benzene chlorination

Petrochemical Solvents

401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	benzene chlorination	Uncertainty	Type	Standard Deviation	95%	General Comment	monochlorobenzene, at plant	o-dichlorobenzene, at plant	p-dichlorobenzene, at plant
662	493	403	Location	Infrastructure	Process	Unit	benzene chlorination	Uncertainty	Type	Standard Deviation	95%	General Comment	monochlorobenzene, at plant	o-dichlorobenzene, at plant	p-dichlorobenzene, at plant	
493			Infrastructure													
403			Unit													
allocated products	-	2	monochlorobenzene, at plant	RER	0	kg	1.05E-1							100	0	0
products	-	2	o-dichlorobenzene, at plant	RER	0	kg	4.39E-3							0	100	0
products	-	2	p-dichlorobenzene, at plant	RER	0	kg	5.95E-3							0	0	100
products	-	2	hydrochloric acid from benzene chlorination, at plant	RER	0	kg	4.09E-2							0	0	0
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.88E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	water, deionised, at plant	CH	0	kg	9.52E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	benzene, at plant	RER	0	kg	1.00E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	chlorine, liquid, production mix, at plant	RER	0	kg	8.38E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	2.37E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.66E-2	1	2.09	(4,5,na,na,na,na); Standard distance 100km			67.265	2.805	3.803	
	5	-	transport, freight, rail	RER	0	tkm	8.26E-2	1	2.09	(4,5,na,na,na,na); Standard distance 600km			67.265	2.805	3.803	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	5.03E-3	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.26E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	5	-	chemical plant, organics	RER	1	unit	6.26E-11	1	3.09	(4,5,na,na,na,na); Estimation			67.265	2.805	3.803	
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.81E-3	1	1.07	(1,2,1,1,1,3); Calculated from the electricity input			67.265	2.805	3.803	
emission water, river	-	4	Benzene	-	-	kg	2.80E-3	1	1.51	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	Benzene	-	-	kg	1.10E-2	1	3.00	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	Benzene, chloro-	-	-	kg	1.02E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	Sodium, ion	-	-	kg	1.36E-3	1	5.00	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	Chloride	-	-	kg	2.10E-3	1	3.00	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	o-Dichlorobenzene	-	-	kg	4.93E-4	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004			67.265	2.805	3.803	
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	5.34E-2	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions			67.265	2.805	3.803	
	-	4	COD, Chemical Oxygen Demand	-	-	kg	5.34E-2	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions			67.265	2.805	3.803	
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.69E-2	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions			67.265	2.805	3.803	
	-	4	TOC, Total Organic Carbon	-	-	kg	1.69E-2	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions			67.265	2.805	3.803	

9.16 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 9.11 shows selected LCI results and the cumulative energy demand for the benzene chlorination.

Tab. 9.11 Selected LCI results and the cumulative energy demand for the benzene chlorination

Name		hydrochloric acid from benzene chlorination, at plant	monochlorobenzene, at plant	o-dichlorobenzene, at plant	p-dichlorobenzene, at plant
Location	Unit	RER	RER	RER	RER
Infrastructure	kg	kg	kg	kg	kg
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	429.7	429.7	429.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	18.3	18.3	18.3
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.1	1.1	1.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.2	1.2	1.2
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1	0.1
selected LCI results					
land occupation	resource	m2a	9.5E-2	9.5E-2	9.5E-2
CO ₂ , fossil	air	kg	1.0E+1	1.0E+1	1.0E+1
NM VOC	air	kg	3.3E-2	3.3E-2	3.3E-2
nitrogen oxides	air	kg	2.0E-2	2.0E-2	2.0E-2
sulphur dioxide	air	kg	3.0E-2	3.0E-2	3.0E-2
particulates, <2.5 um	air	kg	9.9E-4	9.9E-4	9.9E-4
BOD	water	kg	3.4E-1	3.4E-1	3.4E-1
cadmium	soil	kg	8.2E-10	8.2E-10	8.2E-10

9.17 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	benzene chlorination
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Benzene chlorination including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Benzolchlorierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "benzene chlorination" delivers the co-products monochlorobenzene, o-dichlorobenzene, p-dichlorobenzene, and hydrochloric acid. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1986
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Chlorination of benzene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

9.18 References

- Friskhnecht et al. 2007 Friskhnecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Overcash 1998-2004 Overcash, M., Chemical life cycle database, Chemical and Biomolecular Engineering, North Carolina State University, Raleigh, NC 27695-7905. 1998 - 2004

Ullmann 2004

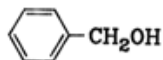
Rossberg, M., Lendle, W., Pfeleiderer, G., Tögel, A., Dreher, E.-L., Langer, E., Rassaerts, H., Kleinschmidt, P., Strack, H., Cook, R., Beck, B., Lipper, K.-A., Torkelson, T., Löser, E., Beutel, K.: Chlorinated Hydrocarbons. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/ueic/articles/>

10 Benzyl alcohol

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

10.1 Introduction

Benzyl alcohol (CAS-No. 100-51-6, C₇H₈OH) is a colorless liquid with a faint aromatic odor and a mildly irritating effect on the mucous membranes.



For this inventory the functional unit is 1 kg of pure benzyl alcohol. The most important chemical and physical properties of benzyl alcohol used in this inventory are given in Table 1.

Tab. 10.1 Chemical and physical properties of benzyl alcohol (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	108.14	g mol ⁻¹	
Specific gravity	1061	kg m ⁻³	at 20 °C
Boiling point	205.4	°C	at normale pressure
Melting point	- 15.4	°C	at normale pressure

10.2 Reserves and Resources

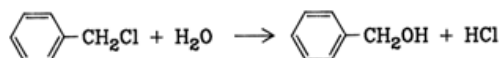
Benzyl alcohol can be produced in many ways. At present there are only two processes of substantial industrial importance: the hydrolysis of benzyl chloride and the hydrogenation of benzaldehyde. Other processes, like the oxidation of toluene, the hydrogenation of benzoic acid esters, the electrochemical reduction of benzoic acid, the hydrolysis of benzyldisulfonic acid, and the decarboxylation of benzyl formate, have no importance in the industrial production of benzyl alcohol (Ullmann 2004).

10.3 Production Technologies and Use

10.3.1 Production Technologies

Hydrolysis of benzyl chloride (Ullmann 2004)

The hydrolysis of benzyl chloride is a reversible reaction which leads to the almost quantitative formation of benzyl alcohol only in the presence of alkaline saponifying agents that combine with the hydrogen chloride as it is formed. It is therefore carried out by heating benzyl chloride with stoichiometric excesses of aqueous solutions of oxides, hydroxides, or carbonates of the alkali or alkaline earth metals. As in the following example, soda is the preferred saponifying agent.



To 610 parts of boiling 10 % soda solution 126.5 parts of benzyl chloride is added with stirring. The reaction mixture is refluxed and stirred until carbon dioxide no longer escapes; this takes an average of

five to six hours. An alkali-resistant steel reactor with brick walls is used. The stirrer and heating coil are made of an alloy with a high nickel content. After the reaction mixture has cooled, the upper layer, consisting of crude benzyl alcohol, is removed. The sodium chloride solution below still contains some soda. Dissolved benzyl alcohol can be obtained from it by adding salt or by extracting with benzene or toluene. The crude benzyl alcohol is carefully fractionated at a reduced pressure in an efficient column. The yield is 85 parts of benzyl alcohol and 10 parts of dibenzyl ether.

In the discontinuous hydrolysis of benzyl chloride the reaction time can be reduced by hydrolyzing the final residue of benzyl chloride with sodium hydroxide solution after the saponification with soda solution has been in progress for three hours. The total saponification time required is then only four hours. Normally 8 – 12 % of dibenzyl ether is formed as an unavoidable byproduct of the alkaline hydrolysis of benzyl chloride. Pure dibenzyl ether is recovered by distillation from the residue in the distillation column. The use of alkali hydroxides instead of carbonates reduces the reaction time but favors the formation of dibenzyl ether, which may then exceed 20 % in the crude benzyl alcohol. The percentage of dibenzyl ether can be reduced if the hydrolysis is performed in the presence of an inert solvent, such as benzene, toluene, or xylene.

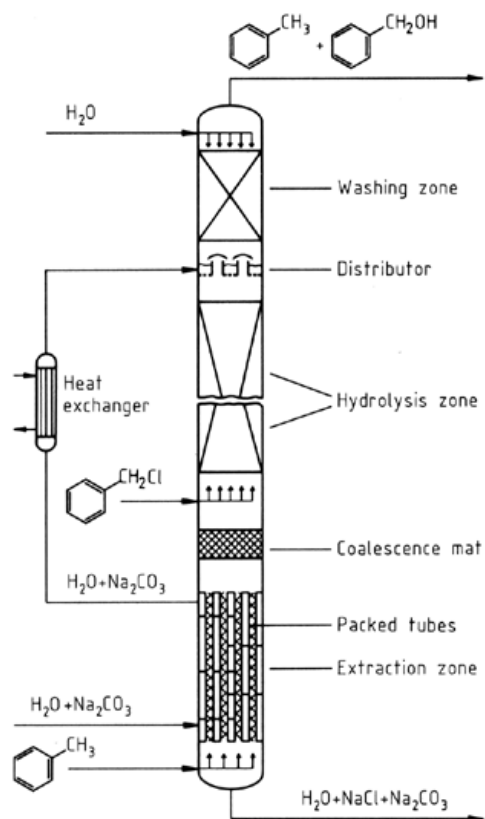


Fig. 10.1 Hydrolysis of benzyl chloride to form benzyl alcohol in a continuous-flow reactor

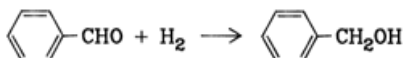
Benzyl alcohol can be produced continuously if benzyl chloride and the aqueous alkaline solution of saponifying agent react in a flow reactor at an elevated temperature (150 – 350 °C) and sufficient pressure to maintain the reactants in the liquid phase. Thorough mixing of the reactants is important; it can be achieved by the use of suitable baffles in the flow reactor or by maintaining an adequate flow rate. The residence time need then be no more than a few minutes. The mixture emerging from the hydrolysis zone is processed in the normal way. Only a small amount of dibenzyl ether is formed and this can be further reduced by using an inert solvent.

In a new continuous process, benzyl chloride, flowing in one direction, and the alkaline saponifying

agent, flowing in the other, are reacted as counter-currents in an inert organic solvent (e.g., toluene, xylene) in a flow reactor. In the extraction zone, the alcohol dissolved in the aqueous alkaline phase is extracted by an inert organic solvent flowing in the opposite direction. Water, likewise as a counter-current, washes the solution of the crude benzyl alcohol in a washing zone. The flow reactor, extracting zone, and washing zone are integrated in a special apparatus (Fig.). The wash water and extract are returned to the hydrolysis zone. This process can be operated at low reaction temperatures (120 – 150 °C) and with a small stoichiometric excess of saponifying agent (5 – 25 %); a small reactor can be used.

In a two-step process, benzyl chloride reacts with sodium acetate to form benzyl acetate, which is then saponified to form benzyl alcohol. In a similar process, benzyl chloride reacts with an alkali or alkaline earth formate in the presence of a catalyst to form benzyl formate. The benzyl alcohol is liberated by subsequent transesterification using an alcohol and a catalyst. Neither of these processes has acquired much importance in the manufacture of benzyl alcohol. However, they could have advantages in the hydrolysis of certain substituted benzyl chlorides that react with soda solution to give dibenzyl ethers and not the desired alcohol.

Hydrogenation and Reduction of Benzaldehyde (Ullmann 2004)



The industrial production of benzyl alcohol by the hydrogenation of benzaldehyde gained importance because substantial quantities of benzaldehyde were available as a byproduct of the production of phenol (Dow process) and -caprolactam (Snia Viscosa process) by the oxidation of toluene.

Depending on the reaction conditions, benzaldehyde can be hydrogenated to form benzyl alcohol, toluene, hydroxymethylcyclohexane, methylcyclohexane, and other products. However, very high yields of benzyl alcohol are obtainable if suitable reaction conditions and catalysts are chosen. Raney nickel doped with transition metals, nickel or platinum metals with phosphines or phosphine oxides, palladium combined with an organic nitrogen or alkali base, water, or another transition metal can serve as the catalyst. If benzaldehyde is hydrogenated at temperatures of 70 – 200 °C and a hydrogen pressure of 1 – 4 MPa, high yields of benzyl alcohol are obtained in short reaction times. In a well-known continuous process, benzaldehyde is hydrogenated to form benzyl alcohol with a high degree of efficiency and a high selectivity in the presence of a platinum–aluminum oxide–lithium oxide catalyst. Stannane, sodium hydride, zinc, and microorganisms can also be used to reduce benzaldehyde to benzyl alcohol, but these processes have no industrial importance. Benzyl alcohol is no longer produced industrially using the Cannizzaro reaction.

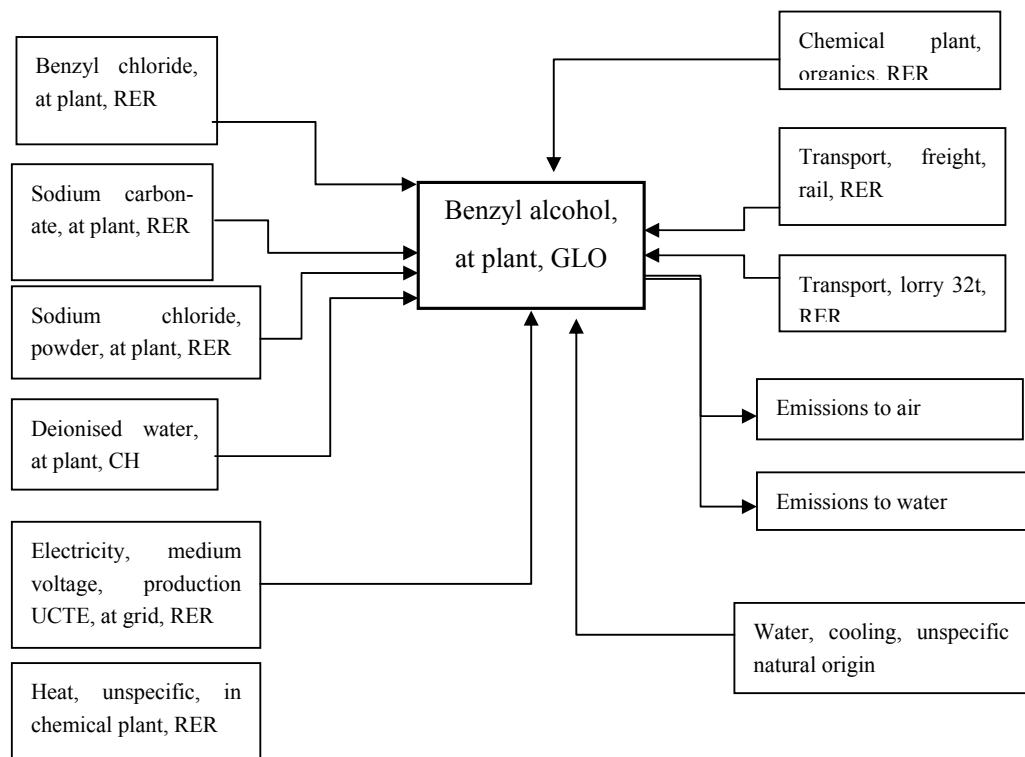
10.3.2 Use

By virtue of its good solvent properties, benzyl alcohol is an important solvent for surface-coating materials and resins. It dissolves cellulose esters and ethers, alkyd resins, acrylic resins, and fats; it is also used as an ingredient in the inks for ball-point pens. It is added in small amounts to surface-coating materials to improve their flow and gloss. In the textile industry, it is used as an auxiliary in the dyeing of wool, polyamides, and polyesters. Because it has only a relatively faint odor, it is used as a solvent and diluting agent in the manufacture of perfumes and flavors. In color photography it is increasingly important as a development accelerator. In pharmacy it is used as a local anesthetic and, because of its microbicidal effect, as an ingredient of ointments and other preparations.

Benzyl alcohol is a starting material for the preparation of numerous benzyl esters that are used as odorants, flavors, stabilizers for volatile perfumes, and plasticizers. Benzyl alcohol is also used in the extractive distillation of m- and p-xylenes and m- and p-cresols.

10.4 System Characterisation

Fig. 10.2 Process flow chart for the production of benzyl alcohol



This report corresponds to the module in the ecoinvent database for the production of **1 kg benzyl alcohol, at plant, in Europe**. For this study the production of benzyl alcohol by means of the hydrolysis of benzyl chloride is considered. All data in the present report are referred to 1 kg benzyl alcohol 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

10.5 Data sources

The data for the benzyl alcohol production are taken from a database, which reports data from US industry production of the years 1998 to 2004 (Overcash 1998-2004). These data are taken from several US production plants and are representative for US production.

10.6 Raw materials and auxiliaries

Tab. 10.2 Consumption of raw materials and auxiliaries for the production of 1 kg benzyl alcohol

Input	Overcash 1998 - 2004
Benzyl chloride (kg)	1.181
Soda (kg)	0.813
Sodium chloride (kg)	0.143
Process water (kg)	1.509
Cooling water (kg)	36.4716

10.7 Energy

Tab. 10.3 Energy consumption for the production of 1 kg benzyl alcohol

Input	Overcash 1998 - 2004	This study
Steam (MJ)	5.30426	4.09146 ²
Electricity (kWh)	0.08225	8.225E-02
Potential recovered energy (MJ)	- 1.2128	

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

10.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 10.4 summarises the total transport amounts for the production of 1 kg benzyl chloride.

Tab. 10.4 Total transport amounts for the production of 1 kg benzyl alcohol

tkm kg ⁻¹ ethyl acetate	lorry	train
Raw materials	2.137E-01	1.2822

10.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of benzyl alcohol plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

10.10 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 8.22E-02 kWh/kg benzyl alcohol is converted to waste heat and that 100% of the waste heat is released to the air.

Overcash (1998 – 2004) reports air emissions from the production of benzyl alcohol.

Tab. 10.5 Process emissions to air from the production of 1 kg benzyl alcohol

Output	Value	Remarks
carbon dioxide, fossil, to air (kg)	2.04E-01	Overcash 1998 - 2004
waste heat (MJ)	2.961E-01	calculated from electricity input

10.11 Emissions to water

Overcash (1998 – 2004) reports several water emissions from the production of benzyl alcohol.

The values for TOC and DOC used in this inventory were calculated from the amount of benzyl alcohol and carbonate in the waste water. For the calculation of the values for DOC the worst case scenario TOC = DOC was used. The value for AOX is calculated from the chloride emission to waste water. A summary of the values used in this inventory is given in Tab. 10.6.

Tab. 10.6 Process emissions to water from the production 1 kg benzyl alcohol

Output	Value	Remarks
benzyl alcohol, to water (kg)	1.0E-02	Overcash (1998 - 2004)
sodium, to water (kg)	3.39E-01	Overcash (1998 - 2004)
chloride, to water (kg)	4.14E-01	Overcash (1998 - 2004)
carbonate, to water (kg)	2.53E-01	Overcash (1998 - 2004)
TOC, DOC (kg)	5.83E-03	calculated from water emissions
AOX (kg)	4.14E-01	calculated from water emissions

10.12 Life cycle inventory and Data Quality Considerations

Tab. 10.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of benzyl alcohol. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 10.7 Unit process raw data for the production of benzyl alcohol

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	benzyl alcohol, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401			Location				RER			
662			InfrastructureProcess				0			
493			Unit				kg			
403										
resource, in water technosphere	-	0	benzyl alcohol, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.65E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	water, deionised, at plant	CH	0	kg	1.51E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	benzyl chloride, at plant	RER	0	kg	1.18E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	soda production, solvay process, at plant	RER	0	kg	8.13E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	sodium chloride, powder, at plant	RER	0	kg	1.43E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	transport, lorry 32t	RER	0	tkm	2.14E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	1.28E+0	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	8.23E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	4.09E+0	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
emission air, high population density	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	2.96E-1	1	1.07	(1,2,1,1,1,3); Calculated from the electricity input
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg	2.04E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Benzyl alcohol	-	-	kg	1.00E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Sodium, ion	-	-	kg	3.39E-1	1	5.00	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Chloride	-	-	kg	4.14E-1	1	3.00	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Carbonate	-	-	kg	2.53E-1	1	3.00	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	5.83E-3	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	5.83E-3	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	4.14E-1	1	1.51	(1,2,1,1,1,3); Calculated from the water emissions

10.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 10.8 shows selected LCI results and the cumulative energy demand for the production of benzyl alcohol.

Tab. 10.8 Selected LCI results and the cumulative energy demand for the production of benzyl alcohol

	Name	benzyl alcohol, at plant	
	Location		
	Unit Infrastructure	Unit	RER kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	90.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	3.8E-1
CO2, fossil	air	kg	3.9E+0
NM VOC	air	kg	4.7E-3
nitrogen oxides	air	kg	6.9E-3
sulphur dioxide	air	kg	9.1E-3
particulates, <2.5 um	air	kg	6.6E-4
BOD	water	kg	1.6E-2
cadmium	soil	kg	2.1E-9

10.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	benzyl alcohol, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of benzyl alcohol including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Benzylalkohol, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "benzyl alcohol, at plant, RER" is modelled for the production of dimethylamine from methanol in Europe. Raw materials, auxiliaries, energy consumptions and emissions are modelled with data from US production. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H8O
	501	StatisticalClassification	
	502	CASNumber	100-51-6
TimePeriod	601	StartDate	1998
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrolysis of benzyl chloride
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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11 Benzyl chloride

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

11.1 Introduction

Benzyl chloride (CAS-No. 100-44-7, $C_6H_5CH_2Cl$; Synonyms: chloromethyl benzene, chlorotoluene) is a colorless liquid which fumes in moist air. It has a pungent odor and is irritating to the mucous membranes and the eyes (i.e., it has a powerful lachrymatory effect). Benzyl chloride may be structurally the simplest side-chain chlorinated derivative of toluene, but economically it is the most important. Benzyl chloride is the starting material for a large number of industrial syntheses. The first preparation of it involved not the chlorination of toluene, however, but the reaction of benzyl alcohol with hydrochloric acid. The solubility of benzyl chloride in water is 0.33 g/L at 4 °C, 0.49 g/L at 20 °C, and 0.55 g/L at 30 °C. Benzyl chloride is freely soluble in chloroform, acetone, acetic acid esters, diethyl ether, and ethyl alcohol. The solubility of chlorine in 100 g of benzyl chloride is 8.0 g at 30 °C, 5.4 g at 50 °C, and 2.1 g at 100 °C.

For this inventory the functional unit is 1 kg of pure benzyl chloride. The most important chemical and physical properties of benzyl chloride used in this inventory are given here.

Tab. 11.1 Chemical and physical properties of benzyl chloride (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	126.58	$g\ mol^{-1}$	
Specific gravity	1100.4	$kg\ m^{-3}$	at 20 °C
Boiling point	179.4	°C	at normale pressure
Melting point	- 39.2	°C	at normale pressure

11.2 Reserves and Resources

Most of the benzyl chloride is manufactured by chlorination of toluene.

11.3 Production Technologies and Use

11.3.1 Production Technologies

Substitution of chlorine for hydrogen in the aliphatic side-chain occurs by way of a radical chain mechanism; by contrast, chlorine substitution on the aromatic ring takes place according to an electrophilic polar mechanism.

This chlorination is highly exothermic (96 – 105 kJ/mol chlorine). In view of the high rate of chlorine radical formation and of hydrogen displacement, it is not surprising that, depending on the substrate and reaction conditions, radical chain lengths of 103 to 106 have been found. Because the mechanisms of side-chain chlorination and nuclear chlorination are fundamentally different, selectivity can be readily achieved. The prerequisites for high side-chain chlorination efficiency are as follows:

- 1) Achievement and maintenance of an optimal radical concentration
- 2) Elimination of components that might impart an electrophilic course to the reaction
- 3) Elimination of components capable of terminating the radical chains

4) Elimination of components conducive to other side reactions

5) In general, taking those precautions, which would encourage radical reactions or suppress electrophilic reactions

The discussion which follows considers each of these points separately.

1) Chlorine radical formation can be promoted by the addition of a radical-forming agent such as 2,2'-azobis(isobutyronitrile) (AIBN), benzoyl peroxide, or hexaphenylethane. Such compounds are consumed in the reaction, however, and thus have to be added repeatedly. For this reason, two other methods, singly or in combination, are of greater importance, particularly in industrial chlorinations: irradiation (ultraviolet light, γ -radiation) and the use of an elevated temperature (100 – 200 °C). In both cases, the effect is a result of excitation of chlorine molecules.

Irradiation is generally carried out with mercury vapor lamps that emit light in the wavelength range 300 – 500 nm, corresponding to the region in which the absorption bands of chlorine lie (hence, its yellowish-green color). The effects of varying the wavelength have been investigated in detail. A solution containing 0.1 % chlorine absorbs 90 % of the available light energy over a distance of only 12 cm. At a chlorine concentration of 3 – 4 mol/L, the corresponding distance is reduced to 0.01 cm. Consequently, thorough mixing of the reactor contents is advantageous, as is the use of reactors of relatively small diameter. These considerations apply regardless of whether the sources of radiation are inside the reactor or external to it. In the case of bubble column reactors, the height of the reactor is also important in view of the velocity of the rising gas bubbles, which is ca. 20 cm/s.

2) Friedel-Crafts catalysts favor nuclear chlorination. For this reason, particular attention must be paid to ensure, for example, that the starting toluene is free of dissolved iron salts and that all rust particles have been filtered. By extension, steel, including stainless steel, is unsuitable as a reactor material. The preferred reactor materials are glass, enamel, and polytetrafluoroethylene.

Extensive purification of technical-grade starting materials is very expensive, however. For this reason, numerous suggestions have been made for additives that might inhibit the undesirable effects of heavy metals. Most such additives are assumed to complex with, e.g., iron cations. Some additives are reported to accelerate the side-chain chlorination, thereby inhibiting in a relative sense nuclear chlorination. The following are examples of additives which have been recommended at concentrations of 0.1 – 2 %: pyridine and phenylpyridines, alkylene polyamines, hexamethylenetetramine, phosphoramides, acid amides, alkyl and dialkyl acid amides, ureas, phosphorus chlorides, alkyl phosphites and phosphines, phosphorus trichloride and trialkyl phosphates, cyclic thioureas, lactams, red phosphorus, and aminoethanol.

3) Oxygen, a well-known radical scavenger, greatly reduces the consumption of chlorine; its presence is therefore undesirable. Toluene can be freed of oxygen, for example, purging with an inert gas. The use of distilled chlorine, likewise, is an effective means of preventing oxygen from being present at significant concentrations.

It has also been reported, however, that even chlorine which contains up to 2 % oxygen is suitable for side-chain chlorination at relatively low temperatures (40 – 50 °C) provided that phosphorus trichloride (1 – 3 %) is present.

4) The presence of water leads to the formation of aqueous hydrochloric acid (indicated by cloudiness) and possibly to the hydrolysis of the chlorinated toluene; in addition, hydrolysis products which arise may be further altered by chlorination. The hydrogen chloride that is inevitably formed during chlorination is thought to be responsible for the fact that nuclear chlorination can never be entirely suppressed.

5) Various authors have drawn attention to the adverse effects caused by excessive chlorine concentrations (especially nuclear chlorination, both addition and substitution). Besides ensuring that the level of irradiation is adequate for rapid reaction, it may be useful to introduce an inert gas (N₂, HCl) into the chlorine stream. Dividing the reactor into smaller units is apparently also advantageous, particularly in the case of a continuous process. Much attention has also been paid to the development of ki-

netic models, to the design of reactors, to the effects of agitators, to the distribution of radiation, and to the effects of inhibitors.

In the course of radical chlorination of toluene, all three hydrogen atoms of the side-chain are successively replaced by chlorine. As a result, mixtures of the three expected compounds are obtained: benzyl chloride, benzal chloride, and benzotrichloride.

Numerous attempts have been made to produce benzyl chloride that is as free as possible of the secondary products benzal chloride and benzotrichloride. One possibility is to restrict the chlorination to only 30 – 40 % of the toluene input and then to separate by distillation the resulting mixture, which still has a very high toluene content. Here, the principal disadvantage is the high cost of the distillation process.

Pure benzyl chloride can be produced by chlorination in the vapor phase, provided the following equipment and procedure are used: The lower end of a packed column is provided with a bottom flask with a heater and an overflow. At the upper end is placed a condenser and a gas outlet. The feed point for fresh toluene and for toluene returning from the condenser is situated in the upper third of the column. The feed point for the chlorine (comprised of nozzles or frits) is located within the lower third. The temperature of the zone in which chlorination occurs is held at 125 °C during operation. This ensures that a temperature range is maintained within which benzyl chloride is a liquid whereas toluene is a gas. Under these conditions, benzyl chloride is rapidly removed from the chlorination section, flowing into the bottom flask; entrained toluene is returned from here to the chlorination section by evaporation. Hydrogen chloride formed during the chlorination, as well as toluene vapors, leave the column at its upper end. The toluene is liquefied in the condenser and returns to the chlorination section where it mixes with fresh toluene. The column is operated continuously. The bottom product is claimed to consist of 0.9 % toluene, 93.6 % benzyl chloride, and 5.5 % distillation residue. Several risks are inherent in this procedure. For example, the ignition temperature of toluene in chlorine gas is 185 °C. In addition, toluene–chlorine and benzyl chloride–chlorine mixtures are explosive over wide ranges (4 – 50 vol%, and 6 – 60 vol% respectively).

Today, most side-chain chlorination of toluene to produce benzyl chloride, together with benzal chloride, is apparently carried out in a continuous fashion. The silver- or lead-lined reaction columns described in the past represent an obsolete technology since they are subject to considerable corrosion. For photochlorination, it is now customary to use reactors of enamel or glass, e.g., of the loop type. The operating techniques have remained fundamentally unchanged, however.

Dibenzyl ether is formed as a byproduct in the alkaline hydrolysis of benzyl chloride to benzyl alcohol. This ether can be re-converted to benzyl chloride by cleavage with hydrogen chloride at a temperature below 100 °C. Zinc chloride, iron chloride, and antimony chloride, are recommended as catalysts. Only a moderately high degree of conversion is achieved (60 – 70 %). Higher yields are claimed to be obtained if the metal halogenide catalysts are replaced by pyridine, pyridine derivatives, or alkylbenzylammonium chloride. Another route to benzyl chloride is the chloromethylation of benzene, although this approach is of no commercial significance.

11.3.2 Use

Benzyl chloride is used mainly to produce plasticizers (e.g., benzyl butyl phthalate), benzyl alcohol, and phenylacetic acid via benzyl cyanide (used in the production of synthetic penicillin). On a smaller scale, it is used to produce quaternary ammonium salts (for disinfectants and phase-transfer catalysts), benzyl esters (benzyl benzoate and benzyl acetate for the flavors and perfumes industry), dyes of the triphenylmethane series, dibenzyl disulfide (antioxidant for lubricants), benzylphenol, and benzylamines.

11.4 System Characterisation

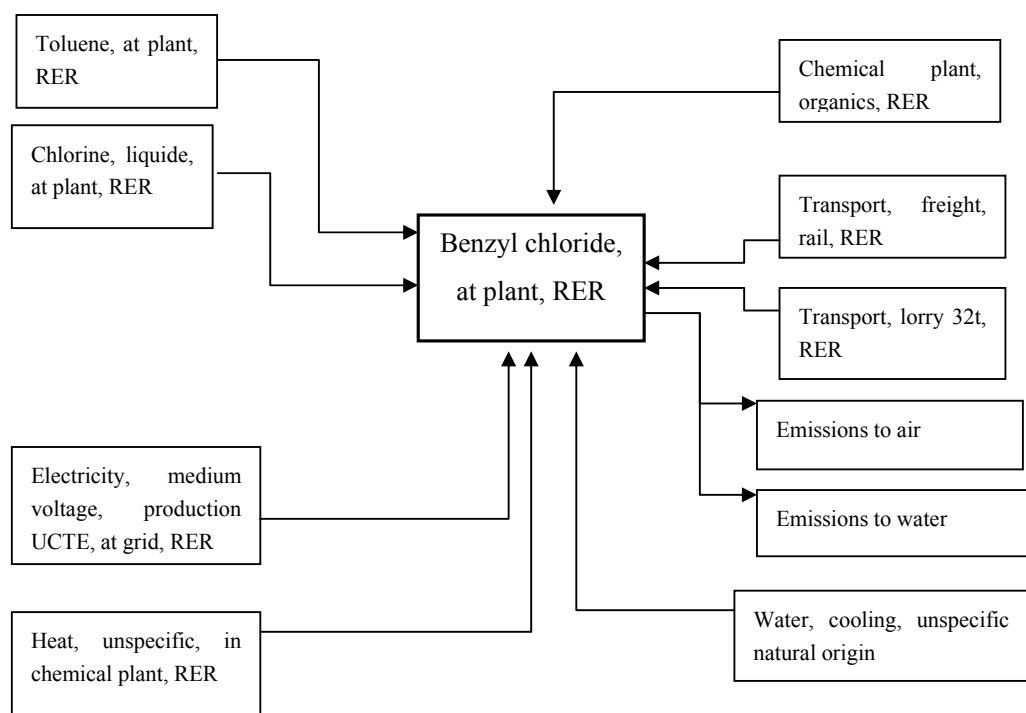


Fig. 11.1 Process chain for the production of benzyl chloride

This report corresponds to the module in the ecoinvent database for the production of **1 kg benzyl chloride, at plant, in Europe**. For this study the production of benzyl chloride by means of the chloriantion of toluene is considered. All data in the present report are referred to 1 kg benzyl chloride 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

11.5 Raw materials and auxiliaries

As there was no data available for the consumption of the raw materials for the production of benzyl chloride, the values are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the benzyl chloride production.

Tab. 11.2 Consumption of raw materials and auxiliaries for the production of 1 kg benzyl chloride

Input	Values	Remarks
Toluene (kg)	0.769	calculated with a yield of 95 %
Chlorine (kg)	0.585	calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

11.6 Energy

Gwalior (2005) gives data for the energy consumption of benzyl chloride production in an Indian plant.

Tab. 11.3 Energy consumption for the production of 1 kg benzyl chloride

Input	Gwalior 2005
Steam (MJ)	4.4825
Electricity (kWh)	2.027E-01

11.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the toluene and 50 km by lorry 32t and 100 km by train for the chlorine. Tab. 11.4 summarises the total transport amounts for the production of 1 kg benzyl chloride.

Tab. 11.4 Total transport amounts for the production of 1 kg benzyl chloride

tkm kg ⁻¹ benzyl chloride	lorry	train
Total	1.06E-01	5.20E-01

11.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of benzyl chloride plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

11.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of benzyl chloride. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 11.5.

Tab. 11.5 Process emissions to air from the production of 1 kg benzyl chloride

Output	Value	Remarks
Toluene, to air (kg)	1.54E-03	Estimated as 0.2 % of input
Chlorine, to air (kg)	1.17E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.11E-1	from waster water treatment
Waste heat (MJ)	7.297E-01	calculated from electricity input

11.10 Emissions to water

The remaining amount of unreacted toluene (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for was assumed with 90%. The carbon contained in the removed toluene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of toluene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 11.6.

The remaining amount of the unreacted chlorine (4.8 % of the input) and the chloride contained in the hydrogen chloride byproduct was assumed to leave the the production process as chloride emission in the waste water. The value for AOX is calculated from the chloride emissions.

Tab. 11.6 Process emissions to water from the production 1 kg benzyl chloride

Output	Value	Remarks
Toluene, to water (kg)	3.69E-03	calculated from mass balance
Chloride, to water (kg)	3.01E-01	calculated from mass balance
COD, BOD (kg)	1.11E-02	calculated from water emissions
TOC, DOC (kg)	3.37E-03	calculated from water emissions
AOX (kg)	3.01E-01	calculated from water emissions

11.11 Life cycle inventory and Data Quality Considerations

Tab. 11.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of benzyl chloride. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 11.7 Unit process raw data for the production of benzyl chloride

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	benzyl chloride, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	benzyl chloride, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	toluene, liquid, at plant	RER	0	kg	7.69E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	chlorine, liquid, production mix, at plant	RER	0	kg	5.85E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry 32t	RER	0	tkm	1.06E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	5.20E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.03E-1	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	4.48E+0	1	1.14	(1,3,1,5,1,3); Gwalior 2005
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	7.30E-1	1	1.14	(1,3,1,5,1,3); Calculated from the electricity input
	-	4	Toluene	-	-	kg	1.54E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Chlorine	-	-	kg	1.17E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.11E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Toluene	-	-	kg	3.69E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Chlorine	-	-	kg	3.01E-1	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.11E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.11E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.37E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.37E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	3.01E-1	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

11.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 11.8 shows selected LCI results and the cumulative energy demand for the production of benzyl chloride.

Tab. 11.8 Selected LCI results and the cumulative energy demand for the production of benzyl chloride

Name	benzyl chloride, at plant		
	Location		
	Unit Infrastructure	Unit	RER kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	65.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	7.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.7E-1
CO2, fossil	air	kg	2.3E+0
NMVOC	air	kg	3.6E-3
nitrogen oxides	air	kg	3.9E-3
sulphur dioxide	air	kg	5.2E-3
particulates, <2.5 um	air	kg	3.6E-4
BOD	water	kg	1.3E-2
cadmium	soil	kg	1.0E-9

11.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	benzyl chloride, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of benzyl chloride including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Benzylchlorid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "benzyl chloride, at plant, RER" is modelled for the production of benzyl chloride from toluene in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H7Cl
	501	StatisticalClassification	
	502	CASNumber	100-44-7
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Chlorination of toluene
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

11.14 References

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12 Butane oxidation

Author: Jürgen Sutter, ETH S+U, Zurich
Review: Roland Hirschier, EMPA, St. Gallen
Last changes: 2007

12.1 Introduction

The liquid-phase oxidation of hydrocarbons is an important process to produce acetic acid, formic acid or methyl acetate. In the United States, butane is used as the hydrocarbon. About 43 kg of formic acid is produced per ton of acetic acid. Unreacted hydrocarbons, volatile neutral constituents, and water are separated first from the oxidation product. Formic acid is separated in the next column; azeotropic distillation is generally used for this purpose. The entrainers preferred in this process are benzene or chlorinated hydrocarbons. The formic acid contains about 2 wt % acetic acid, 5 wt % water, and 3 wt % benzene. Formic acid with a content of about 98 wt % can be produced by further distillation.

12.2 System Characterisation

In this study the oxidation of butane is described. All data in the present report are referred to 1 kg butane. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the oxidation of butane, formic acid, acetic acid, acetone, methyl ethyl ketone, methyl acetate, and ethyl acetate are derived as co-products. Therefore an allocation between the six products was made.

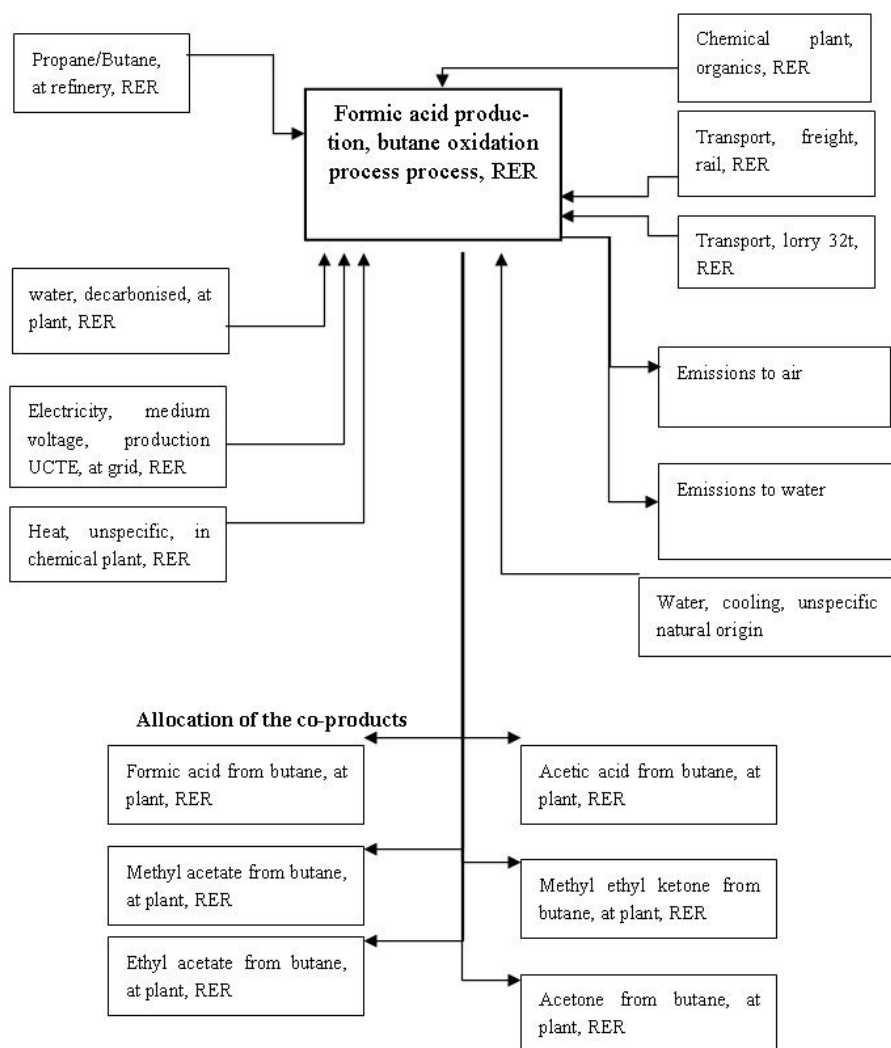


Fig. 12.1 Process chain for the liquid phase oxidation of butane

12.3 Raw materials and auxiliaries

The data for the input of raw materials and water are taken from Chauvel (1989), which reports data from French production plants in the late 1980ies.

The values in Chauvel 1989 are based on the production of acetic acid. In this inventory the values are converted to the base of butane with the yields of the diverse co-products from Emanuel (1963).

The n-butane input is modelled with the ecoinvent modul “propane/butane, at refinery, RER”.

Tab. 12.1 Consumption of raw materials and auxiliaries for the butane oxidation

Input	Chauvel 1989
n-Butane (kg)	1
Process water (kg)	4.65
Cooling water (kg)	242

12.4 Energy

The data for the energy inputs are taken from Chauvel (1989), which reports data from French production plants in the late 1980ies.

The values in Chauvel 1989 are based on the production of acetic acid. In this inventory the values are converted to the base of butane with the yields of the diverse co-products from Emanuel (1963).

Tab. 12.2 Energy consumption for the butane oxidation

Input	Chauvel 1989
Steam (MJ)	9.21
Electricity (kWh)	0.27

12.5 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 12.3 summarises the total transport amounts for the butane oxidation.

Tab. 12.3 Total transport amounts for the butane oxidation

tkm kg ⁻¹ butane	lorry	train
Raw materials	0.1	0.6

12.6 Infrastructure and land use

No information was readily available about infrastructure and land-use of butane oxidation plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

12.7 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the butane oxidation. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 12.4.

Tab. 12.4 Process emissions to air from the butane oxidation

Output	Value	Remarks
Butane, to air (kg)	2E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	3.78E-01	from wastewater treatment
Waste heat (MJ)	5.6	calculated from electricity input

12.8 Emissions to water

The other byproducts are assumed to leave the production process with the waste water. These assumptions lead to a pollution of the waste water with 0.016 kg acetaldehyde, 0.016 kg methanol, 0.023 kg ethanol, 0.01 kg butanol, and 0.013 kg butyl acetate per kg butane (calculated with the yields from Emanuel 1963).

Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for the emissions was assumed with 90%. The carbon contained in the removed byproducts was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of the emissions in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 12.5.

Tab. 12.5 Process emissions to water from the butane oxidation

Output	Value	Remarks
Ethanol, to water (kg)	2.3E-03	calculated from mass balance
Acetaldehyde, to water (kg)	1.6E-03	calculated from mass balance
Methanol, to water (kg)	1.6E-03	calculated from mass balance
Butanol, to water (kg)	1E-03	calculated from mass balance
Butyl acetate, to water (kg)	1.3E-03	calculated from mass balance
COD, BOD (kg)	1.52E-02	calculated from water emissions
TOC, DOC (kg)	4.2E-03	calculated from water emissions

12.9 Allocation of the co-products of the butane oxidation

Within the butane oxidation, acetic acid, methyl ethyl ketone, acetone, ethyl acetate, and methyl acetate are obtained as co-products. As allocation factor the weight of the two products was used. The yields and the used allocation factors of the diverse co-products are given in Tab. 12.6.

Tab. 12.6 Yields and allocation of the co-products from butane oxidation

Process	Butane oxidation, RER		
	Unit	per kg butane	Allocation factor
Formic acid from butane, at plant, RER	kg	0.040	0.0249
Acetic acid from butane, at plant, RER	kg	0.93	0.5795
Methyl ethyl ketone from butane, at plant, RER	kg	0.216	0.1346
Ethyl acetate from butane, at plant, RER	kg	0.182	0.1134
Methyl acetate from butane, at plant, RER	kg	0.175	0.1090
Acetone from butane, at plant, RER	kg	0.062	0.0386

12.10 Data Quality Considerations

Tab. 6.15 shows shows the life cycle inventory and the data quality indicators for the butane oxidation.

The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 12.7 Unit process raw data for the butane oxidation

401	InputGroup OutputGroup	Name	Location	InfrastructureProcesses	Unit	formic acid production, butane oxidation process	Uncertainty Type	Standard Deviation %	GeneralComment	formic acid from butane, at plant	acetic acid from butane, at plant	methyl ethyl ketone from butane, at plant	ethyl acetate from butane, at plant	methyl acetate, at plant	acetone from butane, at plant	
662		Location				RER				RER	RER	RER	RER	RER	RER	
493		InfrastructureProcess				0				0	0	0	0	0	0	
403		Unit				kg				kg	kg	kg	kg	kg	kg	
allocated products products products products resource, in water technosphere	-	2	formic acid from butane, at plant	RER	0	kg	4.00E-2			100	0	0	0	0	0	
	-	2	acetic acid from butane, at plant	RER	0	kg	9.30E-1			0	100	0	0	0	0	
	-	2	methyl ethyl ketone from butane, at plant	RER	0	kg	2.16E-1			0	0	100	0	0	0	
	-	2	ethyl acetate from butane, at plant	RER	0	kg	1.82E-1			0	0	0	100	0	0	
	-	2	methyl acetate, at plant	RER	0	kg	1.75E-1			0	0	0	0	100	0	
	-	2	acetone from butane, at plant	RER	0	kg	6.20E-2			0	0	0	0	0	100	
emission air, high population density	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.42E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	propane/ butane, at refinery	CH	0	kg	1.00E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	water, deionised, at plant	CH	0	kg	4.65E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	transport, freight, rail	RER	0	tkm	6.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.70E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	9.21E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	2.492	57.944	13.458	11.340	10.903	3.863
	5	-	chemical plant, organics	RER	1	unit	6.42E-10	1	3.09	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Heat, waste	-	-	MJ	9.71E-1	1	1.51	(1,3,5,3,1,3); Calculated from the electricity input	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Carbon dioxide, fossil	-	-	kg	1.51E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment	2.492	57.944	13.458	11.340	10.903	3.863
emission water, river	-	4	Butane	-	-	kg	2.00E-3	1	1.62	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Ethanol	-	-	kg	2.30E-3	1	1.62	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Acetaldehyde	-	-	kg	1.60E-3	1	1.62	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Methanol	-	-	kg	1.60E-3	1	3.09	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Butanol	-	-	kg	1.00E-3	1	1.62	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	Butyl acetate	-	-	kg	1.30E-3	1	1.62	(4,5,na,na,na,na); Estimation	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.51E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.51E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4.20E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	2.492	57.944	13.458	11.340	10.903	3.863
	-	4	TOC, Total Organic Carbon	-	-	kg	4.20E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	2.492	57.944	13.458	11.340	10.903	3.863

12.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 12.8 shows selected LCI results and the cumulative energy demand for the butane oxidation.

Tab. 12.8 Selected LCI results and the cumulative energy demand for the butane oxidation

Name			acetic acid from butane, at plant	acetone from butane, at plant	ethyl acetate from butane, at plant	formic acid from butane, at plant	methyl acetate, at plant	methyl ethyl ketone from butane, at plant
Location	Unit		RER	RER	RER	RER	RER	RER
Infrastructure			kg 0	kg 0	kg 0	kg 0	kg 0	kg 0
LCIA results								
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	45.0	45.0	45.0	45.0	45.0	45.0
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	1.8	1.8	1.8	1.8	1.8	1.8
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.3	0.3	0.3	0.3	0.3	0.3
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3	0.3	0.3	0.3	0.3	0.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0	0.0	0.0	0.0	0.0
selected LCI results								
land occupation	resource	m2a	9.3E-2	9.3E-2	9.3E-2	9.3E-2	9.3E-2	9.3E-2
CO ₂ , fossil	air	kg	1.1E+0	1.1E+0	1.1E+0	1.1E+0	1.1E+0	1.1E+0
NMVOG	air	kg	3.7E-3	3.7E-3	3.7E-3	3.7E-3	3.7E-3	3.7E-3
nitrogen oxides	air	kg	2.5E-3	2.5E-3	2.5E-3	2.5E-3	2.5E-3	2.5E-3
sulphur dioxide	air	kg	2.2E-3	2.2E-3	2.2E-3	2.2E-3	2.2E-3	2.2E-3
particulates, <2.5 µm	air	kg	2.0E-4	2.0E-4	2.0E-4	2.0E-4	2.0E-4	2.0E-4
BOD	water	kg	1.1E-2	1.1E-2	1.1E-2	1.1E-2	1.1E-2	1.1E-2
cadmium	soil	kg	3.8E-10	3.8E-10	3.8E-10	3.8E-10	3.8E-10	3.8E-10

12.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	formic acid production, butane oxidation process
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Butane oxidation including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ameisensäure Produktion, Butanoxidierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "formic acid production, butane oxidation process" delivers the co-products formic acid, methyl acetate, ethyl acetate, acetic acid, methyl ethyl ketone, and acetone. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1989
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Oxidation of butane
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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13 Butane-1,4-diol

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

13.1 Introduction

Butane-1,4-diol ($\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$, $\text{C}_4\text{H}_{10}\text{O}_2$, CAS-No. 110-63-4, Synonym: 1,4-butanediol, butylene glycol) is a colorless, almost odorless, hygroscopic liquid, readily soluble in water, alcohols, ketones, glycol ethers, and glycol ether acetates, less soluble in diethyl ether and esters, and not miscible with aliphatic and aromatic hydrocarbons and chlorinated hydrocarbons. For this inventory the functional unit is 1 kg of pure butane-1,4-diol. The most important chemical and physical properties of butane-1,4-diol used in this inventory are given here.

Tab. 13.1 Chemical and physical properties of butane-1,4-diol (Ullmann 2001)

Property	Value	Unit	Remarks
Molecular weight	90,12	g mol^{-1}	
Specific gravity	1017	kg m^{-3}	at 20 °C
Boiling point	230,5	°C	at 101,3 kPa
Melting point	20,3	°C	at normale pressure
Heat of combustion	2,585	MJ kg^{-1}	

13.2 Reserves and Resources

Butane-1,4-diol is usually produced by hydrogenation of 2-butyne-1,4-diol. The synthesis of 2-butyne-1,4-diol, based on the work of Reppe and his coworkers, is carried out by reaction under pressure of acetylene and an aqueous solution of formaldehyde, catalyzed by copper acetylide. In 1984 over 95 % of the world production of butane-1,4-diol was based on the Reppe synthesis (Chauvel 1989).

13.3 Production Technologies and Use

13.3.1 Production Technologies

One step Process (Ullmann 2001)

1,4-Butanediol is made on a large industrial scale by continuous hydrogenation of the 2-butyne-1,4-diol over modified nickel catalysts. The one-stage flow process is carried out at 80 – 160 °C and 300 bar according to .

An aqueous solution of 2-butyne-1,4-diol (30 – 50 %), together with carbon monoxide-free hydrogen and recycled reaction mixture, which acts as medium for dissipation of heat, is lead over a reduced nickel – copper – manganese catalysts on silica gel strands. The initial temperature in the reactor is 80 °C; the temperature must not exceed 170 °C. In order to obtain a better distribution of the liquid, hydrogen is also circulated.

The raw product contains methanol, propanol, and butanol as byproducts as well as traces of 2-methyl-

butane-1,4-diol, hydroxybutyraldehyde, acetals, and triols. The reactor effluent is worked up to pure butane-1,4-diol by fractional distillation.

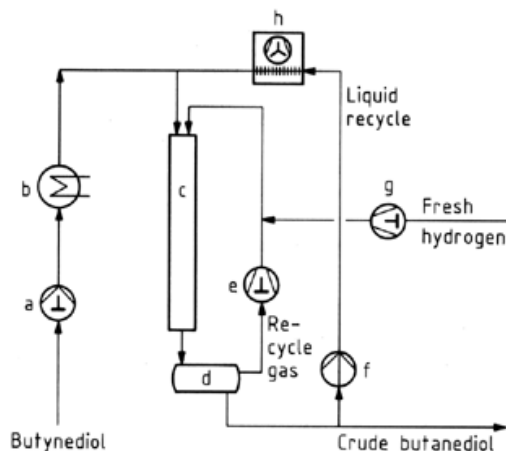


Fig. 13.1 Hydrogenation of 2-butyne-1,4-diol (taken from Ullmann 2001, Figure 3)

a) Piston pump; b) Preheater; c) Hydrogenation reactor; d) Gas – liquid separator; e) Recycle compressor; f) Canned motor pump; g) Hydrogen compressor; h) Air cooler

Two step process (Ullmann 2001)

The hydrogenation of butynediol can also be carried out in two process steps. In the first stage, for example, at 40 bar, mainly 2-butene-1,4-diol is obtained, which is then completely hydrogenated at 300 bar in the second stage.

A further variant is the low-pressure hydrogenation at ca. 20 bar over a suspended Raney nickel catalyst, followed by hydrogenation at 120 – 140 °C and 140 – 210 bar on a fixed-bed contact.

Other Processes (Ullmann 2001)

1,4-Butanediol can also be made from raw materials other than acetylene. Allyl alcohol and synthesis gas ($\text{CO} + \text{H}_2$) can be converted to butane-1,4-diol according to a process invented by Daicel.

Mitsubishi uses a three-step process: (1) the catalytic reaction of butadiene and acetic acid yields 1,4-diacetoxy-2-butene; (2) subsequent hydrogenation gives 1,4-diacetoxybutane; and (3) hydrolysis leads to butane-1,4-diol:

BASF also has patented a similar process, in which acetic acid first adds to butadiene; the product then isomerizes to 1,4-diacetoxy-2-butene.

According to a process practiced by Toyo Soda chlorine first adds to butadiene to form a mixture of 1,4-dichloro-2-butene and 3,4-dichloro-1-butene. This mixture reacts with sodium acetate to form 1,4-diacetoxy-2-butene, which is subsequently hydrogenated directly to butane-1,4-diol.

In a patent described by Shell butadiene reacts with a peroxide to diperoxybutene, which is then converted to butane-1,4-diol by hydrogenation.

13.3.2 Use

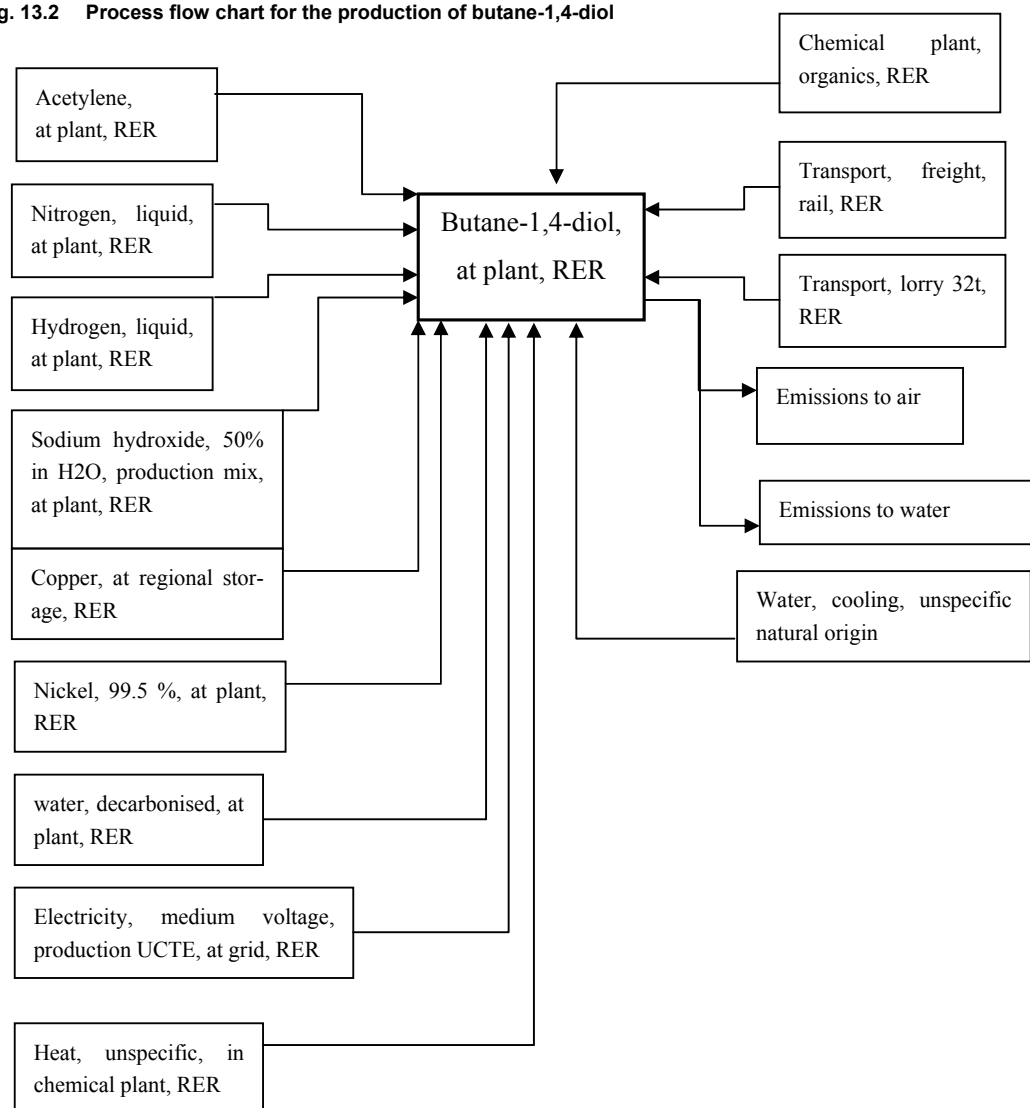
1,4-Butanediol is a starting material for the production of tetrahydrofuran which can be made directly from butadiene according to a process developed by Chevron: 1,2-epoxy-3-butene is made from butadiene and a peroxide and subsequent isomerization yields 2,5-dihydrofuran, which is hydrogenated to tetrahydrofuran.

Another product derived from butane-1,4-diol is butyrolactone. In addition, both butane-1,4-diol and its derivatives are used in the textile, leather, photographic, food, and pharmaceutical industries.

The worldwide capacity for butane-1,4-diol has been ca. 512 000 t/a in 1995. European producers are BASF and GAF/Chemische Werke Hüls; producers in the United States are Du Pont, GAF, and BASF Wyandotte. Whereas all these producers use acetylene as starting material, a plant at Mitsubishi Chemical Industries has been making butane-1,4-diol from butadiene since 1982.

13.4 System Characterisation

Fig. 13.2 Process flow chart for the production of butane-1,4-diol



This report corresponds to the dataset in the ecoinvent database for the production of 1 kg **butane-1,4-**

diol, at plant, in Europe. For this study the production of butane-1,4-diol by hydrogenation of 2-butyne-1,4-diol from acetylene is considered. All data in the present report are referred to 1 kg butane-1,4-diol 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

13.5 Data sources

The production process for butane-1,4-diol was assessed with data from Chauvel 1989, which are calculated for the production of butane-1,4-diol by Reppe synthesis in France in 1986, and from Cooper 2001. These data are calculated for the production of butane-1,4-diol by Reppe synthesis in Midwest USA in 2001.

13.6 Raw materials and auxiliaries

Tab. 13.1 shows the values for the use of raw materials and auxiliaries, which are taken from Chauvel 1989 and Cooper 2001.

Tab. 13.2 Consumption of raw materials and auxiliaries for the production of 1 kg butane-1,4-diol

Input	Chauvel 1989	Cooper 2001	This Inventory
Acetylene (kg)	0.32		0.32
Formaldehyde (kg)	0.7326		7.326E-01
H ₂ (kg)	5.4E-02	4.89E-02	5.4E-02
N ₂ (kg)	3.125E-02		3.125E-02
NaOH (kg)	3E-03		3-E-03
Copper (kg)		2.04E-04	2.04E-04
Nickel (kg)		2.14E-04	2.14E-04
Cooling water (kg)	420	490	490
Process water (kg)	1		1

13.7 Energy

Data for the energy inputs for the production of butane-1,4-diol are taken from Chauvel (1989).

Tab. 13.3 Energy consumption for the production of 1 kg butane-1,4-diol

Input	Values
Steam (MJ)	24.75
Electricity (kWh)	1.8E-01

13.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 50 km by lorry 32t and 100 km by train for the hydrogen and the nitrogen, and 100 km by lorry 32t and 600 km by train for the other raw materials and the auxiliaries. Tab. 13.4 summarises the total transport amounts for the production of 1 kg butane-1,4-diol.

Tab. 13.4 Total transport amounts for the production of 1 kg butane-1,4-diol

tkm kg ⁻¹ butane-1,4-diol	lorry	train
Raw materials	2.344E-01	1.39

13.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of 1,4-butanediol plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

13.10 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 1.8E-01 kWh/kg butane-1,4-diol is converted to waste heat and that 100% of the waste heat is released to the air.

The process emissions to air for the production of butane-1,4-diol are taken from Cooper 2001. The assumed values for the emissions to air are given in Tab. 13.5.

Tab. 13.5 Process emissions to air from the production of 1 kg butane-1,4-diol

Output	Value	Remarks
Acetylene, to air (kg)	4.61E-05	taken from Cooper 2001
Formaldehyde, to air (kg)	5.7E-05	taken from Cooper 2001
Butanol, to air (kg)	3.1E-05	taken from Cooper 2001
Organic compounds, to air (kg)	1.29E-03	taken from Cooper 2001
Copper, to air, (kg)	2.26E-06	taken from Cooper 2001
Nickel, to air (kg)	1.11E-08	taken from Cooper 2001
Waste heat (MJ)	6.48E-01	calculated from electricity input

13.11 Emissions to water

The process emissions to water for the production of butane-1,4-diol are taken from Cooper 2001. The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of formaldehyde and butanol in the waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 13.6.

Tab. 13.6 Process emissions to water from the production 1 kg butane-1,4-diol

Output	Value	Remarks
Formaldehyde, to water (kg)	2.98E-05	taken from Cooper 2001
Copper, to water (kg)	1.34E-06	taken from Cooper 2001
Nickel, to water (kg)	7.7E-07	taken from Cooper 2001
Butanol, to water (kg)	1.1E-05	taken from Cooper 2001
COD, BOD (kg)	5.78E-05	calculated from water emissions
TOC, DOC (kg)	1.91E-05	calculated from water emissions

13.12 Life cycle inventory of butane-1,4-diol production and data Quality Considerations

Tab. 13.7 shows the life cycle inventory and the data quality indicators of the production of butane-1,4-diol. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 13.7 Unit process raw data for the production of butane-1,4-diol

InputGroup	OutputGroup	Name	Location	Infrastructure	Process	Unit	butane-1,4-diol, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
401	0	butane-1,4-diol, at plant	RER	0	kg	1.00E+0				
662										
493										
403										
resource, in water technosphere	4	- Water, cooling, unspecified natural origin	-	-	m3	4.90E-1		1	1.09	(1,3,1,1,1,3); Cooper 2001
	5	- water, deionised, at plant	CH	0	kg	1.00E+0		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- acetylene, at regional storehouse	CH	0	kg	3.20E-1		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- formaldehyde, production mix, at plant	RER	0	kg	7.33E-1		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- hydrogen, liquid, at plant	RER	0	kg	5.40E-2		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- nitrogen, liquid, at plant	RER	0	kg	3.13E-2		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	3.00E-3		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- copper, at regional storage	RER	0	kg	2.04E-4		1	1.09	(1,3,1,1,1,3); Cooper 2001
	5	- nickel, 99.5%, at plant	GLO	0	kg	2.14E-4		1	1.09	(1,3,1,1,1,3); Cooper 2001
	5	- transport, lorry >16t, fleet average	RER	0	tkm	2.34E-1		1	2.09	(4,5,na,na,na,na); Standard distances
	5	- transport, freight, rail	RER	0	tkm	1.39E+0		1	2.09	(4,5,na,na,na,na); Standard distances
	5	- electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.80E-1		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- heat, unspecific, in chemical plant	RER	0	MJ	2.48E+1		1	1.51	(1,3,5,3,1,3); Chauvel 1989
	5	- chemical plant, organics	RER	1	unit	4.00E-10		1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4 Heat, waste	-	-	MJ	6.48E-1		1	1.51	(1,3,5,3,1,3); Calculated from the electricity input
	-	4 Ethyne	-	-	kg	4.61E-5		1	1.51	(1,3,1,1,1,3); Cooper 2001
	-	4 Formaldehyde	-	-	kg	5.70E-5		1	1.51	(1,3,1,1,1,3); Cooper 2001
	-	4 Butanol	-	-	kg	3.10E-5		1	1.09	(1,3,1,1,1,3); Cooper 2001
	-	4 NMVOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	1.29E-3		1	1.51	(1,3,1,1,1,3); Cooper 2001
	-	4 Copper	-	-	kg	2.26E-6		1	5.01	(1,3,1,1,1,3); Cooper 2001
	-	4 Nickel	-	-	kg	1.11E-8		1	5.01	(1,3,1,1,1,3); Cooper 2001
	-	4 Formaldehyde	-	-	kg	2.98E-5		1	3.01	(1,3,1,1,1,3); Cooper 2001
	-	4 Copper, ion	-	-	kg	1.34E-6		1	3.01	(1,3,1,1,1,3); Cooper 2001
	-	4 Nickel, ion	-	-	kg	7.70E-7		1	5.01	(1,3,1,1,1,3); Cooper 2001
emission water, river	-	4 Butanol	-	-	kg	1.10E-5		1	1.09	(1,3,1,1,1,3); Cooper 2001
	-	4 BOD5, Biological Oxygen Demand	-	-	kg	5.78E-5		1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4 COD, Chemical Oxygen Demand	-	-	kg	5.78E-5		1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4 DOC, Dissolved Organic Carbon	-	-	kg	1.91E-5		1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4 TOC, Total Organic Carbon	-	-	kg	1.91E-5		1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-									
	-									

13.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 13.8 shows selected LCI results and the cumulative energy demand for the production of butane-1,4-diol.

Tab. 13.8 Selected LCI results and the cumulative energy demand for the production of butane-1,4-diol

Name		butane-1,4-diol, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	86.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	2.1E-1
CO ₂ , fossil	air	kg	4.1E+0
NM VOC	air	kg	7.1E-3
nitrogen oxides	air	kg	6.6E-3
sulphur dioxide	air	kg	8.6E-3
particulates, <2.5 µm	air	kg	5.5E-4
BOD	water	kg	4.0E-3
cadmium	soil	kg	1.1E-9

13.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	butane-1,4-diol, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 1,4-butanediol including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Butan-1,4-diol, ab Werk
	491	Synonyms	butylene glycol
	492	GeneralComment	The process "c1,4-butanediol, at plant, RER" is modelled for the production of 1,4-butanediol from acetylene in Europe. Raw materials, Energy consumptions and emissions are modelled with literature data. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H10O2
	501	StatisticalClassification	
	502	CASNumber	110-63-4
TimePeriod	601	StartDate	1989
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrigenation of butynediol from acetylene
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

13.15 References

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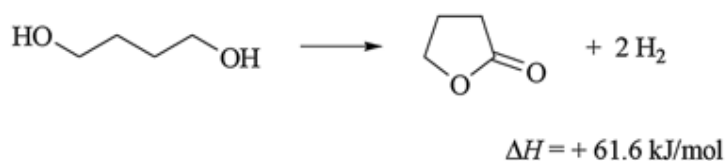
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14 Butane-1,4-diol dehydrogenation

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

14.1 Introduction

The Reppe process is the most important process to produce butyrolactone. It involves the endothermic dehydrogenation of 1,4-butanediol in the gas phase. This process is used by BASF, ISP, and Lyondell.



Preheated 1,4-butanediol vapor is introduced into a hot stream of circulating hydrogen and passed at atmospheric pressure through a bed of copper catalyst at temperatures between 180 and 240°C. The yield of butyrolactone is approximately 95 %. The reaction takes place via γ -hydroxybutyraldehyde.

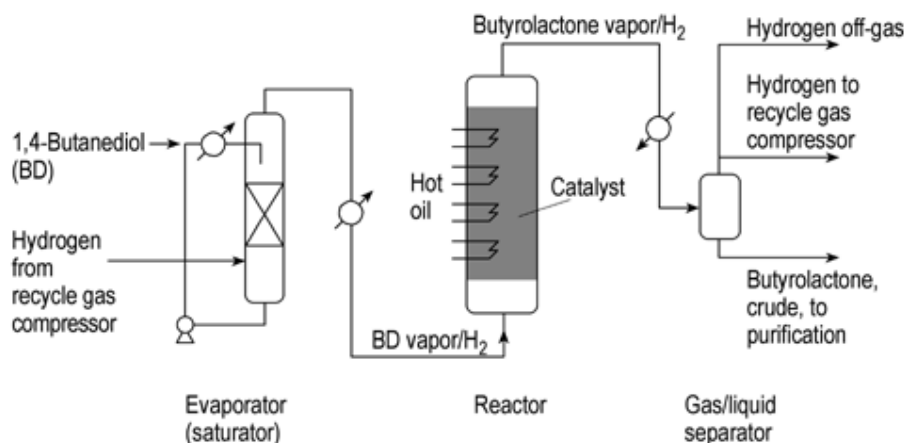


Fig. 14.1 Production of γ -butyrolactone by dehydrogenation of 1,4-butanediol

The byproduct hydrogen off-gas requires only simple purification before reuse (e.g., catalytic methanization of carbon monoxide impurities). The crude butyrolactone separated from the recycle gas stream contains small amounts of byproducts, including 1,4-butanediol, butyric acid, and high boilers, from which butyrolactone is separated by distillation.

Butyrolactone itself is noncorrosive and can be handled in carbon steel apparatus. However, where parts of the synthesis or distillation vessels and pipes come into contact with hot crude product containing butyric acid, they must be made of stainless steel.

14.2 System Characterisation

In this study the Reppe process is considered. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the

manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

Within the Reppe process, butyrolactone and hydrogen are derived as co-products. Therefore an allocation between the two products was made. The following processes were modelled:

- Butanediol dehydrogenation, GLO. Multi output process with butyrolactone, at plant, GLO and hydrogen from butanediol dehydrogenation, at plant, GLO as allocated products

14.3 Butyrolactone

Butyrolactone ($C_4H_6O_2$; CAS 96-48-0) is a colorless, slightly hygroscopic liquid having a faint odor. Butyrolactone is miscible in all proportions with water, alcohols, esters, ethers, ketones, and aromatic hydrocarbons. It has limited miscibility with linear and cyclic aliphatic hydrocarbons. It is an excellent solvent for numerous polymers and a selective solvent for lower hydrocarbons.

Standard-grade butyrolactone is about 99.7 % pure. The main impurities in the Reppe-based process are 1,4-butanediol, butyric acid, and water; in the maleic ester based processes, dimethyl succinate, 4-hydroxybutyl butyl ether, and water. Electronic-grade butyrolactone is about 99.9 % pure and is subject to stringent restrictions regarding the content of specific ions such as Fe ions, Na^+ , Cl^- , NO_3^- .

The most important chemical and physical properties of butyrolactone used in this inventory are given here.

Tab. 14.1 Chemical and physical properties of butyrolactone (Ullmanns 2004)

Property	Value	Unit
Molecular weight	86.09	$g\ mol^{-1}$
Specific gravity	1130	$kg\ m^{-3}$
Boiling point	204-206	$^{\circ}C$
Melting point	-43.5	$^{\circ}C$

14.4 Use

The main use of butyrolactone is as an intermediate in the synthesis of N-methylpyrrolidone (NMP), pyrrolidone, herbicides, growth regulators, and the rubber additive thiodibutyric acid. Butyrolactone is used as a solvent for polymers and as a polymerization catalyst; in hairwave compositions and sun lotions; and in pharmaceuticals. It is also used in printing inks, e.g., for ink-jet printing, as an extractant in the petroleum industry, as a stabilizer for chlorohydrocarbons and phosphorus-based pesticides, and as a nematocide. More recent applications are in the electronics field as a cosolvent for capacitor electrolytes and as a cosolvent for photoresists.

14.5 Raw materials and auxiliaries

As there was no data available for the Reppe process, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the Reppe process.

Tab. 14.2 Consumption of raw materials and auxiliaries for the Reppe process

Input	Values	Remarks
Butanediol (kg)	1	
Cooling water (kg)	22.8	Estimation

14.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the Reppe process. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used.

Tab. 14.3 Energy consumption for the Reppe process

Input	Per kg butanediol
Heat, natural gas burned in industrial furnace > 100 kW	1.9
Electricity (kWh)	0.316

14.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km. Tab. 14.4 summarises the total transport amounts for the Reppe process.

Tab. 14.4 Total transport amounts for the Reppe process

tkm kg ⁻¹ butanediol	lorry	train
Total transports	1E-01	6E-01

14.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of Reppe process plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

14.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the Reppe process. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the butanediol input.

Tab. 14.5 Process emissions to air from the Reppe process

Output	Value	Remarks
Waste heat (MJ)	1.139	calculated from electricity input
Butanediol, to air (kg)	2E-03	estimated as 0.2% of the input
Carbon dioxide, fossil, to air (kg)	8.45E-02	from waster water treatment

14.10 Emissions to water

The remaining amount of unreacted butanediol (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for butanediol is assumed with 90%. The carbon contained in the removed butanediol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of dimethyl amine and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 14.6 Process emissions to water from the Reppe process

Output	Value	Remarks
Butanediol, to water (kg)	4.8E-03	calculated from mass balance
COD, BOD (kg)	9E-03	calculated from water emissions
TOC, DOC (kg)	2.56E-03	calculated from water emissions

14.11 Allocation of butyrolactone and hydrogen

Within the Reppe process, butyrolactone and hydrogen are obtained as co-products. A stoichiometric allocation of the two products is made. The used allocation factors are given in Tab. 14.7.

Tab. 14.7 Allocation of butyrolactone and hydrogen from the Reppe process

Process	Reppe process, GLO		
	Unit	per kg butanediol	Allocation factor (stoichiometric)
Butyrolactone, at plant, GLO	Kg	0.908	95.556
Hydrogen from Reppe process, at plant, GLO	kg	0.042	4.444

14.12 Life cycle inventory of the Reppe process and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the Reppe process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 14.8 Unit process raw data for the Reppe process

401	InputGroup	OutputGroup	Name	Location	Infrastructure	Processes	Unit	butanediol dehydrogenation	UncertaintyType	StandardDeviation95 %	GeneralComment	butyrolactone	hydrogen, from butanediol dehydrogenation
662	493	403	Location	Infrastructure	Processes	Unit		GLO				GLO	GLO
493			InfrastructureProcess										
403			Unit										
allocated products resource, in water technosphere	-	2	butyrolactone	GLO	0	kg		3.62E-1				100	0
	-	2	hydrogen, from butanediol dehydrogenation	GLO	0	kg		1.54E+0				0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3		2.28E-2	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
	5	-	butane-1,4-diol, at plant	RER	0	kg		1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation	95.556	0.444
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ		1.90E+0	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh		3.16E-1	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
	5	-	transport, lorry >16t, fleet average	RER	0	tkm		1.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	95.556	0.444
	5	-	transport, freight, rail	RER	0	tkm		6.00E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	95.556	0.444
emission air, high population density	5	-	chemical plant, organics	RER	1	unit		3.80E-10	1	3.09	(4,5,na,na,na,na); Estimation	95.556	0.444
	-	4	Heat, waste	-	-	MJ		1.14E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input	95.556	0.444
	-	4	1,4-Butanediol	-	-	kg		2.00E-3	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg		8.45E-2	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
	-	4	1,4-Butanediol	-	-	kg		4.80E-3	1	1.30	(4,5,na,na,na,na); Estimation	95.556	0.444
	-	4	BOD5, Biological Oxygen Demand	-	-	kg		9.00E-3	1	1.62	(4,5,na,na,na,na); from water emissions	95.556	0.444
	-	4	COD, Chemical Oxygen Demand	-	-	kg		9.00E-3	1	1.62	(4,5,na,na,na,na); from water emissions	95.556	0.444
	-	4	DOC, Dissolved Organic Carbon	-	-	kg		2.56E-3	1	1.62	(4,5,na,na,na,na); from water emissions	95.556	0.444
	-	4	TOC, Total Organic Carbon	-	-	kg		2.56E-3	1	1.62	(4,5,na,na,na,na); from water emissions	95.556	0.444

14.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht et al. (2007). It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the Reppe process.

Tab. 14.9 Selected LCI results and the cumulative energy demand for the Reppe process

Name			butyrolactone	hydrogen, from butanediol dehydrogenation
Location Unit Infrastructure		Unit	GLO kg 0	GLO kg 0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	63.1	44.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	8.0	5.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.7	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.2	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1
selected LCI results				
land occupation	resource	m2a	1.6E-1	1.2E-1
CO2, fossil	air	kg	3.1E+0	2.2E+0
NM/OC	air	kg	1.2E-2	8.4E-3
nitrogen oxides	air	kg	5.0E-3	3.5E-3
sulphur dioxide	air	kg	6.4E-3	4.5E-3
particulates, <2.5 um	air	kg	4.2E-4	3.0E-4
BOD	water	kg	8.1E-3	5.7E-3
cadmium	soil	kg	9.0E-10	6.4E-10

14.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	butanediol dehydrogenation
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Butanediol dehydrogenation including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Butanedioldehydrierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "butanediol dehydrogenation " delivers the co-products butyrolactone and hydrogen. The allocation is based on stoichiometric calculation.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	extraction
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Gewinnung
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for the world..
Technology	692	Text	Dehydrogenation of butanediol
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

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Schwarz, W., Schossig, J., Rossbacher, R., Höke, H.: Butyrolactone. Published online: 2002. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/emrw/9783527306732/home>

15 Butene hydration

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

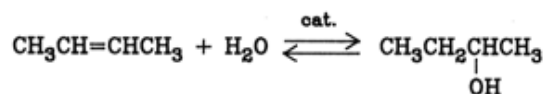
15.1 Introduction

Butene hydration is the most important process to produce 2-butanol (Ullmann 2001).

15.2 Production Technologies

2-Butanol can be obtained by the acid-catalyzed addition of water to 1- or 2-butene. The water adds at the double bond in accordance with the Markovnikov rule. The difference in olefin reactivity can be used for the selective manufacture of 2-butanol or 2-methyl-2-propanol from technical C4-fractions where the two alcohols are prepared in turn and isolated. Hydration of n-butene, on the other hand, requires 75 – 80 % H₂SO₄. Because of the tendency of the olefins to polymerize readily, the requisite reaction temperatures must not be exceeded.

In addition to this indirect hydration another process variant is known to industry in which the olefins are hydrated directly at elevated temperatures (about 200 – 300 °C) and higher pressures (10 – 35 MPa, 100 – 350 bar).



Various catalysts have been described for direct hydration, such as tungsten oxide and aqueous aluminum hydroxide gel suspensions. For the hydration of 1-butene and 2-butene the best results are obtained at temperatures between 220 and 300 °C or in the case of 2-methylpropene, at 190 – 245 °C. Direct hydration of the olefins has not yet established itself as a serious competitor to indirect hydration.

15.3 System Characterisation

In this study the hydration of butene is described. All data in the present report are referred to 1 kg butene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the hydration of butene, 2-butanol and butane are derived as co-products. Therefore an allocation between the two products was made. The following processes are modelled:

- Butene hydration, RER. Multi output process with 2-butanol, at plant, RER and butanes, from butene hydration, at plant, RER as allocated products

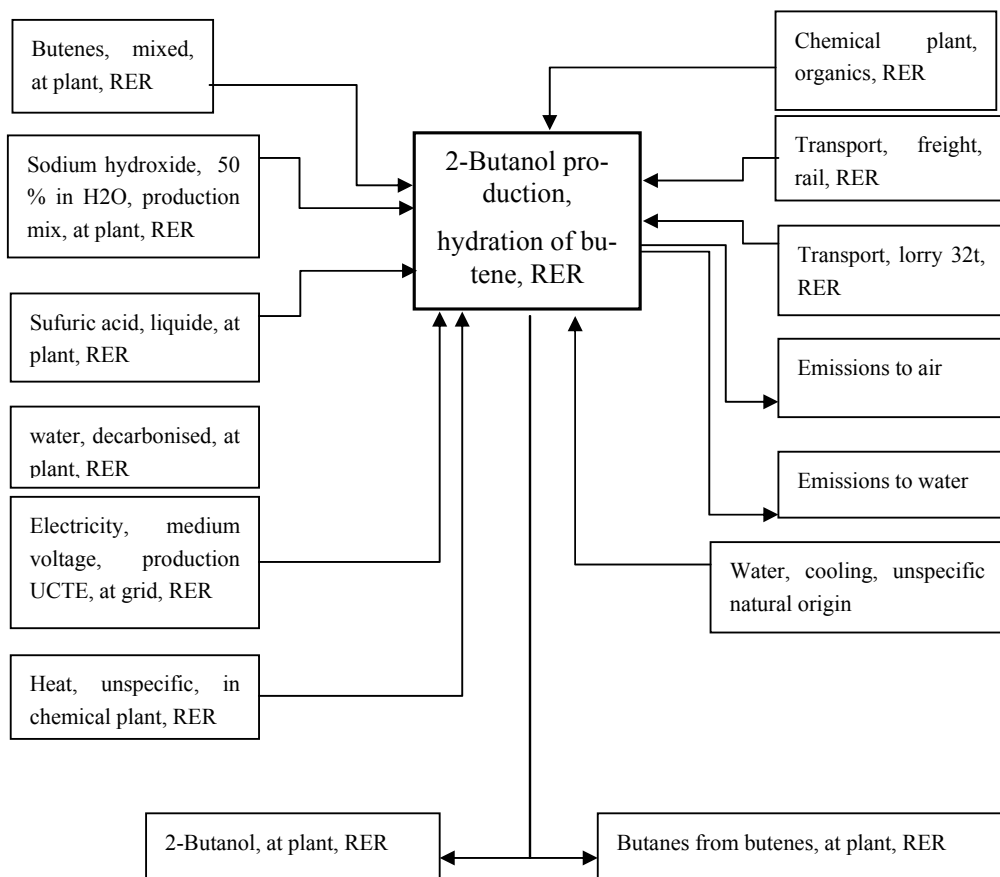


Fig. 1 Process flow chart for the production of 2-butanol

15.4 Raw materials and auxiliaries

The data for the input of raw materials and auxiliaries are taken from Chauvel (1989), which reports data for the indirect hydration of butene from French production plants in the late 1980ies. Schulze (1989) reports data for the direct hydration process.

Tab. 15.1 Consumption of raw materials and auxiliaries for the butene hydration

Input	Chauvel 1989 (per kg butanol)	Schulze 1989 (per kg butanol)	Rudd 1981 (per kg butanol)	This study (per kg butene)
Butene (kg)	1.15	1.09	1.14	1
NaOH (kg)	0.022		0.02	1.91E-02
Sulfuric acid (kg)	0.009		0.01	7.8E-03
Process water (kg)	15	0.25		13
Cooling water (kg)	580	300		504

15.5 Energy

The data for the energy inputs are taken from Chauvel (1989), which reports data for the indirect hy-

dration of butene from French production plants in the late 1980ies. Schulze (1989) reports data for the direct hydration process.

Tab. 15.2 Energy consumption for the butene hydration

Input	Chauvel 1989 (per kg butanol)	Schulze 1989 (per kg butanol)	This study (per kg butene)
Steam (MJ)	30.25	13.75	26.3
Electricity (kWh)	0.14	0.44	1.22E-01

15.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 15.3 summarises the total transport amounts for the production of 1 kg 2-butanol.

Tab. 15.3 Total transport amounts for the butene hydration

tkm kg ⁻¹ 2-butanol	lorry	train
Total	1.05E-01	6.28E-01

15.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of butene hydration plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

15.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the butene hydration. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 15.4.

There was no data available on process emissions to air for the butene hydration. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated as 20 % of the unreacted butene (0.018 kg per kg butene). The values for the emissions to air are given in Tab. 36.4.

Tab. 15.4 Process emissions to air from the butene hydration

Output	Value	Remarks
Butene, to air (kg)	1.8E-02	Estimated as 20 % of the unreacted input
Carbon dioxide, fossil, to air (kg)	2.04E-01	from waster water treatment

Waste heat (MJ)	4.38E-01	calculated from electricity input
-----------------	----------	-----------------------------------

15.9 Emissions to water

The remaining amount of unreacted butene is assumed to leave the production process with the waste water. These assumptions lead to a pollution of the waste water with 0.072 kg butene. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for butene was assumed with 90%. The carbon contained in the removed butene was accounted as CO₂ emissions to air. The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of butene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. The input of sulfuric acid was assumed to leave the system with the waste water as sulfates. A summary of the values used in this inventory is given in Tab. 36.5.

Other assumed emissions to water are sodium (calculated from the NaOH input) and sulphate (calculated from the H₂SO₄ input).

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of butene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 15.5.

Tab. 15.5 Process emissions to water from the butene hydration

Output	Value	Remarks
Butene, to water (kg)	7.2E-03	calculated from mass balance
Sodium, ion, to water (kg)	1.1E-02	calculated from the input
Sulfate, to water (kg)	7.7E-03	calculated from the input
COD, BOD (kg)	2.37E-02	calculated from water emissions
TOC, DOC (kg)	6.17E-03	calculated from water emissions

15.10 Allocation of the co-products of the butene hydration

Within the butene hydration, 2-butanol and butanes are obtained as co-products. According to the data from Chauvel 1989, it was assumed that in average an amount of 0.3 kg butanes per kg 2-butanol is produced. As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 15.6.

Tab. 15.6 Allocation of 2-butanol and the co-product butanes from butene hydration

Process	Butene hydration, RER		
	Unit	per kg butene	Allocation factor
2-Butanol, at plant, RER	kg	8.696E-01	76.923
Butanes from butene, at plant, RER	kg	2.609E-01	23.077

15.11 2-Butanol, at plant, RER

2-Butanol (CAS-No. 78-92-2, C₄H₉OH; Synonym: *sec*-butanol) is a colourless liquid with characteris-

tic odour. It is completely miscible with common organic solvents. Its vapor has an irritant effect on mucous membranes and a narcotic effect in higher concentrations.

For this inventory the functional unit is 1 kg of pure 2-butanol. The most important chemical and physical properties of 2-butanol used in this inventory are given in Tab. 15.7.

Tab. 15.7 Chemical and physical properties of 2-butanol (Ullmann 2001)

Property	Value	Unit	Remarks
Molecular weight	74.12	g mol ⁻¹	
Specific gravity	8065.	kg m ⁻³	at 20 °C
Boiling point	99.5	°C	at normale pressure
Melting point	- 114.7	°C	at normale pressure

15.11.1 Reserves and Resources

The only process for the preparation of 2-butanol with an achieved industrial importance is the *n*-butene hydration

15.11.2 Use

Practically all of the 2-butanol produced is dehydrogenated to 2-butanone (MEK or methyl ethyl ketone) which has found considerable application because of its good solvent properties and its favorable boiling point (79.57 °C). 2-Butanol itself is also used as a solvent: in particular, when mixed with aromatic hydrocarbons it is especially suitable as a solvent for alkyd resins and ethylcellulose lacquers. The ability of 2-butanol to dissolve both water and oils is applied in the manufacture of brake fluids and cleaning agents.

2-Butanol is converted into amines by treatment with ammonia. The acetate is recommended as a good solvent for nitrocellulose; the xanthate is used in ore flotation. 2-Butanol also finds application in the manufacture of perfumes, dyestuffs, fruit essences, and wetting agents.

15.12 Life cycle inventory and Data Quality Considerations

Tab. 15.8 shows the life cycle inventory and the data quality indicators for the inventory of the butene hydration. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 15.8 Unit process raw data for the butene hydration

	401	InputGroup	OutputGro	Name	Location	Infrastructu	reProcess	Unit	butene hydration	Uncertainty	StandardD	eviation95	% GeneralComment	2-butanol, at plant	butanes from butenes, at plant
	662	493	403	Location					RER					RER	RER
				InfrastructureProcess	Unit			0		kg	kg	kg	kg		
allocated products resource, in water technosphere	-	2	2-butanol, at plant	RER	0	kg	8.70E-1							100	0
	-	2	butanes from butenes, at plant	RER	0	kg	2.61E-1							0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	5.04E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	butene, mixed, at plant	RER	0	kg	1.00E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	water, deionised, at plant	CH	0	kg	1.30E+1	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	1.91E-2	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	7.83E-3	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km			76.923	23.077	
	5	-	transport, freight, rail	RER	0	tkm	6.28E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km			76.923	23.077	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.22E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
emission air, high population density	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.63E+1	1	1.51	(1,3,5,3,1,3); Chauvel 1989			76.923	23.077	
	5	-	chemical plant, organics	RER	1	unit	4.52E-10	1	3.09	(4,5,na,na,na,na); Estimation			76.923	23.077	
	-	4	Heat, waste	-	-	MJ	4.38E-1	1	1.51	(1,3,5,3,1,3); Calculated from the electricity input			76.923	23.077	
	-	4	Carbon dioxide, fossil	-	-	kg	2.04E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment			76.923	23.077	
	-	4	Butene	-	-	kg	1.80E-2	1	1.62	(4,5,na,na,na,na); Estimation			76.923	23.077	
	-	4	Butene	-	-	kg	7.21E-3	1	3.09	(4,5,na,na,na,na); Estimation			76.923	23.077	
	-	4	Sodium, ion	-	-	kg	1.10E-2	1	5.10	(4,5,na,na,na,na); Calculated from the input			76.923	23.077	
	-	4	Sulfate	-	-	kg	7.67E-3	1	1.62	(4,5,na,na,na,na); Calculated from the input			76.923	23.077	
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	2.37E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			76.923	23.077	
	-	4	COD, Chemical Oxygen Demand	-	-	kg	6.17E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			76.923	23.077	
emission water, river	-	4	DOC, Dissolved Organic Carbon	-	-	kg	6.17E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			76.923	23.077	
	-	4	TOC, Total Organic Carbon	-	-	kg	6.17E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			76.923	23.077	

15.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht et al. (2007). It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 13.8 shows selected LCI results and the cumulative energy demand for the production of butene hydration.

Tab. 15.9 Selected LCI results and the cumulative energy demand for the butene hydration

Name			2-butanol, at plant	butanes from butenes, at plant
Location	Unit	Unit	RER	RER
Infrastructure			kg 0	kg 0
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	94.6	94.6
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	4.0	4.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.5	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.7	0.7
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0
selected LCI results				
land occupation	resource	m2a	1.1E-1	1.1E-1
CO ₂ , fossil	air	kg	3.6E+0	3.6E+0
NM/VO	air	kg	1.9E-2	1.9E-2
nitrogen oxides	air	kg	5.0E-3	5.0E-3
sulphur dioxide	air	kg	6.6E-3	6.6E-3
particulates, <2.5 µm	air	kg	3.3E-4	3.3E-4
BOD	water	kg	2.4E-2	2.4E-2
cadmium	soil	kg	4.1E-10	4.1E-10

15.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	butene hydration
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Hydration of butene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Buten Hydratation
	491	Synonyms	
	492	GeneralComment	The multioutput-process "butene hydration" delivers the co-products 2-butanol and butanes from butenes. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1981
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydration of butene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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16 Butene hydroformylation

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

16.1 Introduction

Only a few industrial processes are known for the production of pentanols (amyl alcohols), namely, hydroformylation of butenes (oxo synthesis), hydrolysis of chloropentanes, and hydration of pentenes. Production of C5-alcohols from fusel oils by fermentation has been replaced mainly by the oxo synthesis and currently has little industrial importance. Production of pentanols from pentane by the Sharples process (chlorination of pentanes and subsequent hydrolysis) has also been largely replaced by oxo synthesis.

16.2 Production Technologies

Hydroformylation of butenes (1-butene, 2-butene, and isobutene) and subsequent hydrogenation yield 1-pentanol, 2-methyl-1-butanol, and 3-methyl-1-butanol almost exclusively. The remaining C5-alcohol isomers are obtained chiefly by hydration of C5-olefins, hydrogenation of corresponding carbonyl compounds, and other special techniques.

Hydroformylation of butene

The most important process for the production of pentanols is hydroformylation of butenes. The C4-olefins are reacted with a carbon monoxide – hydrogen mixture under the catalytic action of cobalt- or rhodium- containing compounds to yield C5-aldehydes. Hydrogenation of the latter gives the corresponding C5-alcohols. The flow sheet for this process is given in Fig. 16.1.

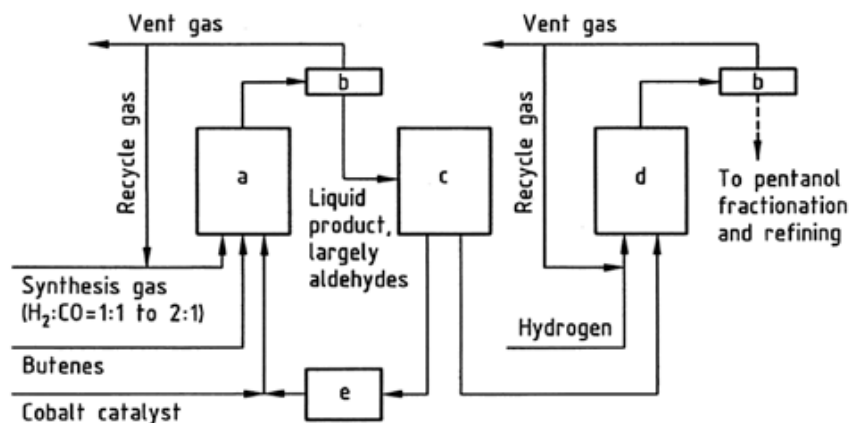


Fig. 16.1 Flow diagram for the production of pentanols

a) Oxo reactor (100 – 200 °C, 10 – 45 MPa); b) Separator; c) Catalyst removal (decobalting); d) Hydrogenation unit (50 – 250 °C, 0.3 – 24 MPa); e) Catalyst recovery

Raffinate I from various cracking processes (after the separation of 1,3-butadiene) is used as raw material and contains ca. 44 – 49 vol % isobutene, 24 – 28 vol % 1-butene, 19 – 21 vol % 2-butene, and 8 – 11 vol % C4-hydrocarbons. Further processing of this C4-raffinate converts isobutene by reaction with methanol to methyl tert-butyl ether. The remaining C4-olefin mixture (raffinate II) is the most important raw material in the industrial production of 1-pentanol and 2-methyl-1-butanol.

Cobalt-catalyzed hydroformylation of n-butenes and subsequent hydrogenation yield a mixture con-

taining ca. 70 % 1-pentanol and 30 % 2-methyl-1-butanol. If rhodium – triphenylphosphane is used as catalyst, a mixture containing up to 90 % 1-pentanol and 10 % 2-methyl-1-butanol is obtained.

2-Butene can be converted to mixtures of C5-aldehydes of varying composition, depending on the rhodium-containing catalyst system and reaction conditions. Thus, a ratio of 1.13:1 of 1-pentanal to 2-methylbutanal is obtained with a rhodium – phosphite catalyst at 130 °C; a ratio of 0.2:1 is obtained by hydroformylation with a rhodium – tricyclohexylphosphane catalyst at 110 °C. The dependence of the n:i ratio on catalyst species and temperature during conversion of 1-butene and 2-butene with a rhodium – triarylphosphane catalyst system is given in.

Hydroformylation of isobutene with rhodium or cobalt catalysts yields mainly 3-methyl-1-butanol; small amounts of 2,2-dimethyl-1-propanal occur chiefly in the cobalt- catalyzed oxo synthesis.

Apart from raffinate II and pure olefins, other C4- olefin mixtures can be used as raw materials for hydroformylation. Thus, hydroformylation of an olefin mixture containing 1-butene, 2-butene, and isobutene (ratio 1:1:2) with $\text{Co}(\text{CO})_4$ as catalyst at 100 °C yields a mixture of 1-pentanol, 2-methyl-1-butanol, and 3-methyl-1-butanol in a ratio of 15:5.2:11. More detailed information on the hydroformylation of butenes is given in.

Catalysts based on $\text{CuO} - \text{ZnO}$, nickel, zirconia, and $\text{Cu} - \text{TiO}_2$ are used industrially for the subsequent hydrogenation of C5-aldehydes.

16.3 System Characterisation

In this study the hydroformylation of butene is described. All data in the present report are referred to 1 kg butene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the hydroformylation of butene, 1-pentanol, 2-methyl-1-butanol, and 3-methyl-1-butanol are derived as co-products. Therefore an allocation between the three products was made.

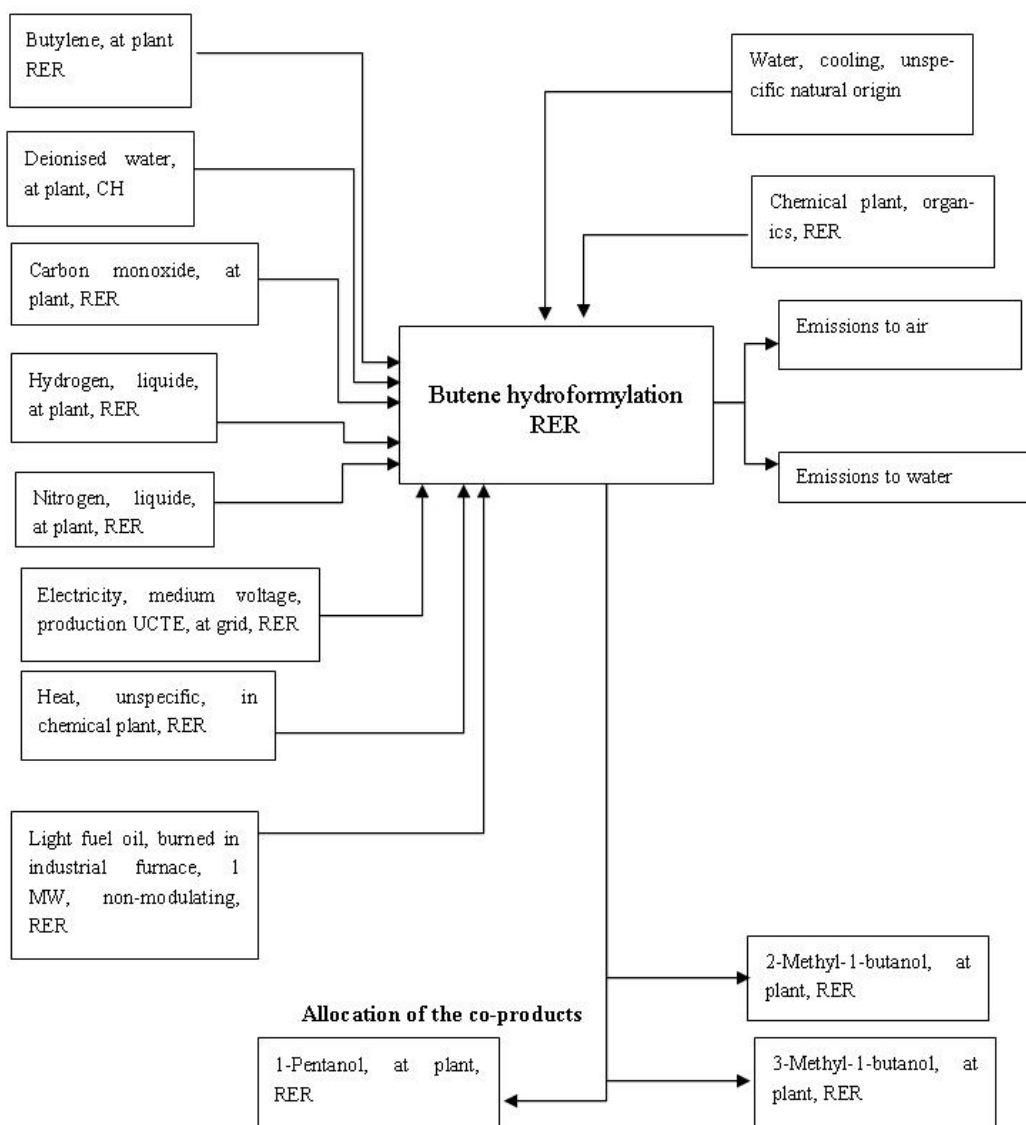


Fig. 16.2 Process flow chart for the butene hydroformylation

16.4 Raw materials and auxiliaries

As there was no data available for the butene hydroformylation, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries was estimated with data from Chauvel 1989, who gives data for the production of isobutanol from the hydroformylation of propylene. This process is considered to be similar to the hydroformylation of butylene.

Tab. 16.1 Consumption of raw materials and auxiliaries for the butene hydroformylation

Input	Values	Remarks
Butene (kg)	1	
Carbon monoxide (kg)	0.311	Calculated with a yield of 95%

H ₂ (kg)	4.44E-02	Calculated with a yield of 95%
N ₂ (kg)	9.666E-03	Estimation with data from similar process
Process water (kg)	1.93	Estimation with data from similar process
Cooling water (kg)	231.9	Estimation with data from similar process

16.5 Energy

As no data for the butene hydroformylation are available, the values for the energy use are estimated with data from Chauvel 1989, who gives data for the production of isobutanol from the hydroformylation of propylene. This process is considered to be similar to the hydroformylation of butylene.

In this study the use of the fuel is modelled with the dataset “light fuel oil, burned in industrial furnace 1 MW, non-modulating” with a energetic efficiency of 74 %, and the use of steam is modelled with the dataset “heat, unspecific, in chemical plant, RER”.

Tab. 16.2 Energy consumption for the butene hydroformylation

Input	Values	Remarks
Steam (MJ)	26.6	Estimation with data from similar process
Electricity (kWh)	0.676	Estimation with data from similar process
Fuel (MJ)	18.4	Estimation with data from similar process

16.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and 50 km by lorry 32t and 100 km by train for the hydrogen and the nitrogen. Tab. 16.3 summarises the total transport amounts for the butene hydroformylation.

Tab. 16.3 Total transport amounts for the butene hydroformylation

tkm kg ⁻¹ 1-pentanol	lorry	train
Total	1.37E-01	8.19E-01

16.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of butene hydroformylation plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

16.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the butene hydroformylation. As approxi-

mation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the butene and the carbon monoxide input and 5% of the hydrogen input. The remaining CO is oxidated to CO₂. The assumed values for the emissions to air are given in Tab. 16.4.

Tab. 16.4 Process emissions to air from the butene hydroformylation

Output	Value	Remarks
Butene, to air (kg)	2E-03	Estimated as 0.2 % of input
Carbon monoxide, fossil, to air (kg)	6.22E-04	Estimated as 1 % of input
Hydrogen, to air (kg)	4.44E-02	Estimated as 1 % of input
Carbon dioxide, fossil, to air (kg)	5.73E-1	from waster water treatment
Waste heat (MJ)	2.43	calculated from electricity input

16.9 Emissions to water

The remaining amount of unreacted butene (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for butene was assumed with 90%. The carbon contained in the removed butene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of butene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 16.5.

Tab. 16.5 Process emissions to water from the butene hydroformylation

Output	Value	Remarks
Butene, to water (kg)	4.8E-03	calculated from mass balance
COD, BOD (kg)	1.58E-02	calculated from water emissions
TOC, DOC (kg)	4.11E-03	calculated from water emissions

16.10 Allocation of the co-products of the butene hydroformylation

Within the butene hydroformylation, 1-pentanol, 2-methyl-1-butanol and 3-methyl-1-butanol are obtained as co-products. As allocation factor the weight of the two products was used. The yields of the diverse co-products and the used allocation factors are given in Tab. 16.6.

Tab. 16.6 Allocation of the co-products from the butene hydroformylation

Process	Butene hydroformylation, RER		
	Unit	per kg butene	Allocation factor
1-Pentanol, at plant, RER	kg	0.929	46.1
2-Methyl-1-butanol, at plant, RER	kg	0.406	20.1
3-Methyl-1-butanol, at plant, RER	kg	0.681	33.8

16.11 1-Pentanol

1-Pentanol (CAS-No. 71-41-0, $C_5H_{11}OH$, Synonym: n-amyl alcohol) is a colourless liquid of characteristic odor. It is only sparingly miscible with water at 20 °C. However, it is almost completely miscible with most organic solvents. For this inventory the functional unit is 1 kg of pure 1-pentanol. The most important chemical and physical properties of 1-pentanol used in this inventory are given in Tab. 16.7.

Tab. 16.7 Chemical and physical properties of 1-pentanol (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	88.2	$g\ mol^{-1}$	
Specific gravity	814	$kg\ m^{-3}$	at 20 °C
Boiling point	137.8	°C	at normale pressure
Melting point	- 78.5	°C	at normale pressure

16.11.1 Use

1-Pentanol is particularly useful as a solvent, extracting agent, and starting material for lubricant additives (zinc dipentyl dithiophosphates) and for auxiliaries in flotation (xanthates). It is used as solvent in the reaction of substituted diamines with diisocyanates and for the production of sulfurized olefins, polyacrylates, and pyridine-2,3-dicarboxylic acid derivatives. 1-Pentanol is employed as an extracting agent in the purification of phosphoric acid and the separation of strontium chloride from aqueous metal chloride solutions.

Other industrially important application of 1-pentanol are the synthesis of starting materials for the production of liquid crystals, dyes, flavorings, and special catalysts.

1-Pentanol is used as starting material for several syntheses in the pharmaceutical industry. Examples include the production of isoindolinone derivatives, 2-alkylbenzimidazoles, and condensed pyridine – pyrazine derivatives. In the plant protection sector it is used as starting material in the synthesis of cyclopropanecarboxylates and phenylurea derivatives.

1-Pentanol is employed in the production of many esters that are applied in different fields.

16.12 2-Methyl-1-butanol

2-Methyl-1-butanol (CAS-No. 137-32-6, $C_5H_{11}OH$) is a colourless liquid of characteristic odor. It is only sparingly miscible with water at 20 °C. However, it is almost completely miscible with most organic solvents. The most important chemical and physical properties of 2-methyl-1-butanol used in this inventory are given in Tab. 16.8.

Tab. 16.8 Chemical and physical properties of 2-methyl-1-butanol (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	88.2	$g\ mol^{-1}$	
Specific gravity	816	$kg\ m^{-3}$	at 20 °C
Boiling point	128	°C	at normale pressure

Melting point	- 70	°C	at normale pressure
---------------	------	----	---------------------

16.12.1 Use

2-Methyl-1-butanol, like 1-pentanol, is used mainly as a solvent and an extracting agent and for the production of various esters; examples are given in. Other important applications of 2-methyl-1-butanol include its use in the production of flotation reagents, liquid crystals, plant protection agents, fragrances, and corrosion inhibitors. 2-Methyl-1-butanol is also part of the catalyst system employed in the polymerization of caprolactam; it is used as a fuel additive and in the extraction of crude oil. A mixture of 1-pentanol and 2-methyl-1-butanol is generally used in practical applications instead of either of the pure alcohols.

16.13 3-Methyl-1-butanol

3-methyl-1-butanol (CAS-No. 123-51-3, $C_5H_{11}OH$) is a colourless liquid of characteristic odor. It is only sparingly miscible with water at 20 °C. However, it is almost completely miscible with most organic solvents. The most important chemical and physical properties of 3-methyl-1-butanol used in this inventory are given in Tab. 16.9.

Tab. 16.9 Chemical and physical properties of 3-methyl-1-butanol (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	88.2	$g\ mol^{-1}$	
Specific gravity	812	$kg\ m^{-3}$	at 20 °C
Boiling point	130.5	°C	at normale pressure
Melting point	- 117.2	°C	at normale pressure

16.13.1 Use

3-Methyl-1-butanol is used mainly as a solvent and extracting agent in several practical applications. An important use is in the production of di(3-methylbutyl) ether; the latter is a component of a catalyst system used for the industrial production of polypropylene. Other uses of 3-methyl-1-butanol in catalysis are discussed in.

3-Methyl-1-butanol is used in the production of herbicides based on 1,2,4-triazolecarboxamides, N-methyl- 4-pyridone derivatives, and substituted phenylhydrazones. It is also employed for the synthesis of fragrances and flavors: 3-methyl-1-butanol reacts e.g., with ethylidenenorbornene to give the corresponding norbornyl ether, with bicyclopentadiene to yield tricyclo[5.2.1.0^{2,6}]dec-3-en-8(9)-yl-3-methylbutyl ether, and with vanillin alcohol to give the corresponding vanillin ether derivative.

3-Methyl-1-butanol is also used for the production of various additives, corrosion inhibitors, and azo dyes. It is a component of surfactants.

16.14 Life cycle inventory of butene hydroformylation and data Quality Considerations

Tab. 16.10 shows the life cycle inventory and the data quality indicators for the inventory of the butene hydroformylation. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 16.10 Unit process raw data for the butene hydroformylation

	401	InputGroup	OutputGro	Name	Location	Infrastructu	reProcess	Unit	amyl alcohol production process RER	Uncertainty	Standard Deviation	95 %	GeneralComment	1-pentanol, at	2-methyl-1-	3-methyl-1-
														plant	butanol, at	butanol, at
														RER	RER	RER
662	493	403		Location	Infrastructure	Process	Unit		RER				RER	RER	RER	
				Unit					0				0	0	0	
									kg				kg	kg	kg	
allocated products products resource, in water technosphere	-	2	1-pentanol, at plant	RER	0	kg	7.48E-1							100	0	0
	-	2	2-methyl-1-butanol, at plant	RER	0	kg	3.27E-1							0	100	0
	-	2	3-methyl-1-butanol, at plant	RER	0	kg	5.48E-1							0	0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.32E+2	1	1.45	(4,5,4,3,3,3); Estimation			46.083	20.138	33.779	
	5	-	butene, mixed, at plant	RER	0	kg	1.00E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation			46.083	20.138	33.779	
	5	-	water, deionised, at plant	CH	0	kg	1.93E+0	1	1.45	(4,5,4,3,3,3); Estimation			46.083	20.138	33.779	
	5	-	carbon monoxide, CO, at plant	RER	0	kg	3.11E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation			46.083	20.138	33.779	
	5	-	hydrogen, liquid, at plant	RER	0	kg	4.44E-2	1	1.38	(4,5,1,1,1,5); Estimation			46.083	20.138	33.779	
	5	-	nitrogen, liquid, at plant	RER	0	kg	9.66E-3	1	1.38	(4,5,1,1,1,5); Estimation			46.083	20.138	33.779	
	5	-	transport, lorry 32t	RER	0	tkm	1.37E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km			46.083	20.138	33.779	
	5	-	transport, freight, rail	RER	0	tkm	8.19E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km			46.083	20.138	33.779	
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	6.76E-1	1	1.45	(4,5,4,3,3,3); Estimation			46.083	20.138	33.779	
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	1.84E+1	1	1.45	(4,5,4,3,3,3); Estimation			46.083	20.138	33.779	
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.66E+1	1	1.45	(4,5,4,3,3,3); Estimation			46.083	20.138	33.779	
	5	-	chemical plant, organics	RER	1	unit	3.72E-10	1	3.09	(4,5,na,na,na,na); Estimation			46.083	20.138	33.779	
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.43E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input			46.083	20.138	33.779	
	-	4	Carbon dioxide, fossil	-	-	kg	5.73E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment			46.083	20.138	33.779	
	-	4	Butene	-	-	kg	2.00E-3	1	1.62	(4,5,na,na,na,na); Estimation			46.083	20.138	33.779	
	-	4	Carbon monoxide, fossil	-	-	kg	6.22E-4	1	5.10	(4,5,na,na,na,na); Estimation			46.083	20.138	33.779	
emission water, river	-	4	Hydrogen	-	-	kg	4.44E-2	1	1.62	(4,5,na,na,na,na); Estimation			46.083	20.138	33.779	
	-	4	Butene	-	-	kg	4.80E-3	1	3.09	(4,5,na,na,na,na); Estimation			46.083	20.138	33.779	
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.58E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			46.083	20.138	33.779	
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.58E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			46.083	20.138	33.779	
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4.11E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			46.083	20.138	33.779	
	-	4	TOC, Total Organic Carbon	-	-	kg	4.11E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions			46.083	20.138	33.779	

16.15 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 16.11 shows selected LCI results and the cumulative energy demand for the butene hydroformylation.

Tab. 16.11 Selected LCI results and the cumulative energy demand for the butene hydroformylation

Name			1-pentanol, at plant	2-methyl-1-butanol, at plant	3-methyl-1-butanol, at plant
Location	Unit	RER	RER	RER	RER
Infrastructure		kg 0	kg 0	kg 0	kg 0
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	93.9	93.9	93.9
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.4	6.4	6.4
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.5	0.5	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9	0.9	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1	0.1	0.1
selected LCI results					
land occupation	resource	m2a	8.9E-2	8.9E-2	8.9E-2
CO ₂ , fossil	air	kg	4.2E+0	4.2E+0	4.2E+0
NM VOC	air	kg	3.9E-3	3.9E-3	3.9E-3
nitrogen oxides	air	kg	5.5E-3	5.5E-3	5.5E-3
sulphur dioxide	air	kg	8.1E-3	8.1E-3	8.1E-3
particulates, <2.5 µm	air	kg	4.0E-4	4.0E-4	4.0E-4
BOD	water	kg	1.8E-2	1.8E-2	1.8E-2
cadmium	soil	kg	4.9E-10	4.9E-10	4.9E-10

16.16 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	amyl alcohol production process RER
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Hydroformylation of butene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Amylalkohol Produktionsprozess, RER
	491	Synonyms	
	492	GeneralComment	The multioutput-process "amyl alcohol production process" delivers the co-products 1-pentanol, 2-methyl-1-butanol, and 3-methyl-1-butanol. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1989
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydroformylation of butene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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17 Butyl acetate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

17.1 Introduction

n-Butyl acetate (CAS-No. 123-86-4, $\text{CH}_3\text{COO}(\text{CH}_2)_3\text{CH}_3$, $\text{C}_6\text{H}_{12}\text{O}_2$; Synonyms: butyl ethanoate, n-butyl ester) is a clear, moderately volatile liquid with a characteristic ester odour. It is miscible with all conventional solvents such as alcohols, ketones, aldehydes, ethers, glycols, glycol ethers, aromatic hydrocarbons and aliphatic hydrocarbons. Its solubility in water is limited. It occurs in many fruits and is a constituent of apple aromas.

For this inventory the functional unit is 1 kg of pure butyl acetate. The most important chemical and physical properties of butyl acetate used in this inventory are given here.

Tab. 17.1 Chemical and physical properties of butyl acetate (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	116.16	g mol^{-1}	
Specific gravity	882	kg m^{-3}	at 20 °C
Boiling point	126	°C	at normale pressure
Melting point	- 73.5	°C	at normale pressure

17.2 Reserves and Resources

The production of n-butyl acetate in 1985 was about 81000 t in the USA and 30000 t in Europe (Ullmanns 2004).

17.3 Production Technologies and Use

17.3.1 Production Technologies

The reaction of an alcohol and an organic acid produces an ester in the presence of an acid catalyst. The reaction, esterification, is the most common process for manufacturing acetate esters.

Other industrial processes include the reaction of acetic anhydride with alcohols and the Tischenko reaction. Other methods for synthesizing acetate esters are the addition of acetic acid to olefins, carbonylation of acetic acid or anhydride, and reductive methods involving carboxylic acids or anhydride.

Commercial processes for esterification are usually continuous and are performed in the liquid phase. The reaction requires strong acids such as sulfuric acid or organic sulfonic acids. Another type of catalyst is ion-exchange resins. In contrast, gas-phase technology is available with sulfonated polymer or supported phosphoric acid as catalyst.

Esterification involves an equilibrium between reactants and product. Adding excess reactants or removal of water drive the equilibrium to favor the formation of ester. Azeotroping agents eliminate water and force the reaction to completion. Common agents are cyclohexane, benzene, and toluene.

Esterification with acetic anhydride offers an alternative route for making acetates. Due to its high re-

activity, acetic anhydride readily forms esters with alcohols. The reaction is irreversible. The esterification goes to completion without eliminating products or water.

Additions of acetic acid to olefins present an alternative way to make acetates, especially for branched esters. Synthesis of these hindered esters by esterification is not efficient. The equilibrium does not favor the desirable products. Rhône-Poulenc uses this method to prepare isopropyl acetate from acetic acid and propylene. Catalysts can be strong protic or Lewis acids, like sulfuric acid or $\text{BF}_3 \cdot \text{OEt}_2$. Ethyl acetate can also be produced directly from acetaldehyde.

Adding hydrogen and carbon monoxide to acetic acid or acetic anhydride can produce acetates. Alternatively, hydrogenation of acetic acid yields ethanol and ethyl acetate. Many studies relate to supported heterogeneous catalysts with molybdenum and group VIII metals (Pd, Rh, or Ru). Additionally, homogeneous catalysis with ruthenium can result in high selectivity.

Commercial processes also include the reaction of aldehydes to make esters. The Tischenko reaction uses alkoxide catalysts. Typical catalysts are aluminum alkoxide, boric acid and magnesium or calcium aluminum alkoxides

17.3.2 Use

Butyl acetate finds use as a solvent with excellent solvency for cellulose nitrate, polymers, resins and oils. It is an important solvent in the manufacture of paints, leather and paper and in the chemical and allied industries

17.4 System Characterisation

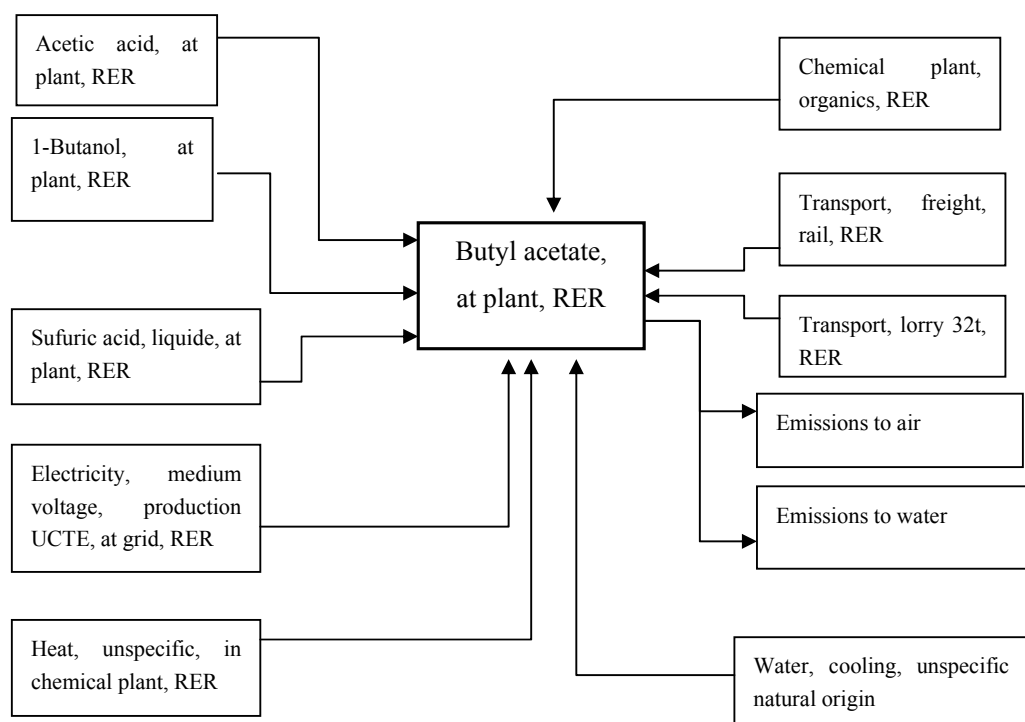


Fig. 17.1 Process flow chart for the production of butyl acetate

This report corresponds to the module in the ecoinvent database for the production of **1 kg butyl ace-**

tate, at plant, in Europe. For this study the production of butyl acetate by means of the esterification of 1-butanol with acetic acid is considered. All data in the present report are referred to 1 kg butyl acetate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

17.5 Raw materials and auxiliaries

As there was no data available for the production of butyl acetate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries, i. e. sulfuric acid was estimated with data from Wells (1991), who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of 1-butanol.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the butyl acetate production.

Tab. 17.2 Consumption of raw materials and auxiliaries for the production of 1 kg butyl acetate

Input	Values	Remarks
Butanol (kg)	6.4453E-01	Calculated with a yield of 95 %
Acetic acid (kg)	5.2218E-01	Calculated with a yield of 95 %
Sulfuric acid (kg)	1.31E-02	estimated
Cooling water (kg)	24	estimated

17.6 Energy

As no data for the production of butyl acetate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of 1-butanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 17.3 Energy consumption for the production of 1 kg butyl acetate

Input	Esterification ¹	This study
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

1 Source: Kim 2003

17.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 17.4 summarises the total transport amounts for the production of 1 kg butyl acetate.

Tab. 17.4 Total transport amounts for the production of 1 kg butyl acetate

tkm kg ⁻¹ butyl acetate	lorry	train
Raw materials	1.229E-01	7.374E-01

17.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of butyl acetate plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

17.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-03 kWh/kg butyl acetate is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of butyl acetate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 17.5.

Tab. 17.5 Process emissions to air from the production of 1 kg butyl acetate

Output	Value	Remarks
Butanol, to air (kg)	1.34E-03	Estimated as 0.2 % of input
Acetic acid, to air (kg)	1.09E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	6.90E-2	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

17.10 Emissions to water

The remaining amount of unreacted acetic acid and butanol (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for butanol and acetic acid was assumed with 90%. The carbon contained in the removed butanol and acetic acid was accounted as CO₂ emissions to air. The sulphate emissions are calculated from the sulphuric acid input.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of butanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC

was used. A summary of the values used in this inventory is given in Tab. 17.6.

Tab. 17.6 Process emissions to water from the production 1 kg butyl acetate

Output	Value	Remarks
Butanol, to water (kg)	3.22E-03	calculated from mass balance
Acetic acid, to water (kg)	2.61E-03	calculated from mass balance
Sulphate, to water (kg)	1.283E-02	calculated from mass balance
COD, BOD (kg)	1.07E-02	calculated from water emissions
TOC, DOC (kg)	3,13E-03	calculated from water emissions

17.11 Life cycle inventory and Data Quality Considerations

Tab. 17.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of butyl acetate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 17.7 Unit process raw data for the production of butyl acetate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	butyl acetate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	butyl acetate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	butanol, 1-, at plant	RER	0	kg	6.72E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	5.44E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	1.31E-2	1	1.51	(1,2,5,1,1,3); Estimation
	5	-	transport, lorry 32t	RER	0	tkm	1.23E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.37E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	8.84E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
	-	4	Butanol	-	-	kg	1.34E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	1.09E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	6.90E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Butanol	-	-	kg	3.22E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.61E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.07E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.07E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.13E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.13E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

17.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 17.8 shows selected LCI results and the cumulative energy demand for the production of butyl acetate.

Tab. 17.8 Selected LCI results and the cumulative energy demand for the production of butyl acetate

Name		butyl acetate, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	98.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	5.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.3E-1
CO2, fossil	air	kg	2.8E+0
NM VOC	air	kg	1.1E-2
nitrogen oxides	air	kg	4.9E-3
sulphur dioxide	air	kg	6.9E-3
particulates, <2.5 um	air	kg	4.5E-4
BOD	water	kg	1.6E-2
cadmium	soil	kg	9.7E-10

17.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	butyl acetate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of butyl acetate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Butylazetat, ab Werk
	491	Synonyms	butyl ethanoate, n-butyl ester
	492	GeneralComment	The process "butyl acetate, at plant, RER" is modelled for the production of butyl acetate from 1-butanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H12O2
	501	StatisticalClassification	
	502	CASNumber	141-78-6
TimePeriod	601	StartDate	1991
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Esterification of 1-butanol
Representative	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

17.14 References

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18 Cyclohexane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

18.1 Introduction

Cyclohexane (CAS-No. 110-82-7, C_6H_{12}) is a clear liquid with a pungent petroleum-like odor. It is essentially insoluble in water but miscible in most organic liquids, and is noncorrosive, easily flammable, and considered to be much less toxic than benzene.

Cyclohexane can exist in two conformations, the chair and boat forms:

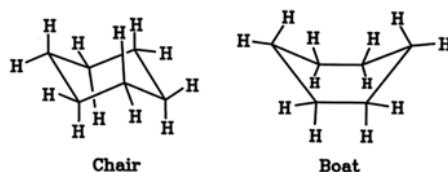


Fig. 18.1 The chair and the boat form of cyclohexane

Transformation from one isomer to the other involves rotation about carbon-carbon single bonds. The energy barriers associated with the transformation are low, but the boat form is estimated to have a potential energy 21 – 25 kJ/mol higher than the chair form. As a result, cyclohexane is almost exclusively in the chair form at room temperature. Stereochemistry has no influence on its use as a raw material for nylon manufacture or in other applications.

The important physical properties of cyclohexane are given here:

Tab. 18.1 Chemical and physical properties of cyclohexane (Ullmann 2001)

Property	Value	Unit	Remarks
Molecular weight	84,157	$g\ mol^{-1}$	
Specific gravity	778,55	$kg\ m^{-3}$	at 20 °C
Boiling point	80,738	°C	at normale pressure
Melting point	6,554	°C	at normale pressure
Heat of combustion, gross caloric value (HHV)	3,922	$MJ\ kg^{-1}$	

18.2 Reserves and Resources

Over 90 % of all cyclohexane is produced commercially by hydrogenation of benzene (Wells 1991). A small amount is produced by superfractionation of the naphtha fraction from crude oil. Naturally occurring cyclohexane can be supplemented by fractionating methylcyclopentane from naphtha and isomerizing it to cyclohexane.

18.3 Production Technologies and Use

18.3.1 Production Technologies

Hydrogenation of Benzene (Ullmann 2001)

Benzene can be hydrogenated catalytically to cyclohexane in either the liquid or the vapor phase in the presence of hydrogen. Several cyclohexane processes, which use nickel, platinum, or palladium as the *catalyst*, have been developed. Usually, the catalyst is supported, e.g., on alumina, but at least one commercial process utilizes Raney nickel.

Hydrogenation proceeds readily and is highly exothermic ($\Delta H_{500\text{K}} = -216.37 \text{ kJ/mol}$). From an equilibrium standpoint, the *reaction temperature* should not exceed 300°C . Above this, the equilibrium begins to shift in favor of benzene so that high-purity cyclohexane cannot be produced. As a result of these thermodynamic considerations, temperature control of the reaction is critical to obtaining essentially complete conversion of benzene to cyclohexane.

Temperature control requires economic and efficient heat removal. This has been addressed in a number of ways by commercial processes. The earlier vapor-phase processes used multistage reactors with recycle of cyclohexane as a diluent to provide a heat sink, staged injection of benzene feed between reactors, and interstage steam generators to absorb the exothermic heat of hydrogenation. In the 1970s processes have been developed that use only one reactor or a combination of a liquid-and a vapor-phase reactor. The objectives of the later processes were to reduce capital cost and improve energy utilization. However, all of the commercial processes have comparably low capital cost and good energy efficiency.

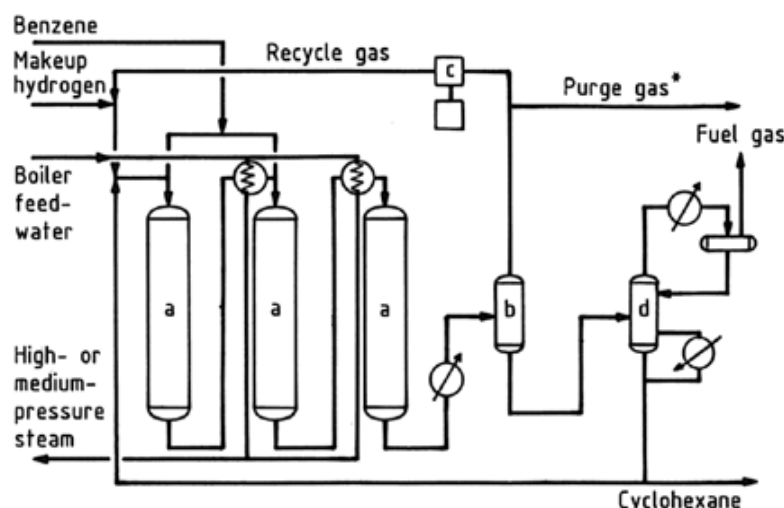


Fig. 18.2 Multistage reactor, vapor-phase process (taken from Ullmann 2001, Figure 1)

a) Reactors; b) Separator drum; c) Compressor; d) Stabilizer. * To hydrogen recovery (optional).

A generalized flow scheme for a *vapor-phase process with multistage reactors* in series is shown in Fig. 18.2. Benzene feed is divided and fed to each of the first two reactors. Recycled cyclohexane is introduced to the first reactor along with hydrogen. The recycled cyclohexane enables higher conversion in the reactors by absorbing part of the heat of hydrogenation. Steam generators between the reactors remove the heat of hydrogenation. The outlet temperature of the last reactor is controlled to achieve essentially 100 % conversion of benzene to cyclohexane. The effluent from the last reactor is cooled, and the vapor and liquid are separated. Part of the hydrogen-rich vapor is recycled to the first reactor, and the rest is purged to fuel gas or hydrogen recovery facilities. The liquid from the separator goes to a stabilizer where the overhead gas is sent to fuel gas; the remaining material is cyclohexane product, part of which is recycled to the first reactor.

A simplified flow scheme for a *process with liquid- and vapor-phase reactors* is provided in Fig. 18.3. Benzene and hydrogen are fed to the liquid-phase reactor, which contains a slurry of finely divided Raney nickel. Temperature is maintained at 180 – 190 °C by pumping the slurry through a steam generator and by vaporization in the reactor. Roughly 95 % of the benzene is converted in this reactor. The vapor is fed to a fixed-bed reactor where the conversion of benzene is completed. The effluent from the fixed-bed reactor is processed as described previously for the vapor-phase process.

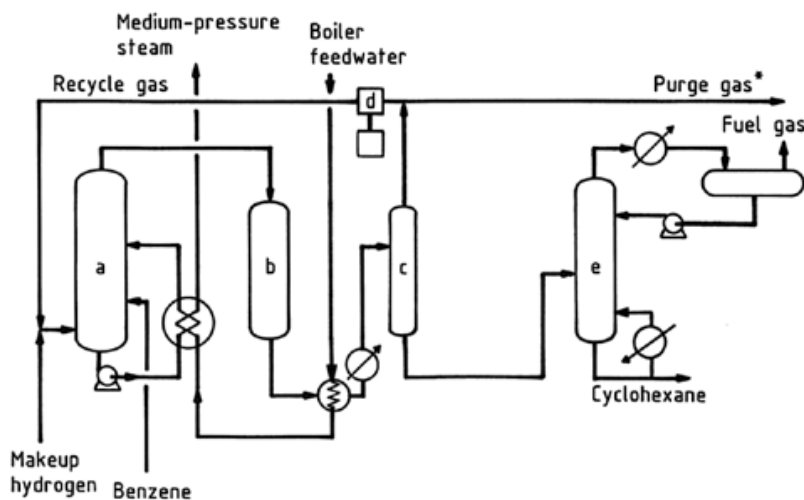


Fig. 18.3 Liquid- and vapor-phase reactor process (taken from Ullmann 2001, Figure 2)

a) Liquid-phase reactor; b) Vapor-phase reactor; c) Separator drum; d) Compressor; e) Stabilizer. * To hydrogen recovery (optional).

Benzene hydrogenation is done typically at 20 – 30 MPa. The maximum reactor temperature is limited to ca. 300 °C so that a typical specification of < 500 mg/kg benzene and < 200 mg/kg methylcyclopentane in the product can be achieved. This is necessary because of the thermodynamic equilibrium between cyclohexane – benzene and cyclohexane – methylcyclopentane. Actually, equilibrium strongly favors methylcyclopentane, but the isomerization reaction is slow enough with the catalysts employed to avoid a problem if the temperature is controlled.

The *hydrogen content* of the makeup hydrogen has no effect on product purity but it does determine the makeup, recycle, and purge gas rates. Streams with as low as 65 vol % hydrogen can be used.

Carbon monoxide and sulfur compounds are *catalyst deactivators*. Both can be present in the hydrogen from catalytic naphtha reformers or ethylene units, which are typical sources of makeup hydrogen. Therefore, the hydrogen-containing stream is usually passed through a methanator to convert carbon monoxide to methane and water. Prior to methanation, hydrogen-containing gas can be scrubbed with caustic to remove sulfur compounds. Commercial benzene contains less than 1 mg/kg sulfur. In some cases, the recycle gas is also scrubbed with caustic to prevent buildup of hydrogen sulfide from the small amount of sulfur in the benzene. With properly treated hydrogen and specification benzene, a catalyst life in excess of three years can be achieved easily in fixed-bed reactors that use noble-metal catalysts supported on a base. The catalyst in the process that uses Raney nickel in suspension is reported to have a typical life of about six months before it must be replaced.

18.3.2 Use

Over 98 % of the cyclohexane produced is used to make nylon intermediates: adipic acid, caprolactam, and hexamethylenediamine, with the first two consuming ca. 95 % of the cyclohexane used in nylon manufacture (Ullmann 2001). Minor miscellaneous uses, such as solvents and polymer reaction dilu-

ents, consume the remainder of the cyclohexane produced.

The USA is the leading manufacturer of cyclohexane, with annual production around 1000000 t in the Year 2004 (Gobi 2003). The whole production in the world is around 3000000 t per year (Gobi 2003). The largest consumption region is Europe, accounting for 39 % of global consumption.

18.4 System Characterisation

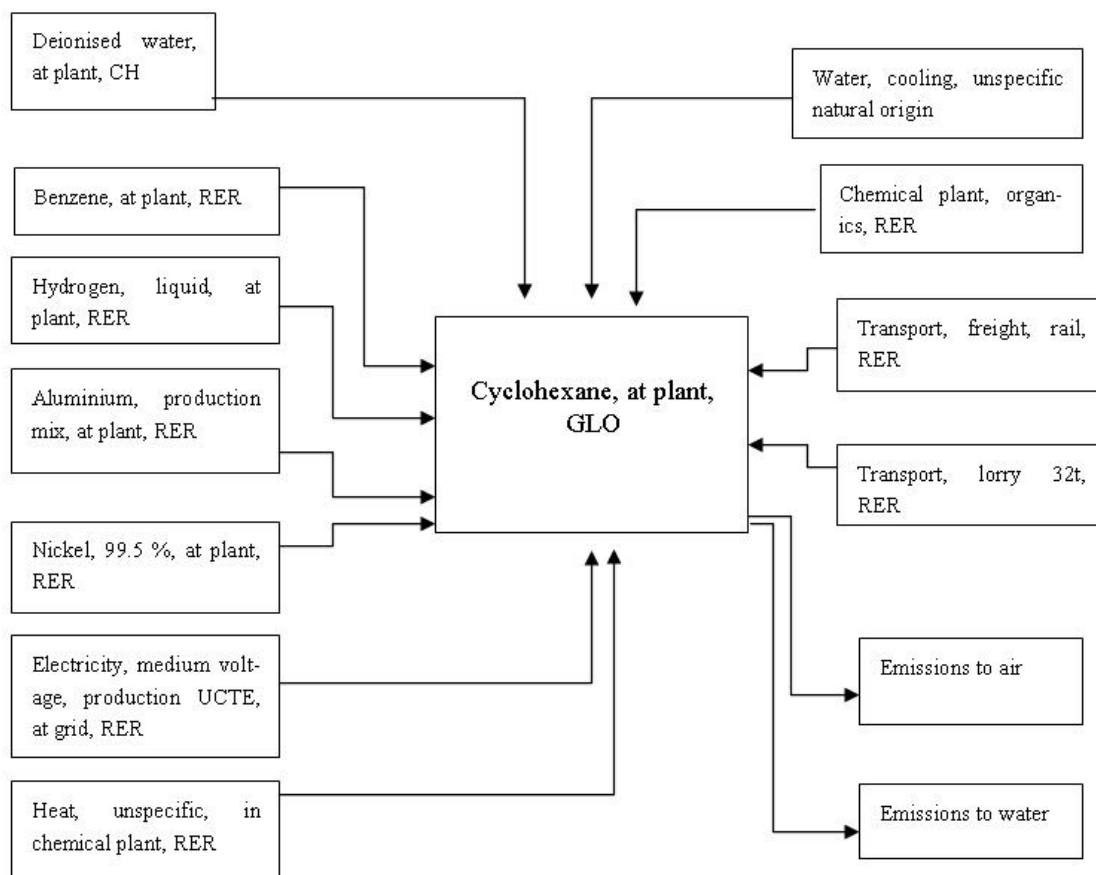


Fig. 18.4 Process flow chart for the production of cyclohexane

This report corresponds to the module in the ecoinvent database for the production of **1 kg cyclohexane, at plant, GLO**. For this study the production of cyclohexane by means of the hydrogenation of benzene is considered. All data in the present report are referred to 1 kg cyclohexane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

18.5 Raw materials and auxiliaries

The values for the input of the raw materials (benzene, hydrogen) and the auxiliaries (cooling water, process water) are calculated as average values of the available data.

The catalyst assumed is Raney nickel (50 % Ni, 50 % Al) with an input of 3.1E-05 kg nickel from the dataset "Nickel, 99.5 %, at plant, RER" and 3.1E-05 kg aluminium from the dataset "Aluminium, production mix, at plant, RER"

Tab. 18.2 Consumption of raw materials and auxiliaries for the production of cyclohexane

Input	Wells 1991	Chauvel 1989 ¹	Chauvel 1989 ²	Rudd 1982	Marcell 1982	Faith 1975	This study
Benzene (kg)	0.93	0.93	0.93	0.94	0.93	0.935	9.325E-01
H ₂ (kg), liquid	0.0655			0.07		0.065	6.683E-02
H ₂ (m ³), vaporous		0.85	0.93		0.3837		
Cooling water (m ³)		0.0035	0.008		0.00757		6.357E-03
Process water (m ³)		0.00095	0.00095		0.000435		7.78E-04
Catalysts (kg)					0.000062	small	6.2E-05

1 Liquid and vapor phase process

2 Vapor phase process

18.6 Energy

The values for the energy inputs are calculated as average values of the available data.

For the electricity input the dataset "electricity, medium voltage, production UCTE, at grid, RER" is used, and for the steam input the dataset "heat, unspecific, in chemical plant, RER".

Tab. 18.3 Energy consumption for the production of cyclohexane

Input	DOE 2002	DOE 2000	Chauvel 1989 ¹	Chauvel 1989 ²	This study
Steam (MJ/kg)	3.7515	3.7055	2.8325	4.2625	3.638
Electricity (kWh/kg)	0.184	0.0969	0.011	0.08	9.298E-02

1 Liquid and vapor phase process

2 Vapor phase process

18.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for benzene, 50 km by lorry 32t and 100 km by train for hydrogen, and 100 km by lorry 32t and 200 km by train for the auxiliaries. Tab. 18.4 summarises the total transport amounts for the production of 1 kg cyclohexane.

Tab. 18.4 Total transport amounts for the production of cyclohexane

t km kg ⁻¹ cyclohexane	lorry	train
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Total transports	9.66E-02	5.66E-01
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18.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of cyclohexane plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

18.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 9.69E-02 kWh/kg, is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of cyclohexane. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the benzene input, and 1 % of the H₂ input. The assumed values for the emissions to air are given in Tab. 18.5.

Tab. 18.5 Process emissions to air from the production of cyclohexane

Output	Value	Remarks
Benzene, to air (kg)	1.87E-03	Estimated as 0.2 % of input
H ₂ , to air (kg)	6.68E-04	Estimated as 1 % of input
Carbon dioxide, fossil, to air (kg)	2.27E-01	from waster water treatment
Waste heat (MJ)	3.35E-01	calculated from electricity input

18.10 Emissions to water

There was no data available on emissions to water for the production of cyclohexane. As approximation the water emissions were estimated to 0.8 % of the benzene input.

Further it was assumed that the waste water is treated in a internal waste water plant. The removal efficiency for benzene was assumed with 90%. The carbon contained in the removed benzene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of benzene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 18.6.

Tab. 18.6 Process emissions to water from the production of cyclohexane

Output	Value	Remarks
Benzene, to water (kg)	7.46E-02	calculated from mass balance
COD, BOD (kg)	2.2E-02	calculated from water emissions
TOC, DOC (kg)	6.9E-03	calculated from water emissions

It was assumed that all spent catalysts are recycled and used otherwise. According to the cut-off methology of the ecoinvent project, no emissions were therefore accounted for the further use or the recovering process of the catalysts.

18.11 Life cycle inventory and Data Quality Considerations

Tab. 18.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of cyclohexane. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 18.7 Unit process raw data for the production of cyclohexane

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	cyclohexane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
resource, in water technosphere	401	0	cyclohexane, at plant	RER	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	6.36E-3	1	1.08	(2,2,1,3,1,1); Chauvel 1989
	493	-	water, deionised, at plant	CH	0	kg	7.78E-1	1	1.05	(1,1,1,1,1,1); Chauvel 1989
	403	-	benzene, at plant	RER	0	kg	9.33E-1	1	1.51	(1,2,5,1,1,3); Stoichiometric calculation
		-	hydrogen, liquid, at plant	RER	0	kg	6.68E-2	1	1.51	(1,2,5,1,1,3); Stoichiometric calculation
		-	aluminium, production mix, at plant	RER	0	kg	3.10E-5	1	1.51	(1,2,5,1,1,3); Faith 1975
		-	nickel, 99.5%, at plant	GLO	0	kg	3.10E-5	1	1.51	(1,2,5,1,1,3); Faith 1975
		-	transport, lorry 32t	RER	0	tkm	9.66E-2	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	5.66E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	9.30E-2	1	1.08	(1,2,1,3,1,3); DOE 2000
		-	heat, unspecific, in chemical plant	RER	0	MJ	3.64E+0	1	1.08	(1,2,1,3,1,3); DOE 2000
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); FAO 2004
emission air, high population density		4	Heat, waste	-	-	MJ	3.35E-1	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
		4	Carbon dioxide, fossil	-	-	kg	1.87E-3	1	1.30	(4,5,na,na,na,na); Calculated from the carbon balance
		4	Benzene	-	-	kg	6.68E-4	1	1.62	(4,5,na,na,na,na); Estimated
		4	Hydrogen	-	-	kg	2.27E-2	1	1.62	(4,5,na,na,na,na); Estimated
emission water, river		4	Benzene	-	-	kg	7.46E-3	1	3.09	(4,5,na,na,na,na); Calculated from mass balance
		4	BOD5, Biological Oxygen Demand	-	-	kg	2.20E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	COD, Chemical Oxygen Demand	-	-	kg	2.20E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	DOC, Dissolved Organic Carbon	-	-	kg	6.90E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	TOC, Total Organic Carbon	-	-	kg	6.90E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

18.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 18.8 shows selected LCI results and the cumulative energy demand for the production of cyclohexane.

Tab. 18.8 Selected LCI results and the cumulative energy demand for the production of cyclohexane

Name		cyclohexane, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	73.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	8.9E-2
CO2, fossil	air	kg	2.0E+0
NM VOC	air	kg	4.5E-3
nitrogen oxides	air	kg	3.6E-3
sulphur dioxide	air	kg	5.2E-3
particulates, <2.5 um	air	kg	2.2E-4
BOD	water	kg	3.2E-3
cadmium	soil	kg	3.3E-10

18.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	cyclohexane, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of cyclohexane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Cyclohexan, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "cyclohexane, at plant, RER" is modelled for the production of cyclohexane from benzene in Europe. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H12
	501	StatisticalClassification	
	502	CASNumber	110-82-7
TimePeriod	601	StartDate	1975
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrgenation of benzene
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 5 Million tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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19 Cyclohexanone

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

19.1 Introduction

Cyclohexanone (CAS-No. 108-94-1, 1-78-6, C₆H₁₀O) is a colorless to pale yellow liquid having an odor similar to that of acetone. It is very soluble in most organic solvents. For this inventory the functional unit is 1 kg of pure cyclohexanone. The most important chemical and physical properties of cyclohexanone used in this inventory are given here:

Tab. 19.1 Chemical and physical properties of cyclohexanone (Ullmanns 2005)

Property	Value	Unit	Remarks
Molecular weight	98.15	g mol ⁻¹	
Specific gravity	945.5	kg m ⁻³	at 20 °C
Boiling point	156.4	°C	at normale pressure
Melting point	- 47	°C	at normale pressure

19.2 Reserves and Resources

Both cyclohexanol and cyclohexanone are produced on a large commercial scale. The vast majority of these compounds and their mixtures is consumed in the production of caprolactam and adipic acid, intermediates in the manufacture of nylon 6 and nylon 66. 95 % of the cyclohexanone worldwide is produced by the oxidation of cyclohexane (Wells 1991)

19.3 Production Technologies and Use

19.3.1 Production Technologies

Cyclohexanol and cyclohexanone were first produced on an industrial scale by the hydrogenation of phenol. This was followed in the 1940s by transition metal-catalyzed liquid-phase air oxidation of cyclohexane, which gave a mixture of cyclohexanol and cyclohexanone, the ratio of which could be partially controlled by the choice of metal catalyst. In the 1950s, Scientific Design developed a boric acid modified liquid-phase air oxidation of cyclohexane. This process, in which the intermediate cyclohexyl hydroperoxide was trapped as the peroxoborate ester, led to a higher yield and a higher ratio of cyclohexanol to cyclohexanone. Around 1990, Asahi commercialized the production of cyclohexanol from cyclohexene, an intermediate obtained from the partial hydrogenation of benzene.

Cyclohexanone can be prepared by dehydrogenation of cyclohexanol; the reverse reaction, hydrogenation of cyclohexanone to cyclohexanol, is also practical although not commercially practiced. Most large-scale cyclohexanol/cyclohexanone facilities employ some form of cyclohexane oxidation technology. The only remaining large phenol-based plants belong to DSM in Holland and Allied in the United States. Asahi has the only commercial cyclohexene-based plant, and there are reports of planned expansions at that facility in Japan as well as building a new plant in China.

Hydrogenation of phenol (Ullmann 2001)

The product of phenol hydrogenation, i.e., cyclohexanol, cyclohexanone, or a mixture of the two, is determined by the metal catalyst. Cyclohexanol can be prepared in high yield by either vapor- or liquid-phase hydrogenation of phenol. Vapor-phase hydrogenation can be carried out by using a supported nickel catalyst or a nickel catalyst containing copper, cobalt, or manganese. The support is usually alumina or silicic acid. Yields as high as 98 % have been reported. A flow diagram of the vapor-phase hydrogenation is shown in Fig. 19.1.

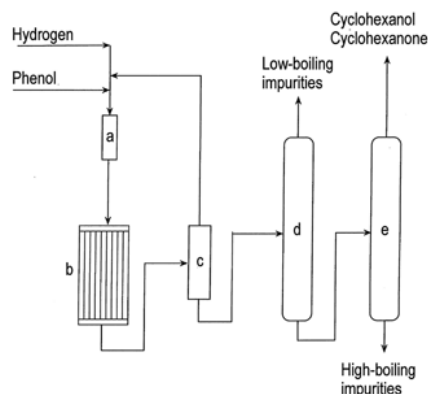


Fig. 19.1 Vapor-phase phenol hydrogenation process

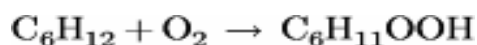
- a) Phenol evaporator b) Hydrogenation reactor c) Condenser d) Low-boiler removal column
e) Cyclohexanol/cyclohexanone recovery column

Hydrogenation of phenol to cyclohexanone can be carried out in the vapor phase with a large variety of supported noble metal catalysts, including palladium, platinum, iridium, ruthenium, rubidium, and osmium. The hydrogenation is normally carried out at 140 – 170 °C and atmospheric pressure. Yields as high as 95 % at 100 % conversion can be obtained.

Commercial liquid-phase hydrogenation of phenol under mild conditions with a palladium on carbon catalyst gives greater than 99 % yield of cyclohexanone at 90 % conversion. By adjusting the operating conditions and changing to a Raney nickel catalyst, 99.9 % selectivity for cyclohexanol can be achieved. Hydrogenation conditions can be adjusted to produce the mixture of cyclohexanol and cyclohexanone that is preferred by adipic acid producers.

Liquid-Phase Oxidation of Cyclohexane

The liquid-phase air oxidation of cyclohexane to cyclohexanol and cyclohexanone was developed in the 1940s. The oxidation is usually carried out, either uncatalyzed or with a soluble cobalt catalyst, in a series of agitated reactors at 140 – 180 °C and at 0.8 – 2 MPa, although a single tower oxidizer can be used. The total residence time in the oxidizers is 15 – 60 min. Some of the initially formed cyclohexyl hydroperoxide is converted in the oxidizers to cyclohexanol, cyclohexanone, and byproducts.



The cyclohexanol to cyclohexanone ratio exiting the air oxidizers can be altered by the choice of catalyst used in the air oxidizer. The normal cobalt-catalyzed air oxidation and “deperoxidation” gives a cyclohexanol to cyclohexanone ratio of about 3.5. If chromium(III) is added to the air oxidizer, it appears to promote dehydration of cyclohexyl hydroperoxide to cyclohexanone and water, as opposed to

the free-radical decomposition mechanism of cobalt, leading to a lower cyclohexanol to cyclohexanone ratio.

Because the intermediate cyclohexyl hydroperoxide and the products, cyclohexanol and cyclohexanone, are more readily oxidized than cyclohexane, the conversion of cyclohexane in the air oxidizers must be kept low (usually under 6 %) to maximize yield. Depending on the conversion, the total yield of alcohol, ketone, and hydroperoxide varies from 70 to 90 %. Byproducts of the oxidation include a wide range of mono- and dicarboxylic acids, esters, aldehydes, and other oxygenated materials.

To maximize overall yield of cyclohexane to the alcohol and ketone, most companies try to maintain a high percentage of the oxidized cyclohexane leaving the oxidizers as cyclohexyl hydroperoxide. To maximize the amount of cyclohexyl hydroperoxide leaving the air oxidizers, the oxidation can be carried out uncatalyzed or with a modified catalyst that minimizes hydroperoxide decomposition in the oxidizer. This permits cyclohexyl hydroperoxide to be decomposed into cyclohexanol and/or cyclohexanone under milder conditions and in higher yields than if it had occurred in the air oxidizer.

For example, the air oxidation product may or may not be washed with water followed by conversion of cyclohexyl hydroperoxide to cyclohexanol and cyclohexanone with a homogeneous or heterogeneous metal catalyst. Another method of decomposing the hydroperoxide is to contact it with an aqueous caustic phase containing a few parts per million of cobalt, which decomposes the hydroperoxide in high yield to the ketone and alcohol. This caustic decomposition process can result in a lower alcohol to ketone ratio in the final product. A flow diagram of this caustic process is shown in Fig. 19.2.

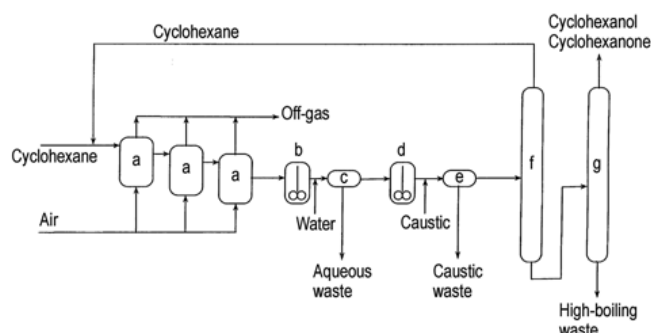


Fig. 19.2 Cyclohexane oxidation/caustic decomposition process

a) Air oxidizers; b) Water wash decanter; c) Decantation; d) Caustic decomposition reactor; e) Caustic decanting; f) Cyclohexane recovery column; g) Cyclohexanol/cyclohexanone recovery column

Boric Acid Modified Oxidation of Cyclohexane

In the 1950s, Scientific Design (now Halcon International) developed a process in which anhydrous metaboric acid was added as a slurry to the first of several staged air oxidation vessels (see Fig. 19.3). The cyclohexyl hydroperoxide formed is trapped as the cyclohexyl perborate ester. The perborate ester then reacts with additional cyclohexane in the air oxidizer to yield the borate ester and additional cyclohexanol. This borate ester is relatively stable in the air oxidizer and is protected from over oxidation to ring-opened products. The ester is subsequently hydrolyzed to cyclohexanol and boric acid. The boric acid is dehydrated to metaboric acid and is recycled to the air oxidizer.

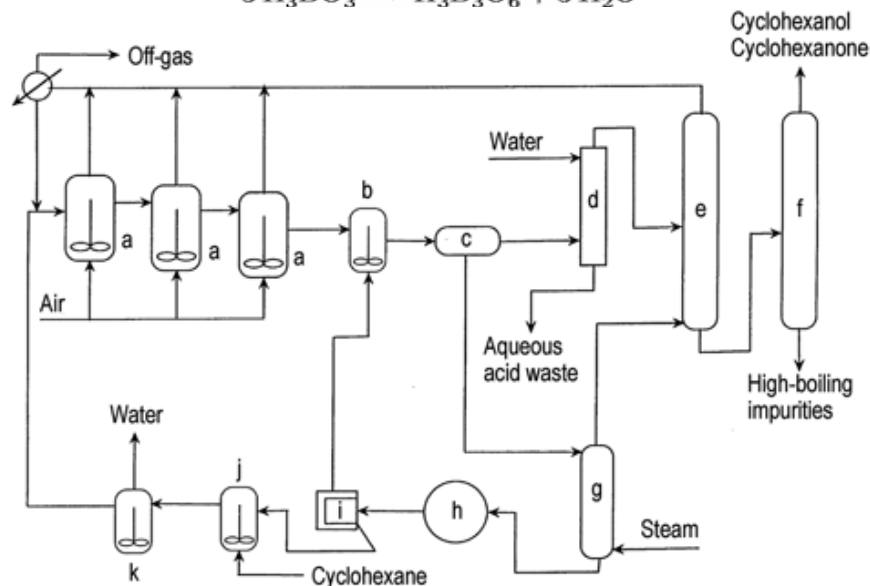
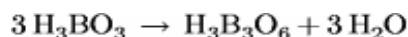
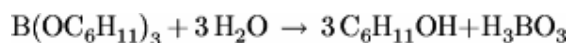


Fig. 19.3 Boric acid modified oxidation of cyclohexane

a) Air oxidizers; b) Ester hydrolysis vessel; c) Decanter; d) Extraction column; e) Cyclohexane recovery column; f) Refining column; g) Stripping column; h) Vacuum crystallizer; i) Boric acid centrifuge; j) Boric acid slurry tank; k) Dehydrator

The chemistry of this process is very similar to the metal-catalyzed oxidation except that the cyclohexyl rings are protected from further attack. Compared to other commercial processes, the boric acid process is characterized not only by a higher investment and a higher operating cost (to recover and recycle the boric acid), but also by a higher yield of cyclohexanol and cyclohexanone (up to 88 %) and an alcohol to ketone ratio as high as 10 : 1. The technology was further developed by Institut Français du Pétrole. The Halcon process has been licensed to several companies around the world. The major producers using this technology today are Solutia, DuPont (UK), Bayer, and Mitsubishi. The original plant at Rhône-Poulenc, based on this technology, was now shut down.

Production and Hydration of Cyclohexene

In 1972 the first patent covering the selective hydrogenation of benzene [71-43-2] to cyclohexene [110-83-8] appeared. Shortly thereafter, two patents described the hydration of cyclohexene to cyclohexanol by using various catalysts. Since then, a large number of patents have appeared worldwide. In 1990, Asahi commercialized this process with a 100 t/a plant in Japan. The process involves three primary steps: the selective hydrogenation of benzene; the separation of the cyclohexene from the unconverted benzene and the over-hydrogenation product cyclohexane; and the hydration of the cyclohexene (Fig. 19.4).

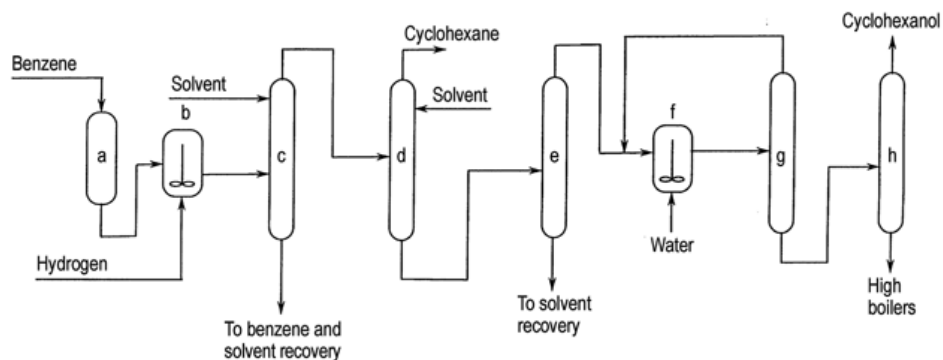


Fig. 19.4 Partial hydrogenation of benzene to cyclohexene, followed by hydration to cyclohexanol

a) Benzene purification; b) Partial hydrogenation; c) Separation of unconverted benzene from cyclohexene and cyclohexane; d) Separation of cyclohexene from cyclohexane; e) Separation of cyclohexene from extractive solvent; f) Hydration of cyclohexene; g) Recycle of unconverted cyclohexene; h) Purification of cyclohexanol

The hydrogenation is carried out with a specially prepared heterogeneous nickel catalyst. The reaction is run to about 50 % conversion, at which point the product consists of about 35 % cyclohexene, 15 % cyclohexane, and 50 % unconverted benzene. The selectivity is very sensitive to impurities such as sulfur and iron. For this reason the benzene must be specially purified and the hydrogenation vessel lined with an inert material.

The mixture of cyclohexane, cyclohexene, and benzene can not be distilled but must go through two successive extractive distillations. The best solvent to use in these distillations has been the subject of numerous patents. The recovered cyclohexane could be converted back to benzene, but is probably sold to other caprolactam and adipic acid manufacturers.

The hydration of cyclohexene is also performed with a heterogeneous catalyst in a slurry reactor. The extent of hydration is limited by the equilibrium constant, which would restrict conversion to about 14 %. The use of several additives to increase the conversion of cyclohexene to cyclohexanol has also been patented.

The yield of this process from benzene to cyclohexanol is extremely high, probably greater than 95 %. Although byproducts are not a concern, the investment is significantly higher than that of a conventional cyclohexane oxidation plant. The fact that the product is cyclohexanol, not cyclohexanone, could make it unattractive as the basis for a caprolactam plant.

Dehydrogenation of Cyclohexanol

Cyclohexanol can be dehydrogenated to cyclohexanone (the desired starting material for caprolactam manufacture) without catalyst by passing the vapors through a tube furnace at 400 – 450 °C. The vapors are cooled, the hydrogen separated, and the cyclohexanone, which contains cyclohexene and water, is purified in a distillation column. The purity of the resulting cyclohexanone is 98 – 99 %.

The catalytic dehydrogenation, which is generally practiced, can be carried out under milder conditions and in better yield over a variety of catalysts, including chromium oxide – copper, copper chromate, nickel, zinc sulfide, and zinc – iron catalyst, cobalt carbonate, and other metals.

19.3.2 Use

Over 97 % of all cyclohexanone produced is oxidized to adipic acid or converted to cyclohexanone oxime, which is then rearranged with sulfuric acid to caprolactam. Other uses for cyclohexanone in-

clude thinner or solvent for synthetic resins, polymers, and lacquers. It is also used as a starting material in the synthesis of many insecticides, herbicides, and pharmaceuticals, as well as being a good solvent for many insecticides.

19.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg cyclohexanone, at plant, in Europe**. For this study the production of cyclohexanone by means of the boric acid modified oxidation of cyclohexane is considered. All data in the present report are referred to 1 kg cyclohexanone 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

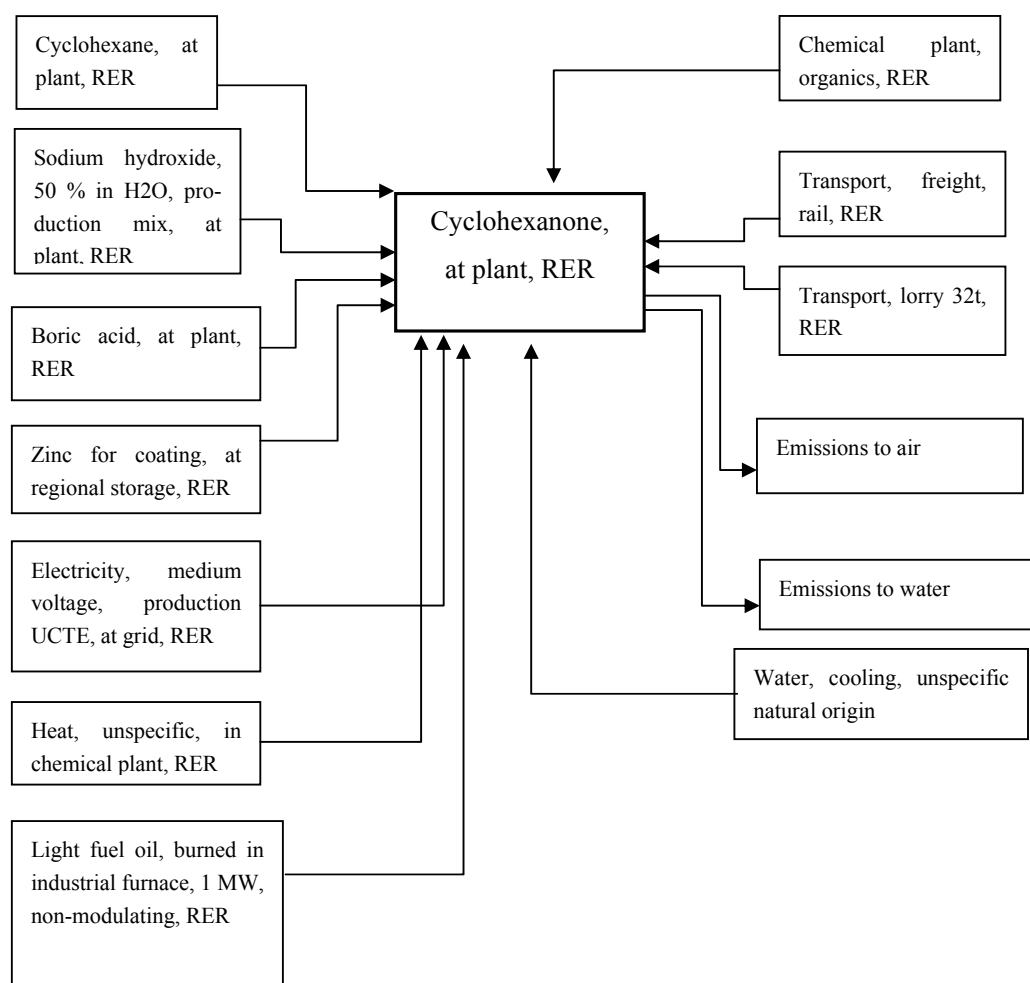


Fig. 19.5 Process flow chart for the production of cyclohexanone

19.5 Raw materials and auxiliaries

Wells 1991 reports data from US plants for the cyclohexanone production. The zinc oxide is calculated as zinc with the dataset “zinc for coatings, at regional storage, RER”.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the cyclohexanone production.

Tab. 19.2 Consumption of raw materials and auxiliaries for the production of 1 kg cyclohexanone

Input	Wells 1991	This study
Cyclohexane (kg)	0.925	9.25E-01
Boric acid (kg)	0.005	5E-03
NaOH 50 % (kg)	0.033	3.3E-02
Zinc oxide (kg)	0.00075	7.5E-04
Cooling water (kg)		24

19.6 Energy

DOE 2001 also reports a consumption of 720.24 kJ per kg cyclohexanone of auxiliary fuel requirement for incineration. These energy input is modelled as “Light fuel oil, burned in industrial furnace, 1 MW, non-modulating, RER”.

The value for the steam input used in this inventory is the difference between the hot utility requirement and the hot utility credit which are reported by DOE 2001.

Tab. 19.3 Energy consumption for the production of 1 kg cyclohexanone

Input	DOE 2001	This study
Steam (MJ)	24	20.3
Hot utility credit (MJ)	- 3.7	
Electricity (kWh)	0.176	1.76E-01
Light fuel oil (kg)	0.017	1.7E-02

19.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 19.4 summarises the total transport amounts for the production of 1 kg cyclohexanone.

Tab. 19.4 Total transport amounts for the production of 1 kg cyclohexanone

tkm kg ⁻¹ cyclohexanone	lorry	train
------------------------------------	-------	-------

Total	9.97E-02	5.98E-01
-------	----------	----------

19.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of cyclohexanone plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

19.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of cyclohexanone. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 19.5.

Tab. 19.5 Process emissions to air from the production of 1 kg cyclohexanone

Output	Value	Remarks
Cyclohexane, to air (kg)	1.85E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	2.09E-02	from waster water treatment
Waste heat (MJ)	6.336E-01	calculated from electricity input

19.10 Emissions to water

There was no data available on emissions to water for the production of cyclohexanone. As approximation the water emissions were estimated to 0.8 % of the cyclohexane input.

Further it was assumed that the waste water is treated in a internal waste water plant. The removal efficiency for benzene was assumed with 90%. The carbon contained in the removed benzene was accounted as CO₂ emissions to air.

Other assumed emissions to water are boron (calculated from the boric acid input), sodium (calculated from the sodium hydroxide input), and zinc (calculated from the zinc oxide input).

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of ethanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 19.6.

Tab. 19.6 Process emissions to water from the production 1 kg cyclohexanone

Output	Value	Remarks
Cyclohexane, to water (kg)	7.4E-03	calculated from mass balance
Boron, to water (kg)	8.87E-04	calculated from mass balance
Zinc, to water (kg)	6.02E-04	calculated from the mass balance

Sodium, to water (kg)	1.898E-02	calculated from the mass balance
COD, BOD (kg)	2.43E-02	calculated from water emissions
TOC, DOC (kg)	6.34E-03	calculated from water emissions

19.11 Life cycle inventory and Data Quality Considerations

Tab. 19.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of cyclohexanone. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 19.7 Unit process raw data for the production of cyclohexanone

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	cyclohexanone, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER			
							0			
							kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	cyclohexanone, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	cyclohexane, at plant	RER	0	kg	9.25E-1	1	1.51	(1,3,5,3,1,3); Wells 1991
	5	-	boric acid, anhydrous, powder, at plant	RER	0	kg	5.00E-3	1	1.51	(1,3,5,3,1,3); Wells 1991
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	3.30E-2	1	1.51	(1,3,5,3,1,3); Wells 1991
	5	-	zinc for coating, at regional storage	RER	0	kg	6.02E-4	1	1.51	(1,3,5,3,1,3); Wells 1991
	5	-	transport, lorry 32t	RER	0	tkm	9.97E-2	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	5.98E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.76E-1	1	1.09	(1,3,1,3,1,3); DOE 2001
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.03E+1	1	1.09	(1,3,1,3,1,3); DOE 2001
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	1.70E-2	1	1.09	(1,3,1,3,1,3); DOE 2001
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	6.34E-1	1	1.09	(1,3,1,3,1,3); Calculated from the electricity input
	-	4	Cyclohexane	-	-	kg	1.85E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	2.09E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Cyclohexane	-	-	kg	7.40E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Boron	-	-	kg	8.87E-4	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Zinc, ion	-	-	kg	6.02E-4	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Sodium, ion	-	-	kg	1.90E-2	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	2.43E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	2.43E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	6.34E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	6.34E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

19.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 19.8 shows selected LCI results and the cumulative energy demand for the production of cyclohexanone.

Tab. 19.8 Selected LCI results and the cumulative energy demand for the production of cyclohexanone

Name		cyclohexanone, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	102.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	5.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.9E-1
CO ₂ , fossil	air	kg	4.0E+0
NM VOC	air	kg	6.9E-3
nitrogen oxides	air	kg	6.3E-3
sulphur dioxide	air	kg	9.1E-3
particulates, <2.5 µm	air	kg	4.3E-4
BOD	water	kg	3.0E-2
cadmium	soil	kg	7.4E-10

19.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	cyclohexanone, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of cyclohexanone including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Cyclohexanon, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "cyclohexanone, at plant, RER" is modelled for the production of cyclohexanone from cyclohexane in Europe. Raw materials and energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H10O
	501	StatisticalClassification	
	502	CASNumber	108-94-1
TimePeriod	601	StartDate	1991
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Oxidation of cyclohexane
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
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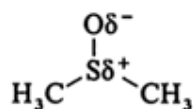
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20 Dimethyl sulfoxide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

20.1 Introduction

Dimethyl sulfoxide ($\text{C}_2\text{H}_6\text{OS}$, CAS-No. 67-68-5; synonym: DMSO) is a colorless, odorless, strongly hygroscopic liquid. It is readily miscible with water and other protic solvents such as alcohols. This property can be attributed to the formation of strong hydrogen bonds between the polarized S–O bond and water molecules. The strong polarity of the S–O bond, with a partial negative charge on oxygen (see formula) is shown by the high dipole moment (4.3 ± 0.1 D at 20 °C) and basicity ($\text{p}K_{\text{B}} = 2.7$) of dimethyl sulfoxide. Pure DMSO forms chain-like polymeric association complexes at room temperature.



For this inventory the functional unit is 1 kg of pure dimethyl sulfoxide. The most important chemical and physical properties of dimethyl sulfoxide used in this inventory are given here.

Tab. 20.1 Chemical and physical properties of dimethyl sulfoxide (Ullmanns 2004)

Property	Value	Unit
Molecular weight	78.13	g mol^{-1}
Specific gravity	1101	kg m^{-3}
Boiling point	189	°C
Melting point	18.5	°C

20.1.1 Use

Dimethyl sulfoxide is an excellent solvent that is widely used for reactions in modern chemistry. Because of its high polarity and good solvation of cations, DMSO dissolves not only a large number of organic compounds, but also many metal salts, particularly those of alkali and alkaline-earth metals. DMSO is also a good solvent for many polymers.

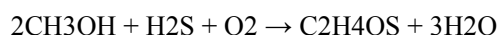
A range of different pharmacological properties and effects have been described for DMSO. For example, it increases the capacity of the skin to absorb drugs, and exhibits both analgesic and bacteriostatic activity. It can be used for the treatment of infection by retroviruses and as an anti-inflammatory agent. Furthermore, DMSO has been described as a preservative and cryo-protective agent for organ and tissue transplants.

In plant protection, DMSO can be used to ward off mammals (e.g., deer) and as a growth stimulant.

20.2 Production Technologies (Ullmanns 2004)

In industry, DMSO is produced by catalytic oxidation of dimethyl sulfide with oxygen or by oxidation with nitrogen dioxide. The oxidation of dimethyl sulfide with oxygen at 7.2 MPa and 105 °C gives good yields of DMSO.

Another way to produce DMSO is an oxidation of methanol with hydrogen sulfide



20.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg N,N-dimethylsulfoxide, at plant, global**. For this study the production of dimethylsulfoxide by means of the oxidation of methanol with hydrogen sulfide is considered. All data in the present report are referred to 1 kg dimethylsulfoxide 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

20.4 Raw materials and auxiliaries

As there was no data available for the production of dimethylsulfoxide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the dimethylsulfoxide production.

Tab. 20.2 Consumption of raw materials and auxiliaries for the dimethylsulfoxide production

Input	Values	Remarks
Methanol (kg)	0.886	Calculated with a yield of 95 %
H ₂ S (kg)	0.471	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

20.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the dimethylsulfoxide production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 20.3 Energy consumption for the dimethylsulfoxide production

Input	per kg dimethylsulfoxide
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

20.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the dimethylsulfoxide production.

Tab. 20.4 Total transport amounts for the dimethylsulfoxide production

tkm kg ⁻¹ dimethylsulfoxide	lorry	train
Total transports	1.36E-01	8.14E-01

20.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of dimethylsulfoxide production plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

20.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of dimethylsulfoxide. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 20.5 Process emissions to air from the production of 1 kg dimethylsulfoxide

Output	Value	Remarks
Methanol, to air (kg)	8.86E-5	Estimated as 0.2 % of input
H ₂ S, to air (kg)	4.71E-5	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	5.47E-2	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

20.9 Emissions to water

The remaining amount of unreacted methanol and hydrogen sulfide (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for methanol is assumed with 90%. The carbon contained in the removed methanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 20.6 Process emissions to water from the production 1 kg dimethylsulfoxide

Output	Value	Remarks
Methanol, to water (kg)	4.42E-03	calculated from mass balance
H ₂ S, to water (kg)	2.35E-02	calculated from mass balance
COD, BOD (kg)	6.36E-03	calculated from water emissions
TOC, DOC (kg)	1.66E-03	calculated from water emissions

20.10 Life cycle inventory of dimethylsulfoxide production and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the dimethylsulfoxide production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 20.7 Unit process raw data for the production of dimethylsulfoxide

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dimethyl sulfoxide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water	-	0	dimethyl sulfoxide, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	methanol, at plant	GLO	0	kg	8.86E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrogen sulphide, H2S, at plant	RER	0	kg	4.71E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.36E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	8.14E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Methanol	-	-	kg	8.86E-5	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen sulfide	-	-	kg	4.71E-5	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	5.47E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Methanol	-	-	kg	4.42E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen sulfide	-	-	kg	2.35E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	6.36E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	6.36E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.66E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	1.66E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

20.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of dimethylsulfoxide.

Tab. 20.8 Selected LCI results and the cumulative energy demand for the dimethylsulfoxide production

Name		dimethyl sulfoxide, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	43.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	3.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.5
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.3E-1
CO ₂ , fossil	air	kg	1.2E+0
NM VOC	air	kg	1.5E-3
nitrogen oxides	air	kg	2.2E-3
sulphur dioxide	air	kg	2.0E-3
particulates, <2.5 um	air	kg	1.9E-4
BOD	water	kg	7.2E-3
cadmium	soil	kg	6.3E-10

20.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dimethyl sulfoxide, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of dimethylsulfoxide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dimethylsulfoxid, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "dimethylsulfoxide, at plant, GLO" is modelled for the production of dimethylsulfoxide from methanol and hydrogen sulfide in Europe. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C2H6SO
	501	StatisticalClassification	
	502	CASNumber	67-68-5
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Sulfoxidation of methanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	49
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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21 Dimethylamine

Author: Jürgen Sutter, ETH S+U, Zurich

Review:

Last changes: 2006

21.1 Introduction

Dimethylamine (CAS-No. 124-40-3, $(\text{CH}_3)_2\text{NH}$; Synonym: N-methylmethamine) is a colourless gas with a distinct, offensive, fishy odor.

For this inventory the functional unit is 1 kg of pure dimethylamine. The most important chemical and physical properties of dimethylamine used in this inventory are given here:

Tab. 21.1 Chemical and physical properties of dimethylamine (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	88.1	g mol^{-1}	
Specific gravity	902	kg m^{-3}	at 20 °C
Boiling point	77.15	°C	at normale pressure
Melting point	- 88.6	°C	at normale pressure

21.2 Reserves and Resources

Dimethylamine is produced by reaction of methanol and ammonia over solid acid catalysts.

21.3 Production Technologies and Use

21.3.1 Production Technologies

Dimethylamine is prepared commercially by the exothermic reaction of methanol with ammonia over an amorphous silica – alumina catalyst at 390 – 430 °C. The reaction proceeds to thermodynamic equilibrium, whose position is governed by the temperature and the nitrogen – carbon ratio

The crude reaction mixture consists essentially of excess ammonia, dimethylamines, reaction water, and unconverted methanol. Purification is generally effected in a train of four to five distillation columns.

In one technique the operating pressures of the columns are chosen in such a way that the top condensers can be cooled by standard 30 °C cooling water, and that the bottom reboilers are fed with saturated steam of about 195 °C. The first column (known as the ammonia column), which operates at 1.7 MPa, removes excess ammonia. The latter is recycled to the synthesis section. In the second column, pure trimethylamine is recovered overhead by using extractive distillation with water and a pressure of 1.0 MPa. Extraction water, reaction water, and unconverted methanol are removed from the bottom of the third column; the dimethylamine is distilled overhead. This mixture is separated in the fourth distillation column (d) at 0.7 MPa, MMA being the overhead and DMA the bottom product.

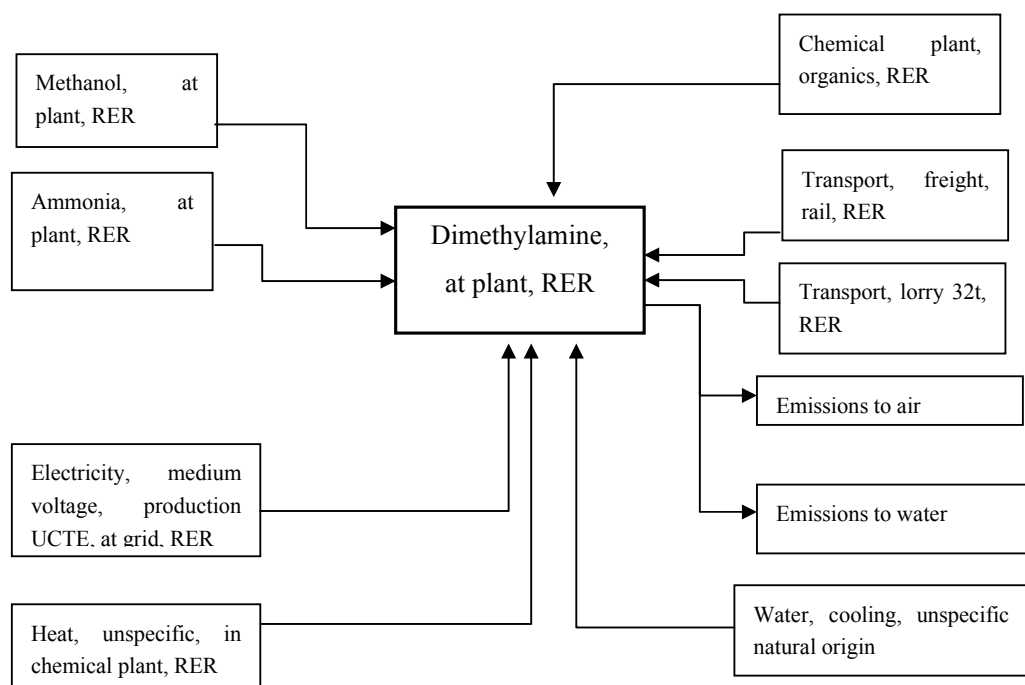
In another technique, a supplementary column operating at 0.7 – 0.9 MPa is inserted after the ammonia column for bottoms removal of reaction water, certain impurities, and unconverted methanol. Most of the extraction water may be recycled; the small remaining quantity of bottom water, containing some impurities and methanol, is disposed of easily either by incineration or biological treatment.

21.3.2 Use

Dimethylamine is a synthetic intermediate. Examples of products based on dimethylamine include fungicides and rubber accelerators, tetramethyl thioperoxydicarbonic diamide, tetramethyl thiodicarbonic diamide, the propellant 1,1-dimethylhydrazine, various pharmaceuticals, monomers such as dimethylaminoethyl methacrylate, solvents, catalysts, the insecticide dimefox, surfactants, and ion-exchange resins.

21.4 System Characterisation

Fig. 21.1 Process flow chart for the production of dimethylamine



This report corresponds to the module in the ecoinvent database for the production of **1 kg dimethylamine, at plant, in Europe**. For this study the production of dimethylamine by means of the reaction of methanol with ammonia is considered. All data in the present report are referred to 1 kg dimethylamine 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

21.5 Raw materials and auxiliaries

As there was no data available for the production of dimethyl amine, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large

chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the dimethylamine production.

Tab. 21.2 Consumption of raw materials and auxiliaries for the production of 1 kg dimethylamine

Input	Values	Remarks
Methanol (kg)	0.749	Calculated with a yield of 95%
Ammonia (kg)	1.11E-01	Calculated with a yield of 95%
Cooling water (kg)	24	Estimation

21.6 Energy

Tab. 21.3 Energy consumption for the production of 1 kg dimethylamine

Input	Kim 2003	This study
Steam (MJ)	3.72	1.76
Electricity (kWh)	0.3583	3.583E-01
Potential recovered energy (MJ)	- 1.96	

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

21.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 21.4 summarises the total transport amounts for the production of 1 kg dimethylamine.

Tab. 21.4 Total transport amounts for the production of 1 kg dimethylamine

tkm kg ⁻¹ dimethylamine	lorry	train
Total	8.6E-02	5.16E-02

21.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of dimethylamine plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

21.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of dimethylamine. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 21.5.

Tab. 21.5 Process emissions to air from the production of 1 kg dimethylamine

Output	Value	Remarks
Methanol, to air (kg)	1.5E-03	Estimated as 0.2 % of input
Ammonia, to air (kg)	7.95E-04	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	4.45E-2	from waster water treatment
Waste heat (MJ)	1.29	calculated from electricity input

21.10 Emissions to water

The remaining amount of unreacted methanol and ammonia (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for methanol was assumed with 90%. The carbon contained in the removed methanol was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of methanol and ammonia in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 21.6.

Tab. 21.6 Process emissions to water from the production 1 kg dimethylamine

Output	Value	Remarks
Methanol, to water (kg)	3.59E-03	calculated from mass balance
Ammonia, to water (kg)	3.97E-01	calculated from mass balance
COD, BOD (kg)	5.17E-03	calculated from water emissions
TOC, DOC (kg)	1.35E-03	calculated from water emissions

21.11 Life cycle inventory and Data Quality Considerations

Tab. 21.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of dimethylamine. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 21.7 Unit process raw data for the production of dimethylamine

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	dimethylamine, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	dimethylamine, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	methanol, at plant	GLO	0	kg	7.49E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	1.11E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry 32t	RER	0	tkm	8.60E-2	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	5.16E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.58E-1	1	1.16	(1,3,1,5,1,4); Kim 2003
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.76E+0	1	1.16	(1,3,1,5,1,4); Kim 2003
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.29E+0	1	1.16	(1,3,1,5,1,4); Calculated from the electricity input
	-	4	Methanol	-	-	kg	1.50E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Ammonia	-	-	kg	7.95E-4	1	1.37	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	4.45E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Methanol	-	-	kg	3.59E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Ammonium, ion	-	-	kg	3.97E-1	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	5.17E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	5.17E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.35E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	1.35E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

21.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 21.8 shows selected LCI results and the cumulative energy demand for the production of dimethylamine.

Tab. 21.8 Selected LCI results and the cumulative energy demand for the production of dimethylamine

Name		dimethylamine, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	38.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.0E-1
CO ₂ , fossil	air	kg	1.2E+0
NM VOC	air	kg	2.7E-3
nitrogen oxides	air	kg	1.9E-3
sulphur dioxide	air	kg	2.2E-3
particulates, <2.5 um	air	kg	2.1E-4
BOD	water	kg	6.5E-3
cadmium	soil	kg	4.7E-10

21.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	dimethylamine, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of dimethylamine including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Dimethylamin, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "dimethylamine, at plant, RER" is modelled for the production of dimethylamine from methanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C2H7N
	501	StatisticalClassification	
	502	CASNumber	204-697-4
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Reaction of methanol with ammonia
Representativeness	722	Percent	10
	724	ProductionVolume	Total production of ethylamines in the world is around 30.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

21.14 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschler R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
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22 Ethyl acetate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

22.1 Introduction

Ethyl acetate (CAS-No. 141-78-6, $\text{CH}_3\text{COOC}_2\text{H}_5$, Synonyms: ethyl acetic ester, acetic ether, acetidin, acetoxyethane, ethyl ethanoate) is a colourless, mobile, volatile liquid with a pleasant fruity odour. It is soluble in ethyl alcohol and ether, and slightly soluble in water. It forms azeotropes with water (8.2 wt % H_2O , bp 70.4 °C), ethanol (30.8 wt % EtOH, bp 71.8 °C), and methanol (44 wt % MeOH, bp 60.2 °C). With water and ethanol, the ester forms a ternary azeotrope (7.8 wt % H_2O , 9.0 wt % EtOH, bp 70.3 °C).

For this inventory the functional unit is 1 kg of pure ethyl acetate. The most important chemical and physical properties of ethyl acetate used in this inventory are given here:

Tab. 22.1 Chemical and physical properties of ethyl acetate (Wells 1991)

Property	Value	Unit	Remarks
Molecular weight	88.1	g mol^{-1}	
Specific gravity	902	kg m^{-3}	at 20 °C
Boiling point	77.15	°C	at normale pressure
Melting point	- 88.6	°C	at normale pressure

22.2 Reserves and Resources

Most of the ethyl acetate manufactured is prepared by the esterification of ethyl alcohol with acetic acid, the catalytic condensation of acetaldehydes with alkoxides (Tischenko reaction) is of minor importance.

22.3 Production Technologies and Use

22.3.1 Production Technologies

From ethyl alcohol and acetic acid by esterification (Wells 1991)

An aqueous solution of acetic acid is mixed with ethanol; then 1% sulfuric acid is added. The mixture is preheated before passing into an esterifying column where it is refluxed and some of the distillate removed overhead. The mixture, containing about 20% of the ester, goes into a second refluxing column where a ternary azeotrope (consisting of 83% ethyl acetate, 9% ethanol and 8% of water) is removed overhead at a temperature of about 70°C.

An equal volume of water is mixed with the distillate after which it separates into two layers. The bottom layer is fed into a separator column and ester is recovered as a ternary azeotrope. Any alcohol remaining in the bottom layer is vaporized and recycled to the esterification column.

The upper layer, containing around 93% ethyl acetate together with some water and ethanol, passes to a refluxing column where it is mixed with sufficient ester to ensure that the water and ethyl alcohol are removed as a ternary azeotrope overhead. The residue in the column, consisting of 95% ethyl acetate, is redistilled to remove any impurities. The process leads to a yield of 99 %.

Tischenko reaction (Wells 1991)

Another process for the production of ethyl acetate of less importance is the condensation of two acetaldehyde molecules (Tischenko reaction). Aluminum ethoxide promoted with chloride ions serve as a catalyst. In practice, acetaldehyde is passed at 0 – 5 °C through a mixture of aluminum fillings and traces of AlCl_3 in ethanol – ethyl acetate; the yield is 95 %.

22.3.2 Use

Ethyl acetate finds use as a solvent in a wide range of applications, across many industries, including:

1. *Surface coating and thinners*: Ethyl acetate is one of the most popular solvents and finds wide use in the manufacture of nitrocellulose lacquers, varnishes and thinners. It exhibits high dilution ratios with both aromatic and aliphatic diluents and is the least toxic of industrial organic solvents.
2. *Pharmaceuticals*: Ethyl acetate is an important component in extractants for the concentration and purification of antibiotics. It is also used as an intermediate in the manufacture of various drugs.
3. *Flavours and essences*: Ethyl acetate finds extensive use in the preparation of synthetic fruit essences, flavours and perfumes.
4. *Flexible packaging*: Substantial quantities of ethyl acetate are used in the manufacture of flexible packaging and in the manufacture of polyester films and BOPP films. It is also used in the treatment of aluminium foils.
5. *Miscellaneous*: Ethyl acetate is used in the manufacture of adhesives, cleaning fluids, inks, nail-polish removers and silk, coated papers, explosiv.

Ethyl acetate is an active solvent that is available in three grades: anhydrous with 85 - 88 % with 12 - 15 % ethanol denaturated with 1 % MIK, 99% and 99.5% (urethane grade). The most popular grade, 99% is mainly used in industrial lacquers and surface coating resins. It is also used as an extraction solvent in the production of pharmaceuticals and food and as a carrier solvent for herbicides. About 60% of demand is in coatings. Process solvents, including pharmaceuticals and organic synthesis, account for 15% of demand, as does printing inks. Miscellaneous uses, including adhesives and cosmetics, account for 10% of consumption.

The total global production for ethyl acetate is estimated to be in the 1.3 million tons in 2002 (Dutia 2004).

22.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg ethyl acetate, at plant, in Europe**. For this study the production of ethyl acetate by means of the esterification of ethanol with acetic acid is considered. All data in the present report are referred to 1 kg ethyl acetate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

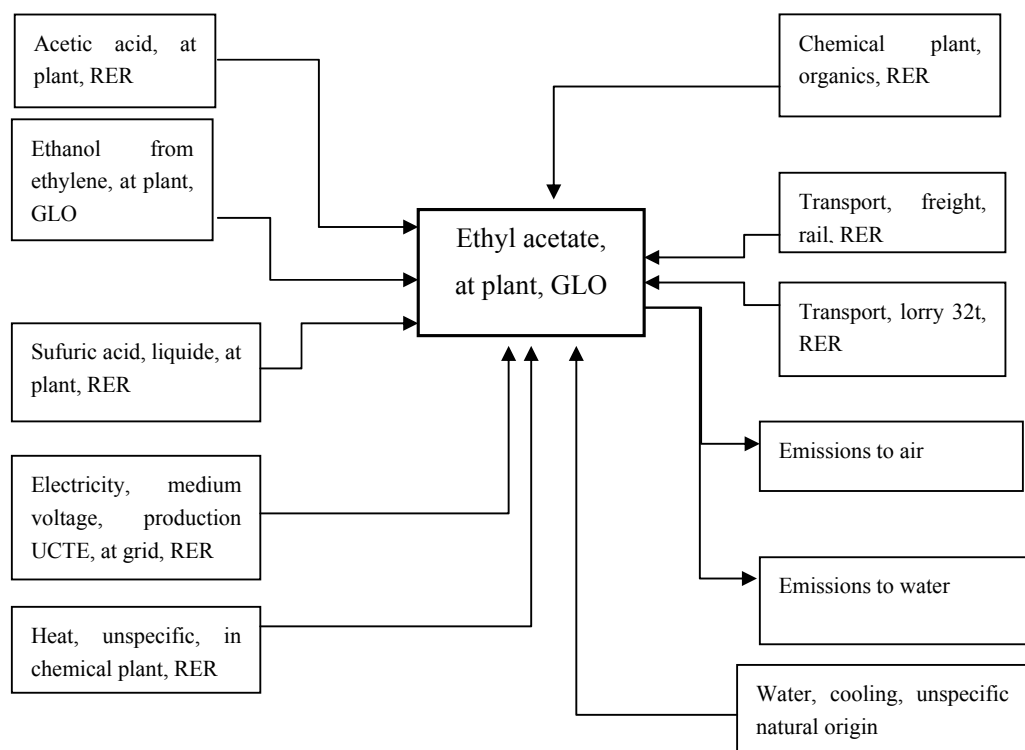


Fig. 22.1 Process flow chart for the production of ethyl acetate

22.5 Raw materials and auxiliaries

The data for the input of raw materials and auxiliaries are taken from Wells (1991), which gives data for US production in the late 1980ies.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the ethyl acetate production.

Tab. 22.2 Consumption of raw materials and auxiliaries for the production of 1 kg ethyl acetate

Input	Esterification ¹	Tischenko ²	This study ³
Ethanol 95 % (kg)	0.62		6.2E-01
Acetic acid (kg)	0.69		6.9E-01
Sulfuric acid (kg)	0.0131		1.31E-02
Acetaldehyde (kg)		1.02	
Cooling water (kg)		150.198	24

¹ Source: Wells 1991

2 Source: Li Xiong 2002

22.6 Energy

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 22.3 Energy consumption for the production of 1 kg ethyl acetate

Input	Esterification ¹	Tischenko ²	This study
Steam (MJ)	10.5	4.125	8.84
Electricity (kWh)	0.00725	0.026014	7.25E-03
Potential recovered energy (MJ)	- 1.66		
Sum of consumed energy (MJ)	8.87		

1 Source: Kim 2003

2 Source: Li Xiong 2002

22.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and 50 km by lorry 32t and 10 km by train the sulphuric acid. Tab. 22.4 summarises the total transport amounts for the production of 1 kg ethyl acetate.

Tab. 22.4 Total transport amounts for the production of 1 kg ethyl acetate

tkm kg ⁻¹ ethyl acetate	lorry	train
Raw materials	1.317E-01	7.873E-01

22.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethyl acetate plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

22.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-03 kWh/kg ethyl acetate is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of ethyl acetate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 22.5.

Tab. 22.5 Process emissions to air from the production of 1 kg ethyl acetate

Output	Value	Remarks
Acetic acid, to air (kg)	1.38E-03	Estimated as 0.2 % of input
Ethanol, to air (kg)	1.24E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	8.54E-3	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

22.10 Emissions to water

There was no data available on process emissions to water for the production of ethyl acetate. As approximation the water emissions were estimated to 0.8% of the raw material input. Further it was assumed that the waste water is treated in a internal waste water plant. The removal efficiency for ethanol and acetic acid was assumed with 90%. The carbon contained in the removed ethanol and acetic acid was accounted as CO₂ emissions to air. The sulphate emissions are calculated from the sulphuric acid input.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of ethanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 22.6.

Tab. 22.6 Process emissions to water from the production 1 kg ethyl acetate

Output	Value	Remarks
Acetic acid, to water (kg)	5.52E-03	calculated from mass balance
Ethanol, to water (kg)	4.96E-03	calculated from mass balance
Sulphate, to water (kg)	1.283E-02	calculated from mass balance
COD, BOD (kg)	1.56E-02	calculated from water emissions
TOC, DOC (kg)	4.79E-03	calculated from water emissions

22.11 Life cycle inventory and Data Quality Considerations

Tab. 22.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of ethyl acetate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 22.7 Unit process raw data for the production of ethyl acetate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	ethyl acetate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	ethyl acetate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	ethanol from ethylene, at plant	RER	0	kg	6.20E-1	1	1.13	(1,2,3,1,1,3); Wells 1991
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	6.90E-1	1	1.13	(1,2,3,1,1,3); Wells 1991
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	1.31E-2	1	1.13	(1,2,3,1,1,3); Wells 1991
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.87E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.13	(1,2,3,1,1,3); Kim 2003
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	8.84E+0	1	1.13	(1,2,3,1,1,3); Kim 2003
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.13	(1,2,3,1,1,3); Calculated from the electricity input
	-	4	Ethanol	-	-	kg	1.24E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	1.38E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	8.54E-3	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Acetic acid	-	-	kg	5.52E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Ethanol	-	-	kg	4.96E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.56E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.56E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4.79E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	4.79E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

22.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 22.8 shows selected LCI results and the cumulative energy demand for the production of ethyl acetate.

Tab. 22.8 Selected LCI results and the cumulative energy demand for the production of ethyl acetate

Name		ethyl acetate, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	76.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.3E-1
CO ₂ , fossil	air	kg	2.6E+0
NM VOC	air	kg	1.1E-2
nitrogen oxides	air	kg	4.7E-3
sulphur dioxide	air	kg	6.2E-3
particulates, <2.5 um	air	kg	4.2E-4
BOD	water	kg	2.1E-2
cadmium	soil	kg	1.0E-9

22.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethyl acetate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of ethyl acetate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Essigester, ab Werk
	491	Synonyms	ethyl acetic ester, acetic ether, acetidin, acetoxyethane, ethyl ethanoate
	492	GeneralComment	The process "ethyl acetate, at plant, RER" is modelled for the production of ethyl acetate from ethanol in Europe. Raw materials and energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H8O2
	501	StatisticalClassification	
	502	CASNumber	123-86-4
TimePeriod	601	StartDate	1991
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Esterification of ethanol
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 1.3 Million tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents
ProofReading	5616	Validator	11
	5615	Details	automatic validation in Excel
	5619	OtherDetails	none

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23 Ethylene glycol diethyl ether

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

23.1 Introduction

Ethylene glycol diethyl ether (C₆H₁₄O₂; CAS 629-14-1; synonym: diethoxyethane) is a colourless liquid. For this inventory the functional unit is 1 kg of pure ethylene glycol diethyl ether. The most important chemical and physical properties of ethylene glycol diethyl ether used in this inventory are given here.

Tab. 23.1 Chemical and physical properties of ethylene glycol diethyl ether (Ullmanns 2004)

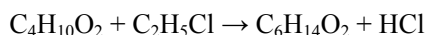
Property	Value	Unit
Molecular weight	118.172	g mol ⁻¹
Specific gravity	841	kg m ⁻³
Boiling point	121.4	°C
Melting point	-74	°C

23.2 Use

Ethylene glycol diethyl ether is mainly used as reaction media and as solvent.

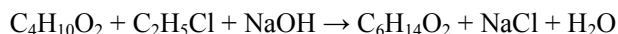
23.3 Production Technologies (Ullmanns 2004)

Ethylene glycol diethyl ether is usually produced by a reaction of ethylene glycol monoethyl ether with monochloroethane.



The hydrochloric acid is neutralized with sodium hydroxide.

Overall reaction:



23.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg ethylene glycol diethyl ether, at plant, global**. For this study the production of ethylene glycol diethyl ether by means of the reaction of ethylene glycol monoethyl ether with monochloroethane is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

23.5 Raw materials and auxiliaries

As there was no data available for the production of ethylene glycol diethyl ether, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). As approximation for monochloroethane the dataset "ethylene dichloride, at plant, RER" is used.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the N ethylene glycol diethyl ether production.

Tab. 23.2 Consumption of raw materials and auxiliaries for the ethylene glycol diethyl ether production

Input	Values	Remarks
Ethylene glycol monoethyl ether (kg)	0.831	Calculated with a yield of 95 %
Monochloroethane (kg) ¹	0.425	Calculated with a yield of 95 %
Sodium hydroxide (kg)	0.351	Estimation
Cooling water (kg)	24	Estimation

1: Approximated with "ethylene dichloride, at plant, RER"

23.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a-1 (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg-1) were used as approximation for the energy consumption of the ethylene glycol diethyl ether production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 23.3 Energy consumption for the ethylene glycol diethyl ether production

Input	per kg Ethylene glycol diethyl ether
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

23.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the ethylene glycol diethyl ether production.

Tab. 23.4 Total transport amounts for the ethylene glycol diethyl ether production

tkm kg ⁻¹ ethylene glycol diethyl ether	lorry	train
Total transports	1.96E-01	1.17

23.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene glycol diethyl ether plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

23.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of ethylene glycol diethyl ether. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 23.5 Process emissions to air from the production of 1 kg ethylene glycol diethyl ether

Output	Value	Remarks
Ethylene glycol monoethyl ether, to air (kg)	8.31E-5	Estimated as 0.2 % of input
Monochloroethane, to air (kg)	4.25E-5	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.1E-1	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

23.10 Emissions to water

The remaining amount of unreacted ethylene glycol monoethyl ether and monochloroethane (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for ethylene glycol monoethyl ether and monochloroethane is assumed with 90%. The carbon contained in the removed ethylene glycol monoethyl ether and monochloroethane is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of ethylene glycol monoethyl ether and monochloroethane in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Another emission to water is sodium chloride from the neutralization of HCl.

Tab. 23.6 Process emissions to water from the production 1 kg ethylene glycol diethyl ether

Output	Value	Remarks
Ethylene glycol monoethyl ether, to water (kg)	4.15E-03	calculated from mass balance
Monochloroethane, to water (kg)	2.12E-03	calculated from mass balance
Sodium, to water (kg)	2.02E-01	calculated from mass balance
Chloride, to water (kg)	3.07E-01	calculated from mass balance
COD, BOD (kg)	1.11E-02	calculated from water emissions
TOC, DOC (kg)	3E-03	calculated from water emissions
AOX (kg)	3.07E-01	calculated from water emissions

23.11 Life cycle inventory of ethylene glycol diethyl ether production and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the ethylene glycol diethyl ether production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 23.7 Unit process raw data for the production of ethylene glycol diethyl ether

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	ethylene glycol diethyl ether, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
resource, in water	401	0	ethylene glycol diethyl ether, at plant	RER	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	ethylene glycol monoethyl ether, at plant	RER	0	kg	8.31E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	ethylene dichloride, at plant	RER	0	kg	4.25E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	3.51E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.96E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	1.17E+0	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
		4	Ethylene glycol monoethyl ether	-	-	kg	8.31E-5	1	1.62	(4,5,na,na,na,na); Estimation
		4	Monochloroethane	-	-	kg	4.25E-5	1	1.62	(4,5,na,na,na,na); Estimation
		4	Carbon dioxide, fossil	-	-	kg	1.09E-1	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river		4	Ethylene glycol monoethyl ether	-	-	kg	4.15E-3	1	1.30	(4,5,na,na,na,na); Estimation
		4	Monochloroethane	-	-	kg	2.12E-3	1	1.30	(4,5,na,na,na,na); Estimation
		4	Sodium, ion	-	-	kg	2.02E-1	1	5.10	(4,5,na,na,na,na); Estimation
		4	Chloride	-	-	kg	3.07E-1	1	3.09	(4,5,na,na,na,na); Estimation
		4	BOD5, Biological Oxygen Demand	-	-	kg	1.11E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	COD, Chemical Oxygen Demand	-	-	kg	1.11E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	DOC, Dissolved Organic Carbon	-	-	kg	3.00E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	TOC, Total Organic Carbon	-	-	kg	3.00E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	3.08E-1	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4		-	-	kg		1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

23.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of ethylene glycol diethyl ether.

Tab. 23.8 Selected LCI results and the cumulative energy demand for the ethylene glycol diethyl ether production

Name		ethylene glycol diethyl ether, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	76.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	11.9
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	3.5E-1
CO ₂ , fossil	air	kg	3.1E+0
NM VOC	air	kg	3.7E-3
nitrogen oxides	air	kg	5.9E-3
sulphur dioxide	air	kg	7.0E-3
particulates, <2.5 um	air	kg	6.0E-4
BOD	water	kg	2.1E-2
cadmium	soil	kg	2.0E-9

23.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethylene glycol diethyl ether, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of ethylene glycol diethyl ether including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ethylenglykoldiethylether, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "ethylene glycol diethyl ether, at plant, GLO" is modelled for the production of ethylene glycol diethyl ether from ethylene glycol monoethyl ether and monochloroethane in the world. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H14O2
	501	StatisticalClassification	
	502	CASNumber	629-14-1
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Ethyl etherification of ethylene glycol monoethyl ether
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	highly pure chemicals

23.14 References

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24 Ethylene glycol dimethyl ether

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

24.1 Introduction

Ethylene glycol dimethyl ether (C₄H₁₀O₂; CAS 110-71-4; synonym: dimethoxyethane) is a colourless liquid. For this inventory the functional unit is 1 kg of pure ethylene glycol dimethyl ether. The most important chemical and physical properties of ethylene glycol dimethyl ether used in this inventory are given here.

Tab. 24.1 Chemical and physical properties of ethylene glycol dimethyl ether (Ullmanns 2004)

Property	Value	Unit
Molecular weight	90.12	g mol ⁻¹
Specific gravity	867	kg m ⁻³
Boiling point	85.2	°C
Melting point	-58	°C

24.2 Use

Ethylene glycol dimethyl ether is mainly used as reaction media and as solvent.

24.3 Production Technologies (Ullmanns 2004)

Ethylene glycol dimethyl ether is usually produced by a reaction of ethylene oxide with dimethyl ether.



24.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg ethylene glycol dimethyl ether, at plant, global**. For this study the production of ethylene glycol dimethyl ether by means of the reaction of ethylene oxide with dimethyl ether is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

24.5 Raw materials and auxiliaries

As there was no data available for the production of ethylene glycol dimethyl ether, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large

chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the N ethylene glycol dimethyl ether production.

Tab. 24.2 Consumption of raw materials and auxiliaries for the ethylene glycol dimethyl ether production

Input	Values	Remarks
Ethylene oxide (kg)	0.515	Calculated with a yield of 98 %
Dimethyl ether (kg)	0.538	Calculated with a yield of 98 %
Cooling water (kg)	24	Estimation

24.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the ethylene glycol dimethyl ether production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 24.3 Energy consumption for the ethylene glycol dimethyl ether production

Input	per kg ethylene glycol dimethyl ether
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

24.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the ethylene glycol dimethyl ether production.

Tab. 24.4 Total transport amounts for the ethylene glycol dimethyl ether production

tkm kg ⁻¹ ethylene glycol dimethyl ether	lorry	train
Total transports	1.05E-01	6.32E-01

24.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene glycol dimethyl ether plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

24.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of ethylene glycol dimethyl ether. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 24.5 Process emissions to air from the production of 1 kg ethylene glycol dimethyl ether

Output	Value	Remarks
Ethylene oxide, to air (kg)	5.15E-05	Estimated as 0.2 % of input
Dimethyl ether, to air (kg)	5.38E-05	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	9.24E-02	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

24.10 Emissions to water

The remaining amount of unreacted ethylene oxide and dimethyl ether (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for ethylene oxide and dimethyl ether is assumed with 90%. The carbon contained in the removed ethylene oxide and dimethyl ether is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of ethylene oxide and dimethyl ether in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 24.6 Process emissions to water from the production 1 kg ethylene glycol dimethyl ether

Output	Value	Remarks
Ethylene oxide, to water (kg)	2.57E-03	calculated from mass balance
Dimethyl ether, to water (kg)	2.68E-03	calculated from mass balance
COD, BOD (kg)	3.55E-03	calculated from water emissions
TOC, DOC (kg)	1.11E-03	calculated from water emissions

24.11 Life cycle inventory of ethylene glycol dimethyl ether production and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the ethylene glycol dimethyl ether production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 24.7 Unit process raw data for the production of ethylene glycol dimethyl ether

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	ethylene glycol dimethyl ether, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
resource, in water	401	0	ethylene glycol dimethyl ether, at plant	RER	0	kg	1.00E+0			
	662	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	493	-	ethylene oxide, at plant	RER	0	kg	5.15E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	403	-	dimethyl ether, at plant	RER	0	kg	5.38E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
		-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
		-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
		-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
		-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
		-								
emission air, high population density		4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
		4	Ethylene oxide	-	-	kg	5.15E-5	1	1.62	(4,5,na,na,na,na); Estimation
		4	Diethyl ether	-	-	kg	5.38E-5	1	1.62	(4,5,na,na,na,na); Estimation
		4	Carbon dioxide, fossil	-	-	kg	9.24E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river		4	Ethylene oxide	-	-	kg	2.57E-3	1	3.09	(4,5,na,na,na,na); Estimation
		4	Dimethyl ether	-	-	kg	2.68E-3	1	1.30	(4,5,na,na,na,na); Estimation
		4	BOD5, Biological Oxygen Demand	-	-	kg	3.55E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	COD, Chemical Oxygen Demand	-	-	kg	3.55E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	DOC, Dissolved Organic Carbon	-	-	kg	1.11E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
		4	TOC, Total Organic Carbon	-	-	kg	1.11E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

24.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of ethylene glycol dimethyl ether.

Tab. 24.8 Selected LCI results and the cumulative energy demand for the ethylene glycol dimethyl ether production

Name		ethylene glycol dimethyl ether, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	69.7
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	5.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.7
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.9E-1
CO ₂ , fossil	air	kg	2.1E+0
NM VOC	air	kg	4.2E-3
nitrogen oxides	air	kg	3.6E-3
sulphur dioxide	air	kg	3.7E-3
particulates, <2.5 um	air	kg	3.2E-4
BOD	water	kg	1.0E-2
cadmium	soil	kg	9.0E-10

24.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethylene glycol dimethyl ether,
Geography	662	Location	at plant
ReferenceFunction	493	InfrastructureProcess	RER
ReferenceFunction	403	Unit	0
DataSetInformation	201	Type	kg
	202	Version	1
	203	energyValues	1.0
	205	LanguageCode	0
	206	LocalLanguageCode	en
DataEntryBy	302	Person	de
	304	QualityNetwork	24
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of ethylene glycol dimethyl ether including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ethylenglykoldimethylether, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "ethylene glycol dimethyl ether, at plant, GLO" is modelled for the production of ethylene glycol dimethyl ether from ethylene oxide in Europe. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H10O2
	501	StatisticalClassification	
	502	CASNumber	110-71-4
TimePeriod	601	StartDate	2000
	602	EndDate	2007
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for the world.
Technology	692	Text	Reaction of ethylene oxide and dimethyl ether
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	49
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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25 Ethylene glycol monoethyl ether

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

25.1 Introduction

Ethylene glycol monoethyl ether (C₄H₁₀O₂; CAS 110-80-5) is a colourless liquid. For this inventory the functional unit is 1 kg of pure ethylene glycol monoethyl ether. The most important chemical and physical properties of ethylene glycol monoethyl ether used in this inventory are given here.

Tab. 25.1 Chemical and physical properties of ethylene glycol monoethyl ether (Ullmanns 2004)

Property	Value	Unit
Molecular weight	90.12	g mol ⁻¹
Specific gravity	929.7	kg m ⁻³
Boiling point	135.1	°C
Melting point	-100	°C

25.2 Use

Ethylene glycol monoethyl ether is mainly used as reaction media and as solvent.

25.3 Production Technologies (Ullmanns 2004)

Ethylene glycol monoethyl ether is usually produced by a reaction of ethylene oxide with ethanol.



25.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg ethylene glycol monoethyl ether, at plant, global**. For this study the production of ethylene glycol monoethyl ether by means of the reaction of ethylene oxide with ethanol is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

25.5 Raw materials and auxiliaries

As there was no data available for the production of ethylene glycol monoethyl ether, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This

value was used in this inventory as approximation for the cooling water consumption of the N ethylene glycol monoethyl ether production.

Tab. 25.2 Consumption of raw materials and auxiliaries for the ethylene glycol monoethyl ether production

Input	Values	Remarks
Ethylene oxide (kg)	0.515	Calculated with a yield of 95 %
Ethanol (kg)	0.538	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

25.6 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the ethylene glycol monoethyl ether production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 25.3 Energy consumption for the ethylene glycol monoethyl ether production

Input	per kg Ethylene glycol monoethyl ether
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

25.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the ethylene glycol monoethyl ether production.

Tab. 25.4 Total transport amounts for the ethylene glycol monoethyl ether production

tkm kg ⁻¹ ethylene glycol monoethyl ether	lorry	train
Total transports	1.05E-01	6.32E-01

25.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene glycol monoethyl ether plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

25.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of ethylene glycol monoethyl ether. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 25.5 Process emissions to air from the production of 1 kg ethylene glycol monoethyl ether

Output	Value	Remarks
Ethylene oxide, to air (kg)	5.15E-05	Estimated as 0.2 % of input
Ethanol, to air (kg)	5.38E-05	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	9.24E-02	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

25.10 Emissions to water

The remaining amount of unreacted ethylene oxide and ethanol (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for ethylene oxide and ethanol is assumed with 90%. The carbon contained in the removed ethylene oxide and ethanol is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of ethylene oxide and ethanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 25.6 Process emissions to water from the production 1 kg ethylene glycol monoethyl ether

Output	Value	Remarks
Ethylene oxide, to water (kg)	2.57E-03	calculated from mass balance
Ethanol, to water (kg)	2.68E-03	calculated from mass balance
COD, BOD (kg)	9.16E-03	calculated from water emissions
TOC, DOC (kg)	2.86E-03	calculated from water emissions

25.11 Life cycle inventory of ethylene glycol monoethyl ether production and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the ethylene glycol monoethyl ether production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 25.7 Unit process raw data for the production of ethylene glycol monoethyl ether

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	ethylene glycol monoethyl ether, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	ethylene glycol monoethyl ether, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	ethylene oxide, at plant	RER	0	kg	5.15E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	ethanol from ethylene, at plant	RER	0	kg	5.38E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Ethylene oxide	-	-	kg	5.15E-5	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Ethanol	-	-	kg	5.38E-5	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	9.24E-2	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Ethylene oxide	-	-	kg	2.57E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Ethanol	-	-	kg	2.68E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	9.16E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	9.16E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.86E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.86E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

25.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of ethylene glycol monoethyl ether.

Tab. 25.8 Selected LCI results and the cumulative energy demand for the ethylene glycol monoethyl ether production

Name		ethylene glycol monoethyl ether, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	62.2
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	5.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.6
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.7E-1
CO ₂ , fossil	air	kg	1.9E+0
NM VOC	air	kg	3.2E-3
nitrogen oxides	air	kg	3.5E-3
sulphur dioxide	air	kg	3.8E-3
particulates, <2.5 um	air	kg	3.0E-4
BOD	water	kg	1.0E-2
cadmium	soil	kg	7.6E-10

25.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethylene glycol monoethyl ether, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of ethylene glycol monoethyl ether including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ethylenglykolmonoethylether, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "ethylene glycol monoethyl ether, at plant, GLO" is modelled for the production of ethylene glycol dimethyl ether from ethylene oxide in Europe. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H10O2
	501	StatisticalClassification	
	502	CASNumber	110-80-5
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Reaction of ethylene oxide and ethanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	49
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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26 Ethylene hydration

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Mirelle Faist Emmenegger, ESU Services, Uster
 Last changes: 2006

Ethylene can be hydrated in two ways: by indirect hydration or by direct hydration.

26.1 Production Technologies and Use

Indirect hydration (Del Borghi 1998)

Gas containing ethylene with a C_2H_4 percentage variable from 35% to 95% reacts in adsorption towers at 55-80°C and 10-35 bars with sulphuric acid with H_2SO_4 percentage varying from 94 to 98%; eventually, the reaction is catalysed by Ag_2SO_4 . The reaction is exothermic and gives mono- and diethylsulphate. Both esters are hydrolysed to ethanol in towers with antiacid coating at 70-100°C. During the hydrolysis at high temperatures, the by-product diethylether is obtained. The sulphuric acid is later concentrated from 50 to 98%. The process efficiency is of 86 %.

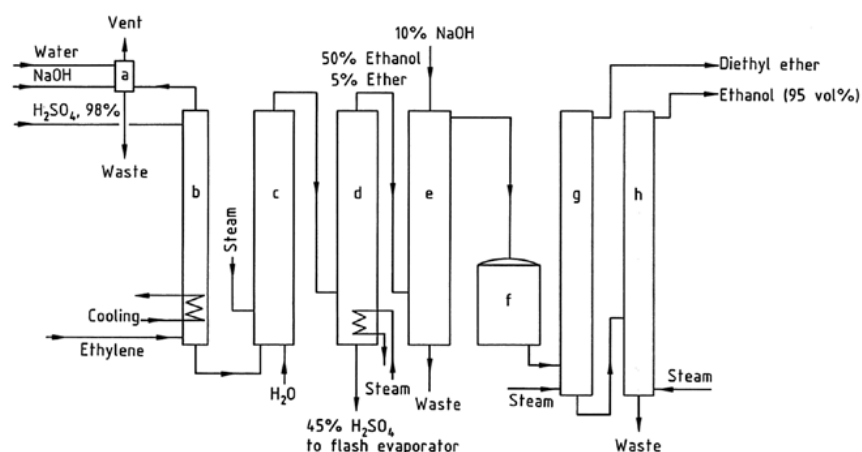
Ethylene is absorbed in two steps:



Hydrolysis involves three steps:



Fig. 26.1 Production of ethanol by means of the indirect hydration of ethanol (taken from Ullmann 2001b: Fig. 4)



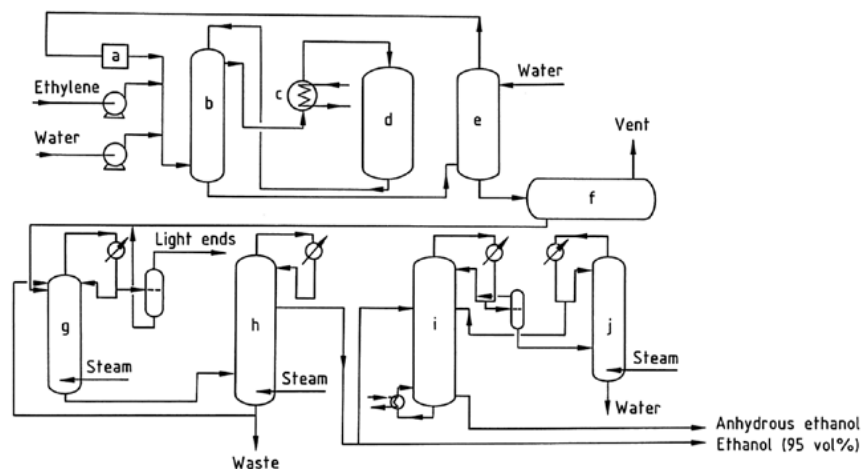
a) Wash column for residual gas; b) Absorber; c) Hydrolyzer; d) Stripping column; e) Wash column for crude ethanol; f) Crude product tank; g) Diethyl ether distillation column; h) Ethanol distillation column

Direct hydration of ethylene (Del Borghi 1998)

The catalytic hydration of ethylene is an heterogeneous process in gaseous phase on acid catalyst, where the catalyst is H_3PO_4 , $T = 300^\circ\text{C}$ and $p = 70$ bars. The ethylene conversion is only of 4% per passage. The reaction gas is recycled many times to increase total efficiency. The ethylene has to be of high purity to avoid inert gases concentration. The reaction gas is cooled to condense the liquid products and separate it from the ethylene to be recycled. Ethanol solution is concentrated and purified by extractive distillation. The ethanol selectivity is about 98%. Co-products of the direct hydration are diethyl ether (1 %), butene (0.6 %), and acetaldehyde (0.2 %). The diethyl ether is purified in the ether distillation column. The other organic byproducts are recovered and fed into the steam generator.



Fig. 26.2 Production of ethanol by means of the direct hydration of ethanol (taken from Ullmann 2001b: Fig. 3)



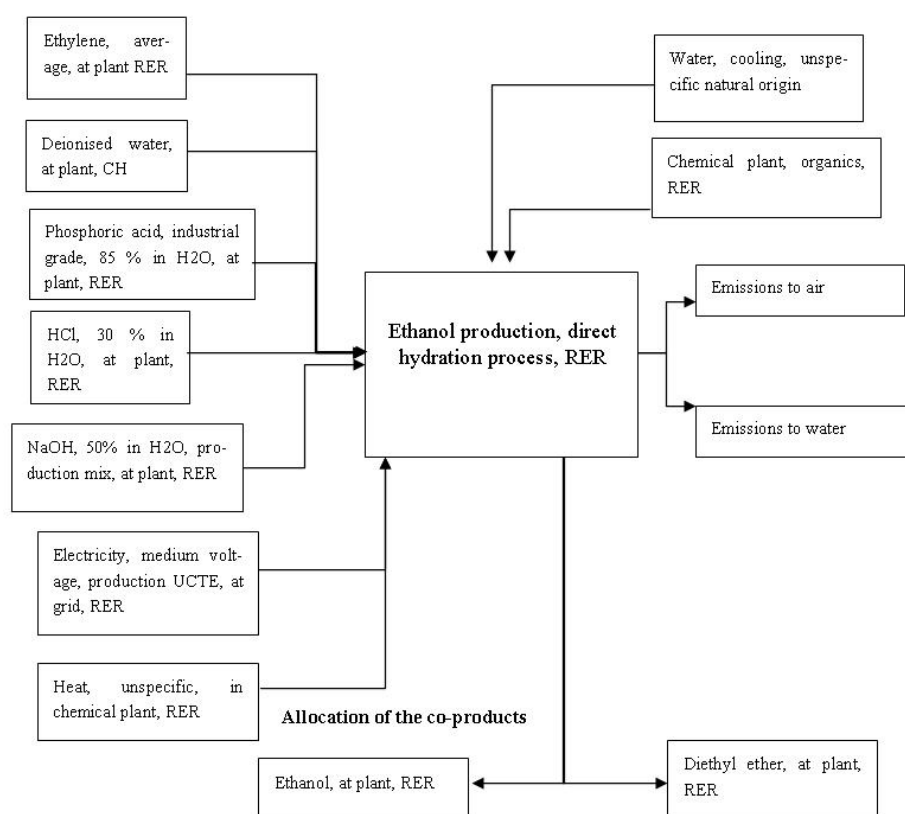
a) Circulation compressor; b) Heat exchanger; c) Superheater; d) Reactor; e) Washer; f) Crude ethanol tank; g) Extractive distillation column; h) Concentration column; i) Dehydration; j) Concentration column

26.2 System Characterisation

In this study the direct hydration of ethylene is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the ethanol production, diethyl ether is derived as co-product. Therefore an allocation between the two products was made. The following processes were modelled:

- Ethanol production, direct hydration process, RER. Multi output process with ethanol, 99.7% in H₂O, at plant, RER and diethyl ether, 99.95% in H₂O, at plant, RER as allocated products; see

Fig. 26.3 Process chain for the production of ethanol



26.3 Raw materials and auxiliaries

Tab. 26.1 shows values for the use of raw materials of the ethylene hydration process from several sources. This study is based on an ethanol yield of 1.59 kg per kg ethylene.

Tab. 26.1 Consumption of raw materials and auxiliaries for the hydration of ethylene

Input	Meyers 1986 (per kg ethanol)	Borghi 1998 (per kg ethanol)	Wells 1991 (per kg ethanol)	Naran Lala 2004 (per kg ethanol)	This study (per kg ethylene)
Ethylene (kg)	0.627		0.627		1.00
Process water (kg)	0.7				1.12
Cooling water (m ³)	0.036	0.013			2.07E-02
H ₃ PO ₄ (100 %) (kg)	0.0007	small	0.0007		1.12E-03
NaOH (100 %) (kg)	0.0018	small			2.87E-03
Entrainer (kg)	0.0002	small			-
HCl (100 %) (kg)	0.0012				1.91E-03
Benzene (kg)				0.00342	5.45E-03 ¹
Ca ₃ PO ₄		small			-

¹ The benzene is used as entrainer.

26.4 Energy

Tab. 26.2 Energy consumption for the hydration of ethylene

Input	Meyers 1986 (per kg ethanol)	Borghi 1998 (per kg ethanol)	Naran Lala 2004 (per kg ethanol)	This study (per kg ethylene)
Steam (MJ)	2.5		2.92	4.66
Electricity (kWh)	0.096		0.017	2.71E-02
Energy (MJ)		11		

The values for the energy consumption are taken from Naran Lala because the data were more up-to-date.

26.5 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for ethylene, H₃PO₄, Benzene, and NaOH, and 100 km by lorry 32t and 200 km by train for HCl. Tab. 14.4 summarises the total transport amounts for the production of 1 kg ethanol.

Tab. 26.3 Total transport amounts for the hydration of ethylene

tkm kg ⁻¹ ethylene	lorry	train
Raw materials	1.00E-01	6.00E-01
Auxiliaries	1.34E-03	5.51E-03
Total transports	1.01E-01	6.06E-01

26.6 Infrastructure and land use

No information was readily available about infrastructure and land-use of ethylene hydration plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

26.7 Emissions to air

Since the whole system is a closed loop and the off gas from the ethylene purification is fed into the steam generator there should be no emissions to air from the process except CO₂ from the burning of the co-products butene and acetaldehyde in the steam generator (Meyers 1986). Nevertheless, in this inventory a fugitive loss of ethylene of 0.2 % of the input is considered. IPPC (2003) considers hydrocarbons emissions from process vents as an important problem.

Tab. 26.4 Process emissions to air from the hydration of ethylene

Output	Value	Remarks
Waste heat (MJ)	9.76E-02	calculated from electricity input
Ethylene, to air (kg)	2.00E-03	estimated as 0.2 % of the input
CO ₂ , fossil, to air (kg)	1.85E-02	from burning of co-products

26.8 Emissions to water

Due to the fact that the co-products usually are recovered from the process system and fed into the steam generator, there are just some emissions of ions to water.

Tab. 26.5 Process emissions to water from the hydration of ethylene

Output	Meyers 1986
PO ₄ ³⁻	1.75E-03
Cl ⁻	3.03E-03
Na ⁺	2.71E-03

26.9 Wastes

Because the wastewater pollutions are listed separately the wastewater is considered as unpolluted.

Tab. 26.6 Waste water to waste water treatment

Output	
Treatment, sewage, unpolluted, to wastewater treatment, class 3	2.08E-02

26.10 Allocation of ethanol and diethyl ether

Within the hydration of ethylene, ethanol and diethyl ether are obtained as co-products. According to the data from Meyers 1986, it was assumed that in average an amount of 0.00815 kg diethyl ether per kg ethanol is produced. As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 14.7.

Tab. 26.7 Allocation of ethanol and the co-product diethyl ether from the hydration of ethylene

Process	Ethanol production, direct hydration process, RER		
	Unit	per kg ethylene	Allocation factor
Ethanol, at plant, RER	Kg	1.59	0.992
Diethyl ether, at plant, RER	kg	1.30E-02	0.008

26.11 Ethanol from ethylene, at plant, RER

Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$, CAS-No. 64-17-5, synonyms: ethyl alcohol, alcohol spirit, spirit of wine, grain alcohol, absolute alcohol, ethyl hydrate) in its pure form (absolute alcohol) is a clear, colorless liquid. It is miscible in all proportions with water and also with ether, acetone, benzene, and some other organic solvents. Anhydrous alcohol is hygroscopic; at a water uptake of 0.3 – 0.4 %, a certain stability does occur. For this inventory the functional unit is 1 kg of pure ethanol. The most important chemical and physical properties of ethanol used in this inventory are given here.

Tab. 26.8 Chemical and physical properties of ethanol

Property	Value	Unit	Remarks
Molecular weight ¹	46.07	g mol^{-1}	
Specific gravity ¹	790	kg m^{-3}	at 20 °C
Boiling point ¹	78.39	°C	at normale pressure
Melting point ¹	-114.15	°C	at normale pressure
Heat of combustion, net caloric value (LHV) ²	23.4	MJ kg^{-1}	
Heat of combustion, gross caloric value (HHV) ²	29.6	MJ kg^{-1}	

¹ Source: Ullmann 2001b

² Source: BIN 2004

26.11.1 Reserves and Resources

Ethanol can be produced in two different ways. Either chemically, by hydration of ethylene, or by fermentation of sugar-containing feeds, starchy feed materials or lignocellulosic materials. Most industrial ethanol is manufactured by the vapour phase hydration of ethylene over a solid phosphoric acid catalyst supported on porous clay beads at around 240 °C and 68 barg (IPPC 2003).

Tab. 26.9 World ethanol production in 2004 (Berg 2004)

Country	Ethanol production (1000 hl)
Brazil	114000
USA	70500
China	29700
EU	20697
Indonesia	17200
Russia	11500
Others	35764
World	299361

26.11.2 Use

Depending on its water content, preparation, and final use, several ethanol products exist on the market. The 99.7% alcohol (often referred to as absolute alcohol) is used extensively for tinctures and pharmaceutical preparations, as a solvent and preservative, as an antiseptic, and in perfume. Ethanol is

an important functional component of alcoholic beverages, which are produced by fermentation of fermentable carbohydrates. The fermentation broth itself may constitute (after processing and aging) a beverage, e.g., in the case of beer or wine, or the alcohol can be concentrated from the broth to produce high-alcohol-containing spirits. If the alcohol is used for purposes other than as a beverage, it is denatured by the addition of substances such as methanol, pyridine, formaldehyde, or sublimate. The denatured alcohol is then used by industry and commerce, principally as a solvent, as a raw material for manufacturing chemicals, or as a fuel.

26.12 Diethyl ether, at plant RER

Diethyl ether ($\text{CH}_3\text{CH}_2\text{--O--CH}_2\text{CH}_3$, CAS-No. 60-29-7, synonyms: ether, ethoxy-ethane) is one of the most important ethers. It is a clear, mobile liquid with a sweetish, slightly pungent, characteristic odor. It is completely miscible with common organic solvents, but only partially miscible with water. The most important properties are given in Tab. 26.10.

Tab. 26.10 Chemical and physical properties of diethyl ether

Property	Value	Unit	Remarks
Molecular weight ¹	74.12	g mol^{-1}	
Specific gravity ¹	710	kg m^{-3}	at 20 °C
Boiling point ¹	34.5	°C	at normale pressure
Melting point ¹	-116.3	°C	at normale pressure

¹ Source: Ullmann 2001a

26.12.1 Production (Ullmann 2001a)

In countries where ethanol is produced synthetically, diethyl ether is produced in sufficient quantities as a co-product to make its synthesis unnecessary. Otherwise, ether is produced from ethanol by dehydration, either with sulfuric acid or catalytically in the vapor phase over alumn. The latter process is simpler, but gives lower yields.

During ethanol synthesis by the direct catalytic hydration of ethylene, about 1 % of the ethylene is converted to diethyl ether as a byproduct (Meyers 1986). Indirect hydration of ethylene with sulfuric acid also yields 10 % ether. The ether yield is matched to the demand by recycling it into the process (for reaction equilibrium, see or by appropriate adjustment of the reaction conditions).

Irrespective of the process used, the diethyl ether must be purified to eliminate small amounts of acetaldehyde, hydrocarbons, and varying amounts of ethanol and water. Removal of the acetaldehyde by distillation leads to loss of ether due to the formation of a binary azeotrope. Removal is also incomplete because part of the acetaldehyde is bound by ethanol as an acetal and is liberated during further processing. Therefore, high-quality ether cannot be obtained by this method.

Acetaldehyde can be removed by hydrogenation or by washing the ether vapor with sodium hydrogen sulfite solution; subsequent washing with sodium hydroxide solution is required to remove any sulfur dioxide present. Water can be removed from the ether azeotrope by passing the vapor over a desiccant such as sodium hydroxide flakes or through a molecular sieve or by high-pressure distillation. At a pressure of ca. 1.1 MPa, a binary azeotrope containing 4.5 wt % water leaves the top of the fractionating column and, after condensation and cooling, separates into two phases. The upper phase is returned to the column, whereas the aqueous phase is recycled into the low-pressure ether column.

For environmental reasons, hydrogenation is preferred for purification and high-pressure distillation for dehydration.

26.12.2 Uses

Two grades of diethyl ether are commercially available: technical-grade ether and anesthetic ether. There are no universally accepted specifications for technical-grade ether, but it is conventionally free of peroxides. Anhydrous technical-grade ether has a maximum water content of 500 mg/kg. Anesthetic ether has a neutral reaction, is free from foreign odors, passes the tests for peroxides and aldehydes, and has a maximum water content of 0.2 %.

Diethyl ether is a good solvent for many oils, fats, resins, alkaloids, odorants, and dyes and, therefore, is widely used as a solvent and as an extractant. When mixed with ethanol, diethyl ether is used to gel nitrocellulose and to dissolve collodion wool.

Ether is used as a reaction medium in the laboratory and in industry due to its chemical stability, low boiling point, and solvent properties for organometallic compounds (Grignard reagents). It is also employed as an anesthetic in medicine, but this use is declining.

26.13 Life cycle inventory of ethylene hydration and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the hydration of ethylene. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 26.11 Unit process raw data for the production of ethanol from ethylene

401	InputGroup	OutputGro	Name	Location	Infrastructu	reProcess	Unit	ethylene hydration process	Uncertainty Standard deviation95	GeneralComment	ethanol from ethylene, at plant	diethyl ether, at plant
662	493	403	Location	Location	Infrastructu	reProcess	Unit	RER			RER	RER
			InfrastructureProcess									
			Unit					0			0	0
								kg			kg	kg
allocated products resource, in water technosphere	-	2	ethanol from ethylene, at plant	RER	0	kg	1.59E+0				100	0
	-	2	diethyl ether, at plant	RER	0	kg	1.30E-2				0	100
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.07E-2	1	1.13 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	5	-	ethylene, average, at plant	RER	0	kg	1.00E+0	1	1.13 (1,2,3,1,1,3);		99.192	0.808
	5	-	water, deionised, at plant	CH	0	kg	1.12E+0	1	1.13 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	5	-	phosphoric acid, industrial grade, 85% in H2O, at plant	RER	0	kg	1.12E-3	1	1.13 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	5	-	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	2.87E-3	1	1.13 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	5	-	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	1.91E-3	1	1.13 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	5	-	benzene, at plant	RER	0	kg	5.45E-3	1	1.14 (1,3,1,5,1,3); Naran Lala 2004		99.192	0.808
	5	-	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	0	m3	2.08E-2	1	1.30 (4,5,na,na,na,na); Calculated from the water input		99.192	0.808
	5	-	transport, lorry 32t	RER	0	tkm	1.01E-1	1	2.09 (4,5,na,na,na,na); Standard distance 100km		99.192	0.808
	5	-	transport, freight, rail	RER	0	tkm	6.06E-1	1	2.09 (4,5,na,na,na,na); Standard distance 600km		99.192	0.808
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.71E-2	1	1.14 (1,3,1,5,1,3); Naran Lala 2004		99.192	0.808
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	4.66E+0	1	1.14 (1,3,1,5,1,3); Naran Lala 2004		99.192	0.808
	5	-	chemical plant, organics	RER	1	unit	4.66E+0	1	3.09 (4,5,na,na,na,na); Estimation		99.192	0.808
emission air, high population density	-	4	Heat, waste	-	-	MJ	9.76E-2	1	1.30 (4,5,na,na,na,na); Calculated from the electricity input		99.192	0.808
	-	4	Carbon dioxide, fossil	-	-	kg	1.85E-2	1	1.13 (1,2,3,1,1,3); from burning of co-products		99.192	0.808
emission water, river	-	4	Ethene	-	-	kg	2.00E-3	1	1.62 (4,5,na,na,na,na); Estimation		99.192	0.808
	-	4	Phosphate	-	-	kg	1.75E-3	1	1.52 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	-	4	Chloride	-	-	kg	3.03E-3	1	3.02 (1,2,3,1,1,3); Meyers 1986		99.192	0.808
	-	4	Sodium, ion	-	-	kg	2.71E-3	1	5.02 (1,2,3,1,1,3); Meyers 1986		99.192	0.808

26.14 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of ethanol from ethylene.

Tab. 26.12 Selected LCI results and the cumulative energy demand for the hydration of ethylene

	Name		diethyl ether, at plant	ethanol from ethylene, at plant
	Location		RER	RER
	Unit	Unit	kg 0	kg 0
Infrastructure				
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	46.4	46.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	1.5	1.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.3	0.3
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.2	0.2
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0
selected LCI results				
land occupation	resource	m2a	5.6E-2	5.7E-2
CO2, fossil	air	kg	1.1E+0	1.1E+0
NMVOc	air	kg	2.8E-3	2.8E-3
nitrogen oxides	air	kg	2.1E-3	2.1E-3
sulphur dioxide	air	kg	2.2E-3	2.2E-3
particulates, <2.5 um	air	kg	1.3E-4	1.3E-4
BOD	water	kg	6.3E-4	6.3E-4
cadmium	soil	kg	2.1E-10	2.1E-10

26.15 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	ethylene hydration process
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Direct hydration of ethylene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ethylenhydratation
	491	Synonyms	
	492	GeneralComment	The multioutput-process "ethylene hydration process" delivers the co-products ethanol 99.7% in H ₂ O and diethyl ether 99.95% in H ₂ O. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1986
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled with data from plants in Europe and India.
Technology	692	Text	Direct hydration of ethylene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Some data are derived from other or unknown plants or have been estimated.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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27 Formic acid

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

27.1 Introduction

Formic acid (CAS-No. 64-18-6, HCOOH , Synonyms: methane acid) is a colorless liquid with a pungent odor, which is completely miscible with water and many polar solvents but only partially miscible with hydrocarbons. Formic acid derived its name from the red ant, *Formica rufa*, in which it was discovered around 1670. Formic acid has been detected in the poison or defense systems of ants, bees, and other insects and also of cnidarians.

For this inventory the functional unit is 1 kg of pure formic acid. The most important chemical and physical properties of formic acid used in this inventory are given here.

Tab. 27.1 Chemical and physical properties of formic acid (Ullmann 2005)

Property	Value	Unit	Remarks
Molecular weight ¹	46.03	g mol^{-1}	
Specific gravity ¹	122	kg m^{-3}	at 20 °C
Boiling point ¹	100.8	°C	at normale pressure
Melting point ¹	8.3	°C	at normale pressure

27.2 Reserves and Resources

The worldwide installed capacity for producing formic acid was about 330 000 t/a in 1988. The installed formic acid processes can be classified in four groups:

1. methyl formate hydrolysis,
2. oxidation of hydrocarbons,
3. hydrolysis of formamide
4. preparation of free formic acid from formates.

The significance of individual processes has shifted considerably in the last decade because the economic data for production and raw products have changed and new processes have been developed. In the past, a large amount of the formic acid utilized was a byproduct in the manufacture of acetic acid. However, the proportion of formic acid produced by dedicated processes has increased.

Formic acid is a byproduct in the production of acetic acid by liquid-phase oxidation of butane or naphtha. For many years, oxidation of hydrocarbons was the most important method of producing acetic acid. Over the past decade, however, the preferred process in the establishment of new acetic acid plants has been the carbonylation of methanol, which is expected to remain the preferred method in the future. As a result, the proportion of formic acid produced worldwide, which results as a byproduct in the manufacture of acetic acid, will continue to decrease.

The production of formic acid by hydrolysis of formamide played an important role in Europe; in 1972, about one-third of the world production was obtained by this process. However, the consumption of ammonia and sulfuric acid, along with the unavoidable production of ammonium sulfate, have in many cases made this process economically inferior. As a result, the direct hydrolysis of methyl

formate is currently preferred.

Another industrial production method involves formation of the free acid from its salts. Sodium formate and calcium formate are used for this purpose.

The economic disadvantages of the methods discussed led to the development of a process specifically dedicated to the production of formic acid, with no undesirable byproducts. In the 1970s, the hydrolysis of methyl formate to methanol and formic acid was developed commercially by various firms into an economically feasible method. This process involves carbonylation of methanol and subsequent hydrolysis of the methyl formate produced. The methanol resulting from this process is returned to the first stage. Formic acid plants based on this process were started up at BASF (Germany) in 1981 and Kemira (Finland) in 1982. Most of the formic acid plants begun in recent years employ this principle of methyl formate hydrolysis.

Table 2 shows the world capacity for various formic acid production processes in 1986. Although other methods for producing formic acid have been patented, they do not appear to have been implemented industrially. Direct synthesis of formic acid from carbon monoxide and water cannot be carried out economically because of the unfavorable position of the equilibrium.

Tab. 27.2 Formic acid production capacity in 1986 (Ullmann 2005)

Process	Capacity (t/a)	Percentage of total capacity (%)
Hydrolysis of methyl formate	160 000	49
Byproduct in hydrocarbon oxidation	74 000	22
Hydrolysis of formamide	10 000	3
Acidolysis of alkali formates	86 000	26
Total	330 000	100

In recent years, various ways of producing formic acid from carbon dioxide and hydrogen have been investigated extensively, but this method has not yet been applied industrially. The oxidation of methyl tert-butyl ether with oxygen to form tert-butyl formate and formic acid has been described

Various methods have been developed for the electrochemical production of formic acid, but they have not been used commercially.

27.3 Use

Formic acid is used primarily in dyeing, in the textile and leather industries; in rubber production; and as an intermediate in the chemical and pharmaceutical industries. The use of formic acid as an aid in the ensilage of green forage has increased sharply.

The worldwide production of formic acid is about 260 000 t/a in 1987 and 390 000 t/a in 1995 (Ullmann 2005).

27.4 System Characterisation

This report corresponds to the module in the ecoinvent database for the production of **1 kg formic acid at plant, in Europe**. Two processes for the formic acid production were considered: the hydrolysis of methyl formate and the hydrocarbon oxidation process. The following processes were modelled:

- Formic acid production, from methyl formate, RER
- Formic acid, at plant, RER. Supply mix from the methyl formate process and the butane oxidation process

All data in the present report are referred to 1 kg formic acid 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

27.5 Formic acid production, from methyl formate, RER

27.5.1 Production Technologies

Methyl formate hydrolysis process (Ullmann 2005)

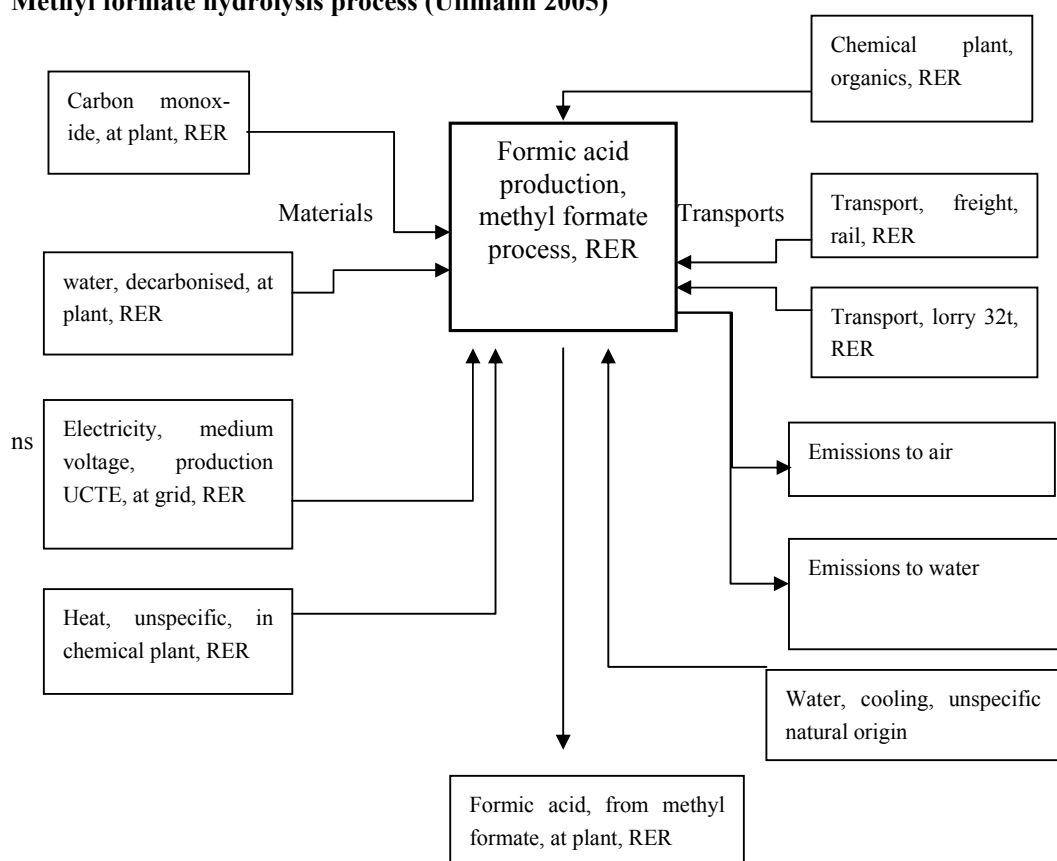


Fig. 27.1 Process flow chart for the production of formic acid (methyl formate process)

Synthesis of formic acid by hydrolysis of methyl formate is based on a two-stage process: in the first stage, methanol is carbonylated with carbon monoxide; in the second stage, methyl formate is hydrolyzed to formic acid and methanol. The methanol is returned to the first stage.

Although the carbonylation of methanol is relatively problem-free and has been carried out industrially for a long time, only recently has the hydrolysis of methyl formate been developed into an economically feasible process. The main problems are associated with work-up of the hydrolysis mixture. Because of the unfavorable position of the equilibrium, reesterification of methanol and formic acid to

methyl formate occurs rapidly during the separation of unreacted methyl formate. Problems also arise in the selection of sufficiently corrosion-resistant materials.

Older publications suggest the reaction of methyl formate and a dicarboxylic acid, with subsequent distillation of formic acid from the higher-boiling esters. Degussa suggested hydrolyzing methyl formate at 200 °C above atmospheric pressure and then reducing the pressure suddenly to less than 0.1 MPa. Separation of the methyl formate–methanol mixture flash-evaporated from the aqueous formic acid can then take place in a liquid separator.

Industrial methods involving carbonylation of methanol and hydrolysis of methyl formate, followed by isolation of the formic acid, were developed by (1) the Leonard Process Company (United States), (2) BASF (Federal Republic of Germany), (3) Halcon – Scientific Design – Bethlehem Steel (United States), and (4) the Scientific Research Institute for the Chlorine Industry of the Ministry of the Chemical Industry (former Soviet Union).

Carbonylation of Methanol

In the four processes mentioned, the first stage involves carbonylation of methanol in the liquid phase with carbon monoxide, in the presence of a basic catalyst:



This reaction was first described by BASF in 1925. As a rule, the catalyst is sodium methoxide. Potassium methoxide has also been proposed as a catalyst; it is more soluble in methyl formate and gives a higher reaction rate. Although fairly high pressures were initially preferred, carbonylation is carried out in new plants at lower pressure. Under these conditions, reaction temperature and catalyst concentration must be increased to achieve acceptable conversion. According to published data, ca. 4.5 MPa, 80 °C, and 2.5 wt % sodium methoxide are employed. About 95 % carbon monoxide, but only about 30 % methanol, is converted under these circumstances. Nearly quantitative conversion of methanol to methyl formate can, nevertheless, be achieved by recycling the unreacted methanol. The carbonylation of methanol is an equilibrium reaction. The reaction rate can be raised by increasing the temperature, the carbon monoxide partial pressure, the catalyst concentration, and the interface between gas and liquid.

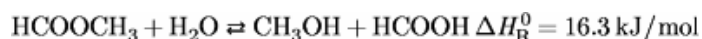
To synthesize methyl formate, gas mixtures with a low proportion of carbon monoxide must first be concentrated; this can be achieved by low temperature distillation, pressure-swing adsorption, prism separators (Monsanto), or the Cosorb process (Tenneco).

In a side reaction, sodium methoxide reacts with methyl formate to form sodium formate and dimethyl ether, and becomes inactivated. The substances used must be anhydrous; otherwise, sodium formate is precipitated to an increasing extent. Sodium formate is considerably less soluble in methyl formate than in methanol. The risk of encrustation and blockage due to precipitation of sodium formate can be reduced by adding poly(ethylene glycol). The carbon monoxide used must contain only a small amount of carbon dioxide; otherwise, the catalytically inactive carbonate is precipitated.

Basic catalysts may reverse the reaction, and methyl formate decomposes into methanol and carbon monoxide. Therefore, undecomposed sodium methoxide in the methyl formate must be neutralized.

Hydrolysis of Methyl Formate

In the second stage, the methyl formate obtained is hydrolyzed:



The equilibrium constant for methyl formate hydrolysis depends on the water: ester ratio. With a mo-

lar ratio of 1, the constant is 0.14, but with a water: methyl formate molar ratio of 15, it is 0.24. Because of the unfavorable position of this equilibrium, a large excess of either water or methyl formate must be used to obtain an economically worthwhile methyl formate conversion. If methyl formate and water are used in a molar ratio of 1 : 1, the conversion is only 30 %, but if the molar ratio of water to methyl formate is increased to 5 – 6, the conversion of methyl formate rises to 60 %. However, a dilute aqueous solution of formic acid is obtained this way, and excess water must be removed from the formic acid with the expenditure of as little energy as possible.

Another way to overcome the unfavorable position of the equilibrium is to hydrolyze methyl formate in the presence of a tertiary amine, e.g., 1-(n-pentyl)imidazole. The base forms a salt-like compound with formic acid; therefore, the concentration of free formic acid decreases and the hydrolysis equilibrium is shifted in the direction of products. In a subsequent step formic acid can be distilled from the base without decomposition.

A two-stage hydrolysis has been suggested, in which a water-soluble formamide is used in the second stage; this forms a salt-like compound with formic acid. It also shifts the equilibrium in the direction of formic acid.

To keep undesirable reesterification as low as possible, the time of direct contact between methanol and formic acid must be as short as possible, and separation must be carried out at the lowest possible temperature. Introduction of methyl formate into the lower part of the column in which lower boiling methyl formate and methanol are separated from water and formic acid, has also been suggested. This largely prevents reesterification because of the excess methyl formate present in the critical region of the column.

The hydrolysis of methyl formate is catalyzed by strong acids, but the efficiency of strong mineral acids is restricted because they also promote the decomposition of formic acid. In the processes described by BASF, Leonhard, and Halcon – Scientific Design – Bethlehem Steel, autocatalysis by formic acid is therefore used. In the Soviet Union, a process has been described in which hydrolysis is carried out in a vertical column reactor divided into two zones; the upper zone is filled with a strongly acidic cation exchanger which partially hydrolyze methyl formate. In the lower part of the reactor, hydrolysis is carried out with autocatalysis by formic acid produced in the upper section.

Dehydration of the Hydrolysis Mixture

Formic acid is marketed in concentrations exceeding 85 wt %; therefore, dehydration of the hydrolysis mixture is an important step in the production of formic acid from methyl formate. For dehydration, the azeotropic point must be overcome. The concentration of formic acid in the azeotropic mixture increases if distillation is carried out under pressure, but the higher boiling point at high pressure also increases the decomposition rate of formic acid. At the same time, the selection of sufficiently corrosion-resistant materials presents considerable problems. A number of entrainers have been proposed for azeotropic distillation.

BASF describes various energy-saving processes for the dehydration. One process involves extractive distillation with N-formylmorpholine. Another method of increasing formic acid concentration in dilute solutions without a considerable input of energy is to extract formic acid from water by liquid – liquid extraction. Secondary amides have been proposed as extractive agents. Formic acid binds weakly to the amide; therefore, water can be distilled to overcome the azeotropic point. In a subsequent column, formic acid can then be distilled from the extractant in vacuo. The energy consumed can be reduced further by optimizing the exchange of heat.

Kemira - Leonard Process

20 000-t/a formic acid plant based on a method developed by the Leonard Process Co. was built at

Kemira in Finland and put into operation in 1982. The process has been developed further by Kemira, and licences for it have been issued in Korea, India and Indonesia.

In the Kemira – Leonard process, Methyl formate and carbonylation is carried out at about 4 MPa and a temperature of approximately 80 °C, with additive-containing alkoxides used as catalyst. Hydrolysis is carried out in two reactors with different operating conditions. Methyl formate and water react in the preliminary reactor in approximately equimolar proportions. The formic acid produced catalyzes the hydrolysis in the main reactor. An excess of methyl formate is employed in the main reactor, and hydrolysis is carried out at ca. 120 °C and 0.9 MPa. The reactor discharge is brought to atmospheric pressure in a flash tank; reesterification is largely prevented by the cooling that occurs during flash-evaporation. methanol are separated under vacuum. Although the reverse reaction is attenuated further as a result, condensation of the low-boiling methyl formate in vacuo presents a problem. The formic acid is dehydrated by distillation. If 85 wt % formic acid is desired, dehydration must be carried out at ca. 0.3 MPa. Even higher concentrations can be achieved by connecting an additional dehydration column downstream under atmospheric pressure, and formic acid concentrations up to ca. 98 wt % can then be drawn off as the distillate.

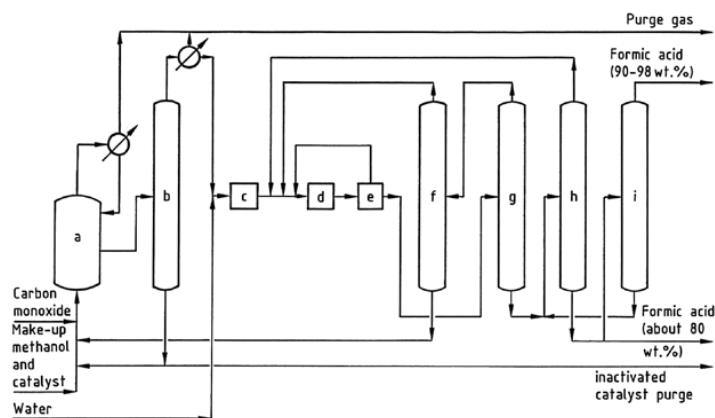


Fig. 27.2 Production of formic acid (Kemira – Leonard process)

a) Methyl formate reactor; b) Methyl formate column; c) Preliminary reactor; d) Main reactor; e) Flash tank; f) Recycle column; g) Acid separation column; h) First product column; i) Second product column

Process Description

Compressed carbon monoxide and methanol are converted into methyl formate in reactor (a). Catalyst is fed into the reactor in a methanol solution. The amount of methanol introduced in this way makes up for methanol losses in the process. The discharge from reactor (a) is flashed and fed into the methyl formate column (b) from which methyl formate is drawn off as the distillate. Methanol and the dissolved catalyst are returned to the reactor; inactivated catalyst (primarily sodium formate) is crystallized and discharged.

Off-gas from column (b) and waste gas from reactor (a) are burned. Methyl formate reacts partially with water in the preliminary reactor (c), and discharge from the preliminary reactor is fed into the main reactor (d) along with recycled methyl formate, methanol, and water. The contact time in this reactor is largely sufficient for equilibrium to be established. Reactor discharge is flashed to approximately atmospheric pressure in the flash tank (e); methyl formate, methanol, and small quantities of formic acid evaporated in this process are recycled to the main reactor (d). Methyl formate and methanol are distilled in vacuo in the acid separation column (g). Contact time is minimized by internals with a small liquid holdup. The distillate is separated into methyl formate and methanol in the recycle

column (f). Formic acid can be concentrated under pressure in a column; if the pressure is ca. 0.3 MPa, 85 % formic acid is drawn off as the bottom product. Distillation in two columns is more industrially controllable. Water is distilled overhead in the first product column (h). The bottom product is concentrated further in the second product column (i), and formic acid with a maximum concentration of ca. 98 wt % is drawn off as distillate. The bottom product is recycled to the first product column.

BASF Process

A 100 000-t/a formic acid plant began operating in Ludwigshafen (Germany) in 1981. In this plant, a technology for the hydrolysis and dehydration was used for the first time.

The production of methyl formate by carbonylation of methanol has been carried out on a large scale for many years at BASF. The carbonylation stage is largely identical to that of the Kemira – Leonard process, but the hydrolysis stage and the dehydration of formic acid are noticeably different.

In the BASF process, hydrolysis is carried out with a large excess of water (about 5 mol of water per mole of methyl formate) to shift the equilibrium in the direction of formic acid. Much of the water is separated by liquid – liquid extraction with a secondary amide.

Process Description

Carbon monoxide and methanol react in the methyl formate reactor (a) in the presence of sodium methoxide. Methyl formate is fed, as a distillate, from the methyl formate column (b) into the formic acid reactor (c) together with recycled methyl formate. Methanol and dissolved catalyst are drawn off from the bottom of column (b) and returned to reactor (a); catalyst decomposition products are discharged by crystallization. In reactor (c), methyl formate is hydrolyzed with excess water at elevated temperature and increased pressure. The reaction product is flashed into the low-boiler column (d). Methyl formate is removed as the distillate, with methanol as a side stream, and dilute aqueous formic acid is drawn off from the bottom into the extraction unit (e). Here, the formic acid and some of the water are extracted by the secondary amide. Most of the water (largely free of formic acid) is recycled to reactor (c). The extract—a mixture of extractant, formic acid, and some water—is distilled in the dehydration column (f). Enough water is distilled via the head for the required formic acid concentration to be obtained in the pure acid column (g). This column is operated in vacuo. The extraction agent is recycled from the bottom of column (g) to the extraction unit.

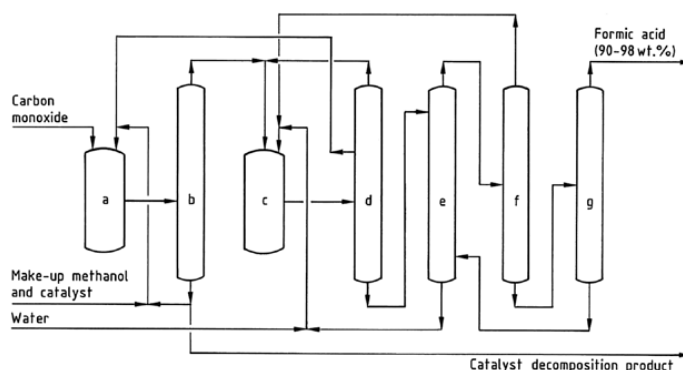


Fig. 27.3 Production of formic acid (BASF process)

a) Methyl formate reactor; b) Methyl formate column; c) Hydrolysis reactor; d) Low-boiler column; e) Extraction unit; f) Dehydration column; g) Pure acid column

27.5.2 Raw materials and auxiliaries

The input of raw materials and auxiliaries are calculated with data from McKetta 1985, who reports data for the US formic acid production.

Tab. 27.3 Consumption of raw materials and auxiliaries for the production of 1 kg formic acid from methyl formate

Input	Herman 1984	McKetta 1985	Inventory ¹
Feed gas, 97 mol % CO (Nm ³)	0.66		
High pressure vent gas, 75 mol % CO (Nm ³)	0.067		
Carbon monoxide (kg)		6.144E-01	6.144E-01
Methanol (kg)		0.04	4E-02
Process water (kg)	0.6		0.6
Cooling water (kg)	600	375.5	375.5

¹ Values used in this inventory

27.5.3 Energy

The input of raw materials and auxiliaries are calculated with data from McKetta 1985, who reports data for the US formic acid production.

Tab. 27.4 Energy consumption for the production of 1 kg formic acid from methyl formate

Input	Herman 1984	Mc Ketta 1985	Inventory ¹
Electricity (kWh)	0.155	1.323E-01	1.323E-01
Steam (MJ)	19.25	19.25	19.25

¹ Values used in this inventory

27.5.4 Transportation

No information is available in the sources consulted concerning transportation of raw materials. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 27.5 summarises the total transport amounts for the production of 1 kg formic acid.

Tab. 27.5 Total transport amounts for the production of 1 kg formic acid from methyl formate

tkm kg ⁻¹ formic acid	lorry	train
Raw materials	8.63E-02	5.178E-01

27.5.5 Infrastructure and land use

No information was readily available about infrastructure and land-use of formic acid plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

27.5.6 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of formic acid. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 1 % of the raw material input and 0.2 % of the intermediate methyl formate. The assumed values for the emissions to air are given in Tab. 27.6.

Tab. 27.6 Process emissions to air from the production of 1 kg formic acid from methyl formate

Output	Value	Remarks
Waste heat (MJ)	4.8E-01	from electricity input
Carbon monoxide, fossil, to air (kg)	6.144E-03	Estimated as 1 % of input
Methyl formate, to air (kg)	2.64E-03	Estimated as 0.2 % of the input
Carbon dioxide, fossil, to air (kg)	1.39E-02	from waste water treatment

27.5.7 Emissions to water

There was no data available on process emissions to water for the production of formic acid. As approximation the water emissions were estimated to 0.8 % of the intermediate methyl formate. These assumption lead to a pollution of the waste water with 10.54 g methyl formate.

Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for acetic acid was assumed with 90% leading to emissions of 1.054 g methyl formate per kg product in the treated water. The carbon contained in the removed methyl formate was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of methyl formate in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 27.7.

Tab. 27.7 Process emissions to water from the production 1 kg formic acid from methyl formate

Output	Value	Remarks
Methyl formate, to water (kg)	1.054E-03	calculated from mass balance
COD, BOD (kg)	1.1E-03	calculated from water emissions
TOC, DOC (kg)	4.2E-04	calculated from water emissions

27.6 Formic acid, at plant, RER

In 1986 49 % of the formic acid production in the world has been made by the hydrolysis of methyl

formate and 22 % by the oxidation of butane (Ulmann 2005). These numbers were applied in order to assess a European mix for the formic acid production, which is representative for 71 % of the production.

Tab. 27.8 Formic acid, at plant, RER. Processes used in the supply mix (RER)

Production process	Unit	Formic acid, at plant, RER
Formic acid from methyl formate, at plant, RER	kg	0.69
Formic acid from butane, at plant, RER	kg	0.31
Total	kg	1

27.7 Data Quality Considerations

Tab. 27.9 shows the life cycle inventory and the data quality indicators for the methyl formate process. Tab. 27.10 shows the life cycle inventory and the data quality indicators for the production mix of formic acid.

The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 27.9 Unit process raw data for the methyl formate process

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	formic acid from methyl formate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
401										
662			Location							
493			InfrastructureProcess							
403			Unit							
resource, in water technosphere	-	0	formic acid from methyl formate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.76E-1	1	1.51	(1,3,5,3,1,3); Mc Ketta 1985
	5	-	water, deionised, at plant	CH	0	kg	6.00E-1	1	1.51	(1,3,5,3,1,3); Herman 1984
	5	-	carbon monoxide, CO, at plant	RER	0	kg	6.14E-1	1	1.51	(1,3,5,3,1,3); Mc Ketta 1985
	5	-	methanol, at plant	GLO	0	kg	4.00E-2	1	1.51	(1,3,5,3,1,3); Mc Ketta 1985
	5	-	transport, lorry 32t	RER	0	tkm	8.63E-2	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	5.18E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.32E-1	1	1.51	(1,3,5,3,1,3); Mc Ketta 1985
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	1.93E+1	1	1.51	(1,3,5,3,1,3); Mc Ketta 1985
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	4.80E-1	1	1.51	(1,3,5,3,1,3); Calculated from the electricity input
	-	4	Carbon dioxide, fossil	-	-	kg	1.39E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	Carbon monoxide, fossil	-	-	kg	6.14E-3	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Methyl formate	-	-	kg	2.64E-3	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Methyl formate	-	-	kg	1.05E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.10E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.10E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	4.20E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	4.20E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

Tab. 27.10 Unit process raw data for the production mix of formic acid

technosphere	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	formic acid, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
	401									
	662		Location				RER			
	493		InfrastructureProcess				0			
technosphere	403		Unit				kg			
	-	0	formic acid, at plant	RER	0	kg	1.00E+0			
	5	-	formic acid from methyl formate, at plant	RER	0	kg	6.90E-1	1	1.06	(1,1,2,1,1,1); Ullmann 2005
technosphere	5	-	formic acid from butane, at plant	RER	0	kg	3.10E-1	1	1.06	(1,1,2,1,1,1); Ullmann 2005

27.8 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 12.8 shows selected LCI results and the cumulative energy demand for the production of formic acid.

Tab. 27.11 Selected LCI results and the cumulative energy demand for the production of formic acid

Name	Location Unit Infrastructure	Unit	formic acid from butane, at plant	formic acid from methyl formate, at plant	formic acid, a plant
			RER	RER	RER
			kg 0	kg 0	kg 0
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	45.0	64.0	58.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	1.8	8.6	6.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.3	0.7	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3	1.3	1.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.1	0.1
selected LCI results					
land occupation	resource	m2a	9.3E-2	1.7E-1	1.4E-1
CO2, fossil	air	kg	1.1E+0	2.9E+0	2.4E+0
NM VOC	air	kg	3.7E-3	4.3E-3	4.2E-3
nitrogen oxides	air	kg	2.5E-3	4.7E-3	4.0E-3
sulphur dioxide	air	kg	2.2E-3	7.9E-3	6.1E-3
particulates, <2.5 um	air	kg	2.0E-4	4.7E-4	3.9E-4
BOD	water	kg	1.1E-2	9.9E-3	1.0E-2
cadmium	soil	kg	3.8E-10	8.9E-10	7.3E-10

27.9 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	formic acid from methyl formate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of formic acid including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Ameisensäure aus Methylformat, ab Werk
	491	Synonyms	methanoic acid
	492	GeneralComment	The process "formic acid from methyl formate, at plant, RER" is modelled for the production of formic acid from methyl formate in Europe. Raw materials and energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	CH2O
	501	StatisticalClassification	
	502	CASNumber	64-18-6
TimePeriod	601	StartDate	1984
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Carbonylation of methanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

ReferenceFunction	401	Name	formic acid, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Formic acid production mix
	404	Amount	1
	490	LocalName	Ameisensäure, ab Werk
	491	Synonyms	methanoic acid
	492	GeneralComment	Formic acid production mix in Europe
	494	InfrastructureIncluded	
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	CH2O
	501	StatisticalClassification	
	502	CASNumber	64-18-6
TimePeriod	601	StartDate	2005
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Production mix
Representativeness	722	Percent	10
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	No extrapolations
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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28 Isobutyl acetate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

28.1 Introduction

Isobutyl acetate (CAS-No. 110-19-0, $C_6H_{12}O_2$; Synonyms: acetic acid isobutyl ester, acetic acid 2-methylpropyl ester, 2-methyl-1-propyl acetate) is a clear, medium-volatility liquid with a characteristic ester odour. It is miscible with all common solvents, e. g. alcohols, ketones, aldehydes, ethers, glycols and glycol ethers. It is readily thinnable with aromatic and aliphatic hydrocarbons. Its miscibility with water, however, is limited.

For this inventory the functional unit is 1 kg of pure isobutyl acetate. The most important chemical and physical properties of isobutyl acetate used in this inventory are given here:

Tab. 28.1 Chemical and physical properties of isobutyl acetate (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	116.16	$g\ mol^{-1}$	
Specific gravity	871	$kg\ m^{-3}$	at 20 °C
Boiling point	117.2	°C	at normale pressure
Melting point	- 98.6	°C	at normale pressure

28.2 Reserves and Resources

The production of isobutyl acetate in 1985 was about 35000 t in the USA and 50000 t in Germany (Ullmanns 2004).

28.3 Production Technologies and Use

28.3.1 Production Technologies

The reaction of an alcohol and an organic acid produces an ester in the presence of an acid catalyst. The reaction, esterification, is the most common process for manufacturing acetate esters.

Commercial processes for esterification are usually continuous and are performed in the liquid phase. The reaction requires sulfuric acid. Another type of catalyst is ion-exchange resins. In contrast, gas-phase technology is available with sulfonated polymer or supported phosphoric acid as catalyst.

Esterification involves an equilibrium between reactants and product. Adding excess reactants or removal of water drive the equilibrium to favor the formation of ester. Azeotroping agents eliminate water and force the reaction to completion. Common agents are cyclohexane, benzene, and toluene.

Esterification with acetic anhydride offers an alternative route for making acetates. Due to its high reactivity, acetic anhydride readily forms esters with alcohols. The reaction is irreversible. The esterification goes to completion without eliminating products or water.

Additions of acetic acid to olefins present an alternative way to make acetates, especially for branched esters. Synthesis of these hindered esters by esterification is not efficient. The equilibrium does not fa-

vor the desirable products. Rhône-Poulenc uses this method to prepare isopropyl acetate from acetic acid and propylene. Catalysts can be strong protic or Lewis acids, like sulfuric acid or $\text{BF}_3 \cdot \text{OEt}_2$. Ethyl acetate can also be produced directly from acetaldehyde.

Adding hydrogen and carbon monoxide to acetic acid or acetic anhydride can produce acetates. Alternatively, hydrogenation of acetic acid yields ethanol and ethyl acetate. Many studies relate to supported heterogeneous catalysts with molybdenum and group VIII metals (Pd, Rh, or Ru). Additionally, homogeneous catalysis with ruthenium can result in high selectivity.

28.3.2 Use

Isobutyl acetate finds use as a solvent or in fragrances, flavors, cosmetics, and soaps. It is a solvent with good solvent power for cellulose nitrate, polymers, resins, and oils, and is an important solvent in the coatings and chemical downstream industries.

28.4 System Characterisation

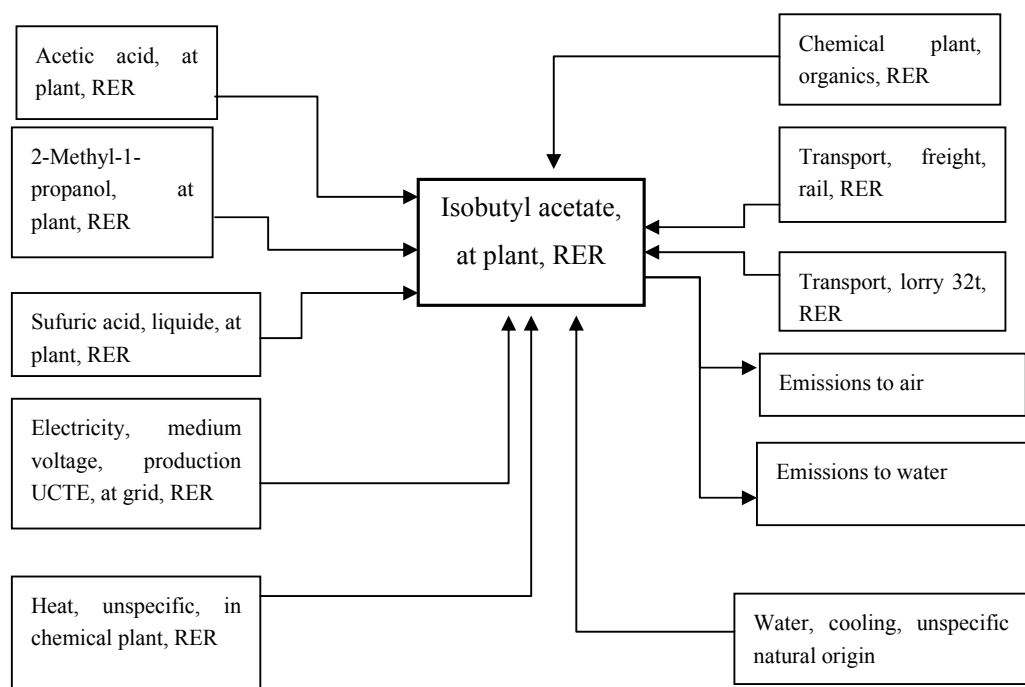


Fig. 28.1 Process flow chart for the production of isobutyl acetate

This report corresponds to the module in the ecoinvent database for the production of **1 kg isobutyl acetate, at plant, in Europe**. For this study the production of isobutyl acetate by means of the esterification of isobutanol with acetic acid is considered. All data in the present report are referred to 1 kg isobutyl acetate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

28.5 Raw materials and auxiliaries

As there was no data available for the production of isobutyl acetate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries, i. e. sulfuric acid was estimated with data from Wells 1991, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of isobutanol.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the isobutyl acetate production.

Tab. 28.2 Consumption of raw materials and auxiliaries for the production of 1 kg isobutyl acetate

Input	Values	Remarks
Isobutanol (kg)	6.715E-01	Calculated with a yield of 95 %
Acetic acid (kg)	5.445E-01	Calculated with a yield of 95 %
Sulfuric acid (kg)	1.31E-02	estimated
Cooling water (kg)	24	estimated

28.6 Energy

As no data for the production of isobutyl acetate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of isobutanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 28.3 Energy consumption for the production of 1 kg isobutyl acetate

Input	Esterification ¹	This study
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

¹ Source: Kim 2003

28.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 28.4 summarises the total transport amounts for the production of 1 kg isobutyl acetate.

Tab. 28.4 Total transport amounts for the production of 1 kg isobutyl acetate

tkm kg ⁻¹ isobutyl acetate	lorry	train
Raw materials	1.229E-01	7.374E-01

28.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of isobutyl acetate plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

28.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-03 kWh/kg isobutyl acetate is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of isobutyl acetate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 28.5.

Tab. 28.5 Process emissions to air from the production of 1 kg isobutyl acetate

Output	Value	Remarks
Isobutanol, to air (kg)	1.34E-03	Estimated as 0.2 % of input
Acetic acid, to air (kg)	1.09E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	6.90E-2	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

28.10 Emissions to water

The remaining amount of unreacted acetic acid and isobutanol (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for isobutanol and acetic acid was assumed with 90%. The carbon contained in the removed isobutanol and acetic acid was accounted as CO₂ emissions to air. The sulphate emissions are calculated from the sulphuric acid input.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of isobutanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 28.6.

Tab. 28.6 Process emissions to water from the production 1 kg isobutyl acetate

Output	Value	Remarks
Isobutanol, to water (kg)	3.22E-03	calculated from mass balance
Acetic acid, to water (kg)	2.61E-03	calculated from mass balance

Sulphate, to water (kg)	1.283E-02	calculated from mass balance
COD, BOD (kg)	1.07E-02	calculated from water emissions
TOC, DOC (kg)	3.13E-03	calculated from water emissions

28.11 Life cycle inventory and Data Quality Considerations

Tab. 28.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of isobutyl acetate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 28.7 Unit process raw data for the production of isobutyl acetate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	isobutyl acetate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	isobutyl acetate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	isobutanol, at plant	RER	0	kg	6.72E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	5.44E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	1.31E-2	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	transport, lorry 32t	RER	0	tkm	1.23E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.37E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	8.84E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
	-	4	2-Methyl-1-propanol	-	-	kg	1.34E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	1.09E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	6.90E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	2-Methyl-1-propanol	-	-	kg	3.22E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.61E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.07E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.07E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.13E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.13E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

28.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 28.8 shows selected LCI results and the cumulative energy demand for the production of isobutyl acetate.

Tab. 28.8 Selected LCI results and the cumulative energy demand for the production of isobutyl acetate

Name		isobutyl acetate, at plant	
Location	Unit	Unit	RER kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	92.6
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.1
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.3E-1
CO ₂ , fossil	air	kg	3.1E+0
NM VOC	air	kg	9.3E-3
nitrogen oxides	air	kg	5.3E-3
sulphur dioxide	air	kg	7.1E-3
particulates, <2.5 um	air	kg	4.4E-4
BOD	water	kg	3.9E-2
cadmium	soil	kg	1.0E-9

28.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	isobutyl acetate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of isobutyl acetate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Isobutylazetat, ab Werk
	491	Synonyms	acetic acid isobutyl ester, acetic acid 2-methylpropyl ester, 2-methyl-1-propyl acetate
	492	GeneralComment	The process "isobutyl acetate, at plant, RER" is modelled for the production of isobutyl acetate from isobutanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C6H12O2
	501	StatisticalClassification	
	502	CASNumber	110-19-0
TimePeriod	601	StartDate	1991
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Esterification of isobutanol
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 85.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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29 Isopropyl acetate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

29.1 Introduction

Isopropyl acetate (CAS-No. 108-21-4, $C_5H_{10}O_2$; Synonyms: sec-propyl acetate, isopropyl ester, isopropyl ethanoate) is a colorless liquid with an aromatic fruity ester odor with moderate solubility in water and a low flash point. It is miscible with most common organic solvents and has properties intermediate between ethyl and isopropyl acetates.

For this inventory the functional unit is 1 kg of pure isopropyl acetate. The most important chemical and physical properties of isopropyl acetate used in this inventory are given here:

Tab. 29.1 Chemical and physical properties of isopropyl acetate (Ulmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	102.13	$g\ mol^{-1}$	
Specific gravity	872	$kg\ m^{-3}$	at 20 °C
Boiling point	90	°C	at normale pressure
Melting point	- 73.4	°C	at normale pressure

29.2 Reserves and Resources

Most of the isopropyl acetate manufactured is prepared by the acetylation of isopropyl alcohol.

29.3 Production Technologies and Use

29.3.1 Production Technologies

Isopropyl acetate is produced by direct acetylation of isopropyl alcohol in the presence of various catalysts (concentrated H_2SO_4 , diethyl sulfate, chlorosulfonic acid, BF_3)

29.3.2 Use

Isopropyl Acetate has good solvency characteristics for resins, such as cellulose, vinyl copolymers, polyesters, polystyrene, acrylics, kauri and manila gums, rosin, dammar, etc. and is commonly used in coatings, cleaning fluids, and as a solvent for fragrances, cosmetics and personal care. One of the main uses is in printing inks where fast evaporation and low hygroscopicity are needed.

Isopropyl Acetate is a fast evaporating, mild odor solvent that is It is an active solvent for many types of synthetic resins such as cellulose, vinyl copolymers, polyesters, polyamides, acrylics, and alkyds. Isopropyl acetate can be used as surfactant in cleaning fluids, coatings, coatings for automotive, coatings for plastic, nail care, printing inks, solvents for cosmetics and personal care, and Solvents for fragrances.

29.4 System Characterisation

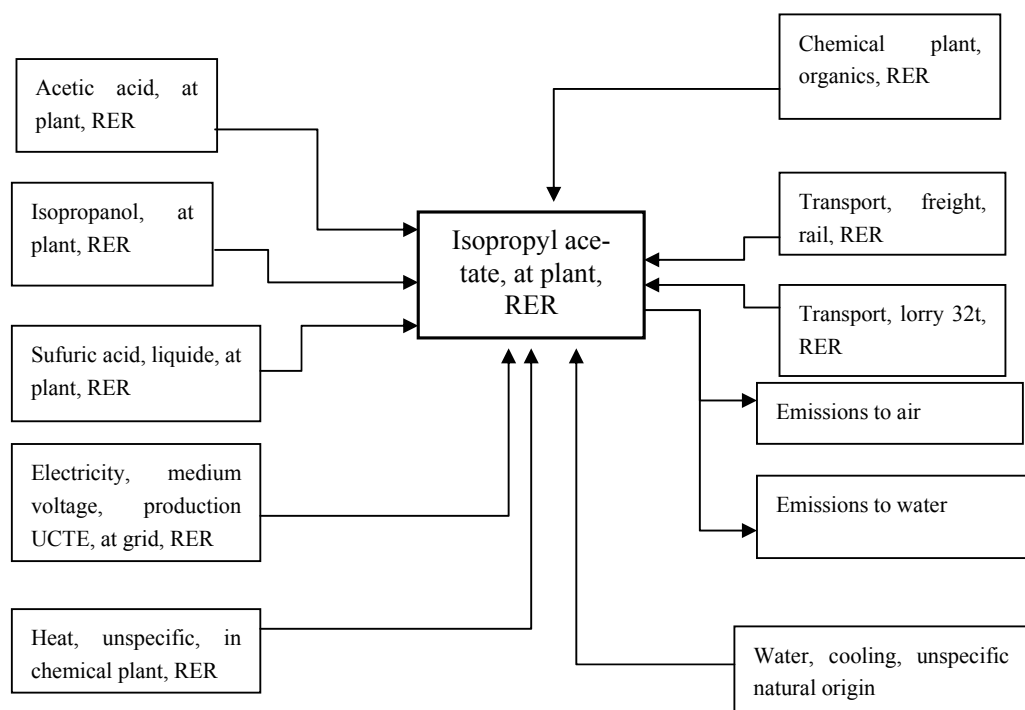


Fig. 29.1 Process flow chart for the production of isopropyl acetate

This report corresponds to the module in the ecoinvent database for the production of **1 kg isopropyl acetate, at plant, in Europe**. For this study the production of isopropyl acetate by means of the esterification of isopropanol with acetic acid is considered. All data in the present report are referred to 1 kg isopropyl acetate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

29.5 Raw materials and auxiliaries

As there was no data available for the production of isopropyl acetate, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries, i. e. sulfuric acid was estimated with data from Wells 1991, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of isopropanol.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the isopropyl acetate production.

Tab. 29.2 Consumption of raw materials and auxiliaries for the production of 1 kg isopropyl acetate

Input	Values	Remarks
Isopropanol (kg)	6.19E-01	Calculated with a yield of 95 %
Acetic acid (kg)	6.19E-01	Calculated with a yield of 95 %
Sulfuric acid (kg)	1.31E-02	estimated
Cooling water (kg)	24	estimated

29.6 Energy

As no data for the production of isopropyl acetate are available, the values for the energy use have been estimated with data from Kim 2003, who gives data for the production of ethyl acetate from ethanol esterification. This process is considered to be similar to the esterification of isopropanol.

The value for the steam input used in this inventory is the difference between the steam and the potential recovered energy, because this recovered energy is usually fed into the heating systems.

Tab. 29.3 Energy consumption for the production of 1 kg isopropyl acetate

Input	Esterification ¹	This study
Steam (MJ)	10.5	8.84
Electricity (kWh)	0.00725	7.25E-03
Potential recovered energy (MJ)	- 1.66	
Sum of consumed energy (MJ)	8.87	

¹ Source: Kim 2003

29.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 29.4 summarises the total transport amounts for the production of 1 kg isopropyl acetate.

Tab. 29.4 Total transport amounts for the production of 1 kg isopropyl acetate

tkm kg ⁻¹ isopropyl acetate	lorry	train
Total	1.25E-01	7.51E-01

29.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of isopropyl acetate plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

29.9 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 7.25E-03 kWh/kg isopropyl acetate is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of isopropyl acetate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 29.5.

Tab. 29.5 Process emissions to air from the production of 1 kg isopropyl acetate

Output	Value	Remarks
Isopropanol, to air (kg)	1.238E-03	Estimated as 0.2 % of input
Acetic acid, to air (kg)	1.238E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	5.88E-02	from waster water treatment
Waste heat (MJ)	2.61E-02	calculated from electricity input

29.10 Emissions to water

The remaining amount of unreacted acetic acid and isopropanol (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for isopropanol and acetic acid was assumed with 90%. The carbon contained in the removed ethanol and acetic acid was accounted as CO₂ emissions to air. The sulphate emissions are calculated from the sulphuric acid input.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of isopropanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 29.6.

Tab. 29.6 Process emissions to water from the production 1 kg isopropyl acetate

Output	Value	Remarks
Isopropanol, to water (kg)	2.97E-03	calculated from mass balance
Acetic acid, to water (kg)	2.97E-03	calculated from mass balance
Sulphate, to water (kg)	1.283E-02	calculated from mass balance
COD, BOD (kg)	9.88E-03	calculated from water emissions
TOC, DOC (kg)	2.97E-03	calculated from water emissions

29.11 Life cycle inventory and Data Quality Considerations

Tab. 29.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of isopropyl acetate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 29.7 Unit process raw data for the production of isopropyl acetate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	isopropyl acetate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
401										
662			Location							
493			InfrastructureProcess							
403			Unit							
resource, in water technosphere	-	0	isopropyl acetate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	isopropanol, at plant	RER	0	kg	6.19E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	acetic acid, 98% in H2O, at plant	RER	0	kg	6.19E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	1.31E-2	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	transport, lorry 32t	RER	0	tkm	1.25E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.51E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	7.25E-3	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	8.84E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	2.61E-2	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
	-	4	2-Propanol	-	-	kg	1.24E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	1.24E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	5.88E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	2-Propanol	-	-	kg	2.97E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Acetic acid	-	-	kg	2.97E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Sulfate	-	-	kg	1.28E-2	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	9.88E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	9.88E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.97E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.97E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4		-	-					

29.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 29.8 shows selected LCI results and the cumulative energy demand for the production of isopropyl acetate.

Tab. 29.8 Selected LCI results and the cumulative energy demand for the production of isopropyl acetate

Name		isopropyl acetate, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	81.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.4
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.4E-1
CO2, fossil	air	kg	2.9E+0
NM VOC	air	kg	1.1E-2
nitrogen oxides	air	kg	4.9E-3
sulphur dioxide	air	kg	7.9E-3
particulates, <2.5 um	air	kg	4.5E-4
BOD	water	kg	1.0E-1
cadmium	soil	kg	1.1E-9

29.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	isopropyl acetate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of isopropyl acetate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Isopropylazetat, ab Werk
	491	Synonyms	sec-propyl acetate, isopropyl ester, isopropyl ethanoate
	492	GeneralComment	The process "isobutyl acetate, at plant, RER" is modelled for the production of isopropyl acetate from isopropanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C5H10O2
	501	StatisticalClassification	
	502	CASNumber	108-21-4
TimePeriod	601	StartDate	1991
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Esterification of isopropanol
Representativeness	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

29.14 References

Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007)

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30 Methyl formate

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

30.1 Introduction

Methyl formate (CAS-No. 107-31-3, C₂H₄O₂) is a colourless, mobile, volatile liquid with a pleasant fruity odour. It is soluble in ethyl alcohol and ether, and slightly soluble in water.

For this inventory the functional unit is 1 kg of pure methyl formate. The most important chemical and physical properties of methyl formate used in this inventory are given here:

Tab. 30.1 Chemical and physical properties of methyl formate (Ullmann 2005)

Property	Value	Unit	Remarks
Molecular weight	60.05	g mol ⁻¹	
Specific gravity	975	kg m ⁻³	at 20 °C
Boiling point	31.7	°C	at normale pressure
Melting point	- 99	°C	at normale pressure

30.2 Reserves and Resources

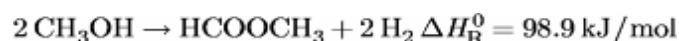
Methyl formate is an intermediate in the production of formic acid and formamide in large manufacturing plants. With increasing interest in the production of synthesis gas by coal gasification, methyl formate will become important as a versatile key intermediate. It can also be used as a starting material in the production of high-purity carbon monoxide. Physical properties, safety data, and marking regulations for methyl formate are summarized below. It is produced by base-catalysed carbonylation of methanol.

30.3 Production Technologies and Use

30.3.1 Production Technologies

Methyl formate is produced by base-catalyzed carbonylation of methanol. Other processes have been developed, but these have not yet been implemented commercially.

The carbonylation of methanol is carried out in the liquid phase with carbon monoxide, in the presence of a basic catalyst:



This reaction was first described by BASF in 1925. As a rule, the catalyst is sodium methoxide. Potassium methoxide has also been proposed as a catalyst; it is more soluble in methyl formate and gives a higher reaction rate. Although fairly high pressures were initially preferred, carbonylation is carried out in new plants at lower pressure. Under these conditions, reaction temperature and catalyst concentration must be increased to achieve acceptable conversion. According to published data, ca. 4.5 MPa, 80 °C, and 2.5 wt % sodium methoxide are employed. About 95 % carbon monoxide, but only about

30 % methanol, is converted under these circumstances. Nearly quantitative conversion of methanol to methyl formate can, nevertheless, be achieved by recycling the unreacted methanol. The carbonylation of methanol is an equilibrium reaction. The reaction rate can be raised by increasing the temperature, the carbon monoxide partial pressure, the catalyst concentration, and the interface between gas and liquid.

To synthesize methyl formate, gas mixtures with a low proportion of carbon monoxide must first be concentrated; this can be achieved by low temperature distillation, pressure-swing adsorption, prism separators (Monsanto), or the Cosorb process (Tenneco).

In a side reaction, sodium methoxide reacts with methyl formate to form sodium formate and dimethyl ether, and becomes inactivated. The substances used must be anhydrous; otherwise, sodium formate is precipitated to an increasing extent. Sodium formate is considerably less soluble in methyl formate than in methanol. The risk of encrustation and blockage due to precipitation of sodium formate can be reduced by adding poly(ethylene glycol). The carbon monoxide used must contain only a small amount of carbon dioxide; otherwise, the catalytically inactive carbonate is precipitated.

Basic catalysts may reverse the reaction, and methyl formate decomposes into methanol and carbon monoxide. Therefore, undecomposed sodium methoxide in the methyl formate must be neutralized.

30.3.2 Use

Most of the methyl formate produced is used as an intermediate for the production of formic acid and formamide. Dimethylformamide is produced by reacting methyl formate with dimethylamine. A new use has been found for methyl formate in the production of foundry molds. The mold is formed from granular refractory material and a binder consisting of phenol – formaldehyde resin which is cured by exposure to methyl formate vapor. This process makes possible the production of rapidly cured foundry molds at low temperature.

Small quantities of methyl formate are used as solvents and insect control agents. With the anticipated extension of C1 chemistry, methyl formate may be used as an intermediate for a number of products

30.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg methyl formate, at plant, in Europe**. For this study the production of methyl formate by means of the carbonylation of methanol is considered. All data in the present report are referred to 1 kg methyl formate 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

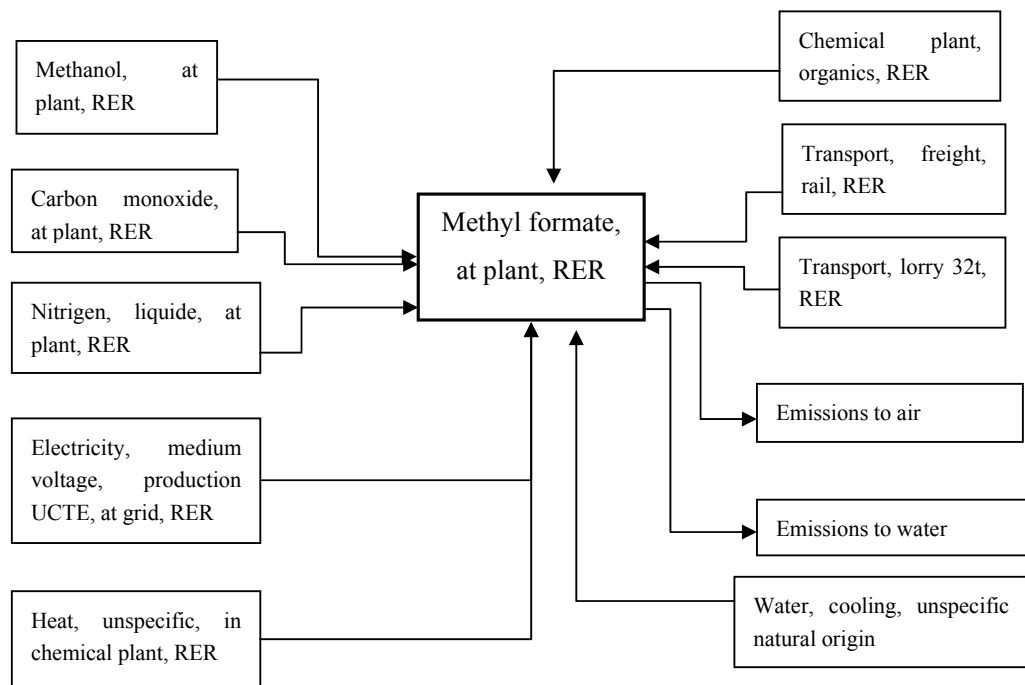


Fig. 30.1 Process flow chart for the production of methyl formate

30.5 Raw materials and auxiliaries

The data for the input of raw materials and auxiliaries are taken from Shouzu (2003), who gives data for methyl formate production in China.

Tab. 30.2 Consumption of raw materials and auxiliaries for the production of 1 kg methyl formate

Input	Values	Remarks
Methanol (kg)	1.09	Shouzu 2003
Carbon monoxide (kg)	4.76E-01	Shouzu 2003
Nitrogen (kg)	5E-01	Shouzu 2003
Cooling water (kg)	300	Shouzu 2003

30.6 Energy

The data for the energy inputs are taken from Shouzu (2003), who gives data for methyl formate production in China.

Tab. 30.3 Energy consumption for the production of 1 kg methyl formate

Input	Shouzu 2003
Steam (MJ)	5.5

Electricity (kWh)	0.48
-------------------	------

30.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and 50 km by lorry 32t and 100 km by train for the nitrogen. Tab. 30.4 summarises the total transport amounts for the production of 1 kg methyl formate.

Tab. 30.4 Total transport amounts for the production of 1 kg methyl formate

tkm kg ⁻¹ methyl formate	lorry	train
Total	1.82E-01	9.9E-01

30.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of methyl formate plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

30.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of methyl formate. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.4% of the methanol input and 2% of the carbon monoxide input. The assumed values for the emissions to air are given in Tab. 30.5.

Tab. 30.5 Process emissions to air from the production of 1 kg methyl formate

Output	Value	Remarks
Methanol, to air (kg)	4.36E-03	Estimated as 0.4 % of input
Carbon monoxide, fossil, to air (kg)	9.52E-03	Estimated as 2 % of input
Carbon dioxide, fossil, to air (kg)	2.16E-2	from waster water treatment
Waste heat (MJ)	1.728	calculated from electricity input

30.10 Emissions to water

The remaining amount of unreacted methanol (1.6 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for methanol was assumed with 90%. The carbon contained in the removed methanol was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of methanol in the treated waste water assuming a carbon conversion of 96% for COD. For the calcula-

tion of the values for BOD and DOC the worst case scenario $BOD = COD$ and $TOC = DOC$ was used. A summary of the values used in this inventory is given in Tab. 30.6.

Tab. 30.6 Process emissions to water from the production 1 kg methyl formate

Output	Value	Remarks
Methanol, to water (kg)	1.74E-03	calculated from mass balance
COD, BOD (kg)	2.5E-03	calculated from water emissions
TOC, DOC (kg)	6.5E-04	calculated from water emissions

30.11 Life cycle inventory and Data Quality Considerations

Tab. 30.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of methyl formate. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 30.7 Unit process raw data for the production of methyl formate

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	methyl formate, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	methyl formate, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.00E-1	1	1.14	(1,3,1,5,1,3); Shouzu 2003
	5	-	methanol, at plant	GLO	0	kg	1.09E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	carbon monoxide, CO, at plant	RER	0	kg	4.76E-4	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	nitrogen, liquid, at plant	RER	0	kg	5.00E-1	1	1.14	(1,3,1,5,1,3); Shouzu 2003
	5	-	transport, lorry 32t	RER	0	tkm	1.82E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	9.90E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	4.80E-1	1	1.14	(1,3,1,5,1,3); Shouzu 2003
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	5.50E+0	1	1.14	(1,3,1,5,1,3); Shouzu 2003
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.73E+0	1	1.14	(1,3,1,5,1,3); Calculated from the electricity input
	-	4	Methanol	-	-	kg	4.36E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon monoxide, fossil	-	-	kg	9.52E-3	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	2.16E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Methanol	-	-	kg	1.74E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	2.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	2.50E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	6.50E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	6.50E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

30.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 30.8 shows selected LCI results and the cumulative energy demand for the production of methyl formate.

Tab. 30.8 Selected LCI results and the cumulative energy demand for the production of methyl formate

Name	methyl formate, at plant		
	Location		
	Unit Infrastructure	Unit	RER kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	80.3
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.2
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	1.6E-1
CO2, fossil	air	kg	2.5E+0
NM VOC	air	kg	6.6E-3
nitrogen oxides	air	kg	4.6E-3
sulphur dioxide	air	kg	6.7E-3
particulates, <2.5 um	air	kg	4.6E-4
BOD	water	kg	8.9E-3
cadmium	soil	kg	9.9E-10

30.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	methyl formate, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of methyl formate including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Methylformat, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "methyl formate, at plant, RER" is modelled for the production of methyl formate from methanol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C2H4O2
	501	StatisticalClassification	
	502	CASNumber	592-84-7
TimePeriod	601	StartDate	2003
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Carbonylation of methanol
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 500.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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31 Methylcyclohexane

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

31.1 Introduction

Methyl cyclohexane (CAS-No. 108-87-2, C₇H₁₄, Synonyms: hexahydrotoluene, cyclohexyl methane) is a colorless transparent liquid with a faint, benzene-like odor. It is soluble in organic solvents as methanol, ethanol, and chloroform. The most important physical properties of methylcyclohexane are given here:

Tab. 31.1 Chemical and physical properties of methyl cyclohexane (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	98.183	g mol ⁻¹	
Specific gravity	769.39	kg m ⁻³	at 20 °C
Boiling point	100.934	°C	at normale pressure
Melting point	- 126.593	°C	at normale pressure
Heat of combustion, gross caloric value (HHV)	4.568	MJ kg ⁻¹	

31.2 Reserves and Resources

Methylcyclohexane can be produced easily by hydrogenation of toluene, but no current market exists for large-scale quantities.

31.3 Production Technologies and Use

31.3.1 Production Technologies

The hydrogenation of toluene is similar to the hydrogenation of benzene. The yield is about 90%. (Ullmann 2004)

31.3.2 Use

The primary use of methylcyclohexane is as a component in the feed to catalytic naphtha reformers where it is readily dehydrogenated to toluene, a high-octane motor gasoline component. Some methylcyclohexane is used for solvent applications in which it is one component of a narrow boiling range fraction of naphtha.

31.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg methylcyclohexane, at plant, RER**. For this study the production of methylcyclohexane by means of the hydrogenation of toluene is considered. All data in the present report are referred to 1 kg methylcyclohexane 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation emissions to air and water. It also includes transportation of

the raw materials and wastes. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

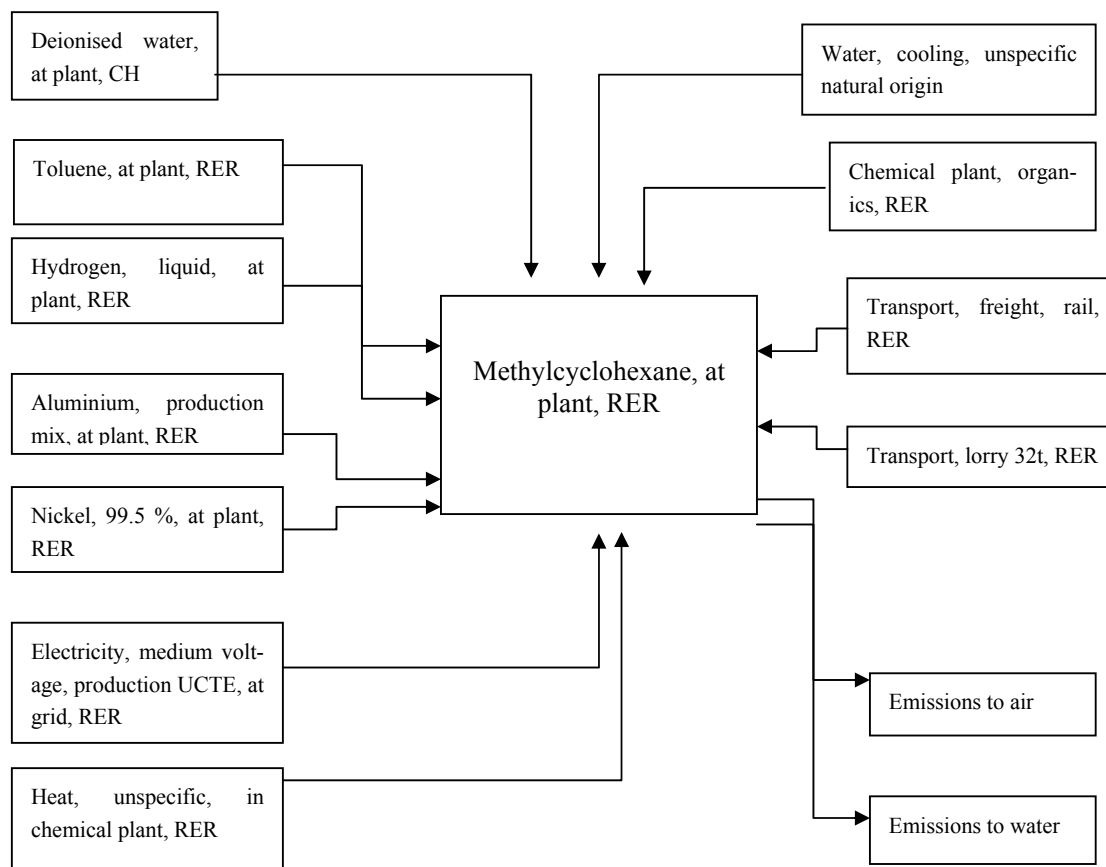


Fig. 31.1 Process flow chart for the production of methylcyclohexane

31.5 Raw materials and auxiliaries

As there was no data available for the production of methylcyclohexane, the values for the consumption of the raw materials are calculated with an assumed yield of 90%. The auxiliaries was estimated with data from Chauvel 1989 and Marcel 1982, who give data for the production of cyclohexane from the hydrogenation of benzene. This process is considered to be similar to the hydrogenation of toluene.

The catalyst assumed is Raney nickel (50 % Ni, 50 % Al) with an input of 3.1E-05 kg nickel from the module “Nickel, 99.5 %, at plant, RER” and 3.1E-05 kg aluminium from the module “Aluminium, production mix, at plant, RER”

Tab. 31.2 Consumption of raw materials and auxiliaries for the production of methylcyclohexane

Input	Value	Remarks
-------	-------	---------

Toluene (kg)	1.043	Calculated with a yield of 90%
H ₂ (kg), liquid	6.8E-02	Calculated with a yield of 90%
Cooling water (kg)	8	Estimated with data from similar process
Process water (kg)	0.95	Estimated with data from similar process
Catalysts (kg)	0.000062	Estimated with data from similar process

31.6 Energy

Wurster (1994) gives data for the electricity input of methylcyclohexane production of a plant in Germany.

As no data for the steam input of the production of methylcyclohexane was available, the value has been estimated with data from Chauvel 1989, who gives data for production of cyclohexane from the hydrogenation of benzene. This process is considered to be similar to the hydrogenation of toluene.

Tab. 31.3 Energy consumption for the production of 1 kg methylcyclohexane

Input	Values	Remarks
Steam (MJ)	28.325	Estimation with data from similar process
Electricity (kWh)	0.242	Wurster 1994

31.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for toluene, 50 km by lorry 32t and 100 km by train for hydrogen, and 100 km by lorry 32t and 200 km by train for the auxiliaries. Tab. 31.4 summarises the total transport amounts for the production of 1 kg cyclohexane.

Tab. 31.4 Total transport amounts for the production of methylcyclohexane

t km kg ⁻¹ methylcyclohexane	lorry	train
Total transports	1.077E-01	6.327E-01

31.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of methylcyclohexane plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

31.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of methylcyclohexane. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 2% of the toluene input and 10% of the hydrogen input. The assumed

values for the emissions to air are given in Tab. 31.5.

Tab. 31.5 Process emissions to air from the production of 1 kg methylcyclohexane

Output	Value	Remarks
Toluene, to air (kg)	2.09E-02	Estimated as 2 % of input
Hydrogen, to air (kg)	6.8E-03	Estimated as 10 % of input
Carbon dioxide, fossil, to air (kg)	2.15E-01	from waster water treatment
Waste heat (MJ)	8.71E-01	calculated from electricity input

31.10 Emissions to water

The remaining amount of unreacted toluene (8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for toluene was assumed with 90%. The carbon contained in the removed toluene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of toluene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 31.6.

Tab. 31.6 Process emissions to water from the production of 1 kg methylcyclohexane

Output	Value	Remarks
Toluene, to water (kg)	8.345E-03	calculated from mass balance
COD, BOD (kg)	2.5E-02	calculated from water emissions
TOC, DOC (kg)	7.61E-03	calculated from water emissions

31.11 Life cycle inventory and Data Quality Considerations

Tab. 31.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of methylcyclohexane. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 31.7 Unit process raw data for the production of methylcyclohexane

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	methylcyclohexane, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	methylcyclohexane, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	8.00E-3	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	water, deionised, at plant	CH	0	kg	9.50E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	toluene, liquid, at plant	RER	0	kg	1.04E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrogen, liquid, at plant	RER	0	kg	6.80E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	aluminium, production mix, at plant	RER	0	kg	3.10E-5	1	1.51	(1,2,5,1,1,3); Estimation
	5	-	nickel, 99.5%, at plant	GLO	0	kg	3.10E-5	1	1.51	(1,2,5,1,1,3); Estimation
	5	-	transport, lorry 32t	RER	0	tkm	1.08E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.33E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	2.42E-1	1	1.16	(1,4,1,3,1,4); Wurster 1994
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.83E+1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	8.71E-1	1	1.16	(1,4,1,3,1,4); Calculated from the electricity input
	-	4	Carbon dioxide, fossil	-	-	kg	2.09E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	Toluene	-	-	kg	6.80E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen	-	-	kg	2.51E-1	1	1.62	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Toluene	-	-	kg	8.34E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	2.50E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	2.50E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	7.61E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	7.61E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

31.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 31.8 shows selected LCI results and the cumulative energy demand for the production of methylcyclohexane.

Tab. 31.8 Selected LCI results and the cumulative energy demand for the production of methylcyclohexane

Name		methylcyclohexane, at plant	
Location	Unit	RER	
Infrastructure		kg	0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	115.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	4.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0
selected LCI results			
land occupation	resource	m2a	1.1E-1
CO ₂ , fossil	air	kg	4.3E+0
NM VOC	air	kg	1.0E-2
nitrogen oxides	air	kg	6.1E-3
sulphur dioxide	air	kg	8.0E-3
particulates, <2.5 um	air	kg	3.9E-4
BOD	water	kg	2.9E-2
cadmium	soil	kg	4.3E-10

31.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	methylcyclohexane, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of methylcyclohexane including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Methylcyclohexan, ab Werk
	491	Synonyms	hexahydrotoluene, cyclohexyl methane
	492	GeneralComment	The process "methylcyclohexane, at plant, RER" is modelled for the production of methylcyclohexane from toluene in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C7H14
	501	StatisticalClassification	
	502	CASNumber	108-87-2
TimePeriod	601	StartDate	1982
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrngenation of toluene
Representative	722	Percent	100
	724	ProductionVolume	No current world market
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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32 N,N-Dimethylformamide

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

32.1 Introduction

N,N-Dimethylformamide (CAS-No. 68-12-2, C_3H_7NO , Synonyms: DMF) is a colorless, high-boiling, nonviscous, highly polar, distillable, hygroscopic liquid with a weak, characteristically amine-like odor. It is miscible in all proportions with water, alcohols, ethers, ketones, esters, and carbon disulfide, as well as with chlorinated and aromatic hydrocarbons. By contrast, it has only limited solubility or complete insolubility in aliphatic hydrocarbons. Many high molecular mass natural products and synthetics are quite soluble in DMF. Examples include poly(vinyl alcohol), poly(vinyl chloride), poly(vinylidene chloride), poly(vinyl acetate) poly(vinyl ether), polyacrylate, polyacrylonitrile, polystyrene, polyester resins, alkyd resins, urea-formaldehyde resins, melamine-formaldehyde resins, phenol-formaldehyde resins, epoxy resins, and natural resins. It is also a good solvent for a number of drying oils, plasticizers, and cellulose derivatives, as well as for both natural and chlorinated rubber. The solubility of polyurethanes in DMF is a function of the conditions under which they are prepared. Most halogenated hydrocarbons are also soluble in DMF.

For this inventory the functional unit is 1 kg of pure N,N-dimethylformamide. The most important chemical and physical properties of N,N-dimethylformamide used in this inventory are given here:

Tab. 32.1 Chemical and physical properties of N,N-dimethylformamide (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	73.09	$g\ mol^{-1}$	
Specific gravity	921.5	$kg\ m^{-3}$	at 20 °C
Boiling point	153.5	°C	at normale pressure
Melting point	- 61	°C	at normale pressure

32.2 Reserves and Resources

N,N-Dimethylformamide is prepared by the reaction of N,N-dimethylamine with carbon monoxide. Two methods dominate present-day commercial preparation of DMF: direct synthesis and a two-step process.

N,N-Dimethylformamide is produced by a number of manufacturers worldwide; total capacity is estimated at 250 000 t per year (Ullmann 2004).

32.3 Production Technologies and Use

32.3.1 Production Technologies

Direct synthesis (Ullmann 2004)

The direct or one-step synthesis of DMF begins with either pure carbon monoxide or a gas stream containing carbon monoxide. This is reacted in a continuous process with N,N-dimethylamine, by using a solution of alkali alkoxide (usually sodium methoxide) in methanol as catalyst. Methyl formate is presumably formed as an intermediate. The reaction mixture passes over an external heat exchanger to

remove the excess heat generated and to ensure thorough mixing of the components.

The reaction is conducted between 0.5 and 11 MPa at 50 – 200 °C. The reaction mixture exits the reactor through a decompression chamber. In addition to N,N-dimethylformamide, the crude product contains methanol, a certain amount of unreacted N,N-dimethylamine, dissolved carbon monoxide, and residual catalyst. The addition of acid or water deactivates any catalyst present, resulting in the formation of sodium formate. Dissolved carbon monoxide, together with inert gases, escapes from the mixture during decompression, and the off-gases are removed by combustion. Preliminary distillation is followed by a second distillation in a separate column; here, dimethylformamide is separated from methanol which contains traces of N,N-dimethylamine. Further distillation results in a product of 99.9 % purity.

Two steps process

The two-step process for the synthesis of N,N-dimethylformamide differs from direct synthesis because methyl formate is prepared separately and introduced in the form of ca. 96 % pure (commercial-grade) material. Equimolar amounts of methyl formate and N,N-dimethylamine are subjected to a continuous reaction at 60 – 100 °C and 0.1 – 0.3 MPa. The resulting product is a mixture of N,N-dimethylformamide and methanol. The purification process involves distillation and is analogous to that described for direct synthesis. However, no separation of salts is required because no catalysts are involved in the process. According to the corrosive properties of both starting materials and products, stainless steel has to be used as material of construction for production facilities.

32.3.2 Use

The principal applications of N,N-dimethylformamide are as a solvent and as an extractant, particularly for salts and compounds with high molecular mass. This role is consistent with its interesting combination of physical and chemical properties: low molecular mass, high dielectric constant, electron-donor characteristics, and ability to form complexes. The use of DMF as a component in synthesis is of relatively minor significance, at least commercially.

The most important use of DMF as a solvent for polymers is in the preparation of polyacrylonitrile solutions for the manufacture of fibrous polyacrylonitrile. Residual DMF can be recovered from the spun fibers with relatively little expenditure of energy. In addition DMF is used as a solvent in the preparation of high-quality surfaces based on polyurethanes, which are otherwise difficult to solubilize, and it is employed as a solvent for coating surfaces with polyamides.

The high boiling point and high solvating power of DMF lead to its further use as a solvent for wire enamel formulated with polyamides, polyesterimides, or polyurethanes. Conversely, its solubilizing ability also makes it an effective component of paint removers, and it is used as a selective solvent in a wide range of manufacturing processes. For example, DMF is useful in separating acetylene from ethylene; butadiene from C4 cuts; and hydrogen chloride, hydrogen sulfide, or sulfur dioxide from CO₂-containing gases. N,N-Dimethylformamide or mixtures containing it are used by oil refiners to extract aromatic compounds from hydrocarbon mixtures.

Solutions of salts in specially purified DMF serve as fillers for electrolytic capacitors, a role made possible by the high dielectric constant of DMF. This substance is also used either alone or with other solvents in the recrystallization of aromatics, heterocyclics, dyes, fluorescent whitening agents, and pigments. The pharmaceutical industry also utilizes its broad solvent and extractant properties

N,N-Dimethylformamide could potentially be employed in the synthesis of aldehydes, acetals, amides, amidines, esters, and heterocycles, but thus far its use has been limited.

32.4 System Characterisation

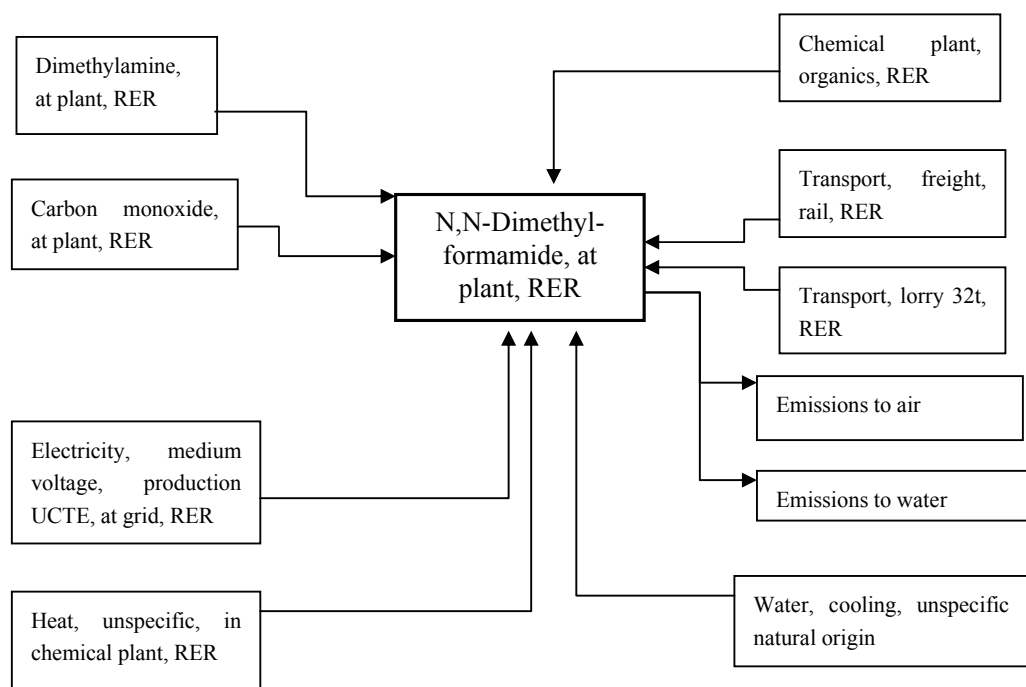


Fig. 32.1 Process flow chart for the production of N,N-dimethylformamide

This report corresponds to the module in the ecoinvent database for the production of **1 kg N,N-dimethylformamide, at plant, in Europe**. For this study the production of N,N-dimethylformamide by means of the direct reaction of carbon monoxide with dimethylamine is considered. All data in the present report are referred to 1 kg N,N-dimethylformamide 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

32.5 Raw materials and auxiliaries

As there was no data available for the production of N,N-dimethylformamide, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007).

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the N,N-dimethylformamide production.

Tab. 32.2 Consumption of raw materials and auxiliaries for the production of 1 kg N,N-dimethylformamide

Input	Values	Remarks
Dimethylamine (kg)	6.49E-01	calculated with a yield of 95%
Carbon monoxide (kg)	4.04E-01	calculated with a yield of 95%
Cooling water (kg)	24	Estimation

32.6 Energy

The energy demand for the N,N-dimethylformamide production is calculated with data from an Indian plant (DSIR 1993).

Tab. 32.3 Energy consumption for the production of 1 kg N,N-dimethylformamide

Input	DSIR 1993
Steam (MJ)	2.34
Electricity (kWh)	0.044

32.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials. Tab. 32.4 summarises the total transport amounts for the production of 1 kg N,N-dimethylformamide.

Tab. 32.4 Total transport amounts for the production of 1 kg N,N-dimethylformamide

km kg ⁻¹ N,N-dimethylformamide	lorry	train
Total	1.05E-01	6.32E-01

32.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of DMF plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

32.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of N,N-methylformamide. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources are estimated to 0.2% of the dimethyl amine and the carbon monoxide input. The remaining CO is oxidated to CO₂.

Tab. 32.5 Process emissions to air from the production of 1 kg N,N-dimethylformamide

Output	Value	Remarks
Dimethylamine, to air (kg)	1.3E-03	Estimated as 0.2 % of input
Carbon monoxide, fossil, to air (kg)	8.07E-04	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	8.53E-2	from waster water treatment and CO oxidation
Waste heat (MJ)	1.584E-01	calculated from electricity input

32.10 Emissions to water

The remaining amount of unreacted dimethylamine (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for dimethylamine was assumed with 90%. The carbon contained in the removed dimethylamine was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of ethanol and acetic acid in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 32.6.

Tab. 32.6 Process emissions to water from the production 1 kg N,N-dimethylformamide

Output	Value	Remarks
Dimethylamine, to water (kg)	3.11E-03	calculated from mass balance
COD, BOD (kg)	7.96E-03	calculated from water emissions
TOC, DOC (kg)	1.66E-03	calculated from water emissions

32.11 Life cycle inventory and Data Quality Considerations

Tab. 32.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of N,N-methylformamide. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 32.7 Unit process raw data for the production of N,N-methylformamide

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	N,N-dimethylformamide, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	N,N-dimethylformamide, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	dimethylamine, at plant	RER	0	kg	6.49E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	carbon monoxide, CO, at plant	RER	0	kg	4.04E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry 32t	RER	0	tkm	1.05E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.32E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	4.40E-2	1	1.14	(1,3,1,5,1,3); DSIR 1993
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	2.34E+0	1	1.14	(1,3,1,5,1,3); DSIR 1993
emission air, high population density	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	1.58E-1	1	1.14	(1,3,1,5,1,3); Calculated from the electricity input
	-	4	Dimethylamine	-	-	kg	1.30E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbon monoxide, fossil	-	-	kg	8.07E-4	1	5.10	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Carbon dioxide, fossil	-	-	kg	8.53E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
	-	4	Dimethylamine	-	-	kg	3.11E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	7.96E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	7.96E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	1.66E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	1.66E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

32.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 32.8 shows selected LCI results and the cumulative energy demand for the production of N,N-methylformamide.

Tab. 32.8 Selected LCI results and the cumulative energy demand for the production of N,N-methylformamide

Name		N,N-dimethylformamide, at plant	
Location	Unit	RER	
Infrastructure		kg	0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	51.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	6.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.9
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	2.0E-1
CO ₂ , fossil	air	kg	1.8E+0
NM VOC	air	kg	3.8E-3
nitrogen oxides	air	kg	3.4E-3
sulphur dioxide	air	kg	4.9E-3
particulates, <2.5 µm	air	kg	3.9E-4
BOD	water	kg	1.7E-2
cadmium	soil	kg	9.8E-10

32.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	N,N-dimethylformamide, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of N,N-dimethylformamide including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	N,N-Dimethylformamid, ab Werk
	491	Synonyms	DMF
	492	GeneralComment	The process "N,N-dimethylformamide, at plant, RER" is modelled for the production of DMF from dimethylamine in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C3H7NO
	501	StatisticalClassification	
	502	CASNumber	68-12-2
TimePeriod	601	StartDate	1993
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Synthesis from N,N-Dimethylamine
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 250.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

32.14 References

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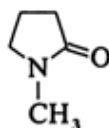
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33 N-Dimethyl-2-pyrrolidone

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

33.1 Introduction

N-Methyl-2-pyrrolidone (C_5H_9NO ; CAS 872-50-4; synonym: NMP) is a colorless liquid with a weakly aminelike odor. It is completely miscible with water and most common organic solvents. NMP is an important, versatile solvent and reaction medium for the chemical industry because of its favorable properties.



Technical-grade NMP has the following specifications (gas chromatography): NMP content, min. 99.5 wt %; methylamine, max. 0.02 wt %; water, max. 0.1 wt % (Karl Fischer titration); color index, APHA 50 max. For special areas of use (e.g., electronics) higher purities are supplied with the following specifications: methylamine, max. 0.01 wt %; water, max. 0.05 wt %.

For this inventory the functional unit is 1 kg of pure N-methyl-2-pyrrolidone. The most important chemical and physical properties of N-methyl-2-pyrrolidone used in this inventory are given here.

Tab. 33.1 Chemical and physical properties of N-methyl-2-pyrrolidone (Ullmanns 2004)

Property	Value	Unit
Molecular weight	99.13	g mol^{-1}
Specific gravity	1028	kg m^{-3}
Boiling point	204.3	$^{\circ}\text{C}$
Melting point	-24.4	$^{\circ}\text{C}$

33.1.1 Reserves and Resources

World production of NMP is currently estimated at 20 000 – 30 000 t/a. The most important producers are BASF and GAF.

33.1.2 Use

NMP is an important solvent because of its low volatility, thermal stability, high polarity, and aprotic, noncorrosive properties. Its favorable toxicological and ecological properties account for the fact that NMP is replacing other solvents such as chlorinated hydrocarbons.

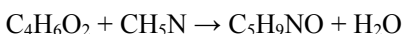
The most important areas of use are:

- Petrochemical processing: acetylene recovery from cracked gas, extraction of aromatics and butadiene, gas purification (removal of CO_2 and H_2S), lube oil extraction
- Engineering plastics: reaction medium for the production of high-temperature polymers such as polyethersulfones, polyamideimides, and polyaramids
- Coatings: solvent for acrylic and epoxy resins, polyurethane paints, waterborne paints or finishes, printing inks, synthesis/diluent of wire enamels, coalescing agent

- Agricultural chemicals: solvent and/or cosolvent for liquid formulations
- Electronics: cleaning agent for silicon wafers, photoresist stripper, auxiliary in printed circuit board technology
- Industrial and domestic cleaning: component in paint strippers and degreasers e.g., removal of oil, fat, and soot from metal surfaces, carbon deposits and other tarry polymeric residues in combustion engines

33.2 Production Technologies (Ullmanns 2004)

Large-scale production of NMP is predominantly carried out by reacting γ -butyrolactone with methylamine in a shaft reactor (high-pressure tube with special baffles) at 200–350 °C and ca. 10 MPa.



Other processes also analogous to those used for pyrrolidone synthesis can also be used, in particular hydrogenation of N-methylsuccinimide or mixtures of maleic or succinic anhydride and methylamine. NMP can also be produced by hydrogenation of N-hydroxymethyl-2-pyrrolidone or by reaction of acrylonitrile with methylamine in the presence of a peroxide radical initiator.

33.3 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg N-methyl-2-pyrrolidone, at plant, global**. For this study the production of N-methyl-2-pyrrolidone by means of the reaction of butyrolactone with methylamine is considered. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

33.4 Raw materials and auxiliaries

Tab. 20.2 shows values for the use of raw materials of the production of N-methyl-2-pyrrolidone. As there was no data available for the production of N-methyl-2-pyrrolidone, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). As approximation for methylamine the dataset "dimethylamine, at plant, RER" is used.

There was no information available on the amount of cooling water used within the plant. In order not to neglect the process cooling water demand this value was approximated with data from a large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf 2000). In this plant in total an average 24 kg water per kg of product were used for. This value was used in this inventory as approximation for the cooling water consumption of the N-methyl-2-pyrrolidone production.

Tab. 33.2 Consumption of raw materials and auxiliaries for the N-methyl-2-pyrrolidone production

Input	Values	Remarks
Butyrolactone (kg)	0.914	Calculated with a yield of 95 %
Methylamine (kg) ¹	0.33	Calculated with a yield of 95 %
Cooling water (kg)	24	Estimation

1: Approximated with "dimethylamine, at plant, RER"

33.5 Energy

There was no information available on the amount of energy used for the production process. In order not to neglect the process energy demand those values were approximated with data from an large chemical plant site in Germany producing 2.05 Mt a⁻¹ (intermediates included) of different chemicals (Gendorf (2000)). The values for the energy consumption per kg of product of this plant (3.2 MJ kg⁻¹) were used as approximation for the energy consumption of the N-methyl-2-pyrrolidone production. This total energy demand contains a split of 50% natural gas, 38% electricity and 12% steam from external energy sources. For this inventory all energy used for heat or steam was assumed to be natural gas. For this inventory an amount of 2 MJ natural gas and 0.333 kW electricity per kg product was used. A summary of the values used is given in Tab. 20.3.

Tab. 33.3 Energy consumption for the N-methyl-2-pyrrolidone production

Input	per kg N-methyl-2-pyrrolidone
Heat, natural gas burned in industrial furnace > 100 kW	2
Electricity (kWh)	0.333

33.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train. Tab. 14.4 summarises the total transport amounts for the N-methyl-2-pyrrolidone production.

Tab. 33.4 Total transport amounts for the N-methyl-2-pyrrolidone production

tkm kg ⁻¹ N-methyl-2-pyrrolidone	lorry	train
Total transports	1.24E-01	7.46E-01

33.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of N-methyl-2-pyrrolidone plants. Therefore, in this study, the infrastructure is estimated based on the module "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

33.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of N-methyl-2-pyrrolidone. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 2.5.

Tab. 33.5 Process emissions to air from the production of 1 kg N-methyl-2-pyrrolidone

Output	Value	Remarks
Butyrolactone, to air (kg)	1.83E-03	Estimated as 0.2 % of input
Methyl amine, to air (kg)	6.59E-04	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.01E-01	from waster water treatment
Waste heat (MJ)	1.1988	calculated from electricity input

33.9 Emissions to water

The remaining amount of unreacted butyrolactone and methyl amine (4.8 % of the input) is assumed to leave the production process with the wastewater. Further it is assumed that this wastewater is treated in a internal wastewater plant. The removal efficiency for butyrolactone and methyl amine is assumed with 90%. The carbon contained in the removed butyrolactone and methyl amine is accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory are calculated from the amount of butyrolactone and methyl amine in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used.

Tab. 33.6 Process emissions to water from the production 1 kg N-methyl-2-pyrrolidone

Output	Value	Remarks
Butyrolactone, to water (kg)	4.39E-03	calculated from mass balance
Methyl amine, to water (kg)	1.58E-03	calculated from mass balance
COD, BOD (kg)	1.35E-02	calculated from water emissions
TOC, DOC (kg)	3.56E-03	calculated from water emissions

33.10 Life cycle inventory of N-methyl-2-pyrrolidone production and data quality considerations

Tab. 14.8 shows the life cycle inventory and the data quality indicators for the N-methyl-2-pyrrolidone production. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 33.7 Unit process raw data for the production of N-methyl-2-pyrrolidone

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	N-methyl-2-pyrrolidone, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water	-	0	N-methyl-2-pyrrolidone, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	butyrolactone	GLO	0	kg	9.14E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	dimethylamine, at plant	RER	0	kg	3.30E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	heat, natural gas, at industrial furnace >100kW	RER	0	MJ	2.00E+0	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.33E-1	1	1.30	(4,5,na,na,na,na); Estimation
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.24E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.46E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.20E+0	1	1.30	(4,5,na,na,na,na); Calculated from the electricity input
	-	4	Butyrolactone	-	-	kg	1.83E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Methyl amine	-	-	kg	6.59E-4	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.01E-1	1	1.30	(4,5,na,na,na,na); Estimation
emission water, river	-	4	Butyrolactone	-	-	kg	4.39E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Methyl amine	-	-	kg	1.58E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.35E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.35E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	3.56E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	3.56E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

33.11 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 14.9 shows selected LCI results and the cumulative energy demand for the production of N-methyl-2-pyrrolidone.

Tab. 33.8 Selected LCI results and the cumulative energy demand for the N-methyl-2-pyrrolidone production

Name		N-methyl-2-pyrrolidone, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	76.9
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	10.0
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.1
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.5
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	2.7E-1
CO ₂ , fossil	air	kg	3.8E+0
NM VOC	air	kg	1.5E-2
nitrogen oxides	air	kg	6.1E-3
sulphur dioxide	air	kg	7.6E-3
particulates, <2.5 um	air	kg	5.7E-4
BOD	water	kg	2.3E-2
cadmium	soil	kg	1.4E-9

33.12 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	N-methyl-2-pyrrolidone, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of N-methyl-2-pyrrolidone including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	N-Methyl-2-pyrrolidon, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "N-methyl-2-pyrrolidone, at plant, GLO" is modelled for the production of N-methyl-2-pyrrolidone from butyrolactone in Europe. Raw materials are modelled with a stoichiometric calculation, Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C5H9NO
	501	StatisticalClassification	
	502	CASNumber	872-50-4
TimePeriod	601	StartDate	2000
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Methamination of butyrolactone
Representative	722	Percent	100
	724	ProductionVolume	unknown
	725	SamplingProcedure	Standard values
	726	Extrapolations	Energy consumptions, emissions, infrastructure and transports are calculated with standard values.
	727	UncertaintyAdjustments	none
DataGeneratorAn	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	49
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

33.13 References

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34 Propanal

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

34.1 Introduction

Propanal (CAS-No. 123-38-6, $\text{CH}_3\text{CH}_2\text{CHO}$, Synonym: propionaldehyde) is a low-boiling, colorless, flammable liquid with a sharp, suffocating odor. It is completely miscible with organic solvents such as alcohols, ether, and benzene.

For this inventory the functional unit is 1 kg of pure propanal. The most important chemical and physical properties of propanal used in this inventory are given here:

Tab. 34.1 Chemical and physical properties of propanal (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	58.08	g mol^{-1}	
Specific gravity	797	kg m^{-3}	at 20 °C
Boiling point	48	°C	at normale pressure
Melting point	- 80	°C	at normale pressure

34.2 Reserves and Resources

Propanal is generally produced by hydroformylation of ethylene. Annual world production was estimated at 154000 t in 1988 (Ullmann 2004)

34.3 Production Technologies and Use

34.3.1 Production Technologies

Low-pressure rhodium-catalyzed hydroformylation of ethylene with carbon monoxide and hydrogen is the most important process for the production of propanal. Some is still produced in the United States (and all in Europe) by the cobalt-catalyzed hydroformylation process at higher pressure, and a small fraction is obtained in Japan as a byproduct from Wacker oxidation of propene during acetone production.

The low-pressure hydroformylation process for production of propanal uses a soluble rhodium catalyst complex with excess triphenylphosphine. Reaction occurs in the liquid phase at 90 – 130 °C and a total pressure of <2.8 MPa with CO pressure <0.38 MPa and H₂ pressure <1.38 MPa. Selectivities are >90 % because ethylene can lead to only one isomer and impurities can be minimized. Major impurities include 2-methyl-2-pentenal, which arises by aldol condensation; trimers from the Tischenko reaction afford 2-methyl-1,3-propanediol monopropionate, which can in turn disproportionate to a mixture of 2-methyl-1,3-propanediol dipropionate and 2-methyl-1,3-propanediol. Byproduct formation during propanal production can be inhibited by exclusion of oxygen, organic acids, and metal contaminants such as iron, which act as Lewis-acid catalysts for its condensation to higher products.

34.3.2 Use

Propanal has no uses as such; it serves as a source of a propyl group for chemical synthesis. Propanal is primarily converted to 1-propanol, propionic acid, and trimethylolethane (trihydroxymethylethane). Minor applications include the production of pharmaceuticals (e.g., meprobrat), agricultural chemicals, rubber additives, and corrosion inhibitors. Some propanal is used as a solvent.

34.4 System Characterisation

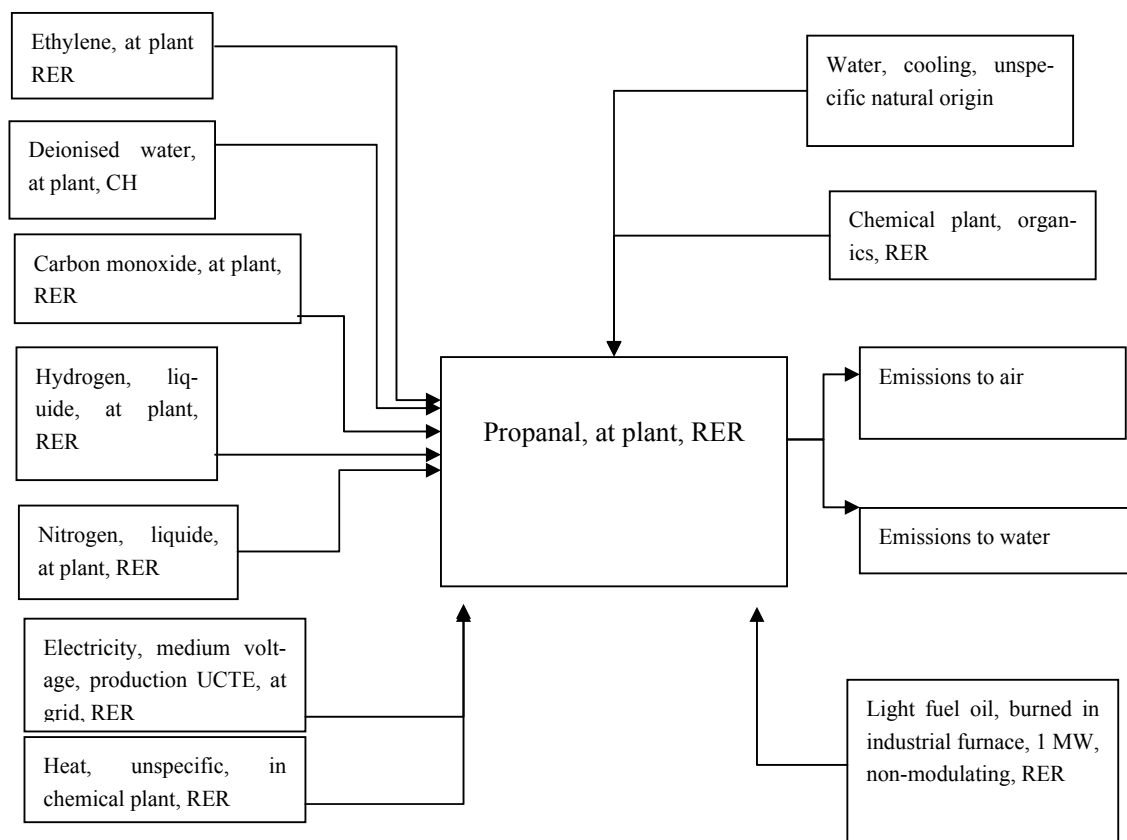


Fig. 34.1 Process flow chart for the production of propanal

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg propanal, at plant, in Europe**. For this study the production of propanal by means of the hydroformylation of ethylene is considered. All data in the present report are referred to 1 kg propanal 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

34.5 Raw materials and auxiliaries

As there was no data available for the production of propanal, the values for the consumption of the raw materials are calculated with an assumed yield of 95 % (Frischknecht et al. 2007). The auxiliaries was estimated with data from Chauvel 1989, who gives data for the production of isobutanol from the hydroformylation of propylene. This process is considered to be similar to the hydroformylation of ethylene.

Tab. 34.2 Consumption of raw materials and auxiliaries for the production of 1 kg 1-propanal

Input	Values	Remarks
Ethylene (kg)	0.508	Calculated with a yield of 95%
Carbon monoxide (kg)	0.508	Calculated with a yield of 95%
H ₂ (kg)	0.036	Calculated with a yield of 95%
N ₂ (kg)	0.005	Estimation with data from similar process
Process water (kg)	1	Estimation with data from similar process
Cooling water (kg)	120	Estimation with data from similar process

34.6 Energy

As no data for the production of propanal are available, the values for the energy use have been estimated with data from Chauvel 1989, who gives data for the production of isobutanol from the hydroformylation of propylene. This process is considered to be similar to the hydroformylation of ethylene.

In this study the use of the fuel is modelled with the dataset “light fuel oil, burned in industrial furnace 1 MW, non-modulating” with a energetic efficiency of 74 %, and the use of steam is modelled with the modul “heat, unspecific, in chemical plant, RER”.

Tab. 34.3 Energy consumption for the production of 1 kg 1-propanal

Input	Values	Remarks
Steam (MJ)	5.5	Estimation with data from similar process
Electricity (kWh)	0.35	Estimation with data from similar process
Fuel (MJ)	9.5	Estimation with data from similar process

34.7 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the ethylene materials and 50 km by lorry 32t and 100 km by train for the carbon monoxide and the hydrogen. Tab. 34.4 summarises the total transport amounts for the production of 1 kg propanal.

Tab. 34.4 Total transport amounts for the production of 1 kg propanal

tkm kg ⁻¹ propanal	lorry	train
Total	7.8E-02	3.59E-01

34.8 Infrastructure and land use

No information was readily available about infrastructure and land-use of propanal plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

34.9 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of propanal. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources are estimated to 0.2% of the ethylene and the carbon monoxide input, and 5% of the hydrogen input. The remaining CO is oxidated to CO₂. The assumed values for the emissions to air are given in Tab. 34.5.

Tab. 34.5 Process emissions to air from the production of 1 kg propanal

Output	Value	Remarks
Ethylene, to air (kg)	1.02E-03	Estimated as 0.2 % of input
Carbon monoxide, fossil, to air (kg)	1.02E-03	Estimated as 0.2 % of input
Hydrogen, to air (kg)	1.81E-03	Estimated as 5 % of input
Carbon dioxide, fossil, to air (kg)	1.09E-1	from waster water treatment and CO oxidation
Waste heat (MJ)	1.26	calculated from electricity input

34.10 Emissions to water

The remaining amount of unreacted ethylene (4.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for ethylene was assumed with 90%. The carbon contained in the removed ethylene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of ethylene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 34.6.

Tab. 34.6 Process emissions to water from the production 1 kg propanal

Output	Value	Remarks
Ethylene, to water (kg)	2.44E-03	calculated from mass balance
COD, BOD (kg)	8.01E-03	calculated from water emissions
TOC, DOC (kg)	2.09E-03	calculated from water emissions

34.11 Life cycle inventory and Data Quality Considerations

Tab. 34.7 shows the life cycle inventory and the data quality indicators for the inventory of the production of propanal. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 34.7 Unit process raw data for the production of propanal

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	propanal, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
							RER 0 kg			
	401									
	662		Location							
	493		InfrastructureProcess							
	403		Unit							
resource, in water technosphere	-	0	propanal, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	1.20E-1	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	water, deionised, at plant	CH	0	kg	1.00E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	ethylene, average, at plant	RER	0	kg	5.08E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	carbon monoxide, CO, at plant	RER	0	kg	5.08E-1	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	hydrogen, liquid, at plant	RER	0	kg	3.63E-2	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	nitrogen, liquid, at plant	RER	0	kg	5.00E-3	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	transport, lorry 32t	RER	0	tkm	7.80E-2	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	3.59E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	3.50E-1	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	5.50E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	9.50E+0	1	1.45	(4,5,4,3,3,3); Estimation
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	1.26E+0	1	1.45	(4,5,4,3,3,3); Calculated from the electricity input
	-	4	Ethene	-	-	kg	1.02E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon monoxide, fossil	-	-	kg	1.02E-3	1	5.10	(4,5,na,na,na,na); Estimation
	-	4	Hydrogen	-	-	kg	1.81E-3	1	1.62	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.09E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	Ethene	-	-	kg	2.44E-3	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	8.01E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	8.01E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	2.09E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	2.09E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

34.12 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 34.8 shows selected LCI results and the cumulative energy demand for the production of propanal.

Tab. 34.8 Selected LCI results and the cumulative energy demand for the production of propanal

	Name	propanal, at plant	
	Location		
	Unit	Unit	RER kg 0
Infrastructure			
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	85.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	8.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.8
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.0
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.1
selected LCI results			
land occupation	resource	m2a	1.5E-1
CO2, fossil	air	kg	3.1E+0
NM VOC	air	kg	3.6E-3
nitrogen oxides	air	kg	5.1E-3
sulphur dioxide	air	kg	7.9E-3
particulates, <2.5 um	air	kg	4.6E-4
BOD	water	kg	1.8E-2
cadmium	soil	kg	8.5E-10

34.13 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	propanal, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of propanal including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Propanal, ab Werk
	491	Synonyms	propione aldehyde
	492	GeneralComment	The process "propanal, at plant, RER" is modelled for the production of propanal from ethylene in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with data from a similar process. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C3H6O
	501	StatisticalClassification	
	502	CASNumber	123-38-6
TimePeriod	601	StartDate	1989
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydroformylation of ethylene
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 154.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard values. Energy consumptions are modelled with a similar process
DataGeneratorAnd	727	UncertaintyAdjustments	none
	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

34.14 References

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35 Propanol

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

Acknowledgement

We thank Michael Overcash for the information provided in a personal communication.

35.1 Introduction

1-Propanol (CAS-No. 71-23-8, C_3H_7OH) is a clear, colourless, flammable liquid with a slight odor resembling that of ethanol. It is completely miscible with water and readily soluble in a variety of common organic solvents (e.g., ethers, esters, acids, ketones, and other alcohols).

For this inventory the functional unit is 1 kg of pure 1-propanol. The most important chemical and physical properties of 1-propanol used in this inventory are given here:

Tab. 35.1 Chemical and physical properties of 1-propanol (Ullmanns 2004)

Property	Value	Unit	Remarks
Molecular weight	60.1	$g\ mol^{-1}$	
Specific gravity	804	$kg\ m^{-3}$	at 20 °C
Boiling point	97.15	°C	at normale pressure
Melting point	- 126.1	°C	at normale pressure

35.2 Reserves and Resources

1-Propanol is manufactured by the hydrogenation of propanal derived from hydroformylation of ethylene. Annual U.S. production of 1-propanol is estimated in 1988 to have been 98000 (Ullmanns 2004).

35.3 Production Technologies and Use

35.3.1 Production Technologies

Hydroformylation of Ethylene

Union Carbide Chemicals and Plastics Company, Inc., and Hoechst Celanese practice the two-step oxo process based on rhodium-substituted phosphine-catalyzed low-pressure hydroformylation technology, while Eastman Kodak Company uses a higher-pressure cobalt-based process for production of the intermediate propanal. Typical oxo conditions employed in the low-pressure rhodium-catalyzed oxo process are 90 – 130 °C, <2.8 MPa total pressure, with CO at <380 kPa, H₂ at <1.4 MPa, and <500 ppm rhodium. Conditions of the cobalt oxo process are 110 – 180 °C, 20 – 30 MPa (200 – 300 atm), 1: 1 to 1: 2 H₂: CO ratios, and 0.1 – 1.0 wt % cobalt based on ethylene. The crude propanal is removed by vapor stripping with excess synthesis gas, and the rhodium – triphenylphosphine catalyst is recycled. Traces of CO are removed in a stripping column from the crude, condensed propanal before hydrogenation to prevent poisoning of the hydrogenation catalyst in the second step. The higher selectivity and milder conditions of the rhodium-catalyzed process provide crude 1-propanol

for hydrogenation containing fewer impurities, such as higher aldol-condensation products and aldehyde trimers (Tischenko esters).

Hydrogenation of propanal

The hydrogenation of propanal is conventionally carried out as a heterogeneous process in either the vapor or liquid phase over a variety of metal catalysts. Heterogeneous vapor-phase processes are effective at ca. 110 – 150 °C, and 0.14 – 1.0 MPa at a 20: 1 mole ratio of hydrogen to propanal. Reductions are accomplished in excess hydrogen, and the heat of reaction is removed by circulating the vapor through external heat exchangers or by cooling the reactor internally.

Hydrogen efficiencies are >90 %, with aldehyde conversions as high as 99.9 % and alcohol yields >99 %. The commonly used commercial hydrogenation catalysts include combinations of copper, zinc, nickel, and chromium compounds. Major impurities at vapor-phase process temperatures are dipropyl ether, ethane, and propyl propionate. In addition to reduced zinc oxide and copper oxide catalysts, selectivity enhancers such as alkali and transition metals can be added to significantly reduce the formation of esters (by Tischenko reaction of the starting aldehyde) and ethers. Addition of 1 – 10 % water to the reactor feed also suppresses ether formation.

The propyl propionate byproduct formed in the catalytic hydrogenation can be separated and hydrolyzed in the presence of reduced CuO – ZnO catalyst at 75 – 300 °C and 9.8 kPa – 9.8 MPa to give 1-propanol as the major product.

Liquid-phase heterogeneous processes are conducted at higher pressures. For example, hydrogenation at 95 – 120 °C and 3.5 MPa of hydrogen gives 1-propanol with >99.9 % purity. Nickel- and copper-based catalysts containing molybdenum, manganese, and sodium promoters are preferred. The catalysts are usually supported on alumina.

35.3.2 Use

1-Propanols is used mainly as solvent for coatings; in antifreeze compositions and household and personal products; and as chemical intermediate for the production of esters, amines, and other organic derivatives.

In 1988 over 75 % of the 1-propanol in the United States was employed in solvent applications, either directly or in the form of acetate ester or glycol ether derivatives. The total use was about 63000 t.

The EC used about 100 000 t of 1-propanol in 1988. In Germany, BASF converts most of its propanal into 1-propanol for printing inks, cosmetics, solvents, and intermediates for propylamines used in pharmaceuticals and pesticides. In Japan, 1000 – 2000 t of 1-propanol was consumed in 1988 for printing inks and paints, all of which was imported.

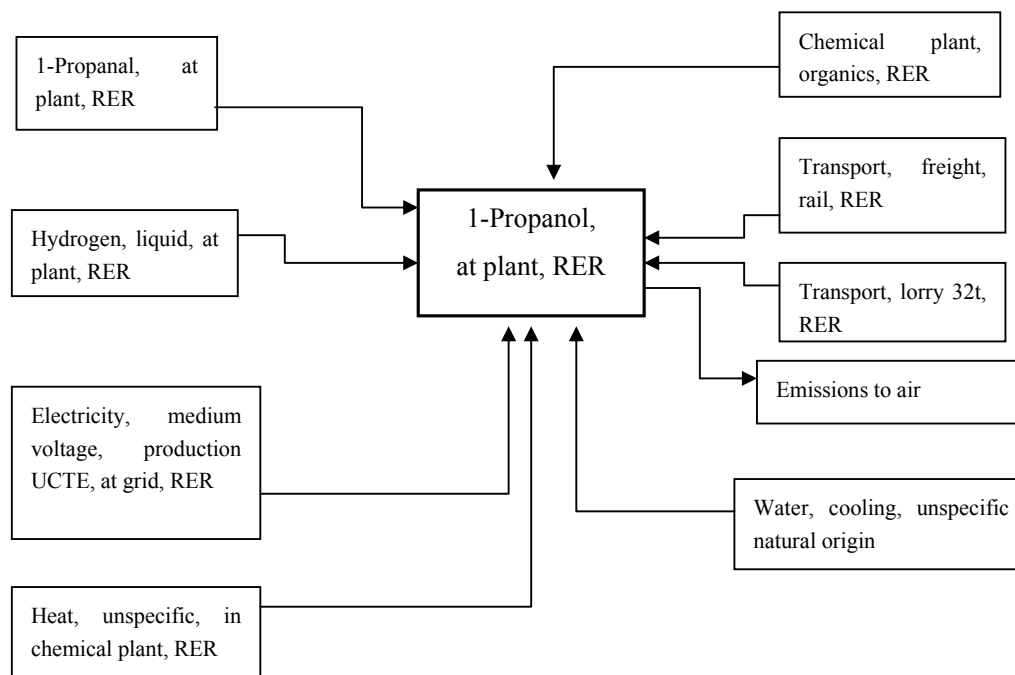
1-Propanol is used commercially to produce glycol ethers. These are characterized by dual functionality, which imparts high solvency, chemical stability, and water compatibility. Glycol ethers such as ethylene glycol monopropyl ether, diethylene glycol monopropyl ether, propylene glycol monopropyl ether, and dipropylene glycol monopropyl ether are marketed by Union Carbide Chemicals and Plastics and Eastman Chemical Products.

35.4 System Characterisation

This report corresponds to the dataset in the ecoinvent database for the production of **1 kg 1-propanol, at plant, in Europe**. For this study the production of 1-propanol by means of the hydrogenation of propanal is considered. All data in the present report are referred to 1 kg 1-propanol 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air. It also includes transportation of the raw materials. For the study

transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

Fig. 35.1 Process chain for the production of 1-propanol



35.5 Data sources

The data for the 1-propanol production are taken from a database, which reports data from US industry production of the years 1998 to 2004 (Overcash 1998-2004). These data are taken from several US production plants and are representative for US production.

35.6 Raw materials and auxiliaries

The values for the raw material input are calculated with data from Overcash (1998-2004).

Tab. 35.2 Consumption of raw materials and auxiliaries for the production of 1 kg 1-propanol

Input	Overcash 1998-2004
Propanal (kg)	9.956E-01
Hydrogen (kg)	3.685E-02
Cooling water (kg)	18.8515

35.7 Energy

The values for the energy inputs are calculated with data from Overcash (1998-2004).

Tab. 35.3 Energy consumption for the production of 1 kg 1-propanol

Input	Overcash 1998-2004
Steam (MJ)	0.394
Electricity (kWh)	0.6264

35.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 35.4 summarises the total transport amounts for the production of 1 kg 1-propanol.

Tab. 35.4 Total transport amounts for the production of 1 kg 1-propanol

tkm kg ⁻¹ 1-propanol	lorry	train
Raw materials	1.03E-01	6.19E-01

35.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of 1-propanol plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

35.10 Emissions to air

It is assumed that 100% of the electricity consumed, i. e. 0.6264 kWh/kg 1-propanol is converted to waste heat and that 100% of the waste heat is released to the air.

Overcash 1998–2004 report air emissions from the production of 1-propanol.

Tab. 35.5 Process emissions to air from the production of 1 kg 1-propanol

Output	Value	Remarks
Propanol, to air (kg)	9.92E-03	Overcash 1998-2004
Propanal, to air (kg)	1.93E-02	Overcash 1998-2004
Hydrogen, to air (kg)	3.19E-3	Overcash 1998-2004
Waste heat (MJ)	2.255	calculated from electricity input

35.11 Emissions to water

According to Overcash 1998-2004 no emissions to water are considered from the production of 1-propanol.

35.12 Life cycle inventory and Data Quality Considerations

Tab. 35.6 shows the life cycle inventory and the data quality indicators for the inventory of the production of 1-propanol. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 35.6 Unit process raw data for the production of 1-propanol

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	1-propanol, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	1-propanol, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	1.89E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	propanal, at plant	RER	0	kg	9.96E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	hydrogen, liquid, at plant	RER	0	kg	3.69E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	transport, lorry 32t	RER	0	tkm	1.03E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	6.19E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	6.26E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	heat, unspecific, in chemical plant	RER	0	MJ	3.94E-1	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
	-	4	Heat, waste	-	-	MJ	2.26E+0	1	1.07	(1,2,1,1,1,3); Calculated from the electricity input
emission air, high population density	-	4	Propanol	-	-	kg	1.93E-2	1	1.07	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Propanal	-	-	kg	9.92E-3	1	1.51	(1,2,1,1,1,3); Overcash 1998 - 2004
	-	4	Hydrogen	-	-	kg	3.19E-3	1	1.51	(1,2,1,1,1,3); Overcash 1998 - 2004

35.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 35.7 shows selected LCI results and the cumulative energy demand for the production of 1-propanol.

Tab. 35.7 Selected LCI results and the cumulative energy demand for the production of 1-propanol

Name		1-propanol, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	93.8
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	11.7
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.2
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	1.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	2.4E-1
CO2, fossil	air	kg	3.6E+0
NM VOC	air	kg	3.3E-2
nitrogen oxides	air	kg	6.2E-3
sulphur dioxide	air	kg	9.4E-3
particulates, <2.5 um	air	kg	6.0E-4
BOD	water	kg	1.8E-2
cadmium	soil	kg	1.3E-9

35.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	1-propanol, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of 1-propanol including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	1-Propanol, ab Werk
	491	Synonyms	
	492	GeneralComment	The process "1-propanol, at plant, RER" is modelled for the production of propanol from propanol in Europe. . Raw materials, auxiliaries, energy consumptions and emissions are modelled with data from US production. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C3H8O
	501	StatisticalClassification	
	502	CASNumber	71-23-8
TimePeriod	601	StartDate	1998
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Hydrngenation of propanol
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 100.000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports are modelled with standard distances.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

35.15 References

- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007) Overview and Methodology. Final report ecoinvent v2.0 No. 1. Swiss Centre for Life Cycle Inventories, Dübendorf, CH, retrieved from: www.ecoinvent.org.
- Overcash 1998-2004 Overcash, M., Chemical life cycle database, Chemical and Biomolecular Engineering, North Carolina State University, Raleigh, NC 27695-7905. 1998 - 2004

Ullmann 2004

Papa, A.: Propanol. In: Ullmann's Encyclopedia of Industrial Chemistry, Seventh Edition, 2004 Electronic Release (ed. Fiedler E., Grossmann G., Kersebohm D., Weiss G. and Witte C.). 7 th Electronic Release Edition. Wiley InterScience, New York, Online-Version under: <http://www.mrw.interscience.wiley.com/ueic/articles/>

36 Propylene ammooxidation

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

36.1 Introduction

The ammooxidation of propylene is the most important process to produce acrylonitrile and acetonitrile.

36.1.1 Production Technologies

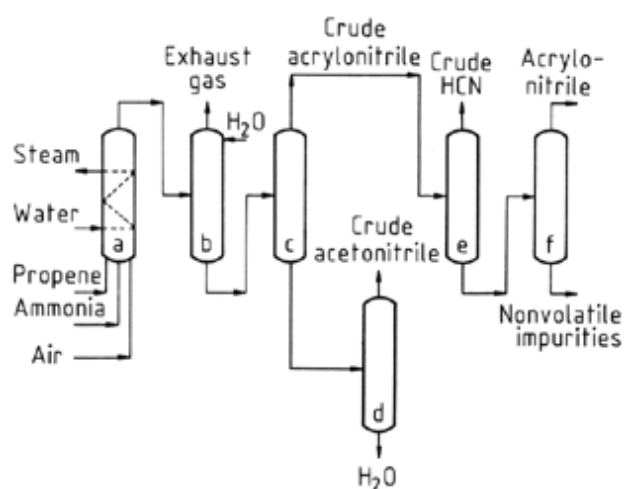
In the Sohio process propene, oxygen (as air), and ammonia are catalytically converted directly to acrylonitrile using a fluidized-bed reactor operated at temperatures of 400 – 500 °C and gauge pressures of 30 – 200 kPa (0.3 – 2 bar):

Complex mechanisms for this reaction have been described.

Approximately stoichiometric amounts of the starting materials are passed through the reactor with residence times of a few seconds. The process is highly selective, requiring no recycling to produce high acrylonitrile yields of approximately 0.8 – 0.9 kg from 1 kg of propene. Acetonitrile (0.02 – 0.11 kg) and hydrogen cyanide (HCN, 0.15 – 0.20 kg) are the principal byproducts (from 1 kg propene). The heat of reaction can be recovered as high-pressure steam.

The catalyst used in the early Sohio process was a bismuth – phosphomolybdate combination. Since that time there has been a continuous search for alternatives and for superior performance, resulting in patents by more than 30 companies. Sohio introduced Catalyst 21 (antimony – uranium) in 1967, Catalyst 41 (ferrobismuth – phosphomolybdate) in 1972, and Catalyst 49 (undisclosed) in 1978. All of these changes were aimed at improved efficiency and reduction in byproducts.

Fig. 36.1 Simplified diagram of the Sohio acrylonitrile process



a) Fluidized-bed reactor; b) Absorber column; c) Extractive distillation column; d) Acetonitrile stripping column; e) Lights fractionation column; f) Product column

The reactor effluent is cooled and scrubbed with water in a countercurrent absorber. The off-gas, consisting primarily of nitrogen, is vented. The reaction products remain in the aqueous phase. Acetonitrile is removed by extractive distillation. Crude acrylonitrile and hydrogen cyanide are distilled overhead while water and acetonitrile are removed from the bottom of the columns. In subsequent distilla-

tions, hydrogen cyanide is separated from wet acrylonitrile, the water content of the product is then reduced, and finally, nonvolatile impurities are removed.

Unused ammonia can be recovered as ammonium sulfate and then disposed of, but it commonly is vented to the atmosphere. Aqueous wastes containing cyanides, sulfates, and various organic byproducts must be disposed of by incineration or deepwell injection or be pretreated for subsequent biological waste treatment.

36.2 System Characterisation

In this study the ammoxidation of propylene is described. All data in the present report are referred to 1 kg propylene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the ammoxidation of propylene, acrylonitrile, acetonitrile, and hydrogen cyanide are derived as co-products. Therefore an allocation between the three products was made.

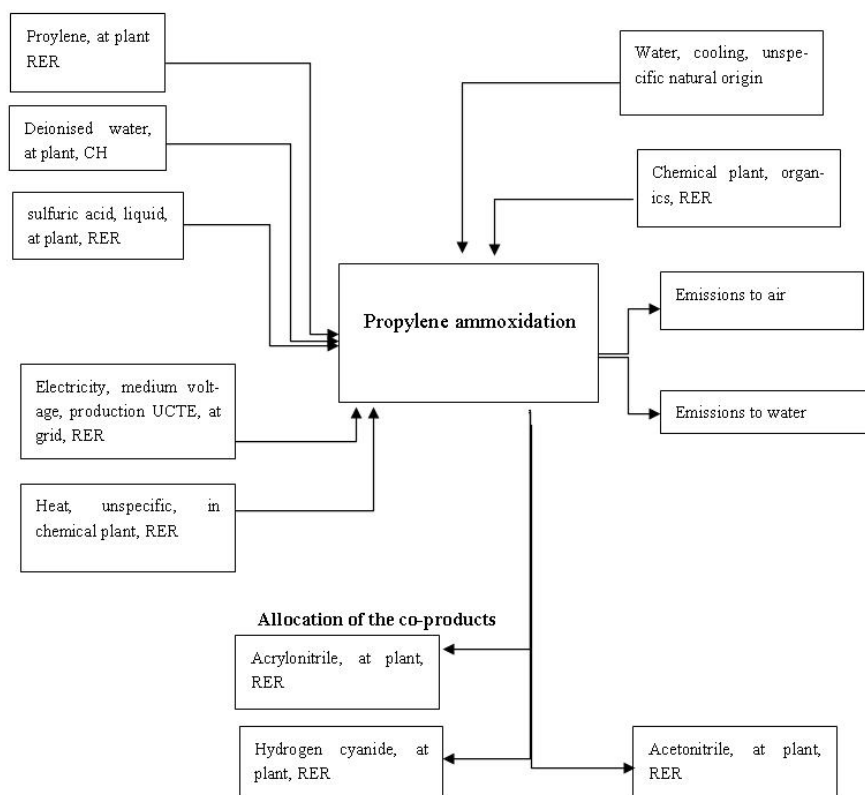


Fig. 36.2 Process flow chart for the production of acetonitrile

36.3 Raw materials and auxiliaries

The data for the input of raw materials and auxiliaries are taken from Chauvel (1989), which reports

data from French production plants in the late 1980ies.

Tab. 36.1 Consumption of raw materials and auxiliaries of the propylene ammoxidation per kg propylene

Input	Inventory
Propylene (kg)	1
Ammonia (kg)	4.238E-01
Sulfuric acid (kg)	4.24E-02
Process water (kg)	4.24
Cooling water (kg)	339

36.4 Energy

The data for the energy inputs are taken from Chauvel (1989), which reports data from French production plants in the late 1980ies.

Tab. 36.2 Energy consumption for the propylene ammoxidation per kg propylene

Input	Inventory
Steam (MJ)	2.33
Electricity (kWh)	2.29E-01

36.5 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for the raw materials and the auxiliaries. Tab. 36.3 summarises the total transport amounts per kg propylene.

Tab. 36.3 Total transport amounts for the propylene ammoxidation per kg propylene

tkm kg ⁻¹ propylene	lorry	train
Total	1.47E-01	8.8E-01

36.6 Infrastructure and land use

No information was readily available about infrastructure and land-use of propylene ammoxidation plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This module assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

36.7 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the propylene ammoxidation. As approxi-

mation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated as 20 % of the unreacted ammonia (0.00383 kg per kg propylene) and 20 % of the unreacted propylene (0.369 kg per kg propylene). The propylene emissions are incinerated with a assumed destruction efficiency of 99,9 % (Inece 2005). The values for the emissions to air are given in Tab. 36.4.

Tab. 36.4 Process emissions to air from the propylene ammoxidation per kg propylene

Output	Value	Remarks
Ammonia, to air (kg)	3.83E-03	Estimated as 20 % of the unreacted input
Propylene, to air (kg)	3.69E-05	Estimated as 0.02 % of the unreacted input
Carbon dioxide, fossil, to air (kg)	5.33E-01	from waste water and off gas treatment
Waste heat (MJ)	8.24E-01	calculated from electricity input

36.8 Emissions to water

The remaining amount of unreacted ammonia and propylene was assumed to leave the production process with the waste water. These assumptions lead to a pollution of the waste water with 0.148 kg ammonia and 0.153 kg propylene. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for propylene was assumed with 90%. The carbon contained in the removed propylene was accounted as CO₂ emissions to air. The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of ammonia and propylene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. The input of sulfuric acid was assumed to leave the system with the waste water as sulfates. A summary of the values used in this inventory is given in Tab. 36.5.

Tab. 36.5 Process emissions to water from the propylene ammoxidation per kg propylene

Output	Value	Remarks
Ammonia, to water (kg)	1.53E-02	calculated from mass balance
Propylene, to water (kg)	1.48E-02	calculated from mass balance
Sulfate, to water (kg)	4.15E-02	calculated from mass balance
COD, BOD (kg)	4.37E-02	calculated from water emissions
TOC, DOC (kg)	1.31E-02	calculated from water emissions

36.9 Allocation of the co-products of the propylene ammoxidation

Within the propylene ammoxidation, acetonitrile, acrylonitrile and hydrogen cyanide are obtained as co-products. The yields of the diverse co-products are given in Tab. 36.6.

As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 36.7.

Tab. 36.6 Yields of the co-products from the propylene ammoxidation (Chauvel 1989)

Co-products	Kg per ton acrylonitrile
-------------	--------------------------

Acrylonitrile	1000
Acetonitrile	120
Hydrogen cyanide	170

Tab. 36.7 Allocation of acetonitrile and the co-products from the Soio process

Process	Propylene ammoxidation		
	Unit	per kg propylene	Allocation factor
Acrylonitrile, at plant, RER	kg	8.47E-01	77.5
Acetonitrile, at plant, RER	kg	1.02E-01	9.3
Hydrogen cyanide from propylene ammoxidation, at plant, RER	kg	1.44E-01	13.2

36.10 Acrylonitrile, at plant, RER

Acrylonitrile (CAS-No. 107-13-1, C_2H_3CN) is a colorless liquid with a slightly pungent odor. It is miscible with numerous organic solvents, including acetone, benzene, carbon tetrachloride, diethyl ether, ethyl acetate, ethylene cyanohydrin, petroleum ether, toluene, some kerosenes, and methanol.

The most important chemical and physical properties of acrylonitrile used in this inventory are given in Tab. 36.8.

Tab. 36.8 Chemical and physical properties of acrylonitrile (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight ¹	53.06	g mol ⁻¹	
Specific gravity ¹	806	kg m ⁻³	at 20 °C
Boiling point ¹	77	°C	at normale pressure
Melting point ¹	- 84	°C	at normale pressure

36.10.1 Use

Acrylic textile fibers are by far the largest end-use product for acrylonitrile. Acrylic fibers find use primarily in wearing apparel and in home furnishings such as carpets and draperies.

36.11 Acetonitrile, at plant, RER

Acetonitrile (CAS-No. 75-05-8, CH_3CN , Synonyms: methylcyanide, cyanomethane, ethanenitrile) is a colourless liquid miscible with water, ethanol and many organic solvents but immiscible with many saturated hydrocarbons (e. g., petroleum fractions).

The most important chemical and physical properties of acetonitrile used in this inventory are given in Tab. 36.9.

Tab. 36.9 Chemical and physical properties of acetonitrile (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight ¹	41.05	g mol ⁻¹	
Specific gravity ¹	713.8	kg m ⁻³	at 20 °C
Boiling point ¹	81.6	°C	at normale pressure
Melting point ¹	- 45	°C	at normale pressure

36.11.1 Use

In 1986 the consumption of acetonitrile amounted to ca. 13 000 t in the United States, Western Europe, and Japan. Along with N-methylpyrrolidone and N,N-dimethylformamide, acetonitrile represents one of the major solvents for extraction of butadiene from crude C4 streams. Acetonitrile is also a valuable reaction solvent, especially in the synthesis of antibiotics. Highly purified acetonitrile finds application as a solvent in high-performance liquid chromatography.

The major use of acetonitrile in organic synthesis is in the production of malononitrile through reaction with cyanogen chloride.

Other applications include the synthesis of trimethyl orthoacetate and acetamidine hydrochloride. The former is used in the synthesis of 3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylic acid (also known as DV-acid) which is a key precursor for pyrethroids. The latter yields N-containing heterocycles such as 4-amino-5-cyano-2-methylpyrimidine used in the synthesis of thiamine.

36.12 Life cycle inventory and Data Quality Considerations

Tab. 36.10 shows the life cycle inventory and the data quality indicators for the inventory of the propylene ammoxidation. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 36.10 Unit process raw data for the propylene ammoxidation

	401	InputGroup	OutputGro	Name	Location	Infrastructu	reProcess	Unit	Sohio process					Uncertainty					Standard deviation95					GeneralComment					acetoneitrile, at plant					acrylonitrile from Sohio process, at plant					hydrogen cyanide from Sohio process, at plant																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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allocated products	662	493	403	-	2	acetonitrile, at plant	RER	0	kg	1.44E-1																100					0					0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
				-	2	acrylonitrile from Sohio process, at plant	RER	0	kg	1.02E-1																0					100					0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
				-	2	hydrogen cyanide from Sohio process, at plant	RER	0	kg	1.44E-1																0					0					100																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
resource, in water technosphere	4	-	Water, cooling, unspecified natural origin	-	-	m3	3.39E-2	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	propylene, at plant	RER	0	kg	1.00E+0	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	water, deionised, at plant	CH	0	kg	4.24E+0	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	ammonia, liquid, at regional storehouse	RER	0	kg	4.24E-1	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	sulphuric acid, liquid, at plant	RER	0	kg	4.24E-2	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	transport, lorry >16t, fleet average	RER	0	tkm	1.47E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	transport, freight, rail	RER	0	tkm	8.80E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
	5	-	electricity, medium voltage, production	UCTE	0	kWh	2.29E-1	1	1.22	(1,3,4,3,1,3); Chauvel 1989					9.305					77.510					13.185																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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36.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 36.11 shows selected LCI results and the cumulative energy demand for the propylene ammoxidation.

Tab. 36.11 Selected LCI results and the cumulative energy demand for the propylene ammoxidation

Name			acetonitrile, at plant	acrylonitrile from Sohio process, at plant	hydrogen cyanide from Sohio process, at plant
Location	Unit	RER	RER	RER	
Infrastructure		kg 0	kg 0	kg 0	
LCIA results					
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	58.9	695.4	83.5
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.5	29.0	3.5
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4	4.9	0.6
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3	3.2	0.4
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.3	0.0
selected LCI results					
land occupation	resource	m2a	9.0E-2	1.1E+0	1.3E-1
CO2, fossil	air	kg	1.9E+0	2.3E+1	2.8E+0
NM VOC	air	kg	1.9E-3	2.3E-2	2.7E-3
nitrogen oxides	air	kg	2.8E-3	3.3E-2	4.0E-3
sulphur dioxide	air	kg	3.8E-3	4.5E-2	5.5E-3
particulates, <2.5 µm	air	kg	3.0E-4	3.6E-3	4.3E-4
BOD	water	kg	3.0E-2	3.6E-1	4.3E-2
cadmium	soil	kg	3.8E-10	4.5E-9	5.4E-10

36.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	Sohio process
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Sohio process including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Sohio Prozess
	491	Synonyms	
	492	GeneralComment	The multioutput-process "Sohio process" delivers the co-products acrylonitrile, hydrogen cyanide, and acetoinitrile. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1989
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Sohio process
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

36.15 References

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- Frischknecht et al. 2007 Frischknecht R., Jungbluth N., Althaus H.-J., Doka G., Dones R., Heck T., Hellweg S., Hirschier R., Nemecek T., Rebitzer G. and Spielmann M. (2007)

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37 Propylene hydroformylation

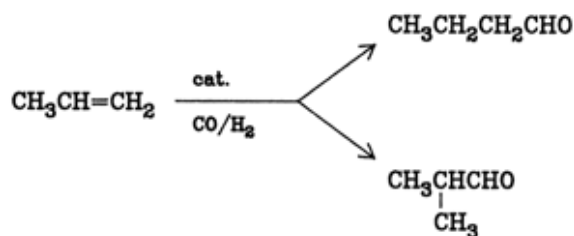
Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

37.1 Introduction

The most important process for the manufacture of 1-butanol and 2-methyl-1-propanol is propylene hydroformylation (oxo synthesis) with subsequent hydrogenation of the aldehydes formed.

37.2 Production Technologies

In the oxo reaction (hydroformylation) carbon monoxide and hydrogen are added to a carbon – carbon double bond in the liquid phase in the presence of catalyst (hydrocarbonyls or substituted hydrocarbonyls of Co, Rh, or Ru). In the first reaction step aldehydes are formed with one more C-atom than the original olefins. For olefins with more than two C-atoms, isomeric aldehyde mixtures are normally obtained. In the case of propylene these consist of 1-butanal and 2-methylpropanal.



There are several variations of the hydroformylation process, the differences being in the reaction conditions (pressure, temperature) as well as the catalyst system used. The classic high-pressure process exclusively used until the beginning of the 1970s operates at pressures of 20 – 30 MPa (200 – 300 bar) CO/H₂ and temperatures of 100 – 180 °C. The catalyst is Co. It leads to about 75 % 1-butanol and about 25 % 2-methyl-1-propanol.

The new process developments of the past few years have led to a clear shift in the range of products. The processes operating at relatively low pressures (1 – 5 MPa, 10 – 50 bar) use modified Rh-catalysts. The isomeric ratios achieved are about 92 : 8 or 95 : 5 1-butanol to 2-methyl-1-propanol. However, by the use of unmodified Rh the percentage of 2-methyl-1-propanol can be increased to about 50 %.

Catalytic hydrogenation of the aldehydes leads to the formation of the corresponding alcohols.

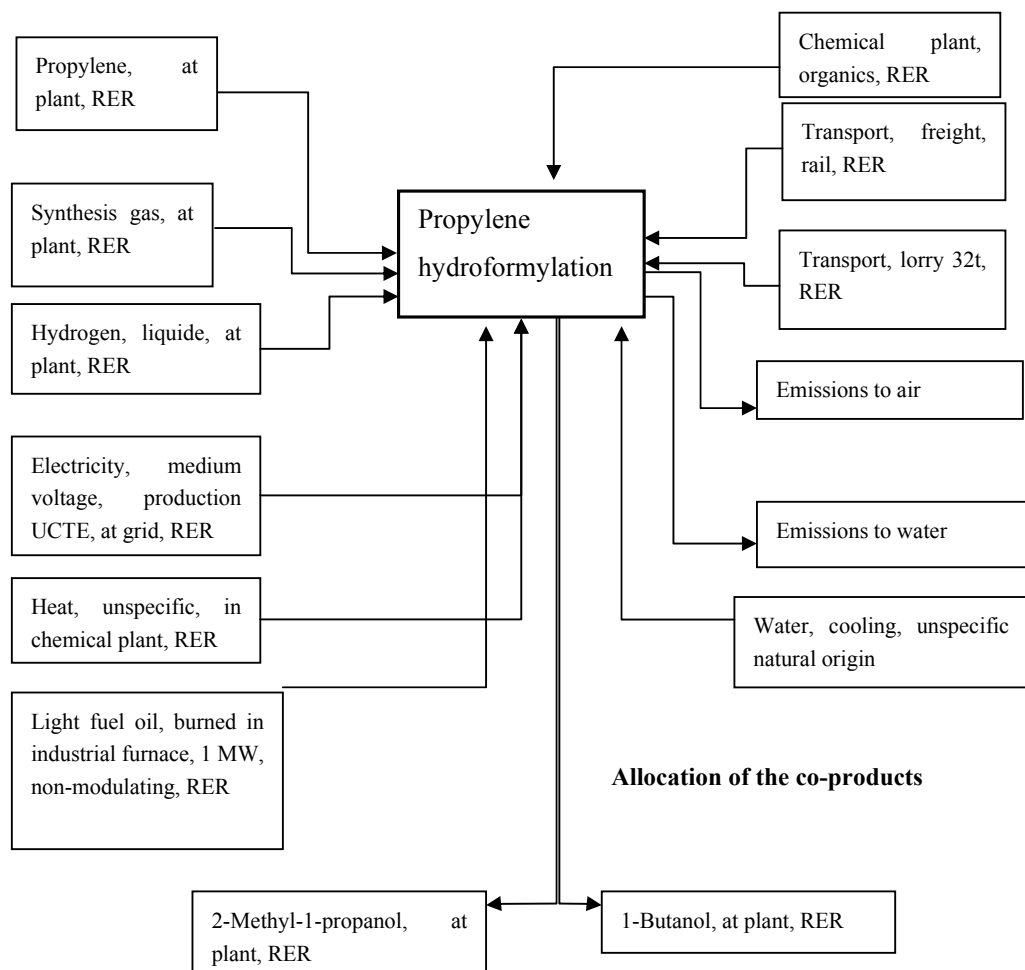
As only primary alcohols can be obtained via the oxo synthesis, it is not possible to produce 2-butanol and 2-methyl-2-propanol by this process.

37.3 System Characterisation

In this study the hydroformylation of propylene is described. All data in the present report are referred to 1 kg propylene. The system includes the process with consumption of raw materials, energy, infrastructure, and land use, as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are lo-

cated in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers. Within the hydroformylation of propylene, 1-butanol and -methyl-1-propanol are derived as co-products. Therefore an allocation between the three products was made.

Fig. 37.1 Process chain for the propylene hydroformylation



37.4 Raw materials and auxiliaries

Chauvel 1989 reports raw material inputs for three different oxo-synthesis processes. In this inventory the values of the Celanese process are used, because this process is the most important in the world today.

The synthesis gas is calculated with 50% carbon monoxide (4.875 Nm³ per kg 2-methyl-1-propanol) and 50% hydrogen (4.875 Nm³ per kg 2-methyl-1-propanol). It is produced from natural gas. From the carbon content of CO and natural gas (CH₄), the amount of natural gas is calculated (6.1 kg CO per kg 2-methyl-1-propanol correspond to 3.48 kg natural gas). The amount of natural gas is calculated with the following properties: density 0.8 kg/Nm³, HHV 36.8 MJ/Nm³. The hydrogen being part of the synthesis gas is included in the natural gas input.

Tab. 37.1 Consumption of raw materials and auxiliaries for the propylene hydroformylation

Input	Ruhrchemie ¹	Shell ²	Celanese ³	This study ⁴
Propylene (kg)	2.83	7.09	9	1
Synthesis gas (Nm ³)	3.67	12.27	9.75	
Natural gas (MJ)				17.8
H ₂ (kg)	0.117	0	0.375	0.042
N ₂ (kg)	0.013	0	0.1125	0.0125
Process water (kg)	3.33	0	12.5	1.39
Cooling water (kg)	400	1364	1375	152.8

1 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.3 kg isobutanol is obtained as co-product.

2 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.11 kg isobutanol is obtained as co-product.

3 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.08 kg isobutanol is obtained as co-product.

4 per kg propylene

37.5 Energy

Chauvel 1989 reports energy inputs for three different oxo-synthesis processes. In this inventory the values of the Celanese process are used, because this process is the most important in the world today.

In this study the use of the fuel is modelled with the dataset “light fuel oil, burned in industrial furnace 1 MW, non-modulating” with a energetic efficiency of 74 %, and the use of steam is modelled with the dataset “heat, unspecific, in chemical plant, RER”.

Tab. 37.2 Energy consumption for the production of propylene hydroformylation

Input	Ruhrchemie ¹	Shell ²	Celanese ³	Inventory ⁴
Steam (MJ)	16.67	20	31.25	3.47
Electricity (kWh)	1.167	1.82	1.875	0.208
Fuel (MJ)	31.67	109.1	112.5	12.5

1 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.3 kg isobutanol is obtained as co-product.

2 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.11 kg isobutanol is obtained as co-product.

3 Source: Chauvel 1989. Functional unit: 1 kg 2-methyl-1-propanol. Per kg 1-butanol 0.08 kg isobutanol is obtained as co-product.

4 per kg propylene

37.6 Transportation

No information is available in the sources consulted concerning transportation of raw materials or auxiliaries. Therefore, the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train for propylene and the natural gas, and 50 km by lorry 32t and 100 km by train for hydrogen and nitrogen. Tab. 37.3 summarises the total transport amounts for the propylene hydroformylation.

Tab. 37.3 Total transport amounts for the propylene hydroformylation

tkm kg ⁻¹ propylene ⁻¹	lorry	train
Total	1.4E-01	8.4E-01

37.7 Infrastructure and land use

No information was readily available about infrastructure and land-use of propylene hydroformylation plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

37.8 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the propylene hydroformylation. As approximation the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated as 0.2 % of the propylene input. Further it is assumed that the remaining hydrogen and 1% of the unreacted carbon monoxide is emitted to air. The remaining CO is oxidized to CO₂. The values for the emissions to air are given in Tab. 37.4.

Tab. 37.4 Process emissions to air from the propylene hydroformylation

Output	Value	Remarks
Propylene, to air (kg)	2E-03	Estimated as 0.2 % of the input
Carbon monoxide, fossil, to air (kg)	1.4E-06	Estimated as 1 % of the unreacted input
Hydrogen, to air (kg)	1.13E-03	calculated from the mass balance
Carbon dioxide, fossil, to air (kg)	4.15E-01	from waste water and off gas treatment
Waste heat (MJ)	0.75	calculated from electricity input

37.9 Emissions to water

The remaining amount of unreacted propylene was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for the propylene was assumed with 90%. The carbon contained in the removed propylene was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of propylene in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 37.5.

Tab. 37.5 Process emissions to water from the propylene hydroformylation

Output	Value	Remarks
Propylene, to water (kg)	1.47E-02	calculated from mass balance

COD, BOD (kg)	4.82E-02	calculated from water emissions
TOC, DOC (kg)	1.26E-02	calculated from water emissions

37.10 Allocation of the co-products

Within the propylene hydroformylation, 2-methyl-1-propanol and 1-butanol are obtained as co-products. According to the data from Chauvel 1989, it was assumed that in average an amount of 0.08 kg 2-methyl-1-propanol per kg 1-butanol is produced. As allocation factor the weight of the two products was used. The used allocation factors are given in Tab. 37.6.

Tab. 37.6 Allocation of 1-butanol and isobutanol from propylene hydroformylation

Process	Propylene hydroformylation, RER		
Co-Product	Unit	per kg propylene	Allocation factor
1-Butanol, at plant, RER	Kg	1.39	92.6
2-methyl-1-propanol, at plant, RER	kg	0.11	7.4

37.11 1-Butanol

1-Butanol (CAS-No. 71-36-3, C_4H_9OH) is a colourless liquid with a characteristic odour. It is completely miscible with common organic solvents. Its vapor has an irritant effect on mucous membranes and a narcotic effect in higher concentrations. The most important chemical and physical properties of 1-butanol used in this inventory are given here:

Tab. 37.7 Chemical and physical properties of 1-butanol (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	74.12	$g\ mol^{-1}$	
Specific gravity	809.8	$kg\ m^{-3}$	at 20 °C
Boiling point	117.7	°C	at normale pressure
Melting point	- 89.3	°C	at normale pressure

37.12 Use

1-Butanol is used principally in the field of surface coating. In the USA, for instance, approx. 85 % of 1-butanol production finds an outlet in this sector of the market. It is used either directly as a solvent for varnishes or it is converted into derivatives which then serve as solvents or monomer components.

1-Butanol cannot be used directly as a solvent in the production of nitrocellulose lacquers; in admixture with toluene, ethanol, or various esters, however, it is an excellent thinner.

The addition of 5 – 10 % of 1-butanol is also valuable in overcoming “blushing” (unwanted white opacity) which may arise when large quantities of thinners, particularly volatile solvents, are used.

1-Butanol is also useful for regulating the viscosity and improving the flow properties of varnishes and for the prevention of “streaking” in paints and lacquers based on spirit-soluble gums and resins.

Although not itself a solvent for substances such as polystyrene and chlorinated rubber, 1-butanol can

be successfully used in quantities up to 20 % as a diluent for the commonly used solvents for these substances, which are mainly the esters of the saturated carboxylic acids; in particular, the acetates.

The acrylic ester of 1-butanol has become increasingly important during the last decade. It has become an essential component of latex paints since such coatings are not only robust but can also be produced relatively cheaply. The 1-butyl esters of phthalic, adipic, sebacic, oleic, azelaic, stearic, and phosphoric acids are used as plasticizers and additives for surface coatings. The most important of these products is di-1-butyl phthalate (DBP). Consumption is, however, stagnating and in some countries it decreased slightly within the last years.

In the USA in particular, but also in the Federal Republic of Germany and Belgium, 1-butanol is used for the manufacture of butylamines.

1-Butanol has numerous applications in the plastics and textile sector. It is used as a coagulation bath for spinning acrylic fibers and in the dyeing of poly(vinyl alcohol) fibers.

37.13 2-Methyl-1-propanol

2-Methyl-1-propanol (CAS-No. 78-83-1, C_4H_9OH ; Synonym: isobutanol) is a colourless liquid with a characteristic odour. It is completely miscible with common organic solvents. Its vapor has an irritant effect on mucous membranes and a narcotic effect in higher concentrations. The most important chemical and physical properties of 2-methyl-1-propanol used in this inventory are given here:

Tab. 37.8 Chemical and physical properties of 2-methyl-1-propanol (Ullmann 2004)

Property	Value	Unit	Remarks
Molecular weight	74.12	g mol^{-1}	
Specific gravity	802.7	kg m^{-3}	at 20 °C
Boiling point	107.9	°C	at normale pressure
Melting point	- 107.9	°C	at normale pressure

37.14 Use

The fields of application of 2-methyl-1-propanol closely resemble those of 1-butanol. Indeed, because of its lower price it can often replace 1-butanol and is accordingly used as a solvent, diluent, and additive for nitrocellulose and synthetic resins; wetting agent; cleaner additive; and component of printing inks and related products. Its capacity for dissolving resins (ketone, phthalate, urea, and melamine – formaldehyde resins) is of commercial importance. As a solvent for phenol – formaldehyde resins, however, 2-methyl-1-propanol is not as effective as 1-butanol.

Several esters exhibit excellent solvent properties; in particular, isobutyl acetate is a very good solvent for fats, chlorinated rubber, polystyrene, and coumarone resins. The esters of phthalic acid, adipic acid, some dicarboxylic acids as well as phosphoric acid are used as plasticizers, particularly for PVC and its copolymers and for cellulose derivatives.

The isobutyl esters of 2,4-dichloro- and 2,4,5- trichlorophenoxyacetic acid exhibit herbicidal activity. For the anti-freezing properties of 2-methyl-1-propanol in gasoline see.

Also of interest is the use of 2-methyl-1-propanol as an extracting agent in the recovery of ammonium phosphate

37.15 Life cycle inventory and Data Quality Considerations

Tab. 37.9 shows the life cycle inventory and the data quality indicators for the inventory of the propylene hydroformylation. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 37.9 Unit process raw data for the propylene hydroformylation

	401	InputGroup	OutputGroup	Name	Location	Infrastructure	refProcess	Unit	propylene hydroformylation	Uncertainty	Standard deviation95 %	GeneralComment	1-butanol, propylene hydroformylation, at plant	isobutanol, at plant
	662			Location					RER				RER	RER
	493			InfrastructureProcess					0				0	0
	403			Unit					kg				kg	kg
allocated products resource, in water technosphere	-	2		1-butanol, propylene hydroformylation, at plant	RER	0		kg	1.39E+0				100	0
	-	2		isobutanol, at plant	RER	0		kg	1.11E-1				0	100
emission air, high population density	4	-		Water, cooling, unspecified natural origin	-	-		m3	1.53E-2	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		propylene, at plant	RER	0		kg	1.00E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		water, deionised, at plant	CH	0		kg	1.39E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		natural gas, high pressure, at consumer	RER	0		MJ	1.78E+1	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		hydrogen, liquid, at plant	RER	0		kg	4.17E-2	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		nitrogen, liquid, at plant	RER	0		kg	1.25E-2	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		transport, lorry >16t, fleet average	RER	0		tkm	1.40E-1	1	2.09	(4,5,na,na,na,na); Standard distance 100km	92.593	7.407
	5	-		transport, freight, rail	RER	0		tkm	8.40E-1	1	2.09	(4,5,na,na,na,na); Standard distance 600km	92.593	7.407
	5	-		electricity, medium voltage, production UCTE, at grid	UCTE	0		kWh	2.08E-1	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0		MJ	3.47E+0	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		heat, unspecific, in chemical plant	RER	0		MJ	1.25E+1	1	1.51	(1,3,5,3,1,3); Chauvel 1989	92.593	7.407
	5	-		chemical plant, organics	RER	1		unit	6.00E-10	1	3.09	(4,5,na,na,na,na); Estimation	92.593	7.407
	-	4		Heat, waste	-	-		MJ	7.50E-1	1	1.51	(1,3,5,3,1,3); Calculated from the electricity input	92.593	7.407
	-	4		Carbon dioxide, fossil	-	-		kg	4.15E-1	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment	92.593	7.407
	-	4		Propene	-	-		kg	2.00E-3	1	1.62	(4,5,na,na,na,na); Estimation	92.593	7.407
emission water, river	-	4		Carbon monoxide, fossil	-	-		kg	1.40E-6	1	5.10	(4,5,na,na,na,na); Estimation	92.593	7.407
	-	4		Hydrogen	-	-		kg	1.13E-3	1	1.62	(4,5,na,na,na,na); Estimation	92.593	7.407
	-	4		Propene	-	-		kg	1.47E-2	1	3.09	(4,5,na,na,na,na); Estimation	92.593	7.407
	-	4		BOD5, Biological Oxygen Demand	-	-		kg	4.82E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	92.593	7.407
	-	4		COD, Chemical Oxygen Demand	-	-		kg	4.82E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	92.593	7.407
	-	4		DOC, Dissolved Organic Carbon	-	-		kg	1.26E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	92.593	7.407
	-	4		TOC, Total Organic Carbon	-	-		kg	1.26E-2	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions	92.593	7.407

37.16 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 37.10 shows selected LCI results and the cumulative energy demand for the propylene hydroformylation.

Tab. 37.10 Selected LCI results and the cumulative energy demand for the propylene hydroformylation

	Name		1-butanol, propylene hydroformylation, at plant	isobutanol, at plant
	Location		RER	RER
	Unit	Unit	kg 0	kg 0
Infrastructure				
LCIA results				
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	77.1	77.1
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	2.6	2.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	0.4	0.4
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	0.3	0.3
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.0	0.0
selected LCI results				
land occupation	resource	m2a	9.2E-2	9.2E-2
CO2, fossil	air	kg	2.0E+0	2.0E+0
NMVOc	air	kg	2.3E-3	2.3E-3
nitrogen oxides	air	kg	3.5E-3	3.5E-3
sulphur dioxide	air	kg	4.1E-3	4.1E-3
particulates, <2.5 um	air	kg	2.2E-4	2.2E-4
BOD	water	kg	3.6E-2	3.6E-2
cadmium	soil	kg	3.9E-10	3.9E-10

37.17 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	propylene hydroformylation
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	5
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Hydroformylation of propylene including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Propen Hydroformylierung
	491	Synonyms	
	492	GeneralComment	The multioutput-process "propylene hydroformylation" delivers the co-products 2-methyl-1-butanol and 1-butanol. The allocation is based on mass balance.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	
	501	StatisticalClassification	
	502	CASNumber	
TimePeriod	601	StartDate	1989
	602	EndDate	2005
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications.
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Direct hydration of ethylene
Representativeness	722	Percent	100
	724	ProductionVolume	na
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	petrochemical solvents

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38 Tetrahydrofuran

Author: Jürgen Sutter, ETH S+U, Zurich
 Review: Roland Hischier, EMPA, St. Gallen
 Last changes: 2007

38.1 Introduction

Tetrahydrofuran (THF, C_4H_8O , CAS-No. 109-99-9, synonyms: tetramethylene oxide, oxolane) is a five-membered cyclic ether with wide application in the chemical industry.



Fig. 38.1 Structural formula of tetrahydrofuran

The pure distilled tetrahydrofuran sold commercially is a colorless, clear, volatile, polar liquid with a characteristic acetone-like odor and a minimum purity of 99.9 wt %. THF is miscible in all proportions with water, alcohols, ethers, and many common solvents. For this inventory the functional unit is 1 kg of pure tetrahydrofuran. The most important chemical and physical properties of THF used in this inventory are given here:

Tab. 38.1 Chemical and physical properties of tetrahydrofuran

Property	Value	Unit	Remarks
Molecular weight	72.1	$g\ mol^{-1}$	
Specific gravity	888	$kg\ m^{-3}$	at 20 °C
Boiling point	66	°C	at normale pressure
Melting point	-108.5	°C	at normale pressure
Heat of combustion	-35.141	$MJ\ kg^{-1}$	

¹ Source: Ullmann 2001

² Source: Simetric 2004

38.2 Reserves and Resources

Tetrahydrofuran is mostly produced by cyclization of 1,4-butanediol by elimination of water. THF production from furfural with pentosans as starting materials has attracted interest. The production of THF from furfural involves catalytic decarbonylation to furan and hydrogenation of the latter to THF.

The projected annual growth rate for tetrahydrofuran production is 4 – 7 % based on anticipated increases in the consumption of polytetrahydrofuran. Major producers of THF are listed in Tab. 38.2, and regional production capacity and consumption estimates are shown in Tab. 38.3.

Tab. 38.2 Major THF producers in 1992 (Ullmanns 2001)

Company	Region	Process	Capacity, $10^3\ t/a$
Du Pont	USA	Reppe	70

BASF	USA, Europe, Japan	Reppe	40
ARCO	USA	Propylene oxide	8
GAF	USA, Europe	Reppe	18
Quaker Oats	USA	Furfural	10
Mitsubishi-Kasei	Japan	Butadiene	16
Toso Corp.	Japan	Reppe	3
Shinwa	Korea	Butane	10
	CIS/China	Furfural/Reppe	10

Tab. 38.3 Worldwide THF production capacity and consumption in 1995 data by geographical area (103 t/a) (Ullmanns 2001)

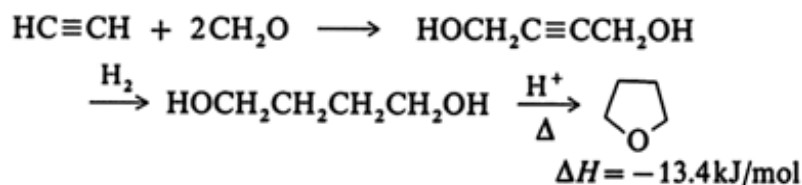
	USA	Western Europe	Southeast Asia	China, CIS	Total
Capacity	133	50	40	5-10	228-233
Consumption	80	70	30	5-10	185-190

38.3 Production Technologies and Use

38.3.1 Production Technologies

Acetylene/Formaldehyde (Ullmann 2001)

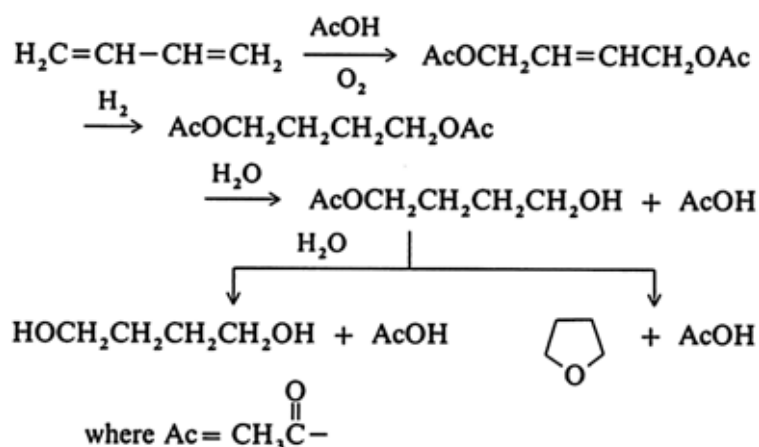
A process developed by Reppe in the 1930s was for many years the preferred synthetic route to 1,4-butanediol and THF, and it is still the most common approach in Europe and the United States. The Reppe process involves a reaction between acetylene and formaldehyde to give 2-butyne-1,4-diol, with subsequent hydrogenation to 1,4-butanediol. The saturated diol is very readily cyclized to THF with elimination of water by acid catalysis above 100 °C. Suitable catalysts include inorganic acids, acidic aluminum silicates, and earth or rare-earth oxides.



Quantitative conversion and a yield of almost 100 % can be achieved in an atmospheric-pressure, continuous process with aluminum oxide as the catalyst provided the THF is continuously distilled out of the reaction mixture as it forms and pure 1,4-butanediol is fed into the reactor at a rate equal to its rate of consumption. The amount of conversion per unit amount of catalyst is very high. Crude butanediol can be converted into THF very selectively and with little added energy by a medium-pressure process described by BASF. The THF – steam mixture obtained from cyclization is first distilled in a rectification column to give the corresponding azeotrope (5.3 wt % water, bp 62.3 °C). Treatment with alkali hydroxide and subsequent distillation provides anhydrous THF. The azeotrope problem can also be circumvented industrially by distillation under pressure.

Butadiene Acetoxylation (Ullmann 2001)

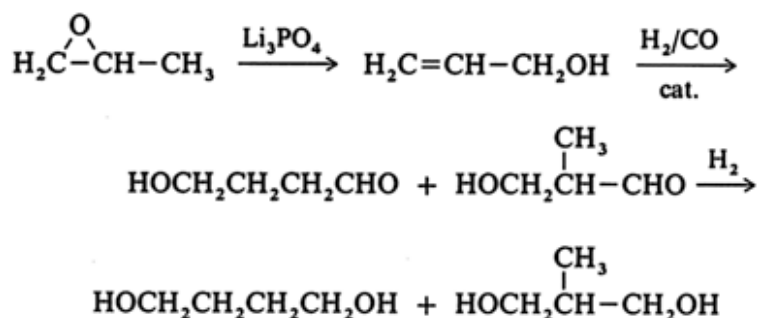
The Mitsubishi-Kasei Corporation in Japan produces 1,4-butanediol and THF in parallel starting from butadiene. The procedure can be described by the following set of equations:



Butadiene is oxidized at 3 MPa/80 °C over a palladium – tellurium catalyst with acetic acid and a nitrogen–oxygen mixture to give 1,4-diacetoxy-2-butene. The olefin is hydrogenated to 1,4-diacetoxybutane, which can be hydrolyzed to butanediol or THF.

Propylene Oxide Process (Ullmann 2001)

Arco produces 1,4-butanediol by the following route:



Propylene oxide is isomerized to allyl alcohol by conventional means at 250 – 300 °C/1 MPa over a trilithium orthophosphate (Li_3PO_4) catalyst. The allyl alcohol is then hydroformylated to 4-hydroxybutyraldehyde and the byproduct 3-hydroxy-2-methylpropionaldehyde by a process developed by Kuraray. Subsequent hydrogenation of the aldehydes gives 1,4-butanediol as the main product together with 2-methyl-1,3-propanediol.

38.3.2 Use

The largest proportion of the 1.4×10^5 t of THF produced worldwide in 1992 was used as a monomer in the manufacture of poly(tetramethylene oxide) (PTMO) also known as poly(tetramethylene ether glycol) (PTMEG) and polytetrahydrofuran (PTHF), important in the production of thermoplastic polyurethanes, elastic fibers, molded elastomers, and copolyesters or copolyamides. A smaller proportion finds use as a solvent for coatings, adhesives, and special varnishes based, for example, on poly(vinyl chloride) (PVC) or polyurethanes; as an extracting agent; and as the preferred medium for or-

ganometallic syntheses.

38.4 System Characterisation

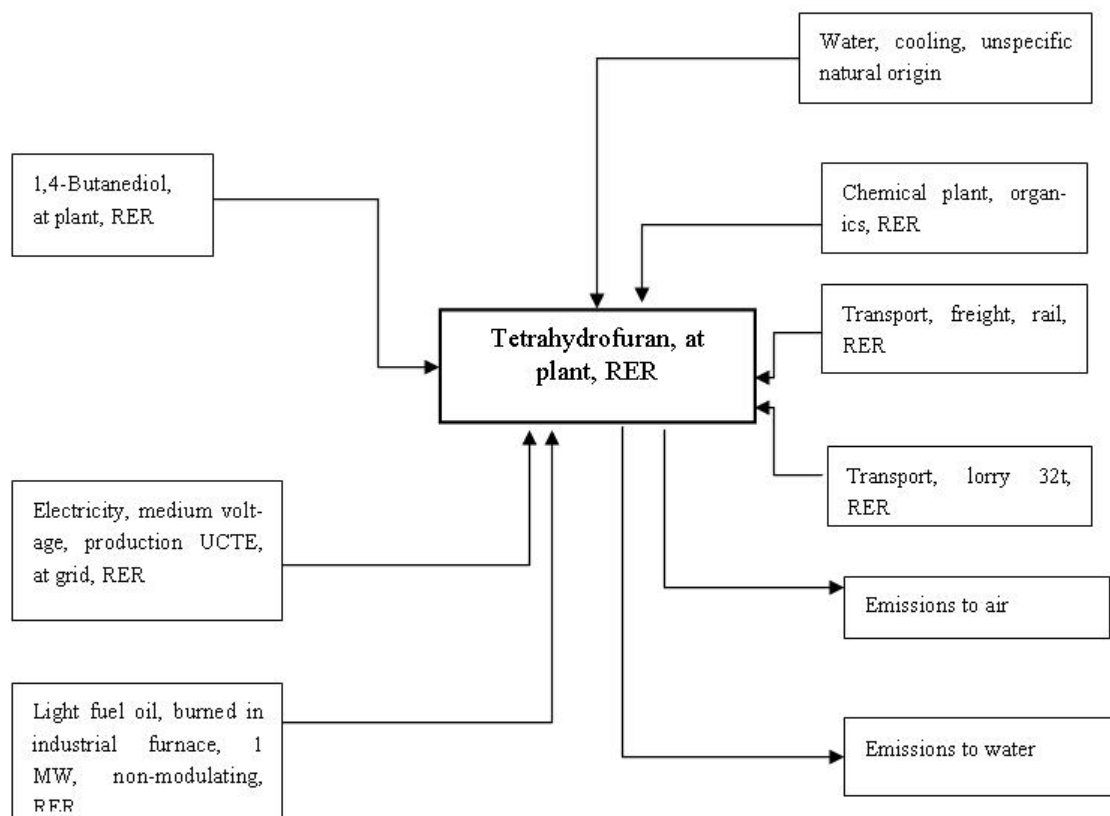


Fig. 38.2 Process chain for the production of tetrahydrofuran

This report corresponds to the dataset in the ecoinvent database for the production of **1-kg tetrahydrofuran, at plant, GLO**. For this study the production of tetrahydrofuran by means of the cyclization of 1,4-butanediol from Reppe synthesis is considered. All data in the present report are referred to 1 kg tetrahydrofuran 100%. The system includes the process with consumption of raw materials, energy, infrastructure and land use as well as the generation of emissions to air and water. It also includes transportation of the raw materials. For the study transient or unstable operations like starting-up or shutting-down, are not included, but the production during stable operation conditions. Storage and transportation of the final product are also not included. It is assumed that the manufacturing plants are located in an urban/industrial area and consequently the emissions are categorised as emanating in a high population density area. The emissions into water are assumed to be emitted into rivers.

38.5 Data sources

The data source used are theoretical knowledge of the chemical process and West Bengal 2004. The authors describe calculated costs of THF plants in West Bengal, India.

38.6 Raw materials and auxiliaries

Ullmann 2001 reports a yield of almost 100 % for the production of Tetrahydrofuran. Nevertheless, in this inventory a yield of 99 % is assumed.

Tab. 38.4 Consumption of raw materials and auxiliaries for the production of 1 kg THF

Input	Value
1,4-Butanediol (kg) ¹	1.263
Cooling water (kg) ²	24

1 Source: stoichiometric calculation with a assumed yield of 99 %.

2 Source: West Bengal 1999

38.7 Energy

West Bengal 2004 reports an energy use of 0.825 l light fuel oil and 0.1125 kWh electricity per kg THF. In this study the use of the light fuel oil is modelled with the dataset "light fuel oil, burned in industrial furnace 1 MW, non-modulating" with a energetic efficiency of 74 % (according to Althaus 2004).

Tab. 38.5 Energy consumption for the production of 1 kg THF (West Bengal 2004)

Input	Value
Light Diesel Oil (kg) ¹	0.693 ¹
Electricity (kWh) ¹	0.1125 ¹

38.8 Transportation

No information is available in the sources consulted concerning transportation of raw materials. Therefore, for the raw materials the following standard distances as defined in Frischknecht et al. (2007) are used: 100 km by lorry 32t and 600 km by train.

Tab. 38.6 summarises the total transport amounts for the production of 1 kg tetrahydrofuran.

Tab. 38.6 Total transport amounts for the production of tetrahydrofuran

tkm kg-1 THF	lorry	train
Raw materials	1.26E-01	7.58E-02

38.9 Infrastructure and land use

No information was readily available about infrastructure and land-use of THF plants. Therefore, in this study, the infrastructure is estimated based on the dataset "chemical plant, organics". This dataset assumes a built area of about 4.2 ha, an average output of 50'000 t/a, and plant life of fifty years. For this study, the estimated value is 4.00 E-10 units per kg of produced chemical.

38.10 Emissions to air

It is assumed that 100% of the electricity consumed is converted to waste heat and that 100% of the waste heat is released to the air.

There was no data available on process emissions to air for the production of THF. As approximation

the air emissions occurring from the purge vent, the distillation vent and fugitive emission sources were estimated to 0.2% of the raw material input. The assumed values for the emissions to air are given in Tab. 38.7.

Tab. 38.7 Process emissions to air from the production of THF

Output	Value	Remarks
1,4-Butanediol, to air (kg)	2.53E-03	Estimated as 0.2 % of input
Carbon dioxide, fossil, to air (kg)	1.78E-02	from waste water treatment
Waste heat (MJ)	4.05E-01	calculated from electricity input

38.11 Emissions to water

The remaining amount of unreacted 1,4-butanediol (0.8 % of the input) was assumed to leave the production process with the waste water. Further it was assumed that this waste water is treated in a internal waste water plant. The removal efficiency for 1,4-butanediol was assumed with 90%. The carbon contained in the removed 1,4-butanediol was accounted as CO₂ emissions to air.

The values for COD, BOD, TOC and DOC used in this inventory were calculated from the amount of 1,4-butanediol in the treated waste water assuming a carbon conversion of 96% for COD. For the calculation of the values for BOD and DOC the worst case scenario BOD = COD and TOC = DOC was used. A summary of the values used in this inventory is given in Tab. 38.8.

Tab. 38.8 Process emissions to water from the production of THF

Output	Value	Remarks
1,4-Butanediol, to water (kg)	1.01E-03	calculated from mass balance
COD, BOD (kg)	1.9E-03	calculated from water emissions
TOC, DOC (kg)	5.4E-04	calculated from water emissions

38.12 Life cycle inventory and data quality considerations

Tab. 38.9 shows the life cycle inventory and the data quality indicators for the inventory of the production of tetrahydrofuran. The simplified approach with a pedigree matrix has been used for calculating the standard deviation.

Tab. 38.9 Unit process raw data for the production of tetrahydrofuran

	InputGroup	OutputGroup	Name	Location	InfrastructureProcess	Unit	tetrahydrofuran, at plant	Uncertainty Type	StandardDeviation95%	GeneralComment
401										
662			Location				RER			
493			InfrastructureProcess				0			
403			Unit				kg			
resource, in water technosphere	-	0	tetrahydrofuran, at plant	RER	0	kg	1.00E+0			
	4	-	Water, cooling, unspecified natural origin	-	-	m3	2.40E-2	1	1.38	(4,5,1,1,1,5); West Bengal 2004
	5	-	butane-1,4-diol, at plant	RER	0	kg	1.26E+0	1	1.38	(4,5,1,1,1,5); Stoichiometric calculation
	5	-	transport, lorry 32t	RER	0	tkm	1.26E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	transport, freight, rail	RER	0	tkm	7.58E-1	1	2.09	(4,5,na,na,na,na); Standard distances
	5	-	electricity, medium voltage, production UCTE, at grid	UCTE	0	kWh	1.13E-1	1	1.38	(4,5,1,1,1,5); West Bengal 2004
	5	-	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	6.93E-1	1	1.38	(4,5,1,1,1,5); West Bengal 2004
	5	-	chemical plant, organics	RER	1	unit	4.00E-10	1	3.09	(4,5,na,na,na,na); Estimation
emission air, high population density	-	4	Heat, waste	-	-	MJ	4.05E-1	1	1.38	(4,5,1,1,1,5); Calculated from the electricity input
	-	4	1,4-Butanediol	-	-	kg	2.53E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	Carbon dioxide, fossil	-	-	kg	1.78E-2	1	1.30	(4,5,na,na,na,na); Calculated from the water treatment
emission water, river	-	4	1,4-Butanediol	-	-	kg	1.01E-3	1	1.30	(4,5,na,na,na,na); Estimation
	-	4	BOD5, Biological Oxygen Demand	-	-	kg	1.90E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	COD, Chemical Oxygen Demand	-	-	kg	1.90E-3	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	DOC, Dissolved Organic Carbon	-	-	kg	5.40E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions
	-	4	TOC, Total Organic Carbon	-	-	kg	5.40E-4	1	1.62	(4,5,na,na,na,na); Calculated from the water emissions

38.13 Cumulative Results and Interpretation

Selected LCI results and values for the cumulative energy demand are presented and discussed in this chapter. Please note that only a small part of the about 1'000 elementary flows is presented here. The selection of the elementary flows shown in the tables is not based on their environmental relevance. It rather allows to show by examples the contributions of the different life cycle phases, or specific inputs from the technosphere to the selected elementary flows. Please refer to the ecoinvent database for the complete LCIs.

The shown selection is not suited for a life cycle assessment of the analysed processes and products. Please use the data downloaded from the database for your own calculations, also because of possible minor deviations between the presented results and the database due to corrections and changes in background data used as inputs in the dataset of interest.

The ecoinvent database also contains life cycle impact assessment results. Assumptions and interpretations were necessary to match current LCIA methods with the ecoinvent inventory results. They are described in Frischknecht 2004. It is strongly advised to read the respective chapters of the implementation report before applying LCIA results.

Tab. 38.10 shows selected LCI results and the cumulative energy demand for the production of tetrahydrofuran.

Tab. 38.10 Selected LCI results and the cumulative energy demand for the production of tetrahydrofuran

Name		tetrahydrofuran, at plant	
Location	Unit	Unit	RER
Infrastructure			kg 0
LCIA results			
cumulative energy demand	non-renewable energy resources, fossil	MJ-Eq	112.4
cumulative energy demand	non-renewable energy resources, nuclear	MJ-Eq	13.6
cumulative energy demand	non-renewable energy resources, primary forest	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, biomass	MJ-Eq	1.5
cumulative energy demand	renewable energy resources, geothermal, converted	MJ-Eq	-
cumulative energy demand	renewable energy resources, solar, converted	MJ-Eq	0.0
cumulative energy demand	renewable energy resources, potential (in barrage water), converted	MJ-Eq	2.1
cumulative energy demand	renewable energy resources, kinetic (in wind), converted	MJ-Eq	0.2
selected LCI results			
land occupation	resource	m2a	3.6E-1
CO ₂ , fossil	air	kg	5.5E+0
NM VOC	air	kg	1.2E-2
nitrogen oxides	air	kg	9.0E-3
sulphur dioxide	air	kg	1.2E-2
particulates, <2.5 µm	air	kg	7.8E-4
BOD	water	kg	7.6E-3
cadmium	soil	kg	1.8E-9

38.14 Appendix: EcoSpold Meta Information

ReferenceFunction	401	Name	tetrahydrofuran, at plant
Geography	662	Location	RER
ReferenceFunction	493	InfrastructureProcess	0
ReferenceFunction	403	Unit	kg
DataSetInformation	201	Type	1
	202	Version	1.0
	203	energyValues	0
	205	LanguageCode	en
	206	LocalLanguageCode	de
DataEntryBy	302	Person	24
	304	QualityNetwork	1
ReferenceFunction	400	DataSetRelatesToProduct	1
	402	IncludedProcesses	Production of tetrahydrofuran including materials, energy uses, infrastructure and emissions.
	404	Amount	1
	490	LocalName	Tetrahydrofuran, ab Werk
	491	Synonyms	tetramethylene oxide, oxolane
	492	GeneralComment	The process "tetrahydrofuran, at plant, RER" is modelled for the production of tetrahydrofuran from 1,4-butanediol in Europe. Raw materials are modelled with a stoichiometric calculation. Energy consumptions are modelled with literature data. The emissions are estimated. Infrastructure and transports are calculated with standard values.
	494	InfrastructureIncluded	1
	495	Category	chemicals
	496	SubCategory	organics
	497	LocalCategory	Chemikalien
	498	LocalSubCategory	Organisch
	499	Formula	C4H8O
	501	StatisticalClassification	
	502	CASNumber	109-99-9
TimePeriod	601	StartDate	1999
	602	EndDate	2006
	603	DataValidForEntirePeriod	1
	611	OtherPeriodText	Time of publications
Geography	663	Text	The inventory is modelled for Europe.
Technology	692	Text	Cyclation of 1,4-butanediol
Representativeness	722	Percent	100
	724	ProductionVolume	Total production in the world is around 230,000 tons.
	725	SamplingProcedure	Literature data
	726	Extrapolations	Transports and emissions are modelled with standard methods.
	727	UncertaintyAdjustments	none
DataGeneratorAnd	751	Person	24
	756	DataPublishedIn	2
	757	ReferenceToPublishedSource	31
	758	Copyright	1
	759	AccessRestrictedTo	0
	760	CompanyCode	
	761	CountryCode	
	762	PageNumbers	solvents

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